TABLES & HIGH-YIELD

Geography

Teaching tables, Confused Pairs, Common Traps, Current Affairs, and other high-yield blocks.

UPPCS + UKPCS Study Library · chapter order follows Subject Notes.

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Topic 1 — Indian Physical Geography: Mountains & Hills

Current Affairs (this topic)

Year	Fact	Why asked	Source
Mar 2024	**Sela Tunnel** (BRO) — all-weather Tezpur–**Tawang** (Arunachal); longest twin-lane tunnel **above ~13,000 ft**	Border tunnel / pass CA	PIB / inauguration
2025	**Z-Morh / Sonamarg Tunnel** — all-weather Srinagar–Ladakh (लद्दाख) axis	Pair with Zoji La (ज़ोजिला)	PIB
2020 / PYQ 2025	Atal Tunnel qualifier only: longest highway tunnel **above 10,000 ft**, not unqualified world longest		PIB / BRO

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q84

The Shiwalik range is primarily composed of which type of material?

A. Igneous rocks B. Consolidated rocks C. Unconsolidated sediments D. Metamorphic rocks

Q. UKPCS Prelims 2025, Q90

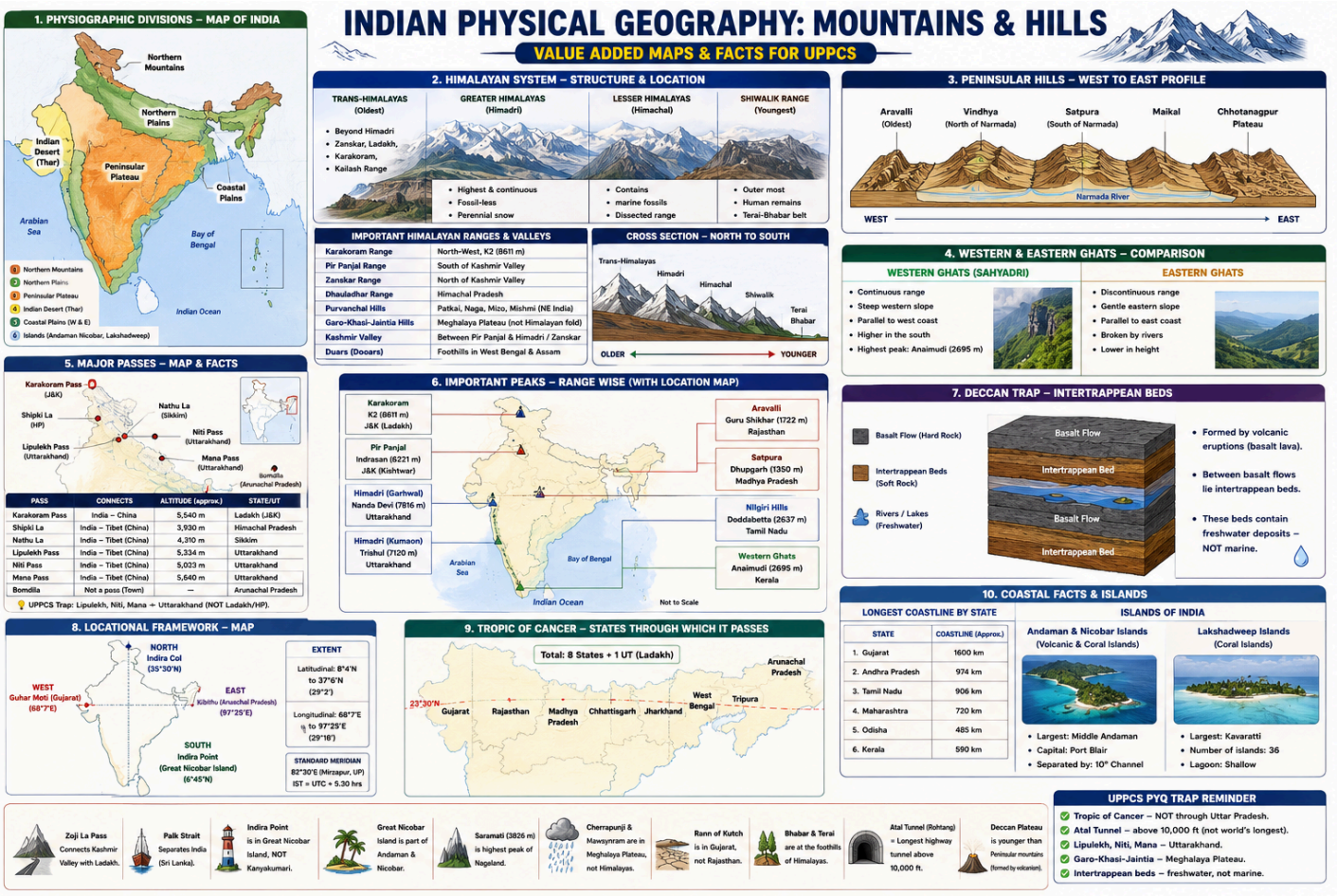
Which is the largest glacier of the Trans-Himalayas?

A. Biafo B. Siachen (सियाचिन) C. Baltoro D. Hispar

Confused Pairs

A	B	Correct	Hindi
Himadri	Himachal	Fossil-less crystalline vs marine fossils	हिमाद्रि / हिमाचल
Himachal	Shiwalik	Marine fossils vs human remains; outer youngest	हिमाचल / शिवालिक
Bhangar	Khadar	Older kankar upland vs newer flood-renewed, more fertile	भांगर / खादर
Bhabar	Terai	Streams sink (pebbles) vs streams re-emerge (marsh/forest)	भाबर / तराई
Western Ghats	Eastern Ghats	Continuous high wall vs discontinuous eroded hills	पश्चिमी घाट / पूर्वी घाट
Vindhya	Satpura	North of Narmada vs south of Narmada	विंध्य / सतपुड़ा
Andaman–Nicobar	Lakshadweep	Volcanic (Bay of Bengal) vs coral atolls (Arabian Sea)	अंडमान / लक्षद्वीप
Deccan Plateau	Deccan Trap	Southern tableland vs basalt lava cover on it	दक्कन पठार / दक्कन ट्रैप
Purvanchal Hills	UP Purvanchal	NE hill system vs eastern UP region	पूर्वांचल पहाड़ियाँ / पूर्वांचल (UP)
Indira Point	Kanyakumari	Southernmost **territory** vs southernmost **mainland**	इंदिरा पॉइंट / कन्याकुमारी
K2	Kanchenjunga	Karakoram (PoK) vs highest peak **fully in India** (Sikkim)	के2 / कंचनजंगा
Dun	Karewa	Shiwalik longitudinal valley vs Kashmir lacustrine terrace	दून / करेवा
Marusthali	Bagar	Sandy desert core vs semi-arid eastern fringe	मरुस्थली / बागड़
10° Channel	9° Channel	Andaman–Nicobar vs Minicoy–rest of Lakshadweep	दस डिग्री / नौ डिग्री
Coast of emergence	Coast of submergence	Coromandel / Kerala raised vs Konkan drowned	उद्गमन / निमज्जन तट
Marwar Plateau	Marwar Plain	**East** of Aravalli vs **west** (Thar)	मारवाड़ पठार / मैदान

A	B	Correct	Hindi
Malnad	Maidan	Karnataka forested hill tract vs rolling granite plain	मलनाड / मैदान
Kangra Valley	Kulu Valley	Strike / longitudinal vs transverse valley (HP)	कांगड़ा / कुल्लू
Standard Meridian	Tropic of Cancer	**82°30′ E** N–S time line (Mirzapur; 5 states) vs **23°30′ N** E–W line (8 states; **not UP**)	मानक याम्योत्तर / कर्क रेखा
Indira Col	Indira Point	Northern extreme (Siachen / Ladakh) vs southernmost **territory** (Great Nicobar)	इंदिरा कोल / इंदिरा पॉइंट



1.0 Physiographic framework

Structural unit	What it is	Note
Peninsular Block	Oldest stable crust; gneiss, granite, schist	Faulted and eroded, **not** recently folded like the Himalaya
Himalayas and extra-peninsular mountains	Young fold belt from India–Eurasia collision	Still rising in zones; includes Karakoram and Purvanchal
Indo–Ganga–Brahmaputra (ब्रह्मपुत्र) Plain	Foredeep filled with Himalayan alluvium	One of the world's largest alluvial (जलोढ़) tracts

1.1 Physiographic Divisions of India

Division	Where	Age	One-line identity
Northern Mountains	Northern frontier	Tertiary (young)	Himalaya + Karakoram + Purvanchal

Division	Where	Age	One-line identity
Northern Plains	Between Himalaya and plateau	Recent	Indus–Ganga–Brahmaputra alluvium
Peninsular Plateau	South of Narmada–Son	Precambrian	Oldest stable block
Indian Desert	Western Rajasthan	Pleistocene–recent	Thar / Marusthali (मरुस्थली), leeward of Aravalli
Coastal Plains	East and west margins	Recent	West **narrow**; East **broad and deltaic**
Islands	Arabian Sea and Bay of Bengal	Recent	Lakshadweep **coral**; A&N **volcanic**

Northern Plains

Sub-region	Location	Fact
Punjab Plain	Punjab, Haryana	Doabs; **chos** gullying on Shiwalik edge
Haryana Tract	Ghaggar–Yamuna	Water-divide; **Ghaggar** (घग्गर) as Saraswati (सरस्वती) successor
Ganga Plain	Delhi to Kolkata	Rohilkhand, Awadh (अवध), Mithila, Magadh
Brahmaputra Plain	Assam	Braided channel with **bils** (ox-bows) and heavy floods

Doab	Between
Bist	Beas (ब्यास) and Satluj
Bari	Beas and Ravi (रावी)
Rachna	Ravi and Chenab (चिनाब)
Chaj	Chenab and Jhelum (झेलम)
Sind (सिंध) Sagar	Jhelum–Chenab and the Indus

Coastal Plains

Western strip	Stretch
Kachchh–Kathiawar	Gujarat
Gujarat Plain	Narmada–Tapi–Mahi (माही)–Sabarmati (साबरमती) fill
Konkan	Daman to Goa
Kanara (Canara)	Goa to Mangaluru
Malabar	Kerala
Eastern strip	Stretch
---	---
Utkal Coast	Odisha coast with the Mahanadi (महानदी) delta and **Chilika**
Northern Circars	Mahanadi to Krishna (कृष्णा)
Carnatic (कर्नाटक)	Krishna to Kaveri (कावेरी)
Coromandel / Payan Ghat	False Divi Point to Kanyakumari

1.2 Himalayan Mountain System

North to south (parallel ranges)

Range	Typical height	Rock / fossil fact
Trans-Himalaya	3000–6500 m	Granitic cold desert **north of the Indus**
Himadri (Greater)	generally **>6000 m**	Crystalline, **fossil-less**, with permanent snow

Range	Typical height	Rock / fossil fact
Himachal (हिमाचल) (Lesser)	3700–4500 m	Sedimentary; **marine fossils**
Shiwalik (शिवालिक) (Outer)	900–1100 m	Youngest unconsolidated rocks; **human remains**

Region	Local name of Shiwalik
Jammu	Jammu Hills
Uttarakhand	Dhang / Dundwa
Nepal	Churia Ghat Hills
Arunachal	Dafla, Miri (मीरी), Abor, Mishmi (outer hills)

West to east (regional sections)

Section (खंड)	River Boundaries (नदी सीमाएं)	Length (लंबाई)	Key Ranges & Peaks (प्रमुख श्रेणियां व चोटियां)
Punjab / Kashmir / Himachal Himalaya	**Indus (सिंधु) to Satluj (सतलज)**	~560 km	Karakoram, Ladakh, Zanskar, Pir Panjal, Dhauladhar; Nanga Parbat
Kumaon Himalaya (कुमाऊँ)	**Satluj (सतलज) to Kali (काली / शारदा)**	~320 km	Nanda Devi, Kamet, Badrinath, Kedarnath, Trishul
Nepal / Central Himalaya (नेपाल)	**Kali (काली) to Teesta (तीस्ता)**	~800 km *(Longest & Highest)*	Mt. Everest, Kanchenjunga, Dhaulagiri, Annapurna, Makalu
Assam / Eastern Himalaya (असम)	**Teesta (तीस्ता) to Dihang (दिहांग / ब्रह्मपुत्र)**	~720 km	Namcha Barwa, Kula Kangri, Kangto

Himalayan peaks (match home)

Peak	m	Note
K2 (Godwin Austin)	8611	**Karakoram** (PoK) — not the main Himalayan arc
Kanchenjunga	8586	Highest peak **fully in India** (Sikkim, Himadri)
Nanda Devi	7816	Uttarakhand; highest entirely in Uttarakhand
Kamet (कामेत)	7756	Uttarakhand
Cho Oyu	8188	Himalaya (Nepal–Tibet) —
Lhotse	8516	Himalaya (Nepal–Tibet) —
Nanga Parbat	8126	Western syntaxial bend (Gilgit–Baltistan / PoK)
Namcha Barwa	7782	**Tibet / China** — **not** in India
Gurla Mandhata	7694	**Tibet**, north of Mansarovar — **not** an Indian peak

1.3 Peninsular Mountains, Hills & Plateaus

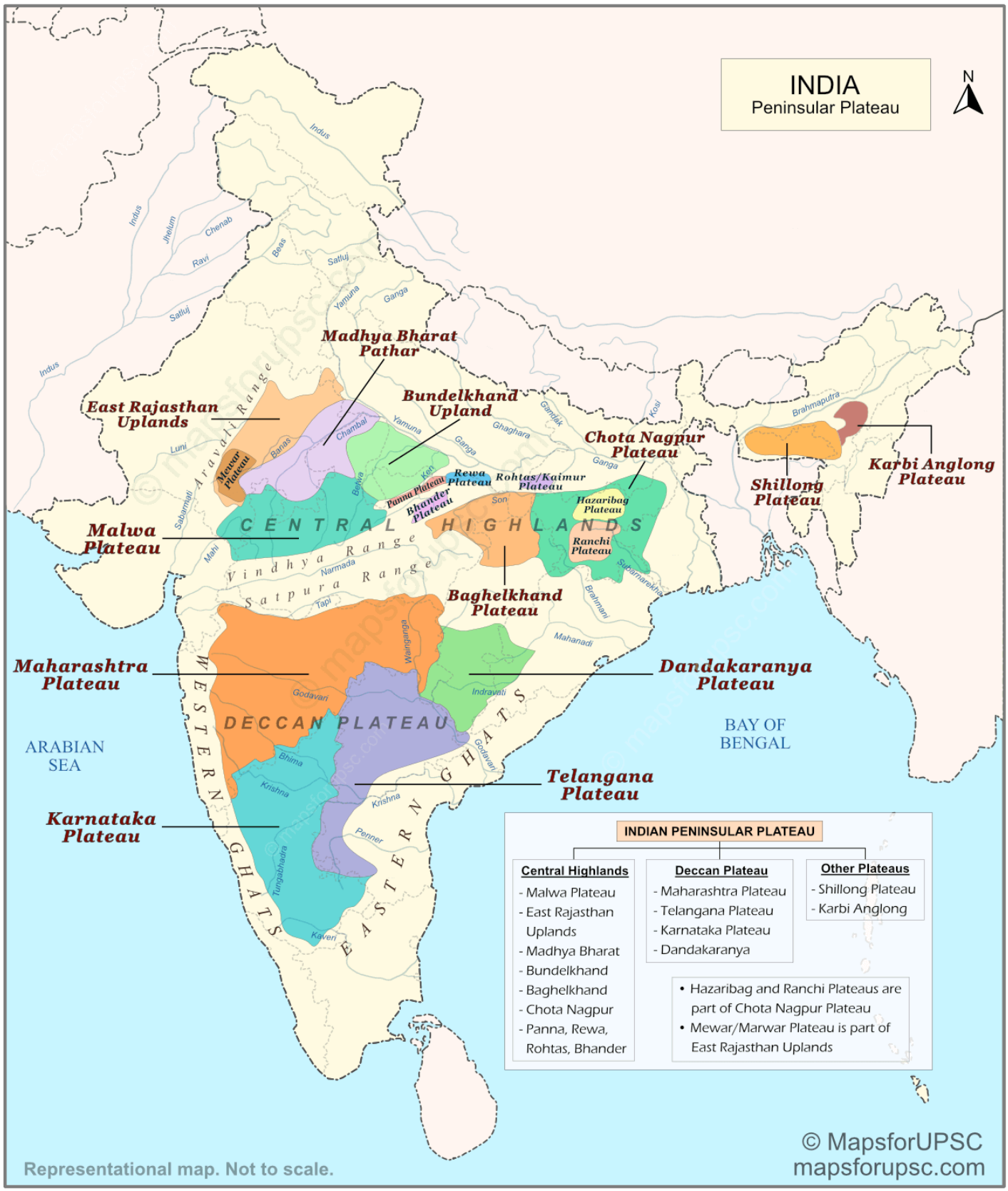
Range	Trend	Location	Highest	Fact
Aravalli	SW–NE	Gujarat–Rajasthan–Delhi	**Guru Shikhar 1722 m**	**Oldest** fold mountains; Rajasthan highest
Vindhya	E–W	**North of Narmada**, MP	Sadbhawna Shikhar	Aryavarta / Dakshinapatha (दक्षिणापथ) divide
Satpura	E–W	**South of Narmada**	**Dhupgarh 1350 m** (Pachmarhi)	Madhya Pradesh highest
Western Ghats (पश्चिमी घाट)	N–S	Tapi gap to Kanyakumari	**Anaimudi 2695 m**	**Continuous**; Sahyadri in Maharashtra

Range	Trend	Location	Highest	Fact
Eastern Ghats	N–S	Odisha to Tamil Nadu	**Jindhagada ~1690 m**	**Discontinuous**
Nilgiri (नीलगिरि)	—	Tamil Nadu–Kerala	**Doddabetta 2637 m**	Junction of both Ghats
Rajmahal	—	Jharkhand	—	Rajmahal Trap (volcanic)
Shevaroy	—	Salem, Tamil Nadu	—	Eastern Ghats; **not** Tirupati

Deccan Plateau and Deccan Trap

Trap layer	Depth fact
Upper	about **450 m**
Middle	about **1200 m**
Lower	about **150 m**
Intertrappean beds	Sediment **between** lava flows; **land / freshwater** fossils — **not** sea plants and animals

Central Highlands and north-eastern plateaus



- The Central Highlands are the **northern** Peninsular Plateau, **north of the Narmada–Son**.
- **Marwar / Mewar (मेवाड़) Plateau** is **east** of the Aravalli (sandstone–shale–limestone, Banas toward Chambal). **Marwar Plain** is the Thar **west** of the Aravalli. Do not swap them.
- **Madhya Bharat Pathar** (Central Highland) is the Chambal basin with ravines / badlands.
- The **Malwa Plateau** is lava-covered and **north-sloping**. It has **dual drainage**: Narmada–Tapi–Mahi to the Arabian Sea, and Chambal–Betwa–Ken to the Yamuna / Bay of Bengal.

- The **Bundelkhand Plateau** lies in south-west Uttar Pradesh and Madhya Pradesh. It is **granite–gneiss**, drought-prone, and includes Jhansi (झांसी), Banda, and Chitrakoot (चित्रकूट). Betwa, Dhasan, and Ken cross it.
- The **Baghelkhand Plateau** lies **east** of Bundelkhand, **north of Maikal**. The Son bounds it on the north. It divides Son drainage from Mahanadi drainage.
- The **Chotanagpur Plateau** lies mainly in Jharkhand (with Odisha–West Bengal edges). Average height is about **700 m**. **Damodar** (दामोदर) flows in a **rift** through Gondwana coalfields.
- **Hazaribagh Plateau** lies north of Damodar. **Ranchi Plateau** lies south of Damodar.
- The **Rajmahal Hills** are the north-eastern basalt edge of Chotanagpur.
- The **Meghalaya Plateau** is Garo–Khasi–Jaintia. The **Malda / Garo–Rajmahal Gap** is a down-fault filled by Ganga–Brahmaputra alluvium.
- **Mikir (Karbi) Hills** are listed with this north-eastern block on the relief map.
- **Jharia** and **Bokaro** are the Damodar-rift coalfields. Iron and mica are the other Chotanagpur matchs.
- Chotanagpur is the easternmost unit of the west-to-east central-India hill sequence.

1.4 Passes, Peaks & Sacred Geography

Mountain passes

Pass	State / UT	Route / note	Trap
Burzil	J&K / Ladakh	Greater Himalaya (pass table)	—
Thaga La	Uttarakhand	Greater Himalaya with Niti / Lipulekh	—
Zoji La (ज़ोजिला)	Ladakh / J&K	Srinagar–Kargil–Leh (लेह)	Pair with Z-Morh tunnel
Banihal	J&K	Jammu–Srinagar; Jawahar Tunnel	Not a Tibet pass
Rohtang La	Himachal Pradesh	Kullu–Lahaul; Atal Tunnel below	Not Karakoram
Bara Lacha La	Himachal Pradesh	Manali–Leh	—
Kunzum La	Himachal Pradesh	Lahaul–Spiti	—
Shipki La	Himachal Pradesh	India–Tibet; Sutlej (सतलुज) valley	—
Niti Pass	**Uttarakhand**	Garhwal (गढ़वाल)–Tibet	2023 Q57 correct pair
Mana Pass	**Uttarakhand**	Near Badrinath (बद्रीनाथ)	**Not HP**
Lipulekh (लिपुलेख)	**Uttarakhand**	Kailash–Mansarovar; India–Nepal–China	**Not Ladakh**
Nathu La	Sikkim	India–Tibet trade (पण्याध्यक्ष) (reopened 2006)	—
Jelep La	Sikkim	Alternate to Nathu La	—
Bomdi La	Arunachal Pradesh	Near **Bomdila town**	Bomdila ≠ pass name
Sela Pass	Arunachal Pradesh	Tezpur–Tawang; **Sela Tunnel** (2024)	—
Bum La	Arunachal Pradesh	Tawang–Tibet	—
Pangsau Pass	Arunachal Pradesh	India–Myanmar (Stilwell Road)	—

Pass	State / UT	Route / note	Trap
Diphu / Aghil	Karakoram	Not Arunachal / not "Ladakh town" traps	2023 wrong options
Khardung La	Ladakh	Leh–Nubra road	High motorable pass; not a Tibet trade pass of the Nathu type

State-wise highest peaks (match-list home)

State / UT	Peak	m	Range
Sikkim	Kanchenjunga	8586	Himadri
Uttarakhand	Nanda Devi	7816	Himadri
Himachal Pradesh	Reo Purgyil	6816	Kinnaur Himalaya
Arunachal Pradesh	Kangto	7090	Eastern Himalaya
J&K / Ladakh	Nun / Kun	7135 / 7077	Zanskar
West Bengal	Sandakphu	3636	Singalila
Nagaland	**Saramati**	3841	Naga Hills
Manipur	Tenipu / Mt Iso	2994	Manipur Hills
Mizoram	Phawngpui	2165	Mizo Hills
Meghalaya (मेघालय)	Shillong Peak	1965	Khasi
Kerala	**Anaimudi** (अनाइमुदी)	2695	Western Ghats
Tamil Nadu	**Doddabetta**	2637	Nilgiri
Karnataka	Mullayanagiri	1930	Western Ghats
Maharashtra	Kalsubai	1646	Western Ghats
Rajasthan (थार)	**Guru Shikhar**	1722	Aravalli
Madhya Pradesh	**Dhupgarh**	1350	Satpura
Andhra Pradesh	Arma Konda	~1680	Eastern Ghats
Odisha (ओडिशा)	Deomali	1672	Eastern Ghats
Gujarat	Girnar	1117	Kathiawar
Jharkhand	Parasnath	1366	Chotanagpur
Uttar Pradesh	Amsot (Sonbhadra)	~941	Kaimur
Chhattisgarh	Gudla Maina	~1270	Maikal extension

Sacred geography

Site	State	Hill / fact
Tirupati (Venkateswara)	Andhra Pradesh	**Tirumala / Mallamalla**, Eastern Ghats — **not Shevaroy**
Kedarnath (केदारनाथ)	Uttarakhand	Garhwal Himalaya; Mandakini (मंदाकिनी)
Badrinath	Uttarakhand	Garhwal Himalaya; Alaknanda (अलकनंदा)
Gangotri	Uttarakhand	Bhagirathi source glacier
Yamunotri	Uttarakhand	Yamuna source glacier
Amarnath	J&K	Lidder Valley (Greater Himalaya, not Pir Panjal)
Vaishno Devi	J&K	**Trikuta Hills**, Jammu
Sabarimala	Kerala	Western Ghats
Palitana (पालिताना)	Gujarat	Shatrunjaya Hills

Site	State	Hill / fact
Parasnath	Jharkhand	Chotanagpur
Girnar	Gujarat	Junagadh (जूनागढ़)
Ajmer Sharif	Rajasthan	Aravalli foothills
Chamundi	Karnataka	Mysuru
Konark (कोणार्क)	Odisha	**Coastal plain** (not a hill shrine)

1.5 Locational Framework

Latitude, longitude, area

Item	Note
Latitude	**8°4′ N – 37°6′ N** (about 29°; about 3214 km N–S)
Longitude	**68°7′ E – 97°25′ E** (about 29°; about 2933 km E–W)
Area	about **3.28 million km²** = **2.4%** of world land
Rank	Conventionally **7th** largest (after Russia, Canada, USA, China, Brazil, Australia)

Standard Meridian

Step	Easy fact
Earth	360° in 24 h means roughly **15°** of longitude per hour, or **1° ≈ 4 minutes**
Offset maths	82.5 ÷ 15 = **5.5 hours** = **5 h 30 min** ahead of GMT
Why this line	It sits near the **middle** of India's longitude span (~68° E to ~97° E), so west (Gujarat) and east (Arunachal / Andaman) share one national clock

State	Place / belt fact	trap
Uttar Pradesh	**Mirzapur** (near Prayagraj belt)	Not Lucknow / Delhi; Tropic does **not** hit UP
Madhya Pradesh	Eastern MP / **Singrauli** belt	Not the whole of MP
Chhattisgarh	Central–east belt (**Bilaspur–Korba** class)	Not Jharkhand
Odisha	**Western** Odisha	Not the Bhubaneswar coast as the "IST city"
Andhra Pradesh	Northern / coastal AP strip	**Not Telangana** (TG lies west of 82.5° E)

Tropic of Cancer (23°30′ N)

Order (west → east)	State on the Tropic
1	**Gujarat**
2	**Rajasthan**
3	**Madhya Pradesh**
4	**Chhattisgarh**
5	**Jharkhand**
6	**West Bengal**
7	**Tripura**
8	**Mizoram**

Extreme points

Extreme	Place	Easy fact
North	**Indira Col**, Siachen / Ladakh	High pass on the northern frontier — not a city
South (territory)	**Indira Point**, Great Nicobar	Southernmost **India** (islands included)
South (mainland)	**Kanyakumari** (Cape Comorin)	Southernmost **peninsula** only
East	**Kibithu**, Anjaw, Arunachal Pradesh	Near the Myanmar–China corner; not Walong as the usual key
West	**Guhar Moti** (also Ghuar Mota), Kutch, Gujarat	Westernmost village belt of Kachchh

Coastline

Rank (longest → shortest)	Coastal state
1	**Gujarat**
2	Andhra Pradesh
3	Tamil Nadu
4	Maharashtra
5	Kerala
6	Odisha
7	Karnataka
8	West Bengal
9	**Goa** (shortest state coast)

Common Traps — Don't Fall For These

1. **Lipulekh is in Ladakh** — FALSE. **Uttarakhand** (India–Nepal–China trijunction).
2. **Mana Pass is in Himachal** — FALSE. **Niti and Mana** are both **Uttarakhand**.
3. **Bomdila is a pass** — FALSE. Bomdila is a **town**; the pass is **Bomdi La** (Arunachal).
4. **Tirupati is on Shevaroy Hills** — FALSE. **Tirumala / Mallamalla (Eastern Ghats)**.
5. **Kanyakumari is India's southernmost point** — FALSE. **Indira Point (Great Nicobar)** = southernmost territory.
6. **Aravalli is the youngest range** — FALSE. **Himadri** is youngest among usual options; Aravalli is **oldest**.
7. **India = 6th largest / wholly tropical** — FALSE. **7th**; **2.4%**; Tropic through the middle; **not** wholly tropical.
8. **Intertrappean beds have sea fossils** — FALSE. **Freshwater/land** fossils.
9. **K2 is in the main Himalaya** — FALSE. **Karakoram**.
10. **Kashmir Valley = Kangra + Dhauladhar** — FALSE. **Pir Panjal + Himadri**.
11. **Annamalai / Sirumalai are Himalayan** — FALSE. **Tamil Nadu peninsular**.
12. **Atal Tunnel is the world's longest highway tunnel (unqualified)** — FALSE. **Pir Panjal**; longest **above 10,000 ft**.
13. **Both Ghats are continuous** — FALSE. **Western = continuous; Eastern = discontinuous**.
14. **Lakshadweep is volcanic like Andaman** — FALSE. Lakshadweep = **coral**; A&N = **volcanic**.
15. **Vindhya lies south of Narmada** — FALSE. **Vindhya = north; Satpura = south**.
16. **Tropic of Cancer crosses UP / Ladakh** — FALSE. **Eight states only** (GJ–RJ–MP–CG–JH–WB–Tripura–Mizoram).
17. **Standard Meridian = Prayagraj / Lucknow / "only eastern states use IST"** — FALSE. Place = **Mirzapur**; path = **UP–MP–CG–Odisha–AP**; IST = **one zone for all India**. Uttarakhand / Jharkhand / Telangana are **not** on the line.
18. **Karewa = dun** — FALSE. Karewa = Kashmir **lacustrine** terrace (saffron). Dun = **Shiwalik** longitudinal valley.
19. **Ten Degree Channel separates Minicoy** — FALSE. **10°** = Andaman–Nicobar. **9°** = Minicoy–rest of Lakshadweep. **8°** = Minicoy–Maldives.

20. **Palghat is the Mumbai–Pune gap** — FALSE. Palghat / Palakkad = **Kerala–Tamil Nadu**. Bhor Ghat = Mumbai–Pune.
21. **Gurla Mandhata is an Indian Trans-Himalayan peak** — FALSE. It is in **Tibet**. Namcha Barwa is also **not** in India.
22. **Whole west coast is submerged** — FALSE. **Konkan** is drowned; **Kerala / Malabar** is emerged.
23. **Marwar Plateau is west of Aravalli** — FALSE. Plateau = **east**; Thar plain = **west**.

Topic 2 — Climate of India

Current Affairs (this topic)

Year	Fact	Why asked	Source
2026	IMD SW monsoon (दक्षिण-पश्चिम) **below normal (~90% of LPA)**	Seasonal rainfall category	IMD LRF Apr/May 2026
2026	ENSO: **neutral** → **El Niño** during Jun–Sep 2026	Weak-monsoon tendency	IMD / MMCFS
2026	**IOD neutral** (no +IOD (धनात्मक) offset of El Niño)	Contrast with 2019 +IOD rescue	IMD Jul 2026
2026	Onset over Kerala **4 June** (slightly after normal 1 June)	Onset date trap	USDA / IMD advance
Ongoing	**Mission Mausam** (MoES/IMD) — improved monsoon & severe-weather forecast	Scheme/institution CA	IMD–MHA

El Niño **tendency** ≠ guaranteed all-India drought (सूखा) — still the 2026 -relevant pairing.

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q96

According to Koppen's climatic classification, the (Amw) climate is found predominantly in which part of India?

A. North-Western India B. Western coastal region C. Thar (थार) desert D. Northern plain

Q. UKPCS Prelims 2025, Q98

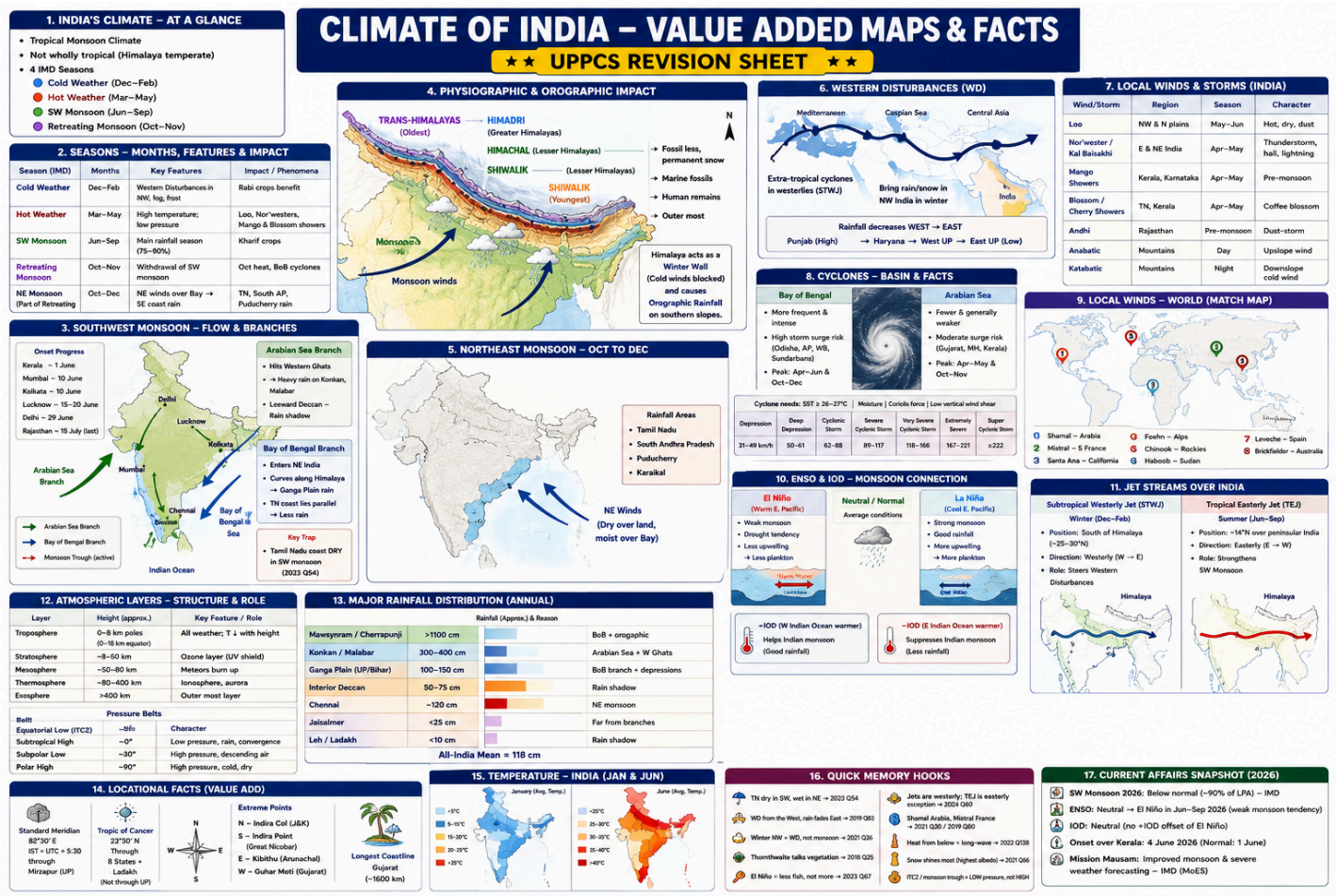
Which local wind affects some parts of Kerala and Tamil Nadu (नाडु) during the pre-monsoon season?

A. Loo B. Mango Shower C. Kal Baisakhi (बैसाखी) D. Mistral

Confused Pairs

A	B	Correct	Hindi
SW monsoon	NE monsoon	Jun–Sep most of India vs Oct–Dec **SE coast only**	दक्षिण-पश्चिम / उत्तर-पूर्व मानसून
Retreating monsoon	Western Disturbance	Withdrawal/Oct heat vs winter extra-tropical NW rain	लौटता मानसून / पश्चिमी विक्षोभ
Loo	Nor'wester	Sustained hot dry wind vs violent pre-monsoon thunderstorm	लू / काल बैसाखी
Anabatic	Katabatic	Day upslope vs night downslope	आरोही / अवरोही पवन
STWJ	TEJ	Winter westerly (WD) vs summer easterly (SW monsoon)	पश्चिमी जेट / उष्णकटिबंधीय पूर्वी जेट
El Niño	La Niña	Warm E Pacific, weak monsoon vs cool E Pacific, strong monsoon	अल नीनो / ला नीना
+IOD	−IOD	Helps Indian monsoon vs suppresses it	धनात्मक / ऋणात्मक IOD
Köppen	Thornthwaite	T+P letter codes vs moisture/vegetation index	कोपेन / थॉर्नथ्वेट
Absolute humidity	Relative humidity	Mass of vapour vs % of saturation (RH falls as T rises)	निरपेक्ष / सापेक्ष आर्द्रता
Active monsoon	Break monsoon	Trough on **Ganga plain** (plains wet) vs trough on **Himalaya** (central India dry)	सक्रिय / विराम मानसून

A	B	Correct	Hindi
Doldrums	Horse latitudes	Rainy ITCZ calm belt vs dry subtropical **high** near 30°	डोलड्रम / हॉर्स अक्षांश
Am (Köppen)	As (Köppen)	Short dry spell / wet SW coast–NE vs **dry summer** Coromandel (NE rain)	Am / As
Willy-willy	Brickfielder	Australian **cyclone** vs Australian **hot local wind**	विली-विली / ब्रिकफील्डर
Bora	Chinook	Adriatic **cold** vs Rockies **warm dry**	बोरा / चिनूक
Harmattan	Khamsin	West Africa dust vs Egypt heat	हरमट्टन / खामसिन



2.1 Climate Basics

Factor	What it does in India
Latitude	Most of India is tropical / subtropical; the far north is cooler
Himalaya	Winter **wall** against Central Asian cold; summer **orographic** rain on southern slopes
Distance from sea	Coasts stay moderate (नरम दल); NW plains and interior Deccan have **continental** extremes
Altitude	Temperature falls about **6.5 °C per 1000 m** (lapse rate)
Relief / orography	Western Ghats → windward wet, leeward Deccan dry; Meghalaya (मेघालय) scarp → extreme rain
Pressure & ITCZ	Summer thermal **low** + north-shifted ITCZ pull the south-west monsoon
Jet streams	Winter STWJ (पश्चिमी जेट) steers Western Disturbances; summer TEJ (उष्णकटिबंधीय पूर्वी जेट) supports SW monsoon
ENSO / IOD	Year-to-year monsoon strength (El Niño weakens; +IOD helps)

Temperature and altitude

Station	January (approx.)	Control
Drass / Ladakh (लद्दाख)	−20 to −40 °C	Altitude plus continentality
Agra (आगरा)	about **16 °C**	Ganga plain —
Darjeeling (about 2000 m)	about **4 °C**	Same latitude as Agra; **lapse rate**
Mumbai / Chennai	24–26 °C	Maritime
Jaisalmer	12–15 °C in January; 40–45 °C in June	Highest **diurnal** range among usual stations

Climate contrast	Main control
Chennai warmer than Kolkata (annual)	**Latitude** (Chennai farther south)
Himalayan snowfall / hill stations cooler	**Altitude**
Rainfall falls West Bengal → Punjab	**Distance from the Bay / moisture loss westward**
Sutlej (सतलुज)–Ganga plain winter rain	**Western Disturbances**

Heat budget, albedo, humidity

Surface	Albedo	Fact
Fresh snow	80–90%	**Highest** among usual surface options
Desert sand	30–40%	Moderate
Prairie (प्रेयरी)	20–25%	—
Paddy / cropland	10–20%	Lower
Forest / water (typical)	under 10–20%	Absorb more; warm more

Pair / Station	Latitude / Location	What differs	Primary Control
Agra vs Darjeeling	Both ~27°N	January: Agra **16 °C** vs Darjeeling **4 °C**	**Altitude & Lapse Rate** (~1 °C per 165 m; Darjeeling ~2,042 m vs Agra ~169 m)
Amritsar vs Shimla	Both ~31°–32°N	Amritsar hot summer / mild winter vs Shimla snow / cool	**Altitude** (Amritsar ~230 m / 756 ft vs Shimla ~2,400 m / 7,866 ft)
Ajmer vs Shillong	Both ~26°–27°N	Ajmer ~587 mm/yr vs Shillong ~5,160 mm/yr	**Distance from Sea & Orographic Relief** (Khasi Hills funnel vs Thar margin)
Chennai Station Profile	13°N (Coromandel)	Mean annual temp **26 °C**, annual rain **63 cm**, annual range **9 °C**	**Tropical Maritime & NE Monsoon dependence**

Traditional Indian Seasons (Hindu Calendar Ritus)

Sr.	Season (*Ritu*)	Hindu Months	Gregorian Months	Key Characteristics
1	**Vasanta** (वसंत - Spring)	Chaitra – Vaishakha	March – April	Mild pleasant weather, blooming flowers, new foliage
2	**Grishma** (ग्रीष्म - Summer)	Jyeshtha – Ashadha	May – June	Intense heat, dry westerly winds (Loo), localized dust storms
3	**Varsha** (वर्षा - Rainy)	Shravana – Bhadrapada	July – August	Peak South-West monsoon, widespread torrential rains, high humidity
4	**Sharad** (शरद - Autumn)	Ashvina – Kartika	September – October	Retreating monsoon, clear skies, October heat, transition to cool
5	**Hemant** (हेमंत - Pre-winter)	Margashirsha – Pausha	November – December	Pleasant cool weather, Coromandel coast rainfall (NE monsoon)

Sr.	Season (*Ritu*)	Hindu Months	Gregorian Months	Key Characteristics
6	**Shishir** (शिशिर - Winter)	Magha – Phalguna	January – February	Coldest months, frost/fog in north, Western Disturbance rains

Temperature inversion

Type	When / where	What happens
Radiation inversion	Clear, calm **winter nights** on the Indo-Gangetic plain	Ground cools fast; fog / smog trap over Delhi, UP, Bihar
Valley / drainage inversion	Kashmir, Doon, hill valleys	Cold air drains to the **floor**; slopes stay milder
Subsidence inversion	Under high-pressure / descending air	Mixing stops; haze can persist

Pressure, wind, Coriolis

Hemisphere	Deflection
Northern (India)	to the **right**
Southern	to the **left**

Spell	Trough position	Rain
Active monsoon (सक्रिय)	Over the **Ganga plain**	Plains wet (UP, Bihar, MP)
Break monsoon (विराम मानसून)	Over the **Himalaya**	Central India dry; foothills wet

Atmosphere, pressure belts, ITCZ

Layer	Height	Fact
Troposphere (क्षोभमंडल)	0–8 km at poles / 0–18 km at the equator	**All weather**; temperature falls with height
Tropopause	Top of the troposphere	Jets fly near here
Stratosphere	to about 50 km	**Ozone UV shield**
Mesosphere (मध्यमंडल)	about 50–80 km	Meteors burn
Thermosphere	about 80–400 km	Ionosphere and aurora
Exosphere	outermost	Space

Step	Process	Surface result
1	Strong heating at the equator	Air rises
2	Rising air reduces surface weight	Equatorial **low** / ITCZ
3	Air aloft moves poleward and sinks near 30°	Subtropical **high** (horse latitudes)
4	Surface air returns from 30° to the equator	**Trade winds**
5	Other surface air from 30° moves toward 60°	**Westerlies**
6	Cold air sinks at the poles	Polar **high**
7	Polar air moves toward 60° and meets westerlies	Subpolar **low**; **polar easterlies**
Cell	Span	Role
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Hadley cell	Equator to ~30°	Rise at ITCZ, sink at 30°, trades return
Ferrel cell	~30° to 60°	Mid-latitude belt of the westerlies
Polar cell	~60° to pole	Polar easterlies; rise near 60°

Step	Process	Surface result
Nickname	Meaning	
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Doldrums	Rainy, weak-surface-wind belt of the ITCZ	
Horse latitudes	Dry, calm subtropical highs near 30°	
Belt	Latitude	Pressure
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Equatorial low / **ITCZ**	Near 0° (shifts with the Sun)	**Low**
Subtropical high	About **30° N / S**	**High**
Subpolar low	About **60° N / S**	**Low**
Polar high	About **90°**	**High**

Season	ITCZ position	Effect on India
Summer (Jun–Sep)	Shifts **north** over South Asia	Forms / guides the monsoon trough; supports SW monsoon
Winter (Dec–Feb)	Shifts **south**	Most of India under dry NE trades
Strong El Niño years	Indian branch often weaker	Monsoon tends to weaken

Wind	Direction	Where	Cause	Effect
Trade winds	East → west (NE trades in NH; SE trades in SH)	Between ~30° highs and the ITCZ	Air leaves subtropical high toward equatorial low; Coriolis turns it	Feed ITCZ rain; after equator crossing they become India's SW monsoon
Westerlies	West → east	About **30°–60°** (both hemispheres)	Air leaves 30° high toward 60° low; Coriolis turns it	Drive mid-latitude storms and help steer Western Disturbances
Polar easterlies	East → west	Polar high toward ~60°	Cold air leaves polar high; Coriolis turns it	Bring cold outbreaks; meet westerlies at the subpolar low

2.2 Monsoon System

System	Months	Where the rain falls	Trap
South-west monsoon	Jun–Sep	Most of India (~75–90% of annual rain)	TN SE coast stays **dry**
Retreating monsoon	Sep–Nov	Withdrawal phase with **October heat** and BoB / Andaman Sea cyclones	Not WD winter rain
North-east monsoon	Oct–Dec	**TN, south AP, SE Karnataka fringe, Puducherry, Karaikal**	Not all-India winter rain
Western Disturbances	Nov–Feb (peak)	NW India rain/snow	Extra-tropical; **not** a monsoon

Monsoon mechanism

Concept	Fact
Thermal	Land heats faster than sea → summer low over India pulls ocean air in
Dynamic (Flohn)	Pressure belts and the ITCZ shift with the Sun; monsoon is largely that seasonal shift
Step	What happens
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Concept	Fact
1	Tibetan Plateau / NW India heat strongly in late spring
2	ITCZ shifts **north** onto India
3	**Mascarene High** strengthens over the southern Indian Ocean
4	SE trades cross the equator roughly between **40° E and 60° E**; Coriolis turns them SW
5	Cross-equatorial flow strengthens as the **Somali Jet**
6	STWJ jumps north of Tibet; **TEJ** sets up near 14° N
7	Onset **burst** at Kerala (~1 June), then the front marches north

Southwest monsoon

Place	Normal onset	Withdrawal (start)
Kerala	about 1 June	Extreme south last (December for NE rains)
Mumbai / Kolkata	about 10 June	Early October
Lucknow (लखनऊ) / eastern UP	about 15–20 June	Late September
Delhi	about **29 June**	Late September
Rajasthan	about **15 July** (last)	September
Chennai	South-west is **weak and dry**	North-east monsoon October–December

Branch	Path	Wet places	Dry / weak places
Arabian Sea branch	Arabian Sea → Western Ghats → inland (also a stream toward central India / Ganga that later meets the Bay branch)	Konkan, Malabar, Goa (गोवा); some rain into Gujarat / western MP	Leeward Deccan and TN rain shadow; **Chhattisgarh basin** is little fed by this branch
Bay of Bengal branch	BoB → NE India → curves west along Himalaya	Assam, Meghalaya, Ganga plain, UP, Bihar	TN coast (sits **parallel** — little orographic lift)

Pattern	Fact
Season	Mostly **June–September** (SW monsoon)
Relief	Orography governs amounts (Western Ghats, Meghalaya scarp)
Distance from sea	Rain generally **declines inland** / with distance from the moisture source
Northern plains	Annual rain **decreases east → west** (humid Bay air loses moisture westward)
Duration	Monsoon season length **decreases south → north** (Kerala longer than the far north-west)
Break	Rainless spells when the trough sits on the **Himalaya**
Rajasthan dryness	Aravalli runs **roughly parallel** to SW winds → little forced uplift on the western Thar

Northeast monsoon and retreating monsoon

Feature	Retreating SW monsoon	North-east monsoon
Months	About **Sep–Nov**	About **Oct–Dec**
Process	SW monsoon withdraws from north to south	Dry NE winds pick up Bay moisture and rain on the SE coast
Signature	October heat, clear spells, and Bay / Andaman Sea cyclones	Coromandel rain
Rain belt	Withdrawal rains + east-coast cyclone rains	TN, south AP, SE Karnataka, SE Kerala fringe, Puducherry, Karaikal

Feature	Retreating SW monsoon	North-east monsoon
North India	Mostly dry	Mostly dry

Indian seasons (IMD)

Season	Months	Main weather
Cold weather	December–February	Western Disturbance rain/snow in the north-west, fog on the plains, and the rabi season
Hot weather	March–May	Loo heat waves, Nor'westers, and mango / blossom showers
South-west monsoon	June–September	Main rainy season for most of India
Retreating	October–November	Withdrawal, October heat, Bay cyclones, then NE monsoon on the SE coast

Western Disturbances

Item	Note
Type	Extra-tropical cyclone
Carrier jet	STWJ branch **south of the Himalaya** (not the northern Tibetan branch)
Peak season	November–February
Rain / snow belt	Punjab, Haryana, Himachal, J&K / Ladakh fringe, Uttarakhand, western UP, northern Rajasthan
Gradient	Decreases **west → east**
Farm link	Helps rabi wheat; also hail, fog, avalanche risk

2.3 Winds, Storms and Cyclones

Local winds and storms of India

Time	Land vs sea	Wind
Day	Land heats faster → low over land	**Sea breeze** (sea → land)
Night	Land cools faster → high over land	**Land breeze** (land → sea)
Wind	When	Direction
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Anabatic	Day	Upslope
Katabatic	Night	Downslope
Foehn / Chinook	After crossing a mountain	Warm and dry downslope

Name	Region	Season	Character
Loo	RJ, PB, HR, UP, Bihar plains	May–June	Hot, dry dust and heat-wave; **not** a thunderstorm
Nor'wester / Kal Baisakhi	WB, Assam, Odisha, Bihar, Jharkhand	April–May	Violent **thunderstorm**; hail, lightning
Mango showers	**Kerala, Karnataka**	April–May	Pre-monsoon; mango ripening
Blossom / cherry showers	TN, Kerala (coffee belt)	April–May	Coffee flowering
Andhi	Rajasthan	Pre-monsoon	Dust-storm wall
Elephanta	Malabar / Kerala	About September	End of SW monsoon; rough sea

Name	Region	Season	Character
Sea / land breeze	Coasts	Daily	Onshore by day; offshore by night
Anabatic	Mountain slopes	**Day**	**Upslope**
Katabatic	Valleys	**Night**	**Downslope** cold drainage

World local winds (full match map)

Family	Wind	Region	trap
Warm dry downslope	**Foehn**	Alps	Not Australia
Warm dry downslope	**Chinook**	Rockies	Not a cold wind
Warm dry downslope	**Zonda**	Andes, Argentina	Not Alps
Warm dry downslope	**Santa Ana**	California	Not Spain
Warm dry downslope	**Berg**	South Africa	—
Cold	**Mistral**	S France, Rhône	**Not Australia**
Cold	**Bora**	Adriatic	Not Chinook
Cold	**Pampero**	Argentina	—
Cold	**Buran / Purga**	Siberia	—
Hot dusty	**Sirocco**	Sahara → Mediterranean	—
Hot dusty	**Khamsin**	Egypt	Not Australia
Hot dusty	**Shamal**	Arabia / Gulf	**Not Austria**
Hot dusty	**Harmattan**	**West** Africa	Not East Africa
Hot dusty	**Samun**	Kurdistan	—
Hot dusty	**Leveche**	Spain	—
Hot dusty	**Ghibli**	Libya	—
Hot dusty	**Brickfielder**	Australia	Not Mistral
Hot dusty	**Haboob**	Sudan	—
Hot dusty	**Black roller**	North America	—
Other	**Yamo**	Japan	—
Other	**Levanter**	Gibraltar	—
Not a local wind	**Willy-willy**	Australia	Cyclone, not Brickfielder

Tropical cyclones

Need	Note
Sea surface temperature	About **26–27 °C or more**
Moisture	Deep moist air
Coriolis	Enough spin — usually **5°–30°** latitude
Vertical wind shear	**Low** (strong SW-monsoon shear suppresses many Bay storms in Jul–Sep)
Part	Meaning
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Eye	Calm centre with sinking air and clearer skies
Eyewall	Ring of strongest winds and rain around the eye
Spiral bands	Outer rain bands

Basin	Frequency / strength	Why	Hit states
Bay of Bengal	**More and stronger**	Warm shallow shelf + river freshwater; storm surge (तूफानी महोर्मि) risk high	Odisha, AP, West Bengal, Sundarbans (सुंदरबन)
Arabian Sea	Fewer, often weaker	Higher salinity; narrower fetch	Gujarat, Maharashtra, Kerala still take hits
IMD class	Wind (km/h)		
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Depression	31–49		
Deep depression	50–61		
Cyclonic storm	62–88		
Severe cyclonic storm	89–117		
Very severe	118–166		
Extremely severe	167–221		
Super cyclone	**222 or more**		
Local storm	Scale	Fact	
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Thunderstorm / Nor'wester	Hours; local convection	**Not** a cyclone	
Tropical cyclone	Days; ocean-scale	Synoptic storm	

Cyclone naming (match bank — next year can swap any pair)

Local / basin name	Where	Note
Hurricane ★	**USA** / Atlantic–Caribbean / E Pacific	2019–2020 PYQ fact
Typhoon ★	NW Pacific — **China** / Philippines / Taiwan class	2020 Q74 "Typhoons–China"
Taifu ★	**Japan**-facing name for NW Pacific storm	Same basin as typhoon; 2019 paired with Japan
Baguio / Baguios ★	**Philippines**	2019–2020 fact
Willy-willies ★	**Australia**	Cyclone — **not** Brickfielder (hot local wind)
Cyclone	North Indian Ocean — **India / BoB / Arabian Sea**	IMD / RSMC New Delhi names
Cordonazo	West Mexico / E Pacific coast	Lucent-class distractor
Hurricane (also)	Caribbean / Gulf of Mexico	Same Atlantic family as USA

2.4 Classification and ocean–atmosphere phenomena

Köppen (India) versus Thornthwaite

Thinker	Method	Note
Köppen	Temperature + precipitation **letter codes**	India's Am / Aw / As / BWh / Cwg map
Thornthwaite	Moisture balance (P minus potential ET)	**"Vegetation is the true index of climate"** — 2018 Q25
Trewartha	Modified Köppen	**Not** Thornthwaite

Letter clue	Meaning	India example
A	Tropical	Kerala, Deccan, Coromandel

Letter clue	Meaning	India example
B	Dry (arid / semi-arid)	Thar, rain-shadow Deccan fringe
C	Temperate / subtropical with distinct seasons	Ganga plain
m	Monsoon (short dry spell)	Kerala / Konkan / NE hills
w	Dry **winter**	Interior Deccan (**Aw**); Ganga plain (**Cwg**); also the second small letter in strings like **Amw**
s	Dry **summer**	**Tamil Nadu As** — dry in SW monsoon; rain in NE monsoon
h	Hot	Thar **BWh**

Köppen Code	Climatic Type	Representative Regions in India
Amw	Monsoon with short dry season	Western coastal plains south of Goa (Malabar/Konkan), parts of Arunachal/Assam
As	Monsoon with dry summer	**Coromandel Coast** (coastal Tamil Nadu and adjoining Andhra Pradesh)
Aw	Tropical Savanna (wet & dry)	Most of the Peninsular plateau south of the Tropic of Cancer
BWhw	Tropical desert (arid)	Western Rajasthan (Thar Desert)
BShw	Semi-arid steppe	Rain-shadow belt east of Western Ghats; parts of Rajasthan, Haryana, Punjab
Cwg	**Humid subtropical with dry winter**	**Great Plains of India (Indo-Gangetic plain), North-East India (including North Bengal and Assam plains), and North Bihar**
Dfc	Cold humid winter with short summer	High-altitude Sikkim and Arunachal Pradesh Himalayas
E / ET	Polar / Tundra	High mountain belts of Jammu & Kashmir, Ladakh, Himachal Pradesh, Uttarakhand

Jet streams

Jet	Season	Direction	Where	Role for India
Polar front jet	Year-round (shifts)	Westerly	High latitudes	Extra-tropical weather belt
STWJ	Winter	Westerly	About 25–30° N, south of Himalaya	Steers Western Disturbances
STWJ north branch	Winter	Westerly	North of Tibet after the plateau splits the jet	Does not drive India's winter rain
STWJ summer position	Summer	Westerly	North of Tibet	Clears space for monsoon setup
TEJ	Summer	Easterly	Near 14° N over the peninsula	Supports SW monsoon circulation
Somali Jet	Summer	Cross-equatorial / SW	Off East Africa → Arabian Sea	Feeds SW monsoon moisture

El Niño, La Niña, and IOD

Name	Ocean	India monsoon	Note
El Niño	Pacific (warm Peru)	Often weaker	Plankton **decrease**
La Niña	Pacific (cool Peru)	Often stronger	Opposite of El Niño
+IOD	Indian Ocean (west warmer)	Helps	Can offset El Niño
−IOD	Indian Ocean (east warmer)	Hurts	Can worsen drought

Common Traps — Don’t Fall For These

1. **TN gets heavy SW monsoon rain** — FALSE. Dry in SW; **NE monsoon** Oct–Dec — 2023 Q54.
2. **Winter NW rain = retreating monsoon** — FALSE. **Western Disturbances** — 2021 Q36.
3. **WD rain increases eastward** — FALSE. Decreases **west → east** — 2019 Q83.
4. **“Vegetation is the true index of climate” = Köppen** — FALSE. **Thornthwaite** — 2018 Q25.
5. **El Niño increases plankton off Peru** — FALSE. **Reduces** upwelling/plankton — 2023 Q67.
6. **Jet streams are easterly** — FALSE as a general rule. Mid-latitude jets = **westerly**; **TEJ** is the summer exception — 2024 Q60.
7. **Shamal = Austria / Mistral = Australia** — FALSE. Shamal = Arabia; Mistral = France.
8. **Atmosphere heated mainly by short-wave sun** — FALSE. **Long-wave terrestrial** — 2022 Q138.
9. **+IOD is bad for Indian monsoon** — FALSE. **+IOD helps**; −IOD hurts.
10. **Ozone UV layer = troposphere** — FALSE. **Stratosphere** — 2018 Q52 / 2023 Q52.
11. **NE monsoon rains all India** — FALSE. **TN, south AP, Puducherry** only.
12. **Mango showers in Punjab** — FALSE. **Kerala/Karnataka**.
13. **Highest albedo = desert sand** — FALSE. **Fresh snow** — 2021 Q66.
14. **ITCZ / monsoon trough = high pressure** — FALSE. Both are **lows**.
15. **Nor’westers = Western Disturbances** — FALSE. Nor’westers = **pre-monsoon thunderstorms**; WD = **winter**.
16. **Harmattan = East African coast** — FALSE. **West Africa** / Gulf of Guinea.
17. **Willy-willy = Brickfielder** — FALSE. Willy-willy = Australian **cyclone**. Brickfielder = Australian **hot local wind**.
18. **Bora is a warm Rockies wind** — FALSE. Bora = **cold Adriatic**. Chinook = warm Rockies.
19. **Active monsoon = trough on Himalaya** — FALSE. Active = trough on **Ganga plain**; break = trough on **Himalaya**.
20. **Köppen As = Kerala / Konkan** — FALSE. **As** = Tamil Nadu dry **summer**; Kerala/Konkan = **Am**.
21. **Retreating monsoon = NE monsoon** — FALSE. Retreating = SW withdrawal + Oct heat + BoB cyclones; NE monsoon = SE-coast rain Oct–Dec.
22. **Monsoon from Hindi / Spanish** — FALSE. Arabic **mausim** (season).
23. **IOD = west Indian Ocean vs eastern Pacific** — FALSE. West vs **east Indian Ocean** (UPSC 2017).
24. **Northern jet branch steers India’s winter rain** — FALSE. Tibet splits the jet; **southern** branch (STWJ south of Himalaya) steers WD.
25. **India lies exactly between the Tropics** — FALSE. Extends to ~**37°6′ N** (subtropical north).
26. **IMD rainy day = any drizzle** — FALSE. **≥ 2.5 mm** in 24 hours.
27. **January tropical/subtropical divide = 21 °C / 15 °C** — FALSE. **18 °C** isotherm.

Topic 3 — Drainage System

Current Affairs (this topic)

Year	Fact	Why asked	Source
Dec 2021 / Dec 2024	Ken–Betwa (बेतवा) Link: Cabinet ₹44,605 cr; PM foundation **25 Dec 2024**	First ILR under implementation	PIB
2025–26	**Only** priority NPP link in implementation (30 identified: 16 peninsular + 14 Himalayan); NWDA nodal	"How many links operational?" trap	Rajya Sabha (राज्यसभा) / Jal Shakti (शक्ति) Aug 2026
2026	Spend ~₹12,382 cr to Jun 2026; target ~Mar 2030; Bundelkhand MP–UP; **Panna TR** forest	Cost / ecology (पारिस्थितिकी) pairing	LS Q Aug 2026
Ongoing	Polavaram includes Godavari–Krishna (Vijayawada) link assistance	Peninsular grid	PIB
2016	National Waterways Act — 106 NWs; core still NW-1 to NW-5	Number vs names	Act

Namami Gange (नमामि गंगे) and GAP are Ganga cleaning programmes. They are not river-origin facts.

🗺️ Visual Cheat Sheet — Complete Indian Drainage System

INDIAN PHYSICAL GEOGRAPHY – DRAINAGE SYSTEM

COMPLETE SOURCE OF TRUTH – UPPCS/UPSC PRELIMS & MAINS

1 QUICK REVISION – RAATA THIS FIRST

India Drainage – 4 Major Himalayan Systems + Peninsular Rivers

- Indus – 3180 km (mostly in Pakistan)
- Ganga – 2525 km; basin ~8.6 lakh km² (largest in India)
- Brahmaputra – 2900 km (Yarlung Tsangpo in Tibet)
- Peninsular Rivers – old, graded, rain-fed

Key Contrasts

HIMALAYAN RIVERS	PENINSULAR RIVERS
Youth, vigorous	Mature, graded
Perennial	Mostly seasonal
V-shaped valleys	Shallow valleys
Meanders, floodplains	Fixed course
Large deltas	Estuaries (west flowing)

IMD Seasons (Hydrology view)

- Cold Weather: Dec – Feb
- Hot Weather: Mar – May
- SW Monsoon / Rainy: Jun – Sep
- Retreating: Oct – Nov
- NE Monsoon (SE coast only): Oct – Dec

2 MAJOR DRAINAGE SYSTEMS OF INDIA

BASIN AREA & LENGTH (High Yield)

River System	Basin Area (approx.)	Length (approx.)
Ganga	~8.6 lakh km² (Largest in India)	2525 km
Godavari	~3.1 lakh km² (Largest Peninsular)	1465 km (Longest Peninsular)
Brahmaputra	~5.8 lakh km²	2900 km
Indus	~3.2 lakh km² (in India part)	3180 km
Krishna	~2.6 lakh km²	1400 km
Mahanadi	~1.4 lakh km²	851 km
Narmada	~0.98 lakh km²	1312 km (Longest West-flowing)
Tapi	~0.65 lakh km²	724 km
Kaveri	~0.81 lakh km²	800 km

TRANS-HIMALAYAN RIVERS: Indus, Sutlej, Brahmaputra (Tsangpo), Zanskar
MCQ = Sutlej (2020 Q52)

3 PANCH PRAYAG (HIMALAYAN CONFLUENCES)

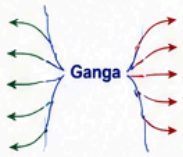
Order (Upstream to Downstream)	Confluence	Rivers Meet	On	Forms
1	Vishnu Prayag	Dhauli Ganga + Alaknanda	Joshimath	
2	Nanda Prayag	Nandakini + Alaknanda	Nandaprayag	
3	Karn Prayag	Pindar + Alaknanda	Karnaprayag	
4	Rudra Prayag	Mandakini + Alaknanda	Rudraprayag	
5	Dev Prayag	Bhagirathi + Alaknanda	Devprayag	GANGA

Memory Trick: V → N → K → R → D
Vishnu → Nanda → Karn → Rudra → Dev
(Dev = Done → Ganga is born) [2023 Q50, 2024 Q74]

4 GANGA SYSTEM – TRIBUTARIES

LEFT BANK TRIBUTARIES (Stand facing downstream)

- Ramganga
- Gomti
- Ghaghra
- Gandak
- Kosi
- Mahananda



RIGHT BANK TRIBUTARIES (Stand facing downstream)

- Yamuna
- Son
- Damodar
- Punpun
- Jalangi

Jonk = Mahanadi tributary NOT Ganga (2021 Q5)

5 YAMUNA TRIBUTARIES (West → East)

Source (Yamunotri) → Meets Yamuna

Chambal (Rajasthan, MP) → Sind (MP) → Betwa (MP, UP) → Ken (MP, UP) → Yamuna

- Order (2020 Q68) = 1, 2, 3, 4 ✓
- Betwa joins Yamuna at Hamirpur (UP) – NOT Prayagraj (2023 Q55)
- Ramganga joins Ganga at Kannauj (UP)

6 ORIGIN OF MAJOR PENINSULAR RIVERS (2023 Q53 = A)

River	Origin
Yamuna	Yamunotri (Uttarakhand)
Krishna	Mahabaleshwar (Maharashtra)
Godavari	Trimbakeshwar, Nashik (Maharashtra)
Mahanadi	Sihawa Hills (Chhattisgarh)
Narmada	Amarkantak (MP)
Tapi (Tapi/Tapti)	Multai (Satpura range, MP) NOT Western Ghats (2018 Q102)
Kaveri	Brahmagiri Hills (Karnataka)
Son	Amarkantak (MP) – radial with Narmada

7 PENINSULAR DRAINAGE – OUTFALL

EAST FLOWING (Bay of Bengal)

Godavari, Krishna, Mahanadi, Kaveri, Pennar, Palar, Damodar (Subarnarekha), Vamsadhara

WEST FLOWING (Arabian Sea)

Narmada, Tapi, Mahi, Periyar (Kerala)

Memory:** Periyar is the only major west flowing river in Kerala (2025 Q116)

8 IMPORTANT TRAPS (High Yield)

- Pranhita → joins Godavari (NOT Mahanadi) (2023 Q49)
- Hemavati → Kaveri tributary
- Srisailam (Krishna) – dam named as per river it is on
- Luni → Rann of Kutch (inland, saline lower course) (2019 Q84)
- Ghaggar → disappears in Thar Desert (inland)
- Barak River → flows to sea (Bay of Bengal), NOT inland (2021 Q65)

9 RIVER FEATURES & PATTERNS

DELTAS (Types)

- Arcuate (Fan shaped) Sundarbans (Ganga-Brahmaputra)
- Bird's foot (Digitate) Mississippi (USA)
- Cuspate Mahanadi Delta
- Estuarine Narmada, Tapi, Mahi, Periyar
- Moribund (Dying) Cauvery Delta (2019 Q85)

DRAINAGE PATTERNS

- Dendritic – on uniform rocks (example: Most Peninsular rivers)
- Trellis – in folded/striped rocks (2024 Q31)
- Radial – streams diverge from a central highland (example: Amarkantak)
- Centripetal – streams flow towards a central basin (example: Imphal basin)

10 RIVER CAPTURE, INLAND & REJUVENATION

RIVER CAPTURE (PIRACY)

✓ Teesta captured by Brahmaputra. A/R both true (2021 Q63)

INLAND DRAINAGE

- Luni → Rann of Kutch (saline)
- Ghaggar → Thar Desert depression

RIVER REJUVENATION

Renewed erosive power after uplift or fall in sea level → forms terraces, gorges, waterfalls.

Gomti in UP = Biological Disaster! (2018 Q99)

11 WATERWAYS, LINKING & MAJOR DAMS

NATIONAL WATERWAYS (NW)

- NW-1 : Ganga – Hooghly (Allahabad to Haldia)
- NW-2 : Brahmaputra (Dhubri to Sadiya)
- NW-3 : West Coast Canal, Kerala (NW-3 Kollam to Kottapuram)
- NW-4 : Krishna – Godavari Canal (Dellang to Puducherry)
- NW-5 : Brahmani – Mahanadi (Talcher to Dhamra)

NEW MULTI-MODAL WATERWAYS

- NW-6 : Barak (Assam)
- NW-7 : Sundarbans Waterways (WB)
- NW-10 : Amba – Arpa – Vainganga, MP

IMPORTANT DAMS (River wise)

Dam	River
Baglihar Dam	– Chenab
Pandoh Dam	– Beas
Srisailam Dam	– Krishna
Hirakud Dam	– Mahanadi
Mettur (Stanley) Dam	– Kaveri
Tehri Dam	– Bhagirathi
Rihand Dam	– Rihand (Son Basin)

RAMGANGA PROJECT (UP) (2024 Q72 = D)

- 198 MW (Garhwal)
- Irrigated area ~5.75 – 5.90 lakh ha (NOT 6 lakh)

★ RAPID FACTS (RAATA)

- ★ Largest basin in India – Ganga
- ★ Largest peninsular basin – Godavari
- ★ Longest peninsular river – Godavari (~1465 km)
- ★ Longest west flowing – Narmada (~1312 km)
- ★ Only river in Kerala flowing to Arabian Sea – Periyar
- ★ Only trellis pattern – in folded mountains
- ★ Radial pattern centre – Amarkantak
- ★ Only moribund delta – Cauvery
- ★ Teesta – captured by Brahmaputra
- ★ Jonk – Mahanadi tributary, NOT Ganga
- ★ Sutlej – Trans-Himalayan (MCQ trap)
- ★ Rann of Kutch – formed by Luni

12 MVET-KNOW TERM COMPARISONS (One-line Difference)

Term	One-line Difference	Hindi
Drainage basin	Land area drained by a river and its tributaries	जल विभाजक क्षेत्र
Watershed / Water divide	Ridge separating two drainage basins	जल विभाजक रेखा
Perennial river	Flows throughout the year (snow-fed + rain-fed)	बारहमासी नदी
Non-perennial / Ephemeral	Flows only after rain; dry channel in dry season	अल्पवाहिक नदी

Term	One-line Difference	Hindi
River capture	Stronger stream steals headwaters of weaker stream (piracy)	नदी अपहरण
Inland drainage	River does not reach sea; ends in lake/plain/desert	अंतःस्थ जल निकास
Antecedent river	Maintains course across rising land (Indus, Brahmaputra)	पूर्वत जल निकास
Superimposed river	Maintains course after burying relief (Chambal)	अध्यारोपित नदी

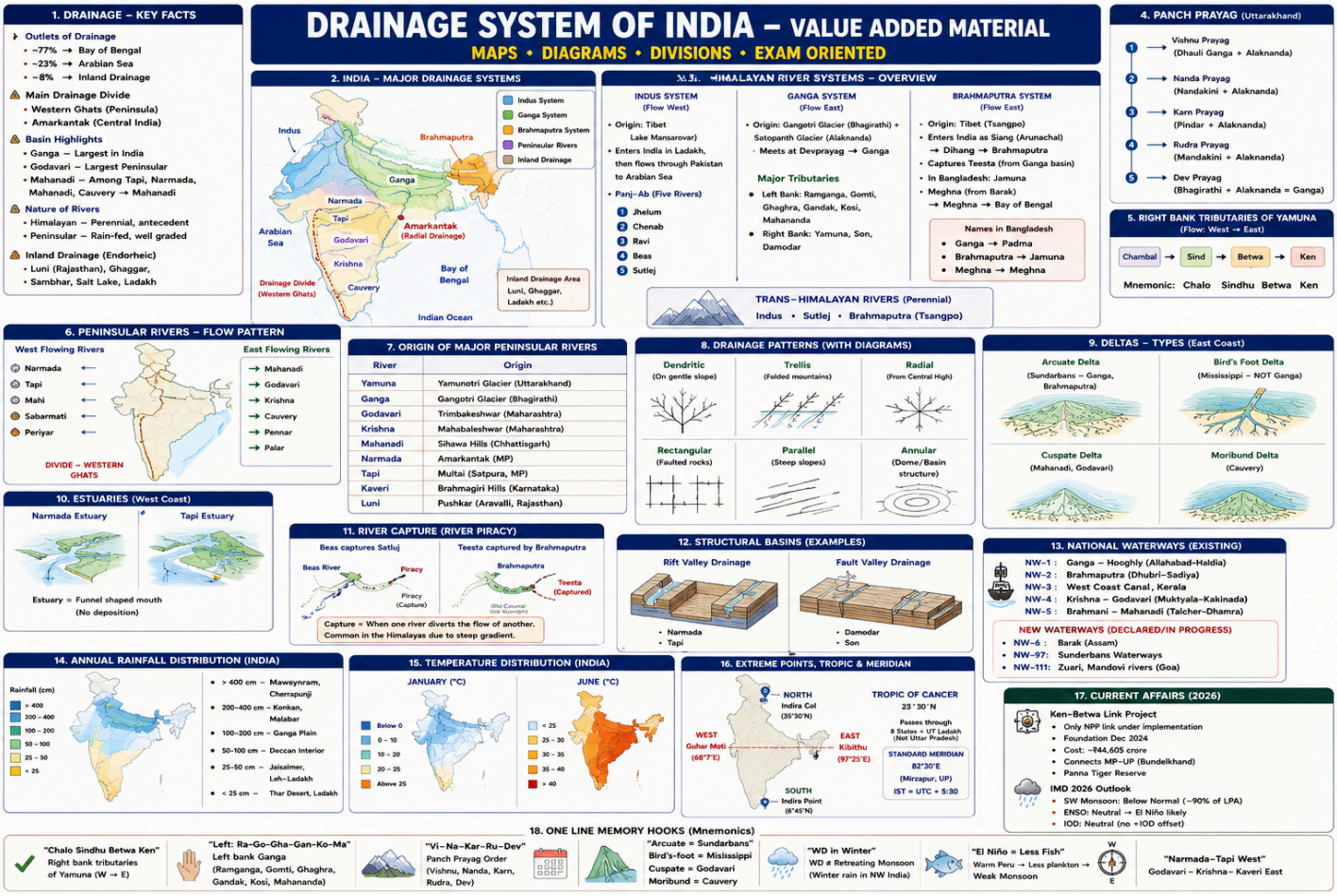
Term	One-line Difference	Hindi
Fault valley river	Course in tectonic rift/graben (Narmada, Tapi, Damodar)	भ्रंश घाटी नदी
Estuary	Drowned river valley; tide-dominated, sediments removed	ज्वारनदमुख
Delta	Deposited sediment at river mouth; progrades into sea	डेल्टा / त्रिभुज
Moribund delta	Old inactive delta lobe no longer receiving sediment	मृत डेल्टा

Term	One-line Difference	Hindi
River rejuvenation	Renewed erosive power after uplift / sea-level fall	नदी पुनः युवा अवस्था
Structural basin	Lowland enclosed by hills; centripetal drainage	संरचनात्मक द्रोणी
Prayag	Sacred Himalayan confluence (Uttarakhand)	प्रयाग
Sangam	Confluence of Ganga + Yamuna at Prayagraj	संगम

UPPCS/UPSC EXAM WEIGHT: ★★★ HIGH | DRAINAGE, TRIBUTARIES, PRAYAG, DELTAS, DAM PAIRS – REPEATED EVERY YEAR | LAST VERIFIED: AUGUST 2026

Confused Pairs

A	B	Correct	Hindi
Ganga basin	Godavari basin	Largest **India** vs largest **peninsular**	गंगा / गोदावरी
Left-bank Ganga	Right-bank Ganga	Himalayan trib vs Yamuna/Son/Damodar	बायाँ / दायाँ तट
Delta	Estuary	East-coast deposition vs west-coast tidal funnel	डेल्टा / ज्वारनदमुख
Arcuate	Bird's-foot	Sundarbans vs **Mississippi**	चापाकार / पक्षी-पाद
Antecedent	Superimposed	Cuts rising Himalaya vs inherited course (Chambal)	पूर्ववर्ती / अध्यारोपित
Trans-Himalayan	Himalayan-origin	Tibet origin + cuts Himalaya (Sutlej) vs rises in Himalaya (Jhelum/Ravi)	पार-हिमालयी
Narmada	Tapi	Amarkantak vs **Multai (Satpura)**	नर्मदा / तापी
Luni	Barak	Inland saline lower vs BoB via Meghna	लूनी / बराक
Padma	Jamuna	Ganga in BD vs Brahmaputra in BD	पद्मा / जमुना
Devprayag	Rudraprayag	+Bhagirathi = **Ganga** vs +Mandakini	देवप्रयाग / रुद्रप्रयाग
Trellis	Dendritic	**Folded** structure vs homogeneous slope	जालनुमा / वृक्षाकार
Consequent	Subsequent	Follows original slope vs later weak-belt tributary	अनुवर्ती / परवर्ती
Concordant	Discordant	Follows structure vs cuts across (antecedent Himalaya)	संगत / असंगत
Pranhita	Jonk	Godavari trib vs Mahanadi trib	प्रणहिता / जोंक
Hyderabad–Musi	Hyderabad–Krishna	City on Musi trib, not Krishna stem	मुसी / कृष्णा
E. Dhauliganga	Alaknanda Dhauli	Kali trib vs Vishnuprayag Dhauli	पूर्वी धौली
Chambal ravines	Damodar rift	Badlands vs fault-valley tag	चंबल / दामोदर
Kosi	Damodar	Sorrow of Bihar vs Sorrow of Bengal	कोसी / दामोदर



3.0 Drainage framework

River basics (must know before systems)

Outlet	Share of area	Typical rivers
Bay of Bengal	about 77%	Ganga, Brahmaputra, Godavari, Krishna, Mahanadi, Kaveri, Pennar, Palar, Vaigai, Damodar, Subarnarekha, Barak
Arabian Sea	about 23%	Indus (in Pakistan), Narmada, Tapi, Mahi, Sabarmati, **Periyar**, Sharavati, Mandovi
Inland	about 8% of area	Luni, Ghaggar, Rajasthan salt (लवणाध्यक्ष) lakes, Ladakh

Type	Meaning	India
Consequent	Follows the original slope of the land	Godavari, Krishna, Kaveri from Western Ghats
Subsequent	Grows later along a weak belt; often joins at right angles	Chambal, Sind, Betwa, Ken, Tons, Son to Yamuna/Ganga
Antecedent	Older than uplift; keeps course by cutting a gorge through rising land	Classic set: **Indus, Sutlej, Brahmaputra**; many PCS keys also tag **Ganga**
Superimposed	Older course inherited onto newer structure; does not follow present slope	**Chambal** (also Son in some notes)
Concordant	Follows structure and slope	Most peninsular rivers
Discordant	Cuts across structure	Himalayan antecedent rivers

3.1 Himalayan rivers

Ganga system

Tributary	Bank	Meets Ganga at / near
Yamuna (यमुना)	Right	**Prayagraj (Sangam)**
Ramganga (रामगंगा)	Left	**Kannauj**
Gomti	Left	Near **Saidpur / Ghazipur** belt (after Lucknow)
Ghaghara	Left	**Chhapra / Revilganj (Saran)**
Gandak	Left	**Sonpur near Patna**
Son	Right	Near **Patna**
Punpun	Right	**Fatuha** (Patna)
Kosi	Left	**Kursela, Katihar**
Mahananda (महानंदा)	Left	West Bengal (last left-bank in India)
Damodar	—	Joins **Hooghly**, not the main Ganga stem

Tributary	Bank of Yamuna	Meets Yamuna at / near
Tons (टौंस)	Left	Near **Kalsi** (Uttarakhand)
Hindon	Left	NCR / western UP belt
Chambal	Right	Near **Etawah**
Sind (सिंध)	Right	Between Chambal and Betwa (Auraiya–Kalpi belt)
Betwa (बेतवा)	Right	**Hamirpur** — **not** Prayagraj
Ken	Right	Near **Banda**

Panch Prayag and Uttarakhand

Prayag	Confluence
Vishnuprayag	Alaknanda + **Dhauli Ganga** (western / Alaknanda Dhauli — not Eastern Dhauliganga of Kali)
Nandaprayag	Alaknanda + **Nandakini**
Karnaprayag	Alaknanda + **Pindar** (पिंडर)
Rudraprayag	Alaknanda + **Mandakini** (Kedarnath belt)
Devprayag	Alaknanda + **Bhagirathi** = **Ganga**

3.4 River features and patterns

Types of deltas — Teaching Card

Type	Meaning	example
Arcuate	Fan-shaped front, many distributaries	Ganga–Brahmaputra (Sundarbans), Nile, Godavari–Krishna
Lobate	Rounded bulge, many mouths	Godavari (often)
Bird's-foot	Long finger-like distributaries	**Mississippi** — **not** Ganga
Cuspate	Pointed front from opposing waves	Ebro; **Mahanadi** often cited for India
Moribund (Bengal subdivision stem)	Dying belt inside a large delta's moribund–mature–active scheme	**Bengal Delta** (UPPCS 2019)

Type	Meaning	example
Moribund (lobe / river match stem)	Abandoned / ageing delta lobe	**Cauvery** in many landform keys
Estuarine mouth	Tidal funnel, little delta build-up	Narmada, Tapi

Origins, cities, dams (one home)

River	Origin
Bhagirathi	Gomukh / Gangotri
Alaknanda	Satopanth (above Badrinath)
Yamuna	Yamunotri (Bandarpunch)
Indus	Tibet, near Kailash / Mansarovar
Sutlej	Rakas Tal, Tibet (Trans-Himalayan; enters near Shipki La)
Brahmaputra	Tibet / Angsi–Chemayungdung (Tsangpo)
Godavari	Nashik / Trimbakeshwar
Krishna	Mahabaleshwar
Mahanadi	Sihawa, Chhattisgarh
Kaveri	Brahmagiri / Talakaveri, Coorg
Narmada	Amarkantak
Tapi	Multai, Satpura
Son	Amarkantak
Chambal	Mhow / Ambedkar Nagar (Malwa)
Luni	Aravalli near Ajmer
Sharavati	Western Ghats (Jog Falls)
Subarnarekha	Chotanagpur (Ranchi / Nagri class)
Jhelum	Verinag (Pir Panjal)
City	River
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Lucknow	Gomti
Ayodhya	Ghaghara / Saryu
Prayagraj	Ganga + Yamuna (Sangam)
Kanpur	Ganga
Varanasi	Ganga
Patna	Ganga
Bhagalpur	Ganga
Surat	Tapi
Hyderabad	Musi
Vijayawada	Krishna
Ahmedabad	Sabarmati
Cuttack	Mahanadi
Bhubaneswar	Not on Mahanadi bank
Ujjain (उज्जैन)	Kshipra

River	Origin
Jabalpur	Narmada
Nasik (नासिक)	Godavari
Nanded	Godavari
Rajahmundry	Godavari
Srinagar	Jhelum
Leh (लेह)	Indus (right bank)
Ludhiana	Old Sutlej belt
Firozpur	Old Sutlej belt
Jamshedpur	Subarnarekha
Guwahati	Brahmaputra
Dibrugarh	Brahmaputra
Kota	Chambal
Gorakhpur	Rapti
Dam / project	River
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Bhakra–Nangal	Sutlej
Pandoh	Beas (**not** Ravi)
Baglihar / Salal	Chenab
Ranjit Sagar	Ravi
Tehri	Bhagirathi
Farakka Barrage	Ganga (Hooghly diversion)
Hirakud	Mahanadi
Nagarjuna Sagar / Srisailam	Krishna (**not** Tungabhadra)
Mettur (Stanley)	Kaveri (**not** Krishna)
Jayakwadi	Godavari (Maharashtra)
Sriram Sagar	Godavari (Telangana)
Polavaram	Godavari (Andhra Pradesh)
Bargi	Narmada (MP, near Jabalpur)
Indira Sagar	Narmada (MP, Khandwa / Punasa)
Omkareshwar	Narmada (MP)
Sardar Sarovar	Narmada (Gujarat, Navagam) — terminal Narmada project
Ukai	Tapi (Gujarat)
Gandhi (गांधी) Sagar	Chambal
Rihand	Rihand (Son system)
Rajghat	Betwa (UP–MP)

Additional City / Town	River	Key Geographic Anchor
Pahalgam	**Lidder**	Starting point of Amarnath Yatra in Kashmir (2,130 m altitude)
Pandharpur	**Chandrabhaga (Bhima)**	Famous pilgrimage centre of Vithoba in Maharashtra
Hampi	**Tungabhadra**	Historic capital of Vijayanagara Empire (not Malaprabha)

Additional City / Town	River	Key Geographic Anchor
Govindghat	**Alaknanda + Lakshman Ganga**	Confluence point en route to Hemkund Sahib and Valley of Flowers
Gaurikund / Guptakashi / Rambara	**Mandakini**	Pilgrim transit halts along the Mandakini gorge to Kedarnath
Bhubaneswar	**Kuakhai** (delta branch)	Southwest of Mahanadi main stem; Cuttack is on Mahanadi
Kanpur	**Ganga**	Largest urban agglomeration (>2.9 million) on the bank of Ganga
Leh	**Indus (Right Bank)**	Situated strictly on the right bank of the Indus in Ladakh

Drainage patterns, rejuvenation, basins — Teaching Card

Pattern	Control	River / ground
Dendritic	Uniform rock	**Ganga** and **Indus** plains; **Godavari**; **Mahanadi**; **Krishna**; **Kaveri**
Trellis	**Folded** ridges and valleys	**Singhbhum** (Chotanagpur)
Radial	Dome or peak	**Narmada** west and **Son** toward the Ganga from **Amarkantak**; Girnar
Centripetal	Closed basin	**Loktak** (Imphal); **Mendha** and **Rupangarh** into **Sambhar**
Parallel	Steep uniform slope	**Sharavati** (Western Ghats)
Rectangular	Joints and faults	Jointed **Vindhyan** streams

3.5 Waterways and river linking

NW	Route
NW-1	Ganga–Bhagirathi–Hooghly (Haldia–Prayagraj) — longest
NW-2	Brahmaputra (Dhubri–Sadiya)
NW-3	Kerala west-coast canal / backwaters
NW-4	Krishna–Godavari canals
NW-5	Brahmani–Mahanadi–Paradip

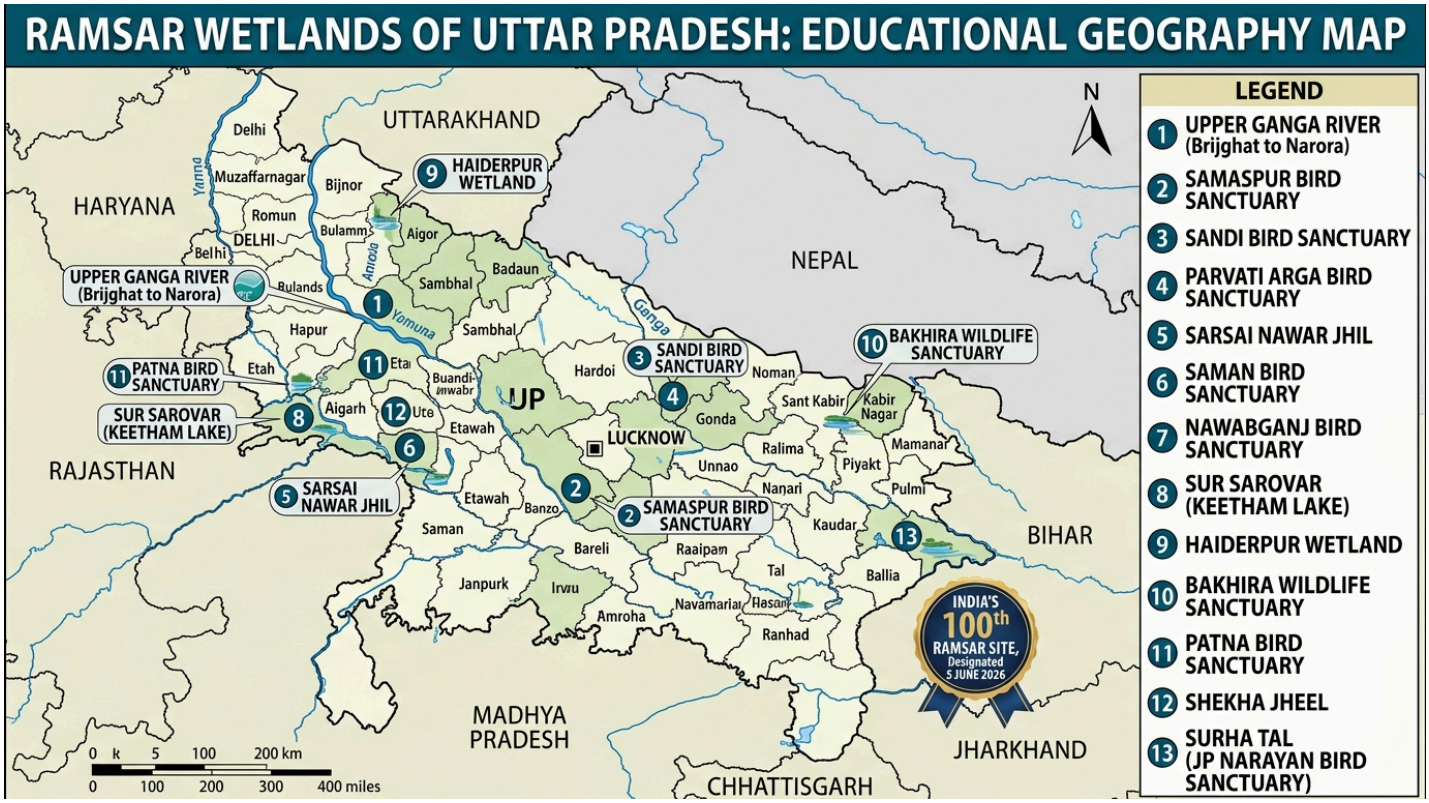
Common Traps — Don't Fall For These

1. **Pranhita → Mahanadi** — FALSE. **Godavari**.
2. **Betwa joins Yamuna at Prayagraj** — FALSE. **Hamirpur**.
3. **Any Prayag forms the Ganga** — FALSE. Only **Devprayag** (Alaknanda + Bhagirathi).
4. **Ganga delta = bird's-foot** — FALSE. **Mississippi**; Ganga is **arcuate**.
5. **Moribund delta subdivision = Cauvery / Krishna–Godavari** — FALSE. UPPCS 2019 keys **Bengal Delta** (moribund / mature / active). Cauvery is the usual trap.
6. **Tapi from Western Ghats** — FALSE. **Multai, Satpura**.
7. **Jonk in Ganga basin** — FALSE. **Mahanadi**.
8. **Srisailam on Tungabhadra / Pandoh on Ravi** — FALSE. **Krishna / Beas**.
9. **Pennar and Palar → Arabian Sea** — FALSE. **Bay of Bengal**; Periyar is west.
10. **Fault-valley tag = Chambal** — FALSE. **Damodar**.
11. **Folded structure = dendritic** — FALSE. **Trellis**.
12. **Trans-Himalayan = Jhelum or Ravi** — FALSE. **Sutlej** among usual options. Also Indus and Brahmaputra are Trans-Himalayan; Ganga is not the pick in that set.

13. **Chambal Sanctuary includes Haryana** — FALSE. **UP, MP, Rajasthan** only (2020).
14. **UP biological disaster = Yamuna** — FALSE. **Gomti**.
15. **Barak = inland saline like Luni** — FALSE. Barak → **BoB**; Luni = fresh upper / saline lower.
16. **Indus / Sutlej / Brahmaputra = superimposed** — FALSE. They are **antecedent**. Superimposed example = **Chambal**.
17. **Hemavati → Mahanadi** — FALSE. **Kaveri**.
18. **Largest share of river *water* = Ganga basin area rank** — FALSE. **Brahmaputra** ~40% of water; Ganga is the largest **Indian basin by area**.
19. **Hyderabad on Krishna** — FALSE. On **Musi** (Krishna tributary).
20. **Northern Assam Brahmaputra tribs = left bank** — FALSE when looking downstream west: north bank = **right bank** (Subansiri, Manas, Sankosh, Teesta).
21. **Eastern Dhauliganga = Alaknanda Prayag set** — FALSE. Eastern Dhauliganga → **Kali / Sharda**.
22. **Gandak flows through UP** — FALSE. Nepal → Bihar → Ganga at Sonpur.
23. **Narmada forms a delta** — FALSE. **Estuary** (rift / hard rock / little silt).
24. **Ludhiana on Ravi** — FALSE. Old **Sutlej** bank belt.

Topic 4 — Lakes, Waterfalls & Water Resources

Current Affairs (this topic)



Year	Fact	Why asked	Source
5 Jun 2026	India designates **100th Ramsar** — **Jai Prakash Narayan Bird Sanctuary (Surha Tal), Ballia, UP** (World Environment (पर्यावरण) Day (विश्व पर्यावरण दिवस))	"How many Ramsar in India / latest UP site?"	MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) / PIB
2025–26	UP Ramsar total **13** — adds **Patna (पटना) BS (Etah)**, **Sheikha/Shekha Jheel (Aligarh (अलीगढ़))**, **Surha Tal (Ballia)**	State-paper list + 2025 Q58 "NOT in UP"	MoEFCC LS annex
2025	India among top Ramsar countries in Asia; proposals in pipeline include Sauj Jheel (UP)	Count vs names trap	MoEFCC
Ongoing	**Jal Jeevan Mission** extended (rural tap water) — water-*supply* overlay, not a dam (दाम -)–river fact	Don't confuse with multipurpose dams	Jal Shakti (शक्ति)
2019–20	Six UP Ramsar added in the Jan 2020 window	Number question	UPPCS 2020

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q93

What is the correct sequence of lakes in descending order of size?

A. Victoria, Superior, Chilka, Michigan B. Superior, Victoria, Michigan, Chilka C. Michigan, Superior, Victoria, Chilka D. Chilka, Michigan, Superior, Victoria

Confused Pairs

Pair	Correct	Trap	Hindi
Chilika vs Sambhar	Chilika = lagoon / often "largest saline"; Sambhar = **inland** saline	Pick Sambhar whenever "saline" appears	चिल्का = तटीय; सांभर = अंतर्देशीय खारा
Chilika vs Vembanad	Chilika = largest **east-coast lagoon**; Vembanad = Kerala largest + India **longest**, **west coast**	Call Vembanad India's overall largest lake	वेम्बनाड = सबसे लंबा, पश्चिम
Chilika vs Pulicat vs Kolleru	Chilika largest E-coast lagoon; Pulicat 2nd; Kolleru freshwater	Kolleru as lagoon king	कोलेरू = मीठा, लैगून नहीं
Wular vs Dal vs Loktak	Wular size king; Dal tourism; Loktak = Keibul Lamjao	Keibul on Wular/Dal	लोकटक = मणिपुर, केइबुल लामजाओ
Kunchikal vs Jog	Kunchikal = UPPCS highest; Jog = famous / **width**	Jog = highest	कुंचिकल ऊँचा; जोग चौड़ा
Jog vs Shivanasamudra	Jog = Sharavati **width**; Shivanasamudra = Cauvery **volume**	Swap crowns	जोग चौड़ा; शिवसमुद्र आयतन
Lodh vs Hundru	Lodh/Budha Ghagh = **Burha**, JH highest; Hundru = Subarnarekha	Lodh on Kanchi/Barakar	लोध = बूढ़ा
Roopkund vs Renuka	Roopkund = UK mystery/skeleton; Renuka = HP Ramsar	Roopkund in HP	रूपकुंड = उत्तराखंड
Kayal vs inland lake	Kayal = Kerala coastal lagoon/backwater	Call Wular a kayal	कायल = केरल
Dudhsagar state	**Goa** (Mandovi)	Karnataka-only	दूधसागर = गोवा
Chachai–Bihad	**Wrong** pair	Treat as correct	चाचई ≠ बिहद
Bhakra vs Pandoh	Bhakra = Sutlej; Pandoh = **Beas**	Pandoh on Ravi/Sutlej	भाखड़ा = सतलुज
SSP vs Indira Sagar	SSP = Narmada **Gujarat**; Indira Sagar/Omkareshwar = Narmada **MP**	Swap states	सरदार सरोवर = गुजरात
Ban Sagar vs Narmada set	Ban Sagar = **Son**	Put Ban Sagar with Bargi/Omkareshwar	बन सागर = सोन
DVC year	**1948** first multipurpose of independent India	1947 / Bhakra-first	DVC = 1948
Sidrapong vs Shivasamudram	Sidrapong **1897** oldest; Shivasamudram **1902** second	Call Shivasamudram oldest	सिद्रापोंग सबसे पुराना
Cauvery dispute parties	**TN + KA + KL + Puducherry**	Add Gujarat / drop PY	कावेरी = चार पक्ष
Gobind Sagar vs Pant Sagar	Gobind = **Bhakra/Sutlej**; Pant = **Rihand/Sonbhadra**	Swap	गोबिंद = भाखड़ा
Chambal vs Gobind/Pant	Chambal = Gandhi/RPS/Jawahar Sagar	Gobind or Pant as Chambal	चंबल ≠ गोबिंद सागर
Tehri confluence	**Bhagirathi + Bhilangana**	Alaknanda / Yamuna alone	टिहरी = भागीरथी–भिलंगना
Telugu Ganga city	Krishna water to **Chennai**	Bangalore / Madurai	तेलुगु गंगा = चेन्नई
NBA / Medha Patkar	Opposed **Sardar Sarovar** height	Tag to Indira Sagar only	मेधा = SSP
Tehri river	**Bhagirathi** (rock-fill)	Alaknanda / Yamuna	टिहरी = भागीरथी
Srisailam vs Tungabhadra Dam	Srisailam = **Krishna**; TB Dam = Tungabhadra	Swap	श्रीशैलम = कृष्णा
Nagarjuna Sagar vs Srisailam	Both Krishna; NS = AP–TS iconic	Put NS on Godavari	नागार्जुन सागर = कृष्णा
Ramganga irrigation	~5.75–5.90 lakh ha	Neat **6 lakh ha**	सिंचाई ≠ 6 लाख हेक्टेयर
Tulbul vs Baglihar	Tulbul = Jhelum/Wular; Baglihar = Chenab	Swap	तुलबुल = झेलम

Pair	Correct	Trap	Hindi
IGC vs Upper Ganga Canal	IGC = longest, **Harike** → RJ; UGC = **Bhimgoda/Haridwar** → W-UP	Mix offtakes	इंदिरा गांधी = सबसे लंबी
IGC vs Gang Canal	**Gang Canal** = oldest developed RJ system (**1927**, Ganga Singh, Hussainiwala)	Call IGC the oldest	गंग नहर = 1927
Harike vs Bhakra source	IGC source = **Harike Barrage** (Sutlej–Beas)	Bhakra / Gandhi Sagar as IGC source	हरिके = IGC
Onega	**Russia**	Canada	ओनेगा = रूस
Oxbow vs lagoon	Oxbow = cut-off meander on a floodplain; lagoon = barred coast	Call Chilika an oxbow	गोखुर ≠ लैगून
Tectonic vs crater	Wular = tectonic basin; Lonar = meteorite in basalt	Lonar as tectonic	वुलर = विवर्तनिक; लोनार = उल्का
Kabartal vs Wular	Kabartal = Bihar oxbow Ramsar; Wular = J&K tectonic freshwater	Swap size kings	कबरताल = बिहार
Sharavati vs Cauvery	Jog is on **Sharavati**, a **west-flowing** Karnataka river	Wrong slip: Sharavati as Cauvery tributary	शरावती ≠ कावेरी
Kunchikal vs Nohkalikai	Kunchikal = highest **total**; Nohkalikai = tallest **plunge**	Swap the two height titles	कुंचिकल कुल; नोहकलिकाइ प्लंज
Kapildhara vs Jog	Kapildhara = **Narmada**; Jog = Sharavati	Kapildhara on Godavari (UPSC 2008 trap)	कपिलधारा = नर्मदा
Shivanasamudra vs Jog	Both Karnataka; Shivanasamudra = **Cauvery**	Put Jog on Cauvery	शिवसमुद्र = कावेरी
Hirakud vs Indira Sagar vs Tehri	Longest earthen vs largest **volume** vs **tallest** dam	Swap the three superlatives	हीराकुड लंबा; इंदिरा सागर आयतन; टिहरी ऊँचा
Idukki vs Mullaperiyar	Both **Periyar**; Idukki = arch hydro	Put Idukki on Cauvery	इडुक्की = पेरियार
Canal vs groundwater irrigation	IG plains = canals + tubewells; peninsula historically **tanks**; GW now >60%	"Dominant source = canals only"	मैदान = नहर; प्रायद्वीप = तालाब
Minor vs major irrigation	Minor CCA ≤ **2000 ha** (~62%); major > **10,000 ha**	Swap shares	लघु ≈ 62%
Dam vs barrage	Dam **stores**; barrage **diverts** (Farakka ~**40,000 cusec** to Hooghly)	Call Farakka a storage dam	बांध जलाशय; बैराज मोड़
Garland vs Visvesvaraya	Garland = **Dastur**; first linking idea often **Visvesvaraya**; later **K.L. Rao**	Tag Garland to Rao alone	गारलैंड = दस्तूर
Jawai Project	**Rajasthan** (Luni tributary)	Tamil Nadu	जवाई = राजस्थान

N.1 Lakes of India

1. Chemistry / size kings (first question type)

If the question says...	Pick	Where	Why this one wins
Largest **lagoon** / east-coast lagoon	**Chilika**	Odisha	Barred from the sea; brackish; biggest east-coast lagoon
Largest **saline** (लवणीय) (**no** word "inland")	**Chilika**	Odisha	Papers often treat Chilika as the saline king by area
Largest **inland saline**	**Sambhar**	Rajasthan	Desert playa; no sea link; evaporation makes it salt
Largest **freshwater**	**Wular**	J&K (Jhelum)	Usual volume fact; tectonic basin

Wrong thought	Correct
"Saline" → always Sambhar	Only if the question says **inland**

Wrong thought	Correct
Chilika = inland Rajasthan lake	Chilika = Odisha **coast**
Wular = saline	Wular = **fresh**
Kolleru = lagoon king	Kolleru = **freshwater** delta (डेल्टा) lake; Chilika is lagoon king
Pulicat = largest lagoon	Pulicat = **second** after Chilika

2. Origin kings (second question type)

Origin	Meaning in one line	Indian fact	Trap
Tectonic	Basin from earth movement / warp	**Wular** (also freshwater king)	Calling Wular a lagoon
Oxbow	Cut-off river meander on a plain	**Kabartal / Kanwar** (Bihar)	Calling it a coastal lagoon
Crater (meteorite)	Hollow from a meteor hit	**Lonar** (Buldhana, MH)	Calling it a volcano or RJ salt pan
Glacial / high mountain	Ice-cut tarn or high closed basin	Chandra Tal, Roopkund, Tsomgo; Pangong / Tso Moriri (high, often brackish)	Calling Pangong India's freshwater size-king
Lagoon (bar)	Sea water trapped behind a sandbar	**Chilika**, Pulicat	Calling Chilika inland saline
Wind / playa	Desert hollow, water evaporates	**Sambhar** belt	Using Sambhar for every "saline" question
Artificial	Reservoir behind a dam	Gobind Sagar, Hirakud, Tehri, Pant Sagar	Treating them as natural glacial lakes

Lake–state match bank

Lake	State / UT	Note
Sala ★2020	**Arunachal Pradesh**	2020 Q56
Badkhal ★2020	**Haryana** (Faridabad belt)	2020 Q56
Loktak ★2020	**Manipur**	Phumdis; Keibul Lamjao
Kaliveli ★2020	**Tamil Nadu**	Coastal wetland
Chilika	**Odisha** (ओडिशा)	Largest E-coast lagoon
Pulicat	**Andhra Pradesh + Tamil Nadu**	Sriharikota bar
Vembanad	**Kerala**	Largest in Kerala; longest in India; **west coast**
Ashtamudi	**Kerala**	Kayal / Ramsar
Sasthamkotta	**Kerala**	Largest freshwater in Kerala (not TN)
Periyar Lake	**Kerala**	Artificial; **not** a lagoon
Kolleru	**Andhra Pradesh**	Freshwater; between Godavari–Krishna
Wular	**J&K**	Largest freshwater (volume)
Dal	**J&K** (Srinagar)	Tourism; not size king
Anchar	**J&K**	Match distractor set
Hokera	**J&K**	**Not** Punjab
Sambhar	**Rajasthan** (थार)	Largest inland saline
Didwana / Kuchaman / Pushkar / Nakki	**Rajasthan**	Nakki = Mount Abu
Pichola / Fateh Sagar	**Rajasthan** (Udaipur (उदयपुर))	Tourism; not Sambhar

Lake	State / UT	Note
Lonar	**Maharashtra** (Buldhana)	Meteorite crater
Kabartal / Kanwar	**Bihar** (Begusarai)	Oxbow Ramsar
Deepor Beel / Chapanala	**Assam**	Deepor = Guwahati Ramsar
Renuka / Nako / Ghepan / Pong	**Himachal Pradesh**	Renuka = Ramsar; Ghepan = Lahaul
Roopkund / Nainital / Bhimtal	**Uttarakhand** (उत्तराखंड)	Roopkund = Mystery Lake (**not** HP)
Surajkund / Sultanpur	**Haryana**	Sultanpur **not** UP
Rudrasagar	**Tripura**	**Not** UP
Pangong / Tso Moriri / Yaya Tso	**Ladakh** (लद्दाख)	High / brackish; Yaya Tso = BHS
Tsomgo / Gurudongmar	**Sikkim**	Glacial tarns
Hussain Sagar / Usman Sagar	**Telangana** (Hyderabad)	Artificial
Chembarambakkam / Kodaikanal	**Tamil Nadu**	Kodaikanal = artificial
Bhojtal (Upper Lake)	**Madhya Pradesh** (Bhopal)	Urban wetland
Nalsarovar	**Gujarat**	Bird sanctuary / Ramsar class
Harike	**Punjab**	Sutlej–Beas meet
Sukhna	**Chandigarh**	Man-made
Phulhar / Gomat Taal	**Uttar Pradesh** (Pilibhit)	Gomti source

World lake–country match bank

Lake	Country	tag / trap
Baikal ★2023	**Russia** (Siberia)	Deepest + oldest freshwater; huge volume
Michigan ★2023	**USA** only	Only Great Lake wholly in USA
Onega ★2023	**Russia**	Europe's 2nd-largest; **not Canada**
Maracaibo ★2023	**Venezuela**	Large brackish; oil belt
Ladoga	**Russia**	Europe's **largest** lake (Onega is 2nd)
Caspian Sea	Russia / Kazakhstan / Turkmenistan / Iran / Azerbaijan	World's **largest lake by area**; saline
Dead Sea	Israel / Jordan	Rift; hypersaline; below sea level
Superior	**USA + Canada**	Largest Great Lake by area
Huron / Erie / Ontario	**USA + Canada**	Shared Great Lakes (unlike Michigan)
Victoria	Uganda / Kenya / Tanzania	Africa's largest by area
Tanganyika	Several E. African states	Deep rift lake
Malawi (Nyasa)	Malawi / Mozambique / Tanzania	Rift lake
Turkana	Kenya (mainly)	Rift; saline
Titicaca	**Peru + Bolivia**	Highest large navigable lake
Great Bear / Great Slave	**Canada**	Not Onega
Loch Ness / Loch Lomond	**Scotland (UK)**	
Eyre	**Australia**	Endorheic salt lake
Chad	Chad / neighbours	Shrinking inland
Aral	Kazakhstan / Uzbekistan	Shrinking saline

Crown	Lake
Largest lake by **area** (world)	**Caspian** (saline)
Largest **freshwater** by **area** (usual set)	**Superior**
Largest / deepest freshwater by **volume**	**Baikal**
Lowest / densest salt	**Dead Sea**

UP wetlands (state paper gold)

Site	District / belt
Upper Ganga River	Brijghat to Narora
Nawabganj	Unnao
Parvati Arga	Gonda
Saman	Mainpuri
Samaspur	Rae Bareli
Sandi	Hardoi
Sarsai Nawar	Etawah
Sur Sarovar	Agra
Haiderpur	Bijnor–Muzaffarnagar (Ganga)
Bakhira	Sant Kabir (कबीर) Nagar
Patna Bird Sanctuary	Etah
Sheikha / Shekha Jheel	Aligarh
Jai Prakash Narayan BS / Surha Tal	**Ballia** (India's 100th Ramsar, 5 Jun 2026)
Name that sounds like UP	Actual state
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Rudrasagar	**Tripura**
Sultanpur	**Haryana**
Surinsar–Mansar	**J&K**

N.2 Waterfalls

Superlative	Fact	Trap
Highest	**Kunchikal** (Varahi, Karnataka; older figure ~455 m)	Jog
Tallest **plunge** (newer World Waterfalls Database)	**Nohkalikai** (Meghalaya (मेघालय), ~340 m)	Using Nohkalikai to beat every Kunchikal height question
Largest by **width**	**Jog / Gersoppa** (Sharavati, Karnataka)	Calling Jog highest
Largest by **volume**	**Shivanasamudra** (Cauvery, Karnataka)	Putting volume crown on Jog
Waterfall	~Height	River
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Kunchikal	older ~455 m; newer DB ~183 m	**Varahi**
Barehipani	~399 / ~217 m by DB revision	Budhabalanga
Nohkalikai	~340 m	Plateau stream

Superlative	Fact	Trap
Nohsngithiang (Seven Sisters) / Mawsmai	~315 m class	—
Dudhsagar	~310–320 m	**Mandovi**
Barkana	~259 m	Sita (सीता) belt
Jog (Gersoppa)	~253 m	**Sharavati**
Khandadhar	~244 m	Korapani Nala → Brahmani
Palani	—	Kullu belt
Hundru	~98 m	**Subarnarekha**
Jonha	—	**Rarhu**
Dasam / Dasong	—	**Kanchi** (कांची)
Lodh / Budha Ghagh	~143 m (468 ft)	**Burha**
Dhuandhar	~30 m	**Narmada**
Kapildhara	—	**Narmada**
Dugdhdhara / Bheraghat belt	—	**Narmada**
Bhalkund (Rahatgarh)	—	**Bina**
Chachai	~130 m	Bihad / Beehar (Tamsa (तमसा))
Chulia	—	**Chambal** (चंबल)
Joranda	~150 m	Budhabalanga
Duduma	—	**Machhkund**
Gokak	—	**Ghataprabha**
Shivanasamudra	~90–98 m	**Cauvery**
Hogenakkal	—	**Cauvery**
Athirappilly	—	Chalakudy
Chitrakote	wide, low (~29 m)	**Indravati**
Amritdhara	—	**Hasdeo (Hasdo)**
Kevti / Rakim Kund	—	Rewa plateau streams

N.3 Dams, canals and hydropower

Star projects

Project	River	Where / type	Reservoir / extra
Bhakra–Nangal	**Sutlej**	HP dam + Punjab headworks; **gravity**; joint **PB–HR–RJ** (command also touches HP)	**Gobind Sagar**; completed **1963**; Nehru's "temple of modern India" tag
Hirakud	**Mahanadi**	Odisha (~15 km from Sambalpur); started **1948**, completed **1957**	Longest earthen-dam narrative (~**4.8 km** main / ~**26 km** with dykes); ~10 lakh ha irrigation note
Sardar Sarovar	**Narmada**	Near **Navagam**, Gujarat; height ~**163 m**; ~**1450 MW**	Beneficiaries **GJ, RJ, MH, MP**; MP takes the largest Narmada water share; **NBA / Medha Patkar** opposed height rise
Indira Sagar (Narmada Sagar)	**Narmada**	Near **Punasa / Khandwa**, MP	Largest reservoir **by volume**; ~1.23 lakh ha + ~**1000 MW** target; **Harsud** town submerged (**2004**)

Project	River	Where / type	Reservoir / extra
Nagarjuna Sagar	**Krishna**	AP–TS border; completed **1967**	Multipurpose; Nalgonda–Khammam–Guntur–Prakasam belt; height ~**125 m**
Tehri	**Bhagirathi** (+ **Bhilangana** confluence at old Tehri)	Uttarakhand; **rock-fill**; THDC	India's **tallest** dam (~**260.5 m**); cleared **1972**, work from **1978**; seismic / displacement protest

Master Table — Major Dams & Multipurpose Projects in India (भारत के प्रमुख बांध एवं बहुउद्देशीय परियोजनाएं)

Project / Dam (बांध / परियोजना)	River (नदी)	State / Union Territory	Key Characteristics & Exam Notes
Tehri Dam (टिहरी बांध)	**Bhagirathi** (+ Bhilangana confluence)	Uttarakhand	**Tallest dam in India** (260.5 m); rock-fill earth dam; THDC
Bhakra-Nangal (भाखड़ा नांगल)	**Sutlej**	Himachal Pradesh & Punjab	**Highest concrete straight gravity dam** (226 m); reservoir: **Gobind Sagar**; joint venture of PB, HR, RJ
Hirakud Dam (हीराकुंड बांध)	**Mahanadi**	Odisha (Sambalpur)	**Longest earthen dam in the world** (main dam 4.8 km; total length with dykes 25.8 km); Gandhi Minar & Nehru Minar
Sardar Sarovar Dam	**Narmada**	Gujarat (Navagam, Kevadia)	Height 163 m; benefits Gujarat, Rajasthan, Maharashtra, MP; Statue of Unity downstream; **Narmada Bachao Andolan** (Medha Patkar)
Indira Sagar Dam (Narmada Sagar)	**Narmada**	Madhya Pradesh (Khandwa)	**Largest water reservoir in India by storage volume** (12.22 BCM); submerged Harsud town
Nagarjuna Sagar Dam	**Krishna**	Andhra Pradesh & Telangana	World's largest masonry dam; built 1955–1967; named after Buddhist scholar Acharya Nagarjuna
Srisailam Dam	**Krishna**	Andhra Pradesh & Telangana	Second largest capacity working hydroelectric station; located in deep gorge of Nallamala hills
Almatti Dam (Lal Bahadur Shastri)	**Krishna**	Karnataka (Bijapur)	Main reservoir of Upper Krishna Irrigation Project
Pandoh Dam (पंडोह बांध)	**Beas**	Himachal Pradesh (Mandi)	Diverts Beas waters into Sutlej via Beas-Sutlej Link for power at Dehar
Pong Dam (Maharana Pratap Sagar)	**Beas**	Himachal Pradesh (Kangra)	Earth-fill embankment dam; wetland Ramsar site
Baglihar Dam (बगलीहार)	**Chenab**	Jammu & Kashmir (Ramban)	Run-of-the-river project; cleared after neutral expert Raymond Lafitte adjudication under Indus Waters Treaty
Salal Dam (सलाल बांध)	**Chenab**	Jammu & Kashmir (Reasi)	Run-of-the-river hydroelectric project on Chenab rock pedestal
Dulhasti Dam (दुलहस्ती)	**Chenab**	Jammu & Kashmir (Kishtwar)	Run-of-the-river power project on Chandra/Chenab river
Pakal Dul & Ratle	**Chenab** & Marusudar	Jammu & Kashmir (Kishtwar)	Major new hydropower projects under execution
Uri Dam (Uri-I & II)	**Jhelum**	Jammu & Kashmir (Baramulla)	Near LoC; run-of-the-river hydro plant
Tulbul Navigation Project	**Jhelum** (Wular Lake outlet)	Jammu & Kashmir	Navigation barrage contested by Pakistan; suspended since 1987
Rihand Dam (Govind Ballabh Pant)	**Rihand** (tributary of Son)	Uttar Pradesh (Pipri, Sonbhadra)	**Largest man-made artificial lake by area in India**; power source for Renukoot aluminium plant

Project / Dam (बांध / परियोजना)	River (नदी)	State / Union Territory	Key Characteristics & Exam Notes
Matatila Dam (माताटीला बांध)	**Betwa**	Uttar Pradesh (Lalitpur)	Multipurpose dam; joint project with MP
Rajghat Dam (Rani Laxmibai)	**Betwa**	UP & MP border (Lalitpur/Chanderi)	Inter-state multipurpose project on Betwa
Parichha Dam (परीछा बांध)	**Betwa**	Uttar Pradesh (Jhansi)	Thermal power station water supply & canal irrigation
Chambal Project (4-stage cascade)	**Chambal**	MP & Rajasthan	1. **Gandhi Sagar** (Mandsaur, MP) 2. **Rana Pratap Sagar** (Rawatbhata, RJ) 3. **Jawahar Sagar** (Kota, RJ) 4. **Kota Barrage** (irrigation canals)
Idukki Dam (इडुक्की बांध)	**Periyar**	Kerala	Concrete **double-curvature arch dam** between Kuravan & Kurathi hills; underground power house at Moolamattom
Mullaperiyar Dam	**Periyar**	Kerala (leased to Tamil Nadu)	Masonry gravity dam built 1895 by British engineer John Pennycuick; administered by TN
Mettur Dam (Stanley Reservoir)	**Cauvery**	Tamil Nadu (Salem)	Built 1934; chief irrigation reservoir for Cauvery delta farmers
Krishna Raja Sagara (KRS)	**Cauvery**	Karnataka (Mandya/Mysuru)	Designed by Sir **M. Visvesvaraya**; Brindavan Gardens located below dam
Ukai Dam (Vallabh Sagar)	**Tapi**	Gujarat (Surat)	Second largest reservoir in Gujarat after Sardar Sarovar
Kakrapar Project	**Tapi**	Gujarat	Weir project for irrigation + Kakrapar Atomic Power Station
Koyna Dam (कोयना बांध)	**Koyna** (tributary of Krishna)	Maharashtra (Satara)	Largest completed hydroelectric power plant in Maharashtra; triggered **1967 Koynanagar earthquake**
Damodar Valley Project (DVC)	**Damodar & Barakar**	Jharkhand & West Bengal	First multipurpose river valley project of independent India (1948); modeled on US **Tennessee Valley Authority (TVA)**; dams at **Tilaiya, Maithon, Belpahari** (on Barakar) and **Panchet** (on Damodar); converted Damodar from "Valley of Sorrow" to "Valley of Plenty"
Tungabhadra Project	**Tungabhadra** (Krishna trib.)	Karnataka & Andhra Pradesh	Major multipurpose joint project (dam at Mallapuram, 2441 m long, 49.3 m high); powerhouses at Mallapuram & Hampi
Pochampad Project (Sri Rama Sagar)	**Godavari**	Telangana (Nizamabad)	Major lifeline irrigation project for North Telangana
Jayakwadi Project	**Godavari**	Maharashtra (Paithan, Chhatrapati Sambhajinagar)	Huge earthen dam across Godavari; reservoir named **Nath Sagar**
Lower Sileru & Upper Sileru	**Sileru** (Godavari basin)	Andhra Pradesh & Odisha	Major hydel stations on the AP-Odisha border
Sabarigiri Project	**Pamba & Kakki**	Kerala (Pathanamthitta)	Second largest hydroelectric project in Kerala
Sharavati Project	**Sharavati**	Karnataka (Jog Falls / Gersoppa)	Major hydroelectric project at Mahatma Gandhi Hydroelectric Works
Ghataprabha & Malaprabha	**Ghataprabha & Malaprabha**	Karnataka (Belagavi / Bagalkote)	Irrigation reservoirs in the Krishna basin
Bhima Project	**Bhima / Pawana**	Maharashtra (Ujjani Dam, Solapur)	Multipurpose irrigation and hydel project
Banasagar Project	**Son**	Madhya Pradesh, UP & Bihar	Joint interstate project (shared cost and benefit 2:1:1 by MP, UP, Bihar) at Deolond (Shahdol)

Project / Dam (बांध / परियोजना)	River (नदी)	State / Union Territory	Key Characteristics & Exam Notes
Thein Dam (Ranjit Sagar Dam)	**Ravi**	Punjab & Jammu & Kashmir	Highest earth-fill gravity dam in Punjab; hydel and irrigation
Mayurakshi Project (Canada Dam)	**Mayurakshi**	West Bengal & Jharkhand	Dam at Massanjore (built with Canadian assistance)
Kangsabati Project	**Kangsabati & Kumari**	West Bengal (Bankura/Purulia)	Earthen dam for flood control and irrigation
Papanasam Project	**Thamirabarani**	Tamil Nadu (Tirunelveli)	Hydroelectric power station
Kundah Hydroelectric Project	**Kundah & Bhavani**	Tamil Nadu (Nilgiris)	Multi-stage cascade hydel system built with Canadian aid
Loktak Hydroelectric Project	**Manipur River / Loktak Lake**	Manipur (Churachandpur)	Power generation and lift irrigation via Loktak Lake trans-basin tunnel
Sharda Sahayak Project	**Ghaghara & Sharda**	Uttar Pradesh	Major canal diversion feeder system for eastern/central UP plains

Multipurpose valley cards (Ghatnachakra)

Project	River / place	Note
Telugu Ganga	Krishna → **Chennai** drinking water	Joint TN–AP–KA–MH
Chukha	Wang Chu / Raidak, **Bhutan**	India-aided; handed to Bhutan **1991**; ~**336 MW**
Mayurakshi	Mayurakshi; Canada / Massanjore dam in JH	Shared **WB–JH**
Babli	Godavari, **Maharashtra**	Disputed with Telangana / AP
Mahakali Treaty	Sharda (शारदा) / Mahakali	India–**Nepal** (**1996**); Sharda, Tanakpur, Pancheshwar
Gandak Project	Gandak Barrage (**1959** treaty)	India–Nepal; **UP + Bihar** (+ Nepal)
Kishau	**Tons** (टौंस) (Yamuna)	HP–UK border; benefits HP/UK (+ others)
Kalpong	Kalpong	First hydro of **Andaman (अंडमान) & Nicobar (निकोबार)**
Rihand / Pant Sagar	Rihand (Son tributary), Pipri, Sonbhadra	Largest UP multipurpose note; Obra power below dam
Rajghat / Rani Lakshmibai (रानी लक्ष्मीबाई)	**Betwa** (बेतवा), Lalitpur	Joint UP–MP
Matatila	Betwa	Joint UP–MP
Pochampad	Godavari	Telangana
Tapovan–Vishnugad	**Dhauliganga** (धौलीगंगा), Chamoli (चमोली) UK	
Dulhasti	**Chenab** (चिनाब), Kishtwar	
Baglihar	**Chenab**, Ramban	IWT / World Bank dispute note
Suil	Suil (Ravi tributary), Chamba HP	
Kol Dam	**Sutlej**, HP	NTPC; silt check for Bhakra note
Meja (RJ)	**Kothari** (Banas), Bhilwara	Irrigation — not UP's Meja

Project	River / place	Note
Kalpasar	Gulf of Khambhat, **Gujarat**	Freshwater reservoir / tidal idea
Teesta hydro	Teesta	Sikkim / West Bengal (TLDP III–IV = WB)
Malampuzha	Malampuzha (Bharathapuzha), Kerala	Largest Kerala reservoir note
Aliyar / Isapur / Kangsabati	—	All are **water reservoirs**

Tribunal	Year / note	Parties (count)
Mahadayi	**2010**	Goa, Karnataka, Maharashtra (**3**)
Godavari (गोदावरी) (Bachawat)	**April 1969**	AP, MP, Chhattisgarh, Odisha, Karnataka (**5**)
Narmada	**6 Oct 1969**	Gujarat, MP, Maharashtra, Rajasthan (**4**)
Mahanadi	**2018**	Odisha, Chhattisgarh (**2**)

River-wise dam map (match fodder)

Indus (सिंधु) system

- **Pandoh** is on the **Beas** (ब्यास). It is not on the Ravi.
- **Pong / Maharana Pratap Sagar** is also on the Beas in Himachal (bird sanctuary reservoir).
- **Ranjit Sagar (Thein)** is on the **Ravi** (रावी).
- **Baglihar**, **Salal** and **Dulhasti** are on the **Chenab**.
- **Nathpa Jhakri** and **Kol Dam** are Sutlej hydro works in Himachal.

Ganga–Yamuna–UP

- **Tehri** and **Koteshwar** stand on the Bhagirathi (Bhilangana joins at the Tehri site).
- **Rihand** (रिहंद) dam makes **Govind Ballabh Pant Sagar** at Pipri, Sonbhadra.
- **Matatila** and **Rajghat (Rani Lakshmibai)** are Betwa projects serving Bundelkhand.
- **Narora** is a Ganga **barrage**, not a high dam.
- **Kalagarh** is the site of the **Ramganga Dam**.

Narmada–Tapi–Chambal

- Narmada cascade in MP includes **Bargi, Indira Sagar, Omkareshwar, Maheshwar**, then **Sardar Sarovar** in Gujarat.
- **Gandhi Sagar**, **Rana Pratap Sagar**, **Jawahar Sagar** and **Kota Barrage** are the Chambal chain (**MP–RJ**).
- **Ukai** and **Kakrapar** are on the **Tapi** (तापी) in Gujarat.
- **Ban Sagar** is on the **Son**, not the Narmada.

East-flowing peninsula

- **Srisailam** and **Almatti** and **Nagarjuna Sagar** are on the **Krishna**.
- **Tungabhadra (तुंगभद्रा) Dam** is on the Tungabhadra (KA–AP joint).
- **Jayakwadi** and **Pochampad** are on the Godavari.
- **Nizam (निजाम) Sagar** is on the **Manjra**, a Godavari tributary.
- **Hirakud** is on the Mahanadi. **Rengali** is on the Brahmani.
- **Mettur** (Stanley Reservoir) is on the Cauvery in Tamil Nadu.
- **Krishna Raja Sagara (KRS)** and **Shivasamudram** are on the Cauvery in Karnataka.

West-flowing Ghats and others

- **Koyna** is on the Koyna (Krishna system) in Maharashtra. The **1967** shock is the textbook **reservoir-induced seismicity** case.
- **Idukki** / **Cheruthoni** are on the **Periyar** in Kerala.

- **Mullaperiyar** is also on the Periyar. Kerala hosts the dam. Tamil Nadu operates it.
- **DVC** dams **Tilaiya, Konar, Maithon, Panchet** sit on the Damodar–Barakar system.

Canals

Canal	Source / offtake	Region	Tag
Gang Canal	Sutlej at **Hussainiwala** (Firozpur)	North Rajasthan / Bikaner command	Among world's **oldest developed** systems; **1927**; Maharaja **Ganga Singh**
Indira Gandhi Canal	**Harike Barrage** (Sutlej–Beas confluence, Punjab)	Ganganagar, Bikaner, Jodhpur, Jaisalmer	India's **longest** (~**649 km** main); foundation **Mar 1958** (G.B. Pant); also draws Beas–Ravi–Sutlej waters (Pong / Beas project support)
Upper Ganga Canal	Ganga at **Bhimgoda, Haridwar** (right bank)	W-UP Doab (Haridwar–Saharanpur–Muzaffarnagar–Meerut (मेरठ)–Ghaziabad–Bulandshahr–Aligarh–Etah–Mathura (मथुरा)–Firozabad–Mainpuri–Agra belt)	Father often tagged **Sir P.T. Cautley**; first discharge **8 Apr 1854**; main ~**298 km**; system ~**6496 km**; also feeds Delhi (~200 cusec) and Harduaganj TPP (~100 cusec)
Lower Ganga Canal	Ganga at **Narora** (Bulandshahr)	Central / lower Doab	Completed **1878**; main ~**99 km**; system ~**8278 km**
Eastern Yamuna Canal	Yamuna left bank at **Tajewala**; **Hathinikund** barrage ~3 km upstream (Haryana)	Hindon–Yamuna Doab: Saharanpur, Muzaffarnagar, Meerut, Ghaziabad	Revived functioning from **1830**; main ~**197 km**
Agra Canal	Yamuna (Okhla)	Agra–Mathura belt	
Sharda Canal	Sharda right bank near **Banbasa** (Nainital / UK border)	Protective irrigation across Rohilkhand–Awadh (अवध)–Purvanchal (पूर्वांचल) (Pilibhit to Prayagraj (प्रयागराज) belt)	Main ~**44 km**; system ~**9961 km** (longest UP canal-system length fact)
Sharda Sahayak / CAD	Sharda command	E / central UP	CADP projects also include Ramganga and Gandak
Betwa Canal	Betwa	Bundelkhand	
Bhakra / Sirhind	Sutlej system	Punjab–Haryana	
Buckingham Canal	East-coast backwaters	AP (Kakinada) → TN (Viluppuram); Coromandel	Freshwater **navigation** canal (~796 km), not IGC
Gandak / **Saran** canals	Gandak barrage near **Triveni / Balmiki Nagar** (~740 m)	NW Bihar	Saran irrigation canal drawn from this system
Ukai Canal	Tapti (ताप्ती) / Ukai	Gujarat	
Ghataprabha Canal	Ghataprabha	Karnataka	
Nira Canal	Nira (Bhima tributary)	Maharashtra	
Jawai Project	Jawai (Luni (लूनी) tributary)	**Rajasthan**	Trap: not Tamil Nadu
Mangalam	Bharathapuzha system	**Palakkad, Kerala**	One of several projects on that river

Irrigation project classes (CCA)

Class	Culturable Command Area	Share of irrigation potential
Minor	Up to **2,000 ha** (wells, tubewells, tanks, lift, drip, sprinkler)	About **62%**
Medium	**2,000–10,000 ha**	Part of the remaining ~**38%** with major
Major	Above **10,000 ha**	Major + medium together about **38%**

Common Traps — Don't Fall For These

1. **Sambhar = largest saline (any wording)** — FALSE unless **inland**. Else often **Chilika**.
2. **Chilika in Rajasthan / freshwater** — FALSE. **Odisha coastal lagoon**.
3. **Kolleru = largest east-coast lagoon** — FALSE. **Chilika**; Kolleru is freshwater.
4. **Keibul Lamjao on Wular / Dal** — FALSE. **Loktak, Manipur**.
5. **Jog = highest waterfall** — FALSE. **Kunchikal** (Varahi, KA) for UPPCS. Jog = renown / **width**; Shivanasamudra = **volume**. 5a. **Lodh / Budha Ghagh on Kanchi or Barakar** — FALSE. On the **Burha** (Latehar). 5b. **Roopkund in Himachal** — FALSE. **Uttarakhand** Mystery / Skeleton Lake. 5c. **Periyar Lake is a lagoon / kayal** — FALSE. It is **artificial**. Kayals are Kerala coastal lagoons. 5d. **Hokera in Punjab / Sasthamkotta in Tamil Nadu** — FALSE. Hokera = **J&K**; Sasthamkotta = **Kerala**.
6. **Dudhsagar = Karnataka-only** — FALSE. The UPPCS answer places it in **Goa**.
7. **Chachai–Bihad is correct** — FALSE. That is the **wrong** pair.
8. **Onega = Canada** — FALSE. **Russia**. Ladoga is also Russia. Michigan = USA-only; Superior is shared USA–Canada.
9. **Rudrasagar / Sultanpur in UP** — FALSE. **Tripura / Haryana**.
10. **Bhakra on Beas / Ravi** — FALSE. **Sutlej**; reservoir **Gobind Sagar**.
11. **Tehri on Alaknanda** — FALSE. **Bhagirathi**, rock-fill.
12. **Ramganga irrigates 6 lakh ha** — FALSE. **~5.75–5.90 lakh ha**.
13. **Sardar Sarovar = Indira Sagar** — FALSE. SSP = Narmada **Gujarat**; Indira Sagar = Narmada **MP**.
14. **Srisailam on Tungabhadra / Pandoh on Ravi** — FALSE. **Krishna / Beas**.
15. **E-UP/Bihar can fully tap Himalayan hydro to their north** — FALSE. Those ranges lie largely in **Nepal**.
16. **Sharavati is a Cauvery tributary** — FALSE. It is **west-flowing** to the Arabian Sea.
17. **Kapildhara is on the Godavari** — FALSE. It is on the **Narmada**.
18. **Jog is India's highest waterfall** — FALSE. **Kunchikal** is highest; Nohkalikai is the tallest plunge.
19. **Hirakud is India's tallest dam** — FALSE. **Tehri** is tallest. Hirakud is the long earthen dam.
20. **Idukki is on the Cauvery** — FALSE. **Periyar**, Kerala, arch dam.
21. **Industry uses most of India's groundwater** — FALSE. **Agriculture ~92%**.
22. **Lonar is a volcanic caldera** — FALSE. It is a **meteorite** crater in Deccan basalt.
23. **Dominant irrigation = canals only** — FALSE. Wells/tubewells dominate (~64% together; tubewell ~48.5% in 2018–19).
24. **IGC source = Bhakra / Gandhi Sagar** — FALSE. **Harike Barrage** (Sutlej–Beas).
25. **Gang Canal = Indira Gandhi Canal** — FALSE. Gang Canal = **1927** Ganga Singh / Hussainiwala; IGC = longest / Harike / 1958.
26. **Lower Ganga offtake = Haridwar** — FALSE. **Narora**; Upper Ganga = Bhimgoda/Haridwar.
27. **Jawai Project = Tamil Nadu** — FALSE. **Rajasthan** (Luni tributary).
28. **Garland Canal = K.L. Rao** — FALSE. **Dastur**; Visvesvaraya/Rao = linking ideas.
29. **Farakka carries 75,000 cusec** — FALSE. About **40,000 cusec**.
30. **Ghaggar feeds IGC** — FALSE.
31. **Minor irrigation = smallest potential share** — FALSE. Minor (~62%) is the largest potential share.
32. **Hariyali = second Green Revolution** — FALSE. Watershed / wasteland–water storage (Jan 2003).
33. **Ban Sagar on Narmada** — FALSE. **Son**.
34. **Harsud submerged by SSP** — FALSE. **Indira Sagar** (2004).
35. **Shivasamudram = oldest hydro** — FALSE when Sidrapong is an option. Sidrapong **1897**; Shivasamudram **1902**.
36. **DVC = 1947 / Bhakra first multipurpose** — FALSE. **DVC 1948** is first of independent India.
37. **Chambal = PB–HR–RJ** — FALSE. **RJ–MP**. Gobind / Pant Sagar are not Chambal.
38. **Tehri on Alaknanda alone** — FALSE. **Bhagirathi** (+ Bhilangana confluence).

39. **Nagarjuna Sagar on Godavari / Cauvery** — FALSE. **Krishna**.
40. **Tilaiya on Konar / Damodar main** — FALSE. **Barakar**.
41. **Telugu Ganga to Bangalore** — FALSE. **Chennai** (Krishna).
42. **Rana Pratap Sagar = Madhya Pradesh** — FALSE. **Rajasthan** (Rawatbhata).
43. **Shivasamudram = Kerala** — FALSE. **Karnataka**.
44. **Babli = Andhra Pradesh project site** — FALSE. Built by **Maharashtra** on Godavari (disputed by AP/TG).

Topic 5 — Soils

Current Affairs (this topic)

Year	Fact	Why asked	Source
19 Feb 2015	Soil Health Card launched at **Suratgarh, Rajasthan**; slogan **Swasth Dharaa, Khet Haraa**; 2015 = International Year of Soils	Scheme–place–slogan	PIB
2022–23 onward	SHC merged into RKVY as **Soil Health and Fertility**	New name vs old scheme	PIB / DA&FW
Jul 2025	**>25 crore** Soil Health Cards distributed; card reports **12 parameters** (N,P,K,S, Zn,Fe,Cu,Mn,B, pH, EC, OC)	Numbers / parameters	PIB Aug 2025
Ongoing	**WDC–PMKSY 2.0** = watershed soil–water conservation (not a soil-type topic)	Scheme vs conservation method	Jal Shakti (शक्ति) / DA&FW
Static	World Soil Day (विश्व मृदा दिवस) **5 December** (UN;)	Date trap vs 5 June	UN / PYQ

Confused Pairs

Pair	Correct	Trap	Hindi
Edaphic vs climatic	Edaphic = **soil**	Call climate "edaphic"	मृदीय = मिट्टी
Urvara vs Usara	Ancient: Urvara = fertile; Usara = sterile	Mix with modern usar chemistry	उर्वरा / ऊसरा
Residual vs transported	Residual / **sedentary** = on rock (regur/red); transported / **azonal** = alluvium	Call alluvium residual	अवशिष्ट / वाहित
Red vs alluvial area	Alluvial is **largest**; red is widespread on the peninsula	Wrong claim: "red is the largest group"	जलोढ़ सबसे बड़ा
NCERT P vs other notes on phosphoric acid	Class 11: alluvium **poor in phosphorus**	Do not copy "phosphoric acid adequate" over the NCERT poor-P fact	फॉस्फोरस कम (NCERT)
Pedocal vs laterite	Pedocal = arid **lime-accumulating**; laterite = wet **leached Fe/Al**	Mix the two climate products	पेडोकल = चूना; लैटेराइट = निक्षालित
Himalaya north vs south slope	Soil on **gentler north-facing** forested slopes; south face often too steep	Soil equally on both Himalayan faces	उत्तर ढाल मिट्टी; दक्षिण तेज
Kari vs Kuttanad vs regur	**Kari** = Kerala peaty (Kottayam–Alappuzha)	Call kari Deccan black	कारी = पीट
Eluviation vs illuviation	E = wash **out** of A/E; I = deposit **in B**	Swap	ऊपरी हानि / निचला जमाव
Khadar vs Bangar	Khadar = new/fine/fertile; Bangar = old/**more kankar**	Khadar has more kankar	खादर नया; बांगर पुराना
Black vs Himalaya	Regur = **Deccan Trap**	Black in Himalaya	काली = दक्कन
Black vs peaty black	Regur = basalt clay; peaty = organic marsh	Kerala marsh = cotton soil	पीट ≠ रेगुर
Red vs laterite	Red = Fe colour on crystalline rock; laterite = hot-wet **leached Fe/Al**	Every red = laterite	लाल ≠ लैटेराइट
Laterite Fe/Al	**Rich** in Fe/Al oxides	"Poor in Fe/Al"	लोहा–एल्युमीनियम प्रचुर
Cotton vs cashew	Cotton = black; cashew/**tapioca**/tea/coffee = laterite	Cotton on laterite as default	कपास = काली
Gypsum vs lime	**Gypsum** = alkali/sodic/usar; **lime** = **acid** only	Lime for usar (Usar BEO trap)	जिप्सम क्षारीय; चूना अम्लीय

Pair	Correct	Trap	Hindi
Splash vs sheet	**Splash** = raindrop first stage; sheet = thin wash	Start ladder at sheet only	स्प्लैश → शीट
Tea vs regur	Tea prefers **slightly acidic**, well-drained	Tea = black cotton default	चाय = अम्लीय
Zinc vs Fe micros	Among Cu/Fe/Mn/Zn, **Zn** is most often deficient in Indian soils	Iron as "most deficient"	जिंक कम
Saline vs sodic	Saline = soluble salts; sodic = exchangeable Na + high pH	One remedy for both	लवणीय / क्षारीय
Shelter belt vs check dam	Shelter belt = **wind**/dunes; check dam = **gully**	Swap agents	रक्षक पट्टी = हवा
World Soil Day	**5 December**	5 June (Environment Day)	मृदा दिवस = 5 दिसंबर
Mitti Bachao	**Hoshangabad, MP**	Darbhanga / Thane / Mysore	मिट्टी बचाओ = होशंगाबाद
Terra Roxa vs regur	Terra Roxa = **Brazil** coffee basaltic	Indian black soil	टेरा रोक्सा = ब्राज़ील
Terra Rossa	Limestone / dolomite weathering (red)	Terra Roxa Brazil coffee	टेरा रोसा
Chernozem vs Chestnut	Steppe humus-rich vs drier less humus	Swap	चेरनोज़ेम / चेस्टनट
Capillary max	**Clay** (tiny pores)	Sand	केशिका
Halophyte	**Saline** soil	Acid / cold soil	हैलोफाइट
Bhabar vs Tarai	Bhabar = pebbles, streams **vanish**; Tarai = streams **re-emerge**, moist fertile	Bhabar = best rice soil	भाबर पत्थर; तराई फसल
Mar vs Parua	Mar/Kabar = Bundelkhand **black**; Parua/Rakar = **red**	Swap colours	मार काली; परुआ लाल
UP soil identity	~90% **alluvial**	UP = black-soil state	UP = जलोढ़
Bangar kankar vs arid kankar	Bangar = old alluvium nodules; arid = **lower-horizon** Ca kankar	Same origin	बांगर / मरु कंकर
Rann salt vs usar reh	Rann crust = SW monsoon salt deposit; canal usar = irrigation+poor drain	Same cause always	कच्छ = मानसून लवण
Gypsum vs lime	**Gypsum** = alkali/sodic/usar; **lime** = **acid** only	Lime for usar (Usar BEO trap)	जिप्सम क्षारीय; चूना अम्लीय
Sorghum vs clover erosion	**Sorghum (Jowar)** = max erosion; **clover** = min erosion	Sorghum protects soil	ज्वार अपरदन अधिकतम; तिपतिया न्यूनतम
Saline leader vs Alkali leader	**Gujarat** = largest saline area; **UP** = largest alkali/sodic wasteland	Confuse saline and alkali state ranks	गुजरात लवण; UP ऊसर
Sand gradient west to east	Sand content **decreases from west to east** in plains	Sand increases eastwards	बालू पश्चिम से पूर्व घटती है
Particle diameter clay vs silt	Clay < **0.002 mm**; Silt = **0.002 to 0.06 mm**	Silt < clay	क्ले < 0.002 mm
Seed modification vs fertility	Seed mod = **yield/pest resistance**, NOT soil fertility	Seed mod improves soil fertility	बीज सुधार ≠ मृदा उर्वरता
Matasi vs Kanhar (CG)	Matasi = **red-yellow**, higher altitude, paddy; Kanhar = lower black clay	Swap	मटासी लाल-पीली; कन्हार काली
Karewa vs alluvium	Karewa = **Kashmir lacustrine** (saffron)	Call it ordinary Gangetic alluvium	करेवा = कश्मीर

N.1 Formation, Profile, Classification

How rock becomes soil (weathering)

Type	Simple meaning	What changes?	India note
Physical	Breaks rock into **smaller pieces**	Size only — chemistry stays the same	Heat cracking, frost, wind abrasion on exposed rock
Chemical	**Changes minerals** inside the rock	New compounds form (hydration, oxidation, carbonation, solution)	Strong under **hot wet monsoon** toward laterite
Biological	Life **mixes and dissolves**	Roots split rock; organic acids + burrowers loosen and enrich	Forest floors, earthworms

Residual vs transported

- **Residual** (mostly peninsular) soils form **in place** on the rock beneath them. They are also called **sedentary** soils.
- Regur on **Deccan Trap basalt** and red soil (लाल मिट्टी) on **granite–gneiss** are residual.
- **Transported** (extra-peninsular plains) soils are carried in by rivers, wind, or ice — **alluvium** is the classic. These are often tagged **azonal** (weak horizons, little relation to the rock underfoot).
- Same climate on different rock can still give different soil. **Basalt** weathers to **black** soil. **Granite–gneiss** weathers to **red** soil.
- Farming, irrigation, and deforestation can strip a profile **faster** than nature rebuilds it.

CLORPT

Factor	Role	India example
Climate	Weathering + leaching intensity	Laterite on hot-wet Ghats
Living organisms	Humus, mixing, nutrient cycling	Forest cover (वन आवरण) in the Himalaya
Relief	Depth, drainage, erosion	Thin soils on steep Himalayan slopes
Parent material	Mineral base of the soil	Deccan Trap weathers to regur
Time	How mature the horizons are	Mature black vs young khadar

Profile (top to bottom)

Horizon	Content	tag
O	Organic litter / humus	Forest floors
A	Topsoil: mineral + humus	Crop root zone
E	Leached zone (if present)	**Eluviation** (wash **out**)
B	Clay / Fe / Al / lime accumulation	**Illuviation** (wash **in**) / subsoil
C	Weathered parent, not fully soil	Transition
R	Unweathered bedrock	Base

ICAR / school classification

Soil	How it forms	Core belt	Star crop / use	Classic trap
Alluvial	River-borne silt (**transported**)	Indo-Gangetic, Brahmaputra (ब्रह्मपुत्र), E-coast deltas (NCERT ~**40%**; other notes ~**46%**)	Wheat, rice, sugarcane, jute	UP is **alluvial**, not a black-soil state
Black (Regur)	On **Deccan Trap basalt** (residual)	MH–MP–GJ–N KA–TG	**Cotton**	**Not** Himalayan

Soil	How it forms	Core belt	Star crop / use	Classic trap
Red & Yellow	On **granite–gneiss** (crystalline)	TN, KA, S MH, Odisha, Chota Nagpur	Millets, pulses	Red ≠ laterite; not always barren
Laterite	**Hot-wet** intense leaching	Ghats, NE hills, plateau edges (**Malabar**/Kerala is the classic home belt)	Cashew, **tapioca**, tea, coffee, cinchona	**Rich** in Fe/Al — not poor; not RJ/UP plains
Desert / Arid	Arid climate + sand	W Rajasthan (Thar)	Millets if irrigated	Not forever barren
Mountain / Forest	Altitude + slope	Himalaya, NE hills	Orchards, terraces	Thin on slopes; ≠ regur
Saline / Usara	Salts + poor drainage	Canal plains, arid belts, coastal deltas, **Rann of Kutch**	Needs reclamation	More water **without** drainage worsens it
Peaty / Marshy	Waterlogged organic pile-up	Kerala (**Kuttanad**), Sundarbans, N Bihar, S Uttarakhand (उत्तराखंड)	Drainage first	Black colour ≠ cotton regur

Soil ↔ state match bank (classic Prelims)

Soil Group (ICAR)	NCERT Area Share	Primary Match State (Commission Anchor)	Alternate / Secondary States (If primary absent)	Why this association? (Geological / Climatic basis)	Classic Prelims Trap / Misconception
Alluvial Soil *(जलोढ़ मृदा)*	**~40%** (Class 11) ~46% (other)	**Uttar Pradesh** (उत्तर प्रदेश)	Bihar, West Bengal, Punjab, Haryana, Coastal deltas (AP/Odisha/TN)	Extensive Indo-Gangetic river floodplains; transported silt (*khadar* & *bangar*)	**Trap:** Bundelkhand (southern UP) is **not** alluvial (it has black *Mar/Kabar* & red *Rakar/Parua*); but UP as a whole holds the single largest alluvial tract (~90% of state area).
Black Soil / Regur *(काली / रेगुर मृदा)*	**~15–16%**	**Maharashtra** (महाराष्ट्र)	Madhya Pradesh (Malwa), Gujarat (Kathiawar), North Karnataka, Telangana	Formed *in situ* by weathering of **Deccan Trap basaltic lava**; rich in titaniferous magnetite; high clay & moisture retention	**Trap:** Regur is strictly peninsular volcanic; **never** match with Himalaya or Gangetic plains. Do not mistake local black patches in eastern India (Rajmahal) as main Deccan regur.
Red & Yellow Soil *(लाल एवं पीली मृदा)*	**~18.5%** *(2nd largest)*	**Andhra Pradesh** (आंध्र प्रदेश) / **Tamil Nadu**	Odisha, Chhattisgarh, Chotanagpur (Jharkhand), Southern Karnataka	Formed on **ancient crystalline & metamorphic granite–gneiss** rocks under low-to-medium rainfall; red from ferric oxide (Fe_2O_3) diffusion	**Trap:** Red soil is **not** laterite! Red soil has good tilth and responds well to fertilizers (pulses, millets, oilseeds); laterite is intensely leached, acidic, and bakes rock-hard.
Laterite Soil *(लेटराइट मृदा)*	**~3.7–4.3%**	**Kerala** (केरल - Malabar Coast)	**Maharashtra** (Western Ghats crests/Konkan if Kerala absent), Karnataka, Meghalaya (Shillong plateau)	Intense **tropical leaching** under alternate wet & dry seasons; silica & lime washed down (*eluviation*), Fe & Al oxides left on top; bakes like brick (*later*)	**Trap:** First identified in **Angadipuram, Kerala (Malabar)** by Francis Buchanan (1807). If Kerala is not in the options, look for **Maharashtra / Karnataka** (Western Ghats). Never pick northern alluvial plains (UP/Punjab) or desert (Rajasthan).
Desert / Arid Soil *(मरुस्थलीय मृदा)*	**~4.3–4.4%**	**Rajasthan** (राजस्थान - Western/Thar)	Northern Gujarat (Kutch fringe), Southern Haryana & Punjab	Hyper-arid to semi-arid climate, mechanical weathering; sandy texture, alkaline pH, high soluble salts, thick subsoil *kankar* (calcareous) pan	**Trap:** Not permanently sterile — highly productive when irrigated (Indira Gandhi Canal proved this for wheat/mustard/cotton). Do not match with wet NE states like Meghalaya!

Soil Group (ICAR)	NCERT Area Share	Primary Match State (Commission Anchor)	Alternate / Secondary States (If primary absent)	Why this association? (Geological / Climatic basis)	Classic Prelims Trap / Misconception
Forest & Mountain Soil *(पर्वतीय / वन मृदा)*	~**8.6%**	**Jammu & Kashmir / Ladakh / Himachal Pradesh / Uttarakhand**	Sikkim, Arunachal Pradesh, higher slopes of Western & Eastern Ghats	Formed under forest cover on steep mountain slopes; thin, skeletal, immature profile (*azonal*); acidic with slow humus decomposition in cold climates	**Trap:** Soil depth varies sharply by relief — thin on steep slopes, deep and fertile in valley bottoms (Kashmir, Doon, Kullu valleys).
Saline & Alkaline Soil *(लवणीय एवं क्षारीय / ऊसर / रेह)*	~**2.1%**	**Uttar Pradesh** (Western & Central canal tracts) / **Gujarat** (Rann)	Punjab, Haryana, Coastal saline tracts (Sundarbans, Chilika)	High evaporation exceeding rainfall + canal over-irrigation without underground drainage → capillary rise of sodium, calcium & magnesium salts	**Trap: Gypsum ($CaSO_4 \cdot 2H_2O$)** is used for alkaline/sodic soils (replaces exchangeable Na with Ca); **Lime ($CaCO_3$)** is for acidic soils (low pH). Never use lime on alkaline/usar soils!
Peaty & Marshy Soil *(पीट एवं दलदली मृदा)*	Localized (~**1%**)	**Kerala** (Kuttanad & Alappuzha/Kottayam - local **Kari** soil)	Sundarbans (West Bengal), Coastal Odisha, Northern Bihar (Koshi floodland), Tarai (Uttarakhand)	Waterlogged conditions in humid tracts with poor drainage; slow anaerobic breakdown of dead vegetation → massive organic accumulation (humus **40–50%**)	**Trap:** Peaty soil is heavy and jet black, but it is **not** regur/black cotton soil! Regur is dry/semi-arid basaltic alkaline soil; Peaty is submerged, acidic, and organic-heavy.

Local / Regional Formation	Exact Match Location	Geological / Pedological Fact	Commission Trap to Avoid
Karewa Soil *(करेवा)*	**Jammu & Kashmir** (Kashmir Valley & Bhadarwah)	Thick lacustrine (lake-bed) silt & clay deposits from Pleistocene glaciation; famous for **Zafran (Saffron)**, almond, and walnut orchards.	Not river alluvium; not volcanic. It is specifically **lacustrine** (झील निक्षेप).
Jharkhand Black Soil Pocket	**Rajmahal Hills** (Santhal Pargana, Jharkhand)	Derived from localized weathering of **Rajmahal basaltic traps** (Jurassic volcanism), forming black clayey patches.	**Major Trap:** Do **not** pick the Damodar or Subarnarekha valley as black soil! Damodar valley is a Gondwana sedimentary rift basin (sandstone/shale/coal), not basalt.
Matasi Soil *(मटासी)*	**Chhattisgarh** (Mahanadi Basin plains)	Yellowish-red sandy-loam soil on slightly elevated drained uplands; rich in silica/iron; traditional staple soil for paddy.	Regional state-exam fact (CGPCS / UPPCS); part of the *Bhata → Matasi → Dorsa → Kanhar* toposequence, not a separate ICAR class.
Bhabar vs Tarai Belt	**Sub-Himalayan Foothills** (Piedmont strip from Indus to Teesta)	**Bhabar:** Porous pebble-gravel-boulder strip along Shiwalik foothills where mountain streams vanish underground (coarse, dry surface, unsuited for farming). **Tarai:** Immediately south of Bhabar; damp, marshy, fine-silted waterlogged belt where lost streams re-emerge; rich in nitrogen & organic matter, excellent for sugarcane and paddy.	Do not invert their spatial order: **Shiwalik → Bhabar (north) → Tarai (south)**. Bhabar streams disappear; Tarai streams re-emerge.

Local Name	Region in UP	Core Soil Group	Distinguishing Characteristics & Crop Fit
Mar (मार)	Bundelkhand (Jalaun, Jhansi, Hamirpur)	**Black Soil** (Regur variant)	Highly clayey, deep black, extremely sticky when wet and cracks wide upon drying (*self-ploughing*); rich in iron & lime; prime for rabi gram, wheat, and oilseeds.
Kabar (काबर)	Bundelkhand (Banda, Hamirpur, Mahoba)	**Black Soil** (Coarse/compact variant)	Dark brown to blackish-grey; contains more sand than Mar; quickly hardens into unworkable stone-like clods if moisture is lost; highly sensitive to plowing window.
Parua / Padwa (परुआ)	Bundelkhand (Hamirpur, Jalaun, Yamuna ravines)	**Red Soil** (Sandy-loam variant)	Light reddish to yellowish-brown sandy loam; well-aerated with good drainage; low organic matter; highly productive when provided with light irrigation (mustard, wheat, jowar).
Rakar (राकर)	Bundelkhand (Sloping terrain, ravines, hill bases)	**Red Soil** (Gravelly/eroded variant)	Coarse, shallow, gravelly red soil formed on eroded granite knolls; low fertility, poor water retention; classified into *Moti Rakar* (gravelly) and *Patli Rakar* (fine); suited only for coarse millets (*bajra*, *sesamum*).
Bhonta / Monta (भोंटा)	Vindhyan tracts (Mirzapur, Sonbhadra, Prayagraj trans-Yamuna)	**Vindhyan Red Sandy Soil**	Coarse, reddish-brown stony soil formed from Vindhyan sandstone and shale weathering; low fertility, drought-prone; supports dry shrubs, pulses, and rainfed coarse grains.
Matiyar / Dhub (मटियार)	Ganga-Yamuna Doab & Eastern UP	**Alluvial Clay-Loam**	Fine clayey-loam alluvium with high moisture-retention capacity; prime paddy, wheat, and sugarcane belt.

Stem to type (decision map)

Diagnostic Clue / Trigger Keyword in Stem	Target Soil / Formation	Key Scientific Rationale & Mechanism	Examiner's Trap / Distractor
"Self-ploughing" *(स्वतः जुताई)* / deep polygonal fissures when dry, swelling & unctuous/sticky when wet	**Black Soil / Regur** *(काली / रेगुर मृदा)*	High content of **montmorillonite** clay mineral; swells substantially on hydration, shrinks and cracks wide on drying, causing natural soil churning.	Not alluvial clay; not peaty soil.
"High moisture retention" / retains subsoil moisture for months; sustains rainfed rabi crops long after rains cease	**Black Soil / Regur** *(काली / रेगुर मृदा)*	Extremely fine, tight argillaceous texture and high capillary water-holding capacity.	Do not pick Alluvial or Red soil (both drain significantly faster).
"Bakes like brick" upon exposure to sun and drying / alternate wet-and-dry tropical leaching	**Laterite Soil** *(लेटराइट मृदा)*	Latin *later* = brick; intense tropical leaching washes away silica and bases (*desilication*), leaving an indurated residual crust of iron and aluminium oxides.	**Trap:** Red soil looks red from iron diffusion, but does **not** bake into rock-hard building bricks.
"River-borne transported sediments" / *khadar* (newer, light, annual flood silt) vs *bangar* (older, clayey, terrace alluvium)	**Alluvial Soil** *(जलोढ़ मृदा)*	Deposited by Himalayan and Peninsular rivers; azonal soil covering ~40% (NCERT) to ~46% of India's plain area.	UP and Bihar are overwhelmingly alluvial; do not assume all plain soils are homogeneous bangar.
"Kankar nodules" *(कंकड़ / Calcareous nodules)* in lower horizons / impure calcium carbonate concretions	**Old Alluvium (Bangar)** OR **Arid Soil**	Downward leaching and precipitation of calcium carbonate ($CaCO_3$) in subsoil under sub-humid to arid regimes.	*Khadar* (new alluvium) is completely devoid of kankar; kankar is characteristic of *Bangar*.
"Thick sandy texture + high soluble salts + impermeable subsoil kankar layer"	**Desert / Arid Soil** *(मरुस्थलीय मृदा)*	Mechanical disintegration dominates over chemical decay under arid conditions; high evaporation draws salts upward; kankar layer restricts root penetration.	Not permanently barren; highly productive when irrigated (e.g., Indira Gandhi Canal command area).
"Thin, skeletal, immature profile on steep slopes" / highly vulnerable to sheet erosion	**Mountain & Forest Soil** *(पर्वतीय मृदा)*	Relief and slope prevent horizon maturation; azonal/skeletal; acidic with raw un-decomposed humus in snow-bound belts; fertile in valley bottoms.	Not regur; not transported plains alluvium.

Diagnostic Clue / Trigger Keyword in Stem	Target Soil / Formation	Key Scientific Rationale & Mechanism	Examiner's Trap / Distractor
"Waterlogged organic mass" / high humus (40–50%) / dark, heavy, acidic / submerged lowlands	**Peaty & Marshy Soil** *(पीट एवं दलदली मृदा)*	Anaerobic conditions in humid wetlands prevent bacterial oxidation of dead plant matter (e.g., Kuttanad 'Kari' in Kerala).	**Trap:** Dark/black color is from decaying organic humus, **not** basaltic volcanic ash or regur titaniferous magnetite!

Diagnostic Clue / Chemical Trait in Stem	Soil Type / Problem Class	Required Soil Amendment / Remedy	Examiner's Trap
White salt efflorescence / crust on surface (*Reh, Kallar, Thur, Luni, Chopan*)	**Saline Soil** *(लवणीय मृदा / Solonchak)*	Leaching with fresh water + installing underground sub-surface tile drainage; salt-tolerant crops (barley, cotton).	Do not add lime or chemical precipitants; physical flushing and drainage are essential.
High exchangeable sodium (Na^+) / pH > 8.5 / black alkali spots (*Usar*)	**Alkaline / Sodic Soil** *(क्षारीय मृदा / Solonetz)*	**Gypsum ($CaSO_4 \cdot 2H_2O$)** or Iron Pyrites (FeS_2) to replace exchangeable Na^+ with Ca^{2+}.	**Major Trap:** Adding Lime ($CaCO_3$) worsens alkaline soil! Lime is exclusively for acidic soils.
Low soil pH (< 5.5) / high acidity / strong leaching of bases	**Acidic Soil** *(अम्लीय मृदा - Laterite / Hill soils)*	**Agricultural Lime** (Quicklime CaO, Slaked lime $Ca(OH)_2$, or Limestone $CaCO_3$), Dolomite, or Wood Ash.	Never prescribe gypsum for acidic soils; gypsum does not raise soil pH significantly.
Rich in Potash and Lime, but universally deficient in Nitrogen, Phosphorus & Humus	**Alluvial & Black Soils of India**	Supplement with NPK fertilizers (ideal Indian ratio **4:2:1**); green manuring with *Dhaincha* or *Sunnhemp*.	Almost all Indian soils (except peaty/forest) are inherently deficient in **Nitrogen, Phosphorus, and Humus**.
Rich in Iron & Aluminium, severely depleted in Silica, Lime, Magnesia, and Potash	**Laterite Soil** *(लेटराइट मृदा)*	Requires heavy doses of manures, lime, and nitrogenous-phosphatic fertilizers to sustain plantation crops.	Candidates confuse Laterite with Black soil; Laterite has been leached of its bases and silica.

Crop / Vegetation Clue in Stem	Direct Target Soil	Typical Geographic Belt
Cotton (कपास) / "Black Cotton Soil" / Rainfed pulses & oilseeds	**Black Soil / Regur**	Deccan Trap: Maharashtra, Malwa (MP), Saurashtra (GJ), North Karnataka
Zafran / Saffron (केसर / जाफरान) / Almond & Walnut orchards	**Karewa Soil** *(करेवा)*	Kashmir Valley (Pampore plateau terraces) & Bhadarwah (J&K)
Cashew nut (काजू), Tapioca, Tea, Coffee, Rubber, Cinchona	**Laterite Soil** *(लेटराइट)*	Western Ghats slopes (Malabar/Kerala, Coorg), Meghalaya hills
Wheat, Rice, Sugarcane, Jute, Mustard (Intensive foodgrain bowl)	**Alluvial Soil** *(जलोढ़)*	Indo-Gangetic Plains (UP, Bihar, WB, Punjab, Haryana)
Bajra, Jowar, Ragi, Pulses, Castor, Groundnut (Rainfed coarse crops)	**Red Soil** *(लाल मृदा)*	Rayalaseema (AP), Tamil Nadu plains, Chotanagpur plateau, Odisha
Below sea-level paddy farming in waterlogged backwater polders	**Kuttanad Kari Soil**	Alappuzha & Kottayam districts (Kerala)

Local Nomenclature / Clue Phrase	Exact Regional Setting	Associated Soil Class & Properties
Mar (मार)	Bundelkhand (Jalaun, Jhansi, Hamirpur)	Deep black clayey regur variant; extremely sticky, highly fertile for gram and wheat.
Kabar (काबर)	Bundelkhand (Banda, Hamirpur, Mahoba)	Coarser black soil variant; sets into rock-hard clods upon drying; plowing must be timed precisely.
Parua / Padwa (परुआ)	Bundelkhand (Hamirpur, Jalaun, Yamuna ravines)	Light reddish/yellowish sandy loam; responds well to light irrigation and fertilizers.

Local Nomenclature / Clue Phrase	Exact Regional Setting	Associated Soil Class & Properties
Rakar (राकर)	Bundelkhand (Sloping edges, ravine fringes, hill tops)	Shallow, gravelly, eroded red soil; low water retention; suitable only for *bajra* or sesamum.
Bhonta / Monta (भोंटा)	Vindhyan hill tracts (Mirzapur, Sonbhadra, Prayagraj trans-Yamuna)	Coarse, stony, reddish sandstone-derived soil; low fertility, rainfed crops.
Bhabar (भाबर)	Sub-Himalayan foothill strip (northern UP / Uttarakhand border)	Porous gravel-and-boulder belt; surface water sinks and flows underground; unsuited for crops.
Tarai (तराई)	South of Bhabar strip	Damp, waterlogged silty belt; underground streams re-emerge; high nitrogen & organic matter; sugarcane and paddy belt.
Bhur (भूर)	Ganga-Yamuna Doab elevated banks	Undulating, elevated sandy mounds formed by wind-blown sand deposits during dry summer months.
Matasi (मटासी)	Chhattisgarh plains (Mahanadi Basin)	Yellowish-red sandy-loam soil on drained higher sites; staple for rainfed rice.

N.1A World soil belts · phytotypes

World Zonal Soils (Dokuchaev Tradition)

Soil Type	Etymology & Literal Meaning	Climate Belt & Biome	Pedological Formation & Characteristics	Classic Prelims Match / Key Trap
Chernozem *(चेरनोज़ेम)*	Russian: *Cherniy* (Black) + *Zemlya* (Earth) = **"Black Earth"**	**Temperate Grasslands** (Eurasian Steppes of Ukraine/Russia, North American Prairies)	Annual death of dense fibrous grass roots; cold winters slow bacterial decay → colossal accumulation of **dark humus (up to 10–16%)**; rich in calcium and moisture.	**The world's most fertile natural soil.** Known as the global "Breadbasket of the World" (wheat belt).
Chestnut *(चेस्टनट / भूरी स्टेपी मृदा)*	Color of chestnut nut (**Dark brown**)	**Semi-Arid Steppe Margins** (drier edge of Chernozem toward deserts)	Similar to Chernozem but receives **lower rainfall** → lower grass biomass → less humus. The color lightens from pitch black to chestnut brown.	Transitional buffer between humid Chernozem and dry Sierozem.
Prairie *(प्रेयरी / Brunizem)*	French: *Prairie* = **Meadow**	**Humid Temperate Grasslands** (US Midwest Corn & Soybean Belt)	Higher rainfall than Chernozem; dense tall-grass growth yields rich humus, but higher moisture causes mild base leaching.	Transitional between Chernozem (grassland) and Podzol (forest). Highly productive.
Podzol *(पोडज़ोल)*	Russian: *Pod* (under) + *Zola* (ash) = **"Under-ash"**	**Cold Temperate Coniferous (Taiga / Boreal)** forests (Canada, Siberia, Scandinavia)	Acidic pine needles + heavy snowmelt and organic acids leach out iron and humus downward, leaving a **bleached, ash-grey silica layer** (E-horizon). Strongly acidic and nutrient-poor.	**Acidic forest soil.** Classic Prelims pairing: Taiga/Coniferous woodland + ash-grey leached soil.
Spodosol *(स्पोडोसोल)*	Greek: *Spodos* (wood ash)	Humid cold temperate climates	**USDA Soil Taxonomy equivalent** for the Podzol family.	Treat Spodosol and Podzol as synonymous in civil services papers.
Laterite *(लेटराइट)*	Latin: *Later* = **Brick**	**Hot-Wet Tropical / Monsoon** belts with alternating wet and dry seasons	Heavy monsoonal downpours wash away silica and bases (*desilication/eluviation*), leaving an insoluble crust of **Iron (Fe) and Aluminium (Al) oxides**. Bakes rock-hard when dry.	**Intensely leached & acidic.** Poor for cereal grains; ideal for **cashew, tapioca, tea, coffee, rubber, cinchona**.
Sierozem *(सिरोज़ेम)*	Russian: *Seriy* (Grey) + *Zemlya* (Earth) = **"Grey Earth"**	**Mid-Latitude Temperate Deserts** and arid scrub (Central Asia, Great Basin)	Rainfall is so low that plant cover is sparse; negligible humus accumulation → pale ash-grey tint. High evaporation draws lime/calcium salts to the surface.	High in mineral salts/lime (*pedocal*), but severely deficient in organic humus and water.

Soil Type	Etymology & Literal Meaning	Climate Belt & Biome	Pedological Formation & Characteristics	Classic Prelims Match / Key Trap
Terra Rossa *(टेरा रोसा)*	Italian: *Terra* (Earth) + *Rossa* (Red) = **"Red Earth"**	**Mediterranean Karst Terrain** (Italy, Greece, Spain, Croatia)	Formed exclusively on **Limestone / Dolomite** bedrock. Rainwater dissolves and carries away soluble $CaCO_3$, leaving behind an insoluble residual **red clay mantle** enriched with ferric iron.	**High-Yield Prelims Pair:** Limestone weathering + Mediterranean climate.

Phytotypes ↔ Ecological Substrate (Plant Habitat Classification)

Plant Label (पादप वर्ग)	Prefix Meaning	Everyday Science Anchor Word	Natural Habitat & Substrate	Typical Examples
Halophyte *(लवणमृद्भिद)*	**Halo** = Salt	**Halogen** *(Chlorine/Iodine form salts like NaCl)*; **Halite** *(rock salt)*	**Saline soils / Coastal salt marshes** *(लवणीय मृदा)*	Mangroves (*Rhizophora*, Sundari), *Salicornia*
Cryophyte *(शीतमृद्भिद)*	**Cryo** = Extreme Cold / Ice	**Cryogenic Engine** *(ISRO's liquid H_2/O_2 rocket engine)*; **Cryotherapy** *(ice therapy)*	**Cold, Polar, Tundra & Snowfields** *(शीत / टुंड्रा क्षेत्र)*	Snow mosses, alpine lichens
Lithophyte *(शैलोद्भिद)*	**Litho** = Stone / Rock	**Lithosphere** *(rocky crust)*; **Paleolithic** *(Old Stone Age)*	**Bare Rock Surfaces & Cliffs** *(चट्टानी धरातल)*	Rock-dwelling lichens, ferns, orchids
Xerophyte *(मरुद्भिद)*	**Xero** = Dry	**Xerox** *(literally "dry copying" without liquid ink)*; **Xerophthalmia** *(dry eyes)*	**Tropical Deserts & Arid Regimes** *(शुष्क मरुस्थल)*	Cactus (*Opuntia*), Acacia (*Babool*), Khejri
Hydrophyte *(जलोद्भिद)*	**Hydro** = Water	**Hydration**, **Hydroelectric**, **Hydrogen**	**Open Water / Aquatic** *(floating or submerged)*	Lotus, Water Lily, *Hydrilla*, *Eichhornia*
Hygrophyte *(आर्द्रतोद्भिद)*	**Hygro** = Moisture / Humidity	**Hygrometer** *(instrument to measure air humidity)*	**Swamps, Marshes & High Humidity** *(दलदली / नम स्थल)*	Marsh grasses, wetland ferns, mosses
Psammophyte *(बालुकोद्भिद)*	**Psammo** = Sand	*Psammology (study of sand)*	**Shifting Sand Dunes & Sand Fields** *(रेतीले टीले)*	Sand binders, marram grass
Chasmophyte *(दरारोद्भिद)*	**Chasmo** = Fissure / Crevice	*Chasm (deep gorge/cleft)*	**Rock Crevices & Fissures** *(चट्टानों की दरारें)*	Saxifrage, cliff crevicular flora
Oxylophyte *(अम्लोद्भिद)*	**Oxy** = Acidic	*Oxygen (historically "acid former")*	**Strongly Acidic Soils** *(अम्लीय मृदा)*	Heather, peat moss (*Sphagnum*)
Calciphyte *(चूनामृद्भिद)*	**Calci** = Lime / Calcium	**Calcium Carbonate** ($CaCO_3$), *Calcification*	**Lime & Calcium-rich Soils** *(चूनायुक्त मृदा)*	Calciphilous limestone flora

N.2 Soil Types

Alluvial

Chemistry	Status
Potash	Generally **rich** (Class 11 fact)
Phosphorus	**Poor** in NCERT Class 11. Some other notes say phosphoric acid is "adequate." Keep the **NCERT: poor P** fact.
Lime / alkalies	Variable; often adequate
Nitrogen, humus	Generally **low**

Belt	Region	Star crops
Upper Ganga	Punjab, Haryana, W-UP	Wheat, sugarcane, rice
Middle–lower Ganga	E-UP, Bihar, WB	Rice, jute, pulses

Belt	Region	Star crops
Brahmaputra	Assam	Rice; tea on nearby terraces
East-coast deltas	Mahanadi (महानदी), Godavari (गोदावरी), Krishna (कृष्णा), Cauvery	Rice, coconut fringe
Peninsular valleys	Narmada, Tapi flats; N Gujarat	Mixed food crops

Black (Regur / cotton soil)

Chemistry	Status
Lime, iron, magnesia, alumina	Rich
Potash	Present / variable
Nitrogen, phosphorus, organic matter / humus	**Deficient** — so "rich in humus" A/R reasons are **false**
Region	Note
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Maharashtra (Vidarbha)	Classic / most widespread cotton regur
MP / Malwa	Deep black / "lava soil" plateau
Gujarat	Black tracts (Saurashtra–Malwa link)
Telangana / Andhra	Trap extensions; deep in upper Godavari–Krishna
N Karnataka	Cotton–jowar (ज्वार)
Parts of Tamil Nadu	Local patches (often on gneiss/schist too)
Rajmahal (Jharkhand)	Trap black
Himalaya (हिमालय)	**Not** a black-soil region

Red & Yellow

Chemistry	Status
Iron oxides (colour); potash often adequate	Present
N, P, humus; sometimes lime	Often deficient
Region	Note
--------	------
Tamil Nadu, Karnataka, AP/TG	Wide red loams
Western Ghat piedmont	Red loamy belt
Southern Maharashtra	Transition with black soils
Odisha, Chota Nagpur, Chhattisgarh	Upland red
Parts of MP / NE / S Middle Ganga	Local red–yellow

Feature	Red / Yellow	Laterite
Parent / climate	Crystalline rocks, often lower rain	Hot-wet **intense leaching**
Colour cause	Fe-oxide diffusion	Fe/Al residue after bases/silica leave
Bases	Variable	Strongly depleted
Typical crops	Millets, pulses	Cashew, tea, coffee, plantations

Laterite

Chemistry	Status
Iron oxide, aluminium compounds	Excess / rich
OM, N, phosphate, calcium, lime, silica	Poor / leached
Item	Note
------	------
Climate	High T + heavy rain + wet–dry
Process	Intense leaching
Building use	Bricks
Crops	**Cashew, tapioca, tea, coffee, rubber, cinchona**
Belt	W Ghats / E Ghats summits, **Rajmahal**, **Vindhya–Satpura–Malwa**, Odisha, WB, Kerala, Karnataka, TN, Assam–Meghalaya hills
Not typical	Rajasthan, Uttar Pradesh plains

Alkaline (sodic / Kallar)

Feature	Saline	Alkaline (sodic)
Main problem	Soluble salts	Exchangeable Na
pH	Near-neutral to moderately high	Typically **high**
Structure	Variable	Dispersed, poor
Surface clue	White **reh**	Often darkish alkali stains
Key remedy	Leach + drain	**Gypsum** + leach

Problem soils (cause and remedy)

Problem	Cause	Crop effect	Remedy
Acidic	Base leaching, high rain (laterite, NE, E Himalayan foothills)	Nutrient fact-up, Al toxicity	**Lime**
Saline	Soluble salts, capillary rise	Germination failure	**Leaching + drainage**
Alkaline	Exchangeable Na, high pH	Structure collapse	**Gypsum + leaching**
Waterlogged	High water table / canal seepage	Root oxygen stress	Drainage; lined canals
Usar mix	Saline + sodic together	Barren patches	Gypsum + leach + drain + organics

N.3 Texture, Fertility, Erosion, Conservation

Texture	Water	Drainage	Tillage	Example
Sandy	Low	Fast	Easy	Desert
Silty	Medium–high	Medium	Moderate (नरम दल)	Some alluvium
Clayey	High	Slow	Hard when wet	Black regur
Loam	Balanced	Balanced	Favourable	Best farm texture

Particle	Diameter
Clay	**less than 0.002 mm**
Silt	About 0.002–0.06 mm
Fine sand	About 0.06–2 mm

Condition	pH tendency	Remedy hint
Laterite / NE humid / peaty	Acidic	Lime
Usar sodic	Strongly alkaline	Gypsum
Healthy alluvial loam	Near neutral	Maintain OM
Tea garden ideal	**Slightly acidic**	Avoid stagnant water + lime excess

Factor	Role
NPK (primary macros, not micros) + secondaries/micros	Direct nutrition
Humus / OM	Structure, water, cation bank
pH	Controls availability
Organisms (earthworms, microbes)	Cycling, porosity
Moisture & aeration	Root function
Drainage	Prevents salinity / waterlogging

Soil Erosion Stages (The Water Erosion Ladder)

Stage / Type	Primary Agent	Physical Mechanism & Diagnostic Feature	Real-World Example & Field Signature	Exam Decision Rule
1. Splash Erosion *(बूंद अपरदन)*	Falling Raindrops	Kinetic impact of heavy raindrops acts like micro-bombs, smashing soil aggregates (crumbs) and detaching fine particles.	Bare plowed soil during the very first monsoon downpours.	**First / Initiating stage** of water erosion.
2. Sheet Erosion *(परत अपरदन)*	Surface Runoff (Overland flow)	Thin, uniform wash of topsoil across a broad slope; strips fertile organic humus layer without creating channels.	Gently sloping cultivated farmland after continuous heavy rains.	**"The Silent Thief"** — widespread, devastating, yet goes unnoticed by farmers until crop yields crash.
3. Rill Erosion *(रिल / अंगुली अपरदन)*	Concentrated Runoff	Runoff gathers into minute, shallow, finger-like incisions (micro-channels) across the field surface.	Sloping fallow fields and orchard floors.	**Key Exam Test:** Small enough that **ordinary tillage/ploughing can easily erase** them.
4. Gully Erosion *(अवनालिका अपरदन)*	Powerful Channelized Runoff	Rills deepen and widen into steep-sided trenches and gullies that cut through subsoil; fragments farmland.	Plateau margins, foothills, upper catchment of Chambal & Yamuna.	**Key Exam Test:** Deep channels that **normal tillage CANNOT erase**; divides land into unusable fragments.
5. Ravine / Badland *(बीहड़ / खड्ड भूमि)*	Advanced Networked Gullies	Dense, intricate labyrinth of deep interconnected ravines and canyons; completely strips soil cover down to parent rock.	**Chambal Basin** (Morena, Bhind, Gwalior in MP; Dholpur in RJ; Agra–Etawah–Jalaun in UP).	Classic **Badland Topography** (*उत्खात भूमि*).
Stream Channel Erosion	Fluvial Streamflow	Under-cutting, scouring, and bank caving along permanent river corridors during peak flood stages.	Floodplains of Ganga, Brahmaputra, Kosi.	Bank erosion and river meandering.

Stage / Type	Primary Agent	Physical Mechanism & Diagnostic Feature	Real-World Example & Field Signature	Exam Decision Rule
Wind / Deflation *(वायु अपरदन)*	High-Velocity Wind	Rolling, saltation, and suspension of dry loose sand; dune migration burying productive lands.	Western **Thar Desert** (Rajasthan, southern Haryana, Kutch).	Prevalent in arid zones where vegetative cover is absent.

Soil Conservation Methods (Categorized for Effortless Recall)

Method (विधि)	Core Mechanism (कार्य सिद्धांत)	Ideal Topographic Setting
Contour Ploughing *(समोच्च रेखीय जुताई)*	Ploughing strictly **across (along) contour lines**, horizontal to the slope (**never** up-down). Furrows act as micro-dams that intercept runoff and let water infiltrate.	**Gentle farm slopes** (up to ~6° slope).
Contour / Graded Bunding *(समोच्च मेड़बंदी)*	Earthen embankments (*bunds*) constructed along contours to impound runoff; graded bunds gently channel excess water safely to drainage outlets.	**Plains and plateau margins** with moderate slopes and permeable soils.
Contour Trenching *(समोच्च खाइयाँ)*	Excavating continuous or staggered trenches along hill contours to catch surface runoff, arrest silt, and recharge subterranean aquifers.	**Barren hill slopes and forest watersheds**.
Terracing / Bench Terracing *(सीढ़ीदार खेत)*	Carving steep hillsides into flat, step-like benches bounded by riser walls; transforms rapid downhill torrents into calm, contained irrigation steps.	**Steep mountains & hill tracts** (Himalaya, Western Ghats). *(Trap: Never select flat plains or Thar desert!).*
Check Dams & Gully Plugs *(रोक बांध / गली प्लग)*	Small rock, brushwood, or concrete barriers erected across gully beds to dissipate water velocity, trap moving silt, and stabilize gully heads.	**Active gullies and ravines** (Chambal, Yamuna ravines).

Method (विधि)	Core Mechanism (कार्य सिद्धांत)	Ideal Farming Setting
Strip Cropping *(पट्टीदार खेती)*	Alternating strips of erosion-permitting crops (e.g., maize, cotton, jowar) with close-growing cover crops (e.g., grasses, pulses, clover) across the slope.	Open undulating plains and windy agricultural slopes.
Cover Cropping & Mulching *(आवरण फसलें एवं पलवार)*	Growing dense ground cover crops or spreading crop residue/straw over bare soil; cushions kinetic raindrop impact and halts sheet wash.	Rainfed farms, orchards, and between crop seasons.
Crop Rotation & Mixed Cropping *(फसल चक्र एवं मिश्रित खेती)*	Alternating tap-rooted and fibrous-rooted crops with leguminous nitrogen fixers; continuously replenishes fertility, preserves crumbs, and avoids seasonal bare exposure.	Intensive cropping systems across India.
Zero / Reduced Tillage + Stubble Retention *(शून्य जुताई)*	Drilling seeds directly into untilled soil through prior crop stubble; eliminates mechanical soil pulverization and conserves deep soil moisture.	Rainfed rabi crops and intensive rice-wheat tracts.
Early-Maturing Crop Varieties	Planting short-duration cultivars that establish rapid ground canopy and harvest before late-monsoon downpours.	Drought-prone and rainfed dryland belts.

Method (विधि)	Core Mechanism (कार्य सिद्धांत)	Ideal Geographical Setting
Shelterbelts / Windbreaks *(रक्षक मेखला / वातरोधी वृक्ष)*	Planting multiple linear rows of dense, deep-rooted trees and shrubs perpendicular to prevailing erosive winds to reduce wind velocity and deflect sandstorms.	**Western Rajasthan (Thar)**, coastal sandy belts, canal banks.
Sand Fences & Dune Stabilization	Erecting porous brushwood fences and planting xerophytic sand-binders (*Calligonum*, *Cenchrus* grass) to immobilize shifting sand dunes.	Active barchan and longitudinal dune fields.

Method (विधि)	Core Mechanism (कार्य सिद्धांत)	Key Focus Region
Afforestation & Reforestation *(वनीकरण)*	Planting native trees with deep root networks that anchor unstable regolith, encourage leaf-litter formation, and regulate watershed hydrology.	Degraded hill catchments and ravine rims.
Regulated Rotational Grazing *(नियंत्रित चराई)*	Limiting livestock numbers and rotating grazing pastures to prevent overgrazing, which strips grass roots and pulverizes topsoil into loose dust.	Hill pastures, grassland tracts, and scrub forests.
Agro-forestry *(कृषि-वानिकी)*	Integrating multipurpose timber, fruit, and fodder trees with agricultural crops and livestock on the same land unit.	Drylands and degraded farm tracts.
Checking Jhum / Shifting Cultivation	Replacing unsustainable slash-and-burn cycles with permanent settled bench terracing and horticultural plantations.	North-Eastern Hill states and parts of the Eastern Ghats.

Problem Condition	Scientific Remedy & Action	Typical Focus Tract
Alkaline / Sodic Soils (*Usar*)	Application of **Gypsum ($CaSO_4 \cdot 2H_2O$)** or Iron Pyrites (FeS_2) to replace toxic exchangeable Na^+ with Ca^{2+}.	Canal-irrigated plains of UP, Punjab, Haryana.
Acidic Soils (Low pH)	Application of **Agricultural Lime ($CaCO_3$)**, Dolomite, or Slaked Lime to neutralize soil acidity and restore base saturation.	Laterite belts (Kerala, Karnataka), humid North-East hills.
Waterlogged & Saline Soils	Installation of subterranean tile drainage + lining of canals to lower elevated water tables and halt capillary salt deposition.	Command areas of major canal irrigation projects.

Key Institutions & Milestones in Soil Conservation

Initiative / Landmark	Core Fact & Prelims Takeaway
Mitti Bachao Andolan *(Save the Soil Movement)*	Started in **1977** at **Hoshangabad (now Narmadapuram), Madhya Pradesh**, against waterlogging and soil salinisation caused by the **Tawa Dam Project**. *(Classic Trap: Not Thane, Mysore, or Darbhanga!).*
Central Soil Conservation Board (CSCB)	Established in **1953** by the Government of India to formulate soil conservation programs across different agro-climatic zones.
World Soil Day	Celebrated annually on **5 December** (birthday of Thailand's King Bhumibol Adulyadej, who championed soil science).

N.4 UP Focus

UP soil belts — expanded teaching map

Group	Where	Character	Crop / tag
Bhabar	Shivalik (शिवालिक) foothill strip (~8–16 km); Saharanpur–Bijnor west	Shallow, pebbly, porous; streams **disappear** into fans	**Poor** for crops
Tarai	South of Bhabar (~15–30 km belt)	Streams **re-emerge**; moist, N + OM rich, silty–clayey	Rice, wheat, sugarcane
Gangetic alluvium	West–east plains	Deep fertile; khadar/bangar/bhur	~90% of state
Bundelkhand	Jhansi (झांसी), Lalitpur, Hamirpur, Jalaun, Banda, Mahoba, Chitrakoot (चित्रकूट)	Mar/Kabar black; Parua/Rakar red	Local-name favourite
Vindhyan / Baghelkhand fringe	Mirzapur (मिर्ज़ापुर), Sonbhadra (सोनभद्र), parts of Prayagraj (प्रयागराज)–Chandauli	Red–yellow with Fe/Al; **Bhonta/Monta**	Pulses, millets, oilseeds
Aravali fringe	Parts of Agra	Reddish, Chambal-brought	Local fringe
Usar / Reh	W–central canal plains	Saline–alkaline crusts	Gypsum + drainage
Local name	Colour family	Trait	Core districts
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Group	Where	Character	Crop / tag
Mar	Black	Fine clayey and OM-rich	Jalaun, Hamirpur, Jhansi, Banda
Kabar	Dark to grey-black	Coarse and extremely sticky	Lalitpur, Banda, Jalaun, Hamirpur
Parua	Yellowish red	Sandy to clayey; well aerated	Throughout Bundelkhand
Rakar (pathari)	Reddish-brown	Coarse on slopes; slightly acidic	Sloping Bundelkhand

Problem	Hotspot	Fix
Usar / reh	W–central canal tracts	Gypsum + drainage + OM
Ravines / gullies	Agra–Etawah–Jalaun	Check dams, afforestation, grazing control
Wind (summer)	Agra–Mathura (मथुरा)–Etawah	Shelter belts / cover
Waterlogging	Tarai + canal seepage	Drainage; canal lining
Fertility decline	Intensive plains	Rotation, legumes, organic returns

Common Traps — Don't Fall For These

1. **Edaphic = climate** — FALSE. Edaphic = **soil**.
2. **Black soil in Himalaya** — FALSE. **Deccan Trap / regur**.
3. **Laterite poor in Fe/Al** — FALSE. **Rich** in Fe/Al.
4. **Every red soil is laterite** — FALSE. Laterite needs hot-wet extreme leaching.
5. **Khadar has more kankar** — FALSE. **Bangar** is the kankar-heavy fact.
6. **Gypsum for acid soils** — FALSE. **Lime** for acid; **gypsum** for alkali/sodic.
7. **Irrigation alone cures usar** — FALSE. Needs **drainage** (else more reh).
8. **Mitti Bachao = Darbhanga** — FALSE. **Hoshangabad, MP**.
9. **World Soil Day = 5 June** — FALSE. **5 December**; 5 June = Environment Day.
10. **UP = black-soil state** — FALSE. ~**90% alluvial**; black mainly Bundelkhand Mar/Kabar.
11. **Mar = Bundelkhand red** — FALSE. Mar/Kabar = **black**; Parua/Rakar = **red**.
12. **Bhabar = best rice soil** — FALSE. Pebbly/poor; streams vanish; **Tarai** re-emerges and is moist fertile.
13. **Shelter belt stops gullies** — FALSE. Shelter belt = **wind**; check dams = gullies.
14. **Terra Roxa = Indian regur** — FALSE. **Brazil** basaltic coffee soil.
15. **Contour bunding = desert margins / flood plains** — FALSE. Classic use = **hill / mountain slopes**.
16. **Earthworms are bad for agriculture** — FALSE. They are **ecosystem engineers** that raise fertility.
17. **Clay pores are large** — FALSE. Clay pores are **tiny**; that is why capillary climb and water holding are high.
18. **Organic matter cuts water-holding capacity** — FALSE. OM usually **raises** water holding; irrigation can still **salinise** land.
19. **Terra Rossa forms on granite** — FALSE. On **limestone / dolomite**.
20. **Halophytes = acidic soils** — FALSE. **Saline** soils.
21. **Earthworms have two pairs of hearts** — FALSE. Fertility statement alone is correct.
22. **Alluvial always rich in phosphorus** — FALSE. Class 11 keys **rich potash, poor phosphorus**.
23. **Karewa = Gangetic bangar** — FALSE. Kashmir **lacustrine**; saffron belt.
24. **Rann salt = only canal usar** — FALSE. Rann crust is largely **SW monsoon** salt deposit.
25. **Red soil is India's largest group** — FALSE. **Alluvial** is largest.
26. **Black soil covers 46 lakh km²** — FALSE. Keep **~16.6%**. The 46-lakh figure cannot fit India.
27. **Alluvium is always rich in phosphorus** — FALSE. Follow **NCERT: poor P**.

28. **Himalayan soil is equal on both faces** — FALSE. **North-facing** gentler slopes hold soil; south faces are often stripped.
29. **IGC only blesses the desert** — FALSE. It can also **create usar** without drainage.
30. **Lime reclaims usar** — FALSE. **Gypsum** for alkali/usar; **lime** for acid soils.
31. **Laterite = Rajasthan / UP plain soil** — FALSE. Hot-wet Ghats / Malabar / plateau edges.
32. **Erosion ladder starts at gully** — FALSE. **Splash → sheet → rill → gully → ravine**.
33. **Black soil rich in humus** — FALSE. Suitable for cotton despite **low** N–P–humus.
34. **Tea = alkaline / regur default** — FALSE. Prefers **slightly acidic**, drained soil.

Topic 6 — Agriculture

Current Affairs (this topic)

Year	Fact	Why asked	Source
May 2026	CCEA raised MSP for **14 Kharif (खरीफ)** crops, MS 2026–27; common paddy **₹2,441/q** (+₹72); Grade A ₹2,461	Latest paddy MSP figure	PIB / CCEA
Static rule	MSP for **22 mandated crops**; CACP recommends, Cabinet decides; target **≥1.5×** all-India cost (Budget 2018–19)	Institution > old rupees	PIB
2025–26 Kharif	Highest MSP hike then: nigerseed, ragi, cotton, sesamum; extra push pulses/oilseeds/**Shree Anna**	Diversification narrative	PIB
Ongoing	**PM-AASHA** (PSS via NAFED/NCCF); **e-NAM**; **PM-KISAN**; **PMFBY**	Scheme names	DA&FW
2025	Delhi **bio-decomposer** (fungal mix, free) for stubble	Free fungal mix for stubble-to-manure	PYQ / GNCTD

Confused Pairs

Pair	Correct	Trap	Hindi
Kharif vs rabi	Wheat = **rabi**; cotton/rice = **kharif**	Wheat kharif / cotton rabi	गेहूँ रबी; कपास खरीफ
NSA vs GCA	Intensity = **GCA/NSA × 100**	Invert the ratio	सकल/शुद्ध
MSP vs FRP	Cereals/oilseeds/pulses = MSP; cane = **FRP**	Call cane MSP	गन्ना = FRP
CACP vs Cabinet	CACP **recommends**; Cabinet **decides**	RBI/NITI recommends	सीएसीपी सिफारिश
Swaminathan vs Kurien	Swaminathan = GR; Kurien = **milk**	Swap	कुरियन = दूध
Borlaug Nobel	**Peace**	Agriculture	शांति नोबेल
Yellow vs Golden	Yellow = **oilseeds**; Golden = **horti + honey**	Swap	पीला = तिलहन
Grey vs Black	Grey = fertiliser; Black = petroleum	Swap	ग्रे = उर्वरक
UP vs MH cane	UP **quantity**; MH **productivity**	Coops explain UP yield (they don't)	UP मात्रा; MH उत्पादकता
WB vs UP jute	Jute = **WB**	UP–Jute	जूट = बंगाल
Kerala rubber vs GJ tea	KL–Rubber **correct**; GJ–Tea / Assam–Wheat **wrong**	Pick GJ tea	रबर = केरल
Ahmedabad vs Mumbai	Ahmedabad in cotton **region**; largest mill centre traditionally **Mumbai**	Ahmedabad = largest centre	कपड़ा केंद्र = मुंबई
Potato CIP	**Agra (Singna)**	Aligarh	आलू केंद्र = आगरा
Sultana / Gulabi / Kali Champa vs grapes	These three names are **guava** varieties	Pick grapes because Sultana is also a grape name	अमरूद ≠ अंगूर
Azolla vs pesticide	Azolla = **biofertiliser** in rice	Insecticide	एजोला = जैव उर्वरक
15 vs 127 vs 20	ACZ **15** / NARP **~127** / AER **~20**	Mix counts	15 कृषि-जलवायु
Sen vs Swaminathan	Food **entitlements** = Amartya Sen	Swaminathan	सेन = एंटाइटलमेंट
Two wheat belts	Ganga–Satluj plains + Deccan black soil	Assam as wheat belt	दो गेहूँ पेटियाँ
White gold vs golden fibre	Cotton = white gold; jute = golden fibre	Swap nicknames	सफेद सोना / सुनहरा रेशा

Pair	Correct	Trap	Hindi
White vs tossa jute	White = *capsularis* (flood-tolerant); tossa = *olitorius* (upland, better fibre)	Swap flood / upland	सफेद / टोसा जूट
Cotton species vs staple	Four species (*arboreum / herbaceum / hirsutum / barbadense*); *hirsutum* = bulk; ELS = *barbadense*	"India grows only short staple"	चार प्रजाति; ELS कम
Sikkim organic	Sikkim = first fully organic state	Call any NE state first	सिक्किम जैविक
Rice rank	Textbook **#2 after China**; 2024–25 GoI **#1 claim**	Mixing the usual rank with the 2025 headline	रैंक अलग रखें
WB rice vs PB yield	WB often **volume** leader; Punjab **yield** leader	Punjab = largest producer always	बंगाल मात्रा; पंजाब उपज
DSR vs transplant	DSR = seed in field, less water; transplant = nursery + puddle	DSR needs more standing water	डीएसआर = कम पानी
Ratoon vs rotation	Ratoon = cane from **stubble**; rotation = change the crop	Call wheat a ratoon crop	रैटून = गन्ना
Mixed vs double vs inter	Mixed = crops+livestock; Double = two crops/year; Inter = same-time rows	Swap definitions	मिश्रित ≠ डबल
Parallel cropping	Wheat + mustard	Potato + rice as parallel	समान्तर = गेहूँ+सरसों
15 ACZ vs 20 AER	Planning ACZ **15**; NBSS AER **20**; Sengupta micro **60**	Swap 15/20	15 जलवायु; 20 पारिस्थितिक
Pantnagar year	First agri uni **1960** (Nehru; later GBPUAT)	1950 / 1970	पंतनगर 1960
Randhawa vs Swaminathan	History of Indian Agriculture = **Randhawa**	Credit Swaminathan	रंधावा पुस्तक
Agmark Act	**1937** grading mark	Call it co-op egg mark	एगमार्क 1937
Moisture 14%	FAO safe storage ≤ **14%**	16 / 18 / 20%	नमी 14%
Cowpea green manure	Highest N% among common options	Pick dhaincha only	लोबिया N%
Fertigation phosphate	Avoid rock / super phosphate in fertigation	"All P fertilisers OK"	फर्टिगेशन P
Coimbatore cane	Sugarcane Breeding Institute	Lucknow only	कोयंबटूर
Pratappur mill	First sugar mill **1903**	Mawana / Balrampur first	प्रतापपुर
Contract farming	Pioneer **Punjab**	Pick Haryana / TN	पंजाब अनुबंध

N.1 Land Use, Farming Types, Seasons

Class	What it means
Forests	Area reported as forest
Barren and unculturable land	Rocky, desert, or otherwise unfit for crops
Permanent pastures and grazing land	Grasslands kept for grazing
Miscellaneous tree crops and groves	Orchards, groves, and tree crops outside net sown
Culturable waste	Land that could be ploughed but is currently unused
Fallow (current + other)	Cultivable land left unsown for a season or longer
Net sown area (NSA)	Land sown with crops **at least once** in the year

Farming types

Type	Recall
Primitive subsistence	Family food; includes **jhum**
Intensive subsistence	Small farms, high labour, rice/wheat
Commercial grain	Market surplus (GR wheat belts)
Plantation	Estate perennial — tea/coffee/rubber
Mixed farming	Crops **and** livestock together
Dairy	Milk commercial
Mediterranean (world)	Winter rain; citrus/olives/vines
Truck farming	Vegetables near cities
Dryland farming	Rainfed millets, pulses
Wetland / irrigated	Standing water or canal/tube-well
Contract farming	**Punjab** pioneer in India

Cropping patterns

Pattern	Recall
Mixed cropping	Two+ crops **together** at once
Intercropping	Fixed **row ratios**
Parallel cropping	Parallel rows; classic **wheat + mustard**
Double cropping	Two+ crops in **one crop year** (sequential)
Crop rotation	Sequential change on same land
Sequential cropping	One after another same year
Relay cropping	Second sown **before** first harvest
Ratoon cropping	From **stubble** (sugarcane)
Multiple cropping	More than one crop / year → intensity

Seasons

Season	Months	Typical crops
Kharif	Jun–Oct (SW monsoon (दक्षिण-पश्चिम))	Rice, maize, jowar, bajra, ragi, cotton, jute, groundnut, soybean, tur, sesame
Rabi	Oct–Mar	Wheat, barley, gram, mustard, peas, linseed
Zaid	Mar–Jun	Watermelon, cucumber, fodder, vegetables
Long / perennial	—	Sugarcane (10–18 months); tea, coffee, rubber

N.2 Food Crops

Rice

Belt	States
Lower Ganga / delta	West Bengal (multi-rice + jute)
Middle Ganga	E-UP, Bihar
Coastal east	Odisha, AP/TG, TN
Irrigated NW	Punjab, Haryana, W-UP

Belt	States
Krishna–Godavari delta	AP — “Rice Bowl of India”
Chhattisgarh plain	“Rice bowl” tag

Maize & millets (nutri-cereals)

Crop	Leading state	Note
Jowar (ज्वार)	**Maharashtra**	Some rabi jowar in the peninsula
Bajra (बाजरा)	**Rajasthan**	Lowest-rainfall cereal
Ragi	**Karnataka**	Ca-rich; red/lateritic soils OK

Pulses

Pulse	Season	Note
Gram / chana	**Rabi**	MP often among leaders
Masoor (lentil)	Rabi	Cool season
Tur / arhar	**Kharif**	Central India; red gram origin often tagged **India**
Moong, urad, moth	Kharif	Short-duration; drylands; urad can be kharif and rabi in the south
Peas	Rabi	With wheat belts

N.3 Cash, Fibre & Plantation

Oilseeds — Yellow Revolution

Crop	Season	Core state
Groundnut	Kharif	**Gujarat**
Mustard / rapeseed	**Rabi**	**Rajasthan**
Soybean	Kharif	**Madhya Pradesh** (oilseed, not pulse)
Sesame	Kharif	Drylands
Sunflower	Flexible	KA/MH/AP
Castor	Drought-hardy	GJ/RJ
Linseed	Rabi	Central/north
Safflower	Cool dry rabi	Deccan
Niger	Minor	Tribal (आदिवासी)/peninsula

Cotton (white gold)

Species	Common tag	What it gives
Gossypium arboreum	Desi / Old World	Short staple; drought- and pest-hardy
Gossypium herbaceum	Desi / Old World	Short–medium; harsh rainfed tracts (GJ / KA pockets)
Gossypium hirsutum	American / Upland	**Bulk of India’s crop**; medium to long staple; most **Bt** hybrids
Gossypium barbadense	Egyptian / Sea Island	**Extra-long staple (ELS)**; small premium pockets in the south

Class	Approx. length	India note
Short staple	≤ about **20 mm**	Desi types; coarse cloth, blending, surgical cotton
Medium staple	about **20.5–24.5 mm**	Large share of domestic mill use

Class	Approx. length	India note
Medium-long staple	about **25–27 mm**	Modern mill demand
Long staple	about **27.5–32 mm**	High-speed spinning; MSP is also quoted for a long-staple grade
Extra-long staple (ELS)	about **≥32.5 mm**	Premium yarn; India still **imports** much ELS (Egypt (मिस्र) / USA style)

Jute (golden fibre) & mesta

Type	Botanical name	Field note
White jute	*Corchorus capsularis*	Stronger flood tolerance; classical lowland / waterlogged delta crop
Tossa jute	*Corchorus olitorius*	Softer, usually better fibre and higher yield; prefers **uplands** and hates prolonged flood
Mesta	*Hibiscus cannabinus* / *H. sabdariffa* (related fibre, not true jute)	Grown on **drier** tracts as a jute substitute; coarser fibre

Tea · Coffee · Rubber · Spices

Crop	Climate	Site	India
Tea	20–30°C; 150–300 cm	Well-drained **slopes** (no stagnant water)	Assam (volume), WB **Darjeeling GI**, Nilgiris/TN–KL; **plantation**
Coffee	15–28°C; 150–250 cm	**Shade**, well-drained	**KA > KL > TN** (Baba Budan / Coorg / Wayanad / Nilgiri)
Rubber	25–35°C; **>200 cm**	Humid equatorial; *Hevea brasiliensis* latex	**Kerala** (also TN, KA, A&N, Tripura pockets)
Pepper / cardamom	Hot wet Ghats	Hills	**Kerala** largest
Chilli	Warm Deccan	—	AP/TG strong — **not Kerala-only**
Turmeric / ginger	Warm	South + NE	Many GIs

Coffee species	Note
Arabica	Higher elevation, milder cup, more pest-sensitive
Robusta	Lower elevation, stronger cup, hardier

N.4 Fruits, Potato, GI

GI / variety	Place
Darjeeling Tea (दार्जिलिंग चाय)	WB hills
Basmati	North India GI belt
Alphonso	Maharashtra
Banganapalle	AP
Gir (गीर) Kesar	Gujarat
Dasheri / Langra	UP
Malabar Pepper	Kerala
Kashmiri saffron	J&K
Nashik grapes	MH
Coorg coffee / Alleppey cardamom	KA / KL GI note

N.5 Agricultural Regions & Agro-climatic Zones

System	Count	Use
Planning Commission **ACZ**	**15** (~72–73 sub-zones)	Broad planning
ICAR **NARP**	**~127**	Research / extension
NBSS&LUP **AER**	**~20**	Soils + LGP (Length of Growing Period)

15 Agro-climatic Zones (Planning Commission)

#	Zone	Core	Agri tag
1	Western Himalayan	J&K, HP, UK	Apple, valley cereals
2	Eastern Himalayan	Assam, NE, Sikkim, N WB	Tea, rice, spices, jhum
3	Lower Gangetic Plains	West Bengal	Rice, jute, mustard
4	Middle Gangetic Plains	E-UP, Bihar	Rice–wheat, maize, cane
5	Upper Gangetic Plains	W/central UP	Wheat–rice–cane (GR)
6	Trans-Gangetic Plains	PB, HR, Delhi, parts RJ	Intensive wheat–rice
7	Eastern Plateau & Hills	CG, JH, OD uplands	Rice, millets, rainfed
8	Central Plateau & Hills	MP, RJ, Bundelkhand edge	Soybean, wheat, gram
9	Western Plateau & Hills	MH upland	Cotton, jowar, pulses
10	Southern Plateau & Hills	Interior AP/TG, KA, TN	Groundnut, millets, cotton
11	East Coast Plains & Hills	Coastal OD–AP–TN	Rice, coconut, cashew
12	West Coast Plains & Ghats	KL, coastal KA, Goa (गोवा), Konkan	Rice, coconut, spices, rubber, coffee
13	Gujarat Plains & Hills	Gujarat	Cotton, groundnut
14	Western Dry Region	W Rajasthan	Bajra, pulses, livestock
15	Islands Region	A&N, Lakshadweep (लक्षद्वीप)	Coconut, horticulture

Agro-ecological regions (NBSS&LUP ≈ 20)

Ecosystem	Example AER tags
Arid	Western Himalayas cold arid; Western Plain–Kachchh; Deccan Plateau (दक्कन पठार) arid
Semi-arid	Upper / Middle Ganga plains; Rajasthan–Gujarat plains; Deccan red–black / red sandy
Sub-humid	Eastern Plateau (Satpura (सतपुड़ा)–Mahanadi (महानदी), Bundelkhand, red–laterite (लेटराइट)); Lower Ganga; warm W Himalaya (हिमालय); Bengal Basin
Humid–perhumid	Assam–N Bengal plains; Eastern Himalayas; Purvanchal (पूर्वांचल) hills (पूर्वांचल पहाड़ियाँ)
Coastal	Eastern Coastal Plains + A&N; Western Ghats (पश्चिमी घाट) coastal plains and hills

N.6 Irrigation & Command Area Development

Type	Best terrain	Risk / note
Perennial canal	Alluvial plains (PB–HR–UP; IGC RJ). Clayey beds **leak less**	Waterlogging / usar
Inundation canal	Flood plains	Unreliable timing

Type	Best terrain	Risk / note
River lift	Where canal flow is irregular; common in **peninsular** belts; **Pattiseema** = Godavari→Krishna	Pumping cost
Tube well	Alluvial aquifers; GR NW + W-UP	Groundwater depletion
Dug well	Local / hard-rock	Limited yield
Tank	**Peninsular** hard rock	Siltation
Drip	Horti, cane, cotton, veg; 40–60% water save; less weed / erosion	High efficiency; capital cost
Sprinkler	Sandy / undulating	Wind drift

N.7 Seeds, ICAR, Universities, Pests & Practices

Schemes (practice)

Scheme	Key point
Soil Health Card	Soil fertility / balanced nutrient advice
PMKSY (incl. more crop per drop)	Irrigation access / water efficiency
PKVY (Paramparagat Krishi Vikas Yojana)	Organic / traditional farming clusters
Kisan Credit Card	Crop credit via commercial, cooperative and RRBs
e-NAM / national agri market push	Better price discovery
NFSM	Rice, wheat, pulses (later oilseeds / millets) productivity

Disease / problem	Cause
Citrus canker	**Bacteria**
Red rot of sugarcane	**Fungus**
Potato “Krishnakant”	**O_2 deficiency**
Wheat “Sahu”	**Insect**
Wheat Karnal bunt	Fungus (*Tilletia indica*)
Wheat rusts	Yellow / brown / black
Paddy Khaira	Zn deficiency ($ZnSO_4$ spray)
Potato late blight	Classic potato disease
Arhar wilt	Wilt disease

N.8 Animal Husbandry

Breed	Animal	Tag
Murrah	Buffalo	Haryana / high yield
Bhadawari	Buffalo	UP / Bundelkhand note
Gir	Cattle	Gujarat
Sahiwal	Cattle	Punjab-region note; high milk
Tharparkar	Cattle	Western Rajasthan border
Deoni	Cattle	Dual-purpose
Jamunapari	Goat	UP renown; high milk

Breed	Animal	Tag
Barbari / Beetal	Goat	Other common breeds

N.9 Jhum, Organic, Precision

System	Core	Where / tool
Jhum (shifting)	Slash-burn, 1–3 yr crop, then fallow	**NE** (jhum); **podu / penda** (AP/Odisha); **bewar / dahiya / mashan** (MP–CG); **kumari** (W Ghats); **waltre** (SE Rajasthan); **poonam** (Kerala); **kuruwa** (Jharkhand); **khil** (Himalayan belt)
Organic	No synthetic agrochemicals (standards)	Compost, rotation, Azolla, Rhizobium; **Sikkim** = first fully organic state (declared **2016**)
Precision	Site-specific inputs	GIS / GPS / sensors / drones — not "stop irrigation"

N.11 Green Revolution & Colour Revolutions

Person	Role
M.S. Swaminathan	GR in India; Evergreen idea
Norman Borlaug	HYV wheat; Nobel **Peace** not Agriculture
Verghese Kurien	White / milk
L.K. Jha	Economic Administration Reforms
Morarji Desai (आमिल)	Bank social control
Amartya Sen	Food **entitlements** — not Swaminathan
Hiralal Chaudhuri	Blue Revolution (fisheries) tag
Revolution	Sector
----------	--------
Green	Food grains (wheat → rice)
White	Milk (Kurien / Flood / NDDB)
Blue	Fisheries / aquaculture
Yellow	Oilseeds
Golden	Horticulture **and honey**
Silver	Eggs / poultry
Pink	Onion / meat / prawn (context)
Grey	Fertilisers
Black	Petroleum
Brown	Sometimes leather / cocoa / biofuel talk — weaker association
Rainbow	Integrated multi-sector
Evergreen	Sustainable productivity

Colour Revolutions — must-score set

Colour	Sector
Golden	Horticulture **and honey**
Grey	Fertilisers
Yellow	Oilseeds

Colour	Sector
Black	Petroleum

UP Focus (compact)

Theme	Fact
Sugarcane	Quantity leader often; productivity < MH; Sugar Bowl tag
Wheat–rice	Western UP GR (Upper Gangetic / Zone 5)
Potato	Leading producer; CIP **Agra** (आगरा) not Aligarh
Mango	Dasheri / Langra; Amrapali hybrid
Dairy	Large bovine base; Jamunapari goat; top milk state
Mentha oil	UP is the classic near-total producer
Not	Jute / tea / rubber leader
SAU	**Pantnagar (first, 1960)**; Kanpur / Meerut

Common Traps — Don't Fall For These

1. **Wheat = kharif / cotton = rabi** — FALSE. Wheat **rabi**; cotton **kharif**.
2. **CACP / RBI / NITI decides MSP** — FALSE. CACP **recommends**; **Cabinet** decides.
3. **Cane price = cereal MSP** — FALSE. Cane = **FRP** (+ possible SAP).
4. **Borlaug Nobel = Agriculture** — FALSE. **Peace**.
5. **Kurien = Green Revolution** — FALSE. Kurien = **White / milk**; Swaminathan = GR.
6. **Yellow = horticulture** — FALSE. Yellow = **oilseeds**; Golden = horticulture **and honey**.
7. **UP–Jute / Gujarat–Tea / Assam–Wheat** — FALSE. Correct pair in that set: **Kerala–Rubber**.
8. **Ahmedabad = India's largest cotton textile centre** — FALSE. In cotton *region* yes; largest mill centre traditionally **Mumbai**.
9. **Maharashtra coops explain UP's lower cane yield** — FALSE. UP leads quantity; Maharashtra leads productivity. Cooperatives do not cause that yield gap.
10. **CIP potato centre = Aligarh** — FALSE. **Agra (Singna)**; UP is potato leader.
11. **Sultana/Gulabi/Kali Champa = grapes** — FALSE. They are **guava** varieties. The name Sultana is the grape mix-up.
12. **ACZ = 127 or 20** — FALSE. Planning ACZ = **15**; NARP ~127; AER ~20.
13. **Azolla = insecticide** — FALSE. **Biofertiliser** in flooded rice (2023).
14. **Food entitlements = Swaminathan** — FALSE. **Amartya Sen**.
15. **Tea = zaid cereal** — FALSE. **Plantation** perennial.
16. **Only one wheat belt** — FALSE. Ganga–Satluj plains **and** Deccan black-soil wheat.
17. **Cotton = golden fibre** — FALSE. Cotton = **white gold**; jute = **golden fibre**.
18. **Any NE state = first organic** — FALSE. **Sikkim** is the first fully organic state.
19. **India is always rice #1** — FALSE. Textbook rank is **#2 after China**. The **2024–25** GoI claim of first place is a separate recent claim.
20. **Punjab is always India's largest rice producer** — FALSE. It is the **yield** king; **West Bengal** often leads **volume**.
21. **DSR needs more standing water than transplanting** — FALSE. DSR is the **water-saving** method.
22. **BGREI is a Punjab second Green Revolution** — FALSE. It is an **eastern** rice-system programme.
23. **India has 20 agro-climatic and 15 agro-ecological regions** — FALSE. Counts are **swapped** (ACZ 15; AER 20).
24. **First agri university = 1950 / Ludhiana** — FALSE. **Pantnagar, 1960** (later GBPUAT).

25. **History of Indian Agriculture = Swaminathan** — FALSE. Author is **M.S. Randhawa**.
26. **Mixed farming = two crops only** — FALSE. Mixed farming = crops **plus livestock**.
27. **Double cropping = two crops at the same time** — FALSE. That is closer to inter / parallel cropping; double = two crops in **one year**.
28. **Agmark = egg co-op** — FALSE. It is a **1937** quality grading mark.
29. **Fertigation works best with rock phosphate** — FALSE. Avoid rock / super phosphate (precipitation).
30. **Blue Revolution = poultry** — FALSE. Blue = **fisheries / pisciculture**.
31. **Sugarcane breeding HQ = Lucknow only** — FALSE. Classic breeding institute is **Coimbatore**.
32. **Coffee Board HQ = Hyderabad** — FALSE. **Bengaluru**.
33. **Contract farming pioneer = Haryana** — FALSE. Classic pioneer is **Punjab**.
34. **FAO safe grain moisture = 18%** — FALSE. About **14%**.
35. **Seed Village = ban buying seed** — FALSE. It trains farmers to produce quality seed for self-use and neighbours at affordable cost.

Topic 7 — Natural Vegetation & Biodiversity Geography

Current Affairs (this topic)

Year	Fact	Why asked	Source
5 Jun 2026	**Surha Tal / JP Narayan BS (Ballia)** = India's **100th Ramsar**; UP **13th**	Count + latest UP site	MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) / PIB
2025–26	UP Ramsar adds Patna (पटना) BS, Sheikha Jheel, Surha Tal (full names in N.6)	State list (राज्य सूची)	MoEFCC
Budget 2023–24	**MISHTI** mangrove restoration — not a 2019 start	Mangrove scheme year	Budget
Static	Olivia (Coast Guard (तटरक्षक बल), Odisha, Nov–May); coral warming 2024 PYQ	Coastal ops	ICG / PYQ
ISFR	Use the **report year in the question**; 2017 = 21.54% is a frozen PYQ	Figure trap	FSI

Lakes-as-wetlands. This file keeps the **geography** facts.

Confused Pairs

Pair	Correct	Trap	Hindi
Thornthwaite vs Köppen	Vegetation = climate **index** → Thornthwaite	Köppen	थॉर्नथ्वेट
Evergreen vs moist deciduous	Evergreen **>200**; moist **100–200**, **largest area**	Evergreen = largest	पर्णपाती सबसे बड़ा
Simlipal	**Odisha**	Madhya Pradesh (2018)	सिमलीपाल = ओडिशा
Nokrek	**Meghalaya**	Manipur (2025)	नोकरेक = मेघालय
Nanda Devi	**Uttarakhand**	Uttar Pradesh	नंदा देवी = उत्तराखंड
Botanical garden	**Ex-situ**	In-situ	वनस्पति उद्यान = वाह्य
Dudhwa district	**Lakhimpur Kheri**	Bahraich / Pilibhit	दुधवा = लखीमपुर
NP vs TR in UP	Only NP = Dudhwa; Pilibhit/Amangarh/Ranipur = **TR**	Extra NPs	केवल एक NP
Chandraprabha	**Chandauli**	Sonbhadra (2023)	चंदप्रभा = चंदौली
Hangul	**Dachigam / Kashmir**	Assam rhino belt	हांगुल = कश्मीर
Keibul Lamjao	**Loktak, Manipur**	Wular/Dal	केइबुल = लोकटक
Diversity vs latitude	**Decreases** equator → poles	Increases toward poles	विविधता ध्रुव की ओर घटती
MISHTI year	Budget **2023–24**	2019	मिश्टी ≠ 2019
Olivia season	Nesting **Nov–May**	August	ऑलिविया ≠ अगस्त
Ramsar NOT UP	Rudrasagar Tripura; Sultanpur HR; Surinsar–Mansar J&K	Mark as UP	रुद्रसागर = त्रिपुरा
Ballia	Near-nil **forest**; **Surha Tal** Ramsar	"No wetland"	बलिया = वन नहीं, रामसर हाँ

Pair	Correct	Trap	Hindi
Forest cover vs RFA	ISFR **cover** ≠ legal recorded forest	Mix the two	आवरण ≠ अधिसूचित
Agasthyamalai	Kerala **and TN**	Only one state as if exclusive	अगस्त्यमलाई = केरल–TN
Nilgiri BR	**First** Indian BR (1986)	Call Nanda Devi first	नीलगिरि पहला
Hotspot count	**Four** Indian hotspots	Mix with 18 BRs	चार हॉटस्पॉट
Hotspot names	Himalaya + WG–Sri Lanka + Indo-Burma + Sundaland (**Nicobar**)	Split Himalaya east/west, or add Andaman as if it were Sundaland	निकोबार = सुंडालैंड
Wet evergreen rain	NCERT **>200 cm**; many notes also use **>250** for wet evergreen	Mix the two numbers in one question	200 / 250
Dry evergreen vs wet evergreen	Dry evergreen = **TN coast**, NE monsoon ~100 cm	Call it Western Ghat rain forest	कोरोमंडल शुष्क सदाबहार
Chir vs deodar belt	Chir = **subtropical pine** 1000–2000 m west; deodar = temperate	Swap belts	चीड़ निचला; देवदार ऊँचा
Keystone vs flagship	Keystone = ecosystem collapse if lost; flagship = public mascot	Call panda a keystone fact for India	कीस्टोन ≠ फ्लैगशिप
Flame of the forest	**Butea monosperma** / Palash / Dhak; UP state flower	Bauhinia / Jacaranda / Teak	पलाश = UP फूल
Katha wood	**Khair**	Sal / babool / saaja	कत्था = खैर
Tree-line E vs W	**Lower** in **Western** Himalaya	Higher in west	पश्चिम ट्री-लाइन नीची
Himalaya aspect	**Southern** slopes denser (more rain + sun)	Northern denser	दक्षिणी ढाल सघन
Vembanad mangrove	**Kerala**	Karnataka	वेम्बनाड = केरल
Mahogany belt	Tropical evergreen (not Himalaya)	Call it Himalayan with fir/spruce	महोगनी ≠ हिमालय
Khejri vs Dhak	Khejri = desert social forestry; Dhak/Palash = flame of forest	Swap uses	खेजड़ी ≠ पलाश

N.1 Natural Vegetation & Forest Types

Type	Rain (cm)	Region	Trees / trait
Tropical evergreen (Selvas)	**>200**	W Ghats, NE, Kerala, WB, Andamans	Multilayer; rosewood, mahogany, ebony, cinchona, rubber, bamboo; **no** complete dry leaf-shed
Semi-evergreen	Slightly less than pure evergreen	W Ghats / NE / Andamans mix zone	Evergreen + deciduous mix
Moist deciduous	**100–200**	Central India, Terai (तराई)–Bhabar (भाबर), E slopes of W Ghats	**Teak, sal**, shisham, neem, mango; **largest area**; sheds in dry season
Dry deciduous	**70–100**	UP, AP, MH, KA, TN; drier peninsula	Rosewood, mango, sandalwood, mahua
Thorn / scrub	**<50–70**	RJ, GJ, SW Punjab–Haryana, rain-shadow / W AP	Babool, ber, khair, khejri, palas, wild date; xerophytes
Montane	altitude	Himalaya; S Indian hills (sholas)	Belts, not plains rainfall ladder; deodar 1500–2500 m west
Littoral / mangrove	tidal	Ganga (गंगा)–Brahmaputra (ब्रह्मपुत्र), Godavari (गोदावरी)–Krishna (कृष्णा) deltas + coasts	Sundari; pneumatophores; saline (लवणीय)

Head	Sub-types
Moist tropical	Wet evergreen · Semi-evergreen · Moist deciduous · Littoral and swamp
Dry tropical	Dry evergreen · Dry deciduous · Thorn
Montane sub-tropical	Broad-leaved hill · Moist hill (pine) · Dry evergreen
Montane temperate	Wet temperate · Himalayan moist temperate · Himalayan dry temperate
Alpine	Sub-alpine · Moist alpine scrub · Dry alpine scrub
Share of forest area (class figures)	Type
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About **37%**	Tropical **moist deciduous** (largest)
About **28%**	Tropical **dry deciduous**
About **8%**	Tropical wet evergreen
About **6%**	Sub-tropical moist hill (pine)
About **4%**	Tropical semi-evergreen

World vegetation matches

Name	Where	Memory
Maquis	Mediterranean coast	Maquis = Mediterranean
Chaparral	California	Both start with C
Matorral	Chile	Both have "l"
Fynbos	South Africa (Cape)	Fynbos = fine bush of the Cape

Plant	Belt	Why
Epiphytes (orchids on trees)	Equatorial rainforest	Need constant wet heat and a canopy
Acacia	Sahara / arid Africa	Thorn tree of dry tracts
Baobab	Savanna	Fat-trunk tree of grassland with scattered trees
Cedars	Mediterranean	Atlas / Lebanon cedar, not equatorial forest

N.2 Type Cards

Evergreen & semi-evergreen

Note	Standard figure	Alternate note you may see
Rain	**>200 cm**	Wet evergreen often written **>250 cm**
Mean annual temperature	Above about **22°C**	Some notes use about **25–27°C**
Humidity	Near **70%**	Some notes write **>77%**
Dry season	Almost none / very short	Short dry season in wet-evergreen notes

Deciduous (monsoon forest)

Type	Rain (usual)	Area	Signature trees
Moist deciduous	**100–200 cm**	Largest share (~**37%**)	**Teak**, **sal**, shisham, neem, mango
Dry deciduous	**70–100 cm** (some notes widen to 100–150)	UP, AP, MH, KA, TN; drier peninsula	Rosewood, mango, sandalwood, mahua

Type	Rain (usual)	Area	Signature trees
Whole deciduous family (wide notes)	Sometimes written **70–200 cm**	Same monsoon-forest family	Split by rain, not by a different "family"

Thorn / scrub (arid / tropical thorn)

Note	Figure / trait
Rain	Often **<75 cm** (NCERT / many notes **<70 cm**)
Humidity	**<50%**
Temperature	About **25–30°C**
Tree height	Only about **6–10 m**, scattered
Dominants	**Acacia** and **Euphorbia**

Montane (altitude elevator)

Height (approx.)	Belt	What you see
Foothills	Tropical / sub-tropical deciduous	Warm lower slopes
About **1000–2000 m**	Wet temperate oak–chestnut	Broad-leaf temperate start
About **2000–3000 m** (southern slopes)	Chir / deodar / pine / spruce / fir	Conifer timber belt
Higher	Alpine juniper / rhododendron / birch	Scrub toward snowline
Highest usable pasture	**Bugyals** (alpine meadows)	Then permanent snow

Point	Eastern / Central	Western / north-western
Tree-line	About **3600–3800 m**	About **3300–3600 m** (**lower**)
Alpine climb	Climbs **higher**	Stops earlier
Why	Nearer the **equator** and the **sea**; wetter	Drier monsoon side
Resource note	Wet forests	Forests are a **major land resource** of the western Himalayan region

Mangrove / tidal (littoral & swamp)

Mangrove pocket	State
Achra–Ratnagiri	**Maharashtra**
Coondapur	**Karnataka**
Pichavaram	**Tamil Nadu**
Vembanad	**Kerala** (not Karnataka)

Fact	Note
Flame of the forest	**Butea monosperma** = Dhak / Palash = UP state flower
Katha	From **khair** wood
Khejri	Multipurpose arid / social-forestry tree
Birch	Himalayas; writing-bark tradition
Fern	Vascular, spore-bearing; **no flowers / no seeds**
Oryx vs Chiru	Oryx = hot arid; Chiru = cold high-mountain steppe (स्टेपी) / semi-desert
Teak–Central India	Major teak expansion in MP–CG–Odisha; MP often leads dense deciduous area
Cinchona	Evergreen / Terai–NE match; **not** Chhattisgarh

N.3 Forest Distribution & ISFR

Active Forest Benchmark (For 2025 & 2026 Exams)

Parameter	Latest Figure (For 2025–26 Exams)	% of India's TGA	Trend / Change
Forest Cover	**7,15,343 sq km**	**21.76%**	Net increase across country
Tree Cover	**1,12,014 sq km**	**3.41%**	Trees outside forest (TOF)
Total Green Cover	**8,27,357 sq km**	**25.17%**	Net gain of **+1,445 sq km**

Metric	Official Figure	% of UP's Total Area	Notes
Forest Cover	**15,045.80 sq km**	**6.24%**	Satellite canopy ≥ 10%
Tree Cover	**8,950.92 sq km**	**3.72%**	Small patches & roadside trees
Total Green Cover	**23,996.72 sq km**	**9.96%**	UP gained **+559 sq km** (#2 in India!)

Metric	Official Figure	% of UK's Total Area	Notes
Forest Cover	**~24,305 sq km**	**45.44%**	Actual satellite canopy mapped
Recorded Forest Area (RFA)	**~38,000 sq km**	**~71.05%**	Legally notified forest in government records
Growing Stock Rank	**Rank #1 in India**	—	Highest increase in forest growing stock nationwide

N.4 Plant Acclimatization & Adaptations

Type	Hindi	Meaning	Example
Xerophyte	मरुद्भिद	Dry-adapted	Thorn forest: spines, succulence, deep roots
Hydrophyte	जलोद्भिद	Water-adapted	Lotus, hydrilla
Mesophyte	समोद्भिद	Moderate moisture	Most crops / deciduous understorey
Halophyte	लवणमृद्भिद	Salt-adapted	Mangroves
Epiphyte	अधिपादप	Grows on another plant	Equatorial orchids

Feature	Hindi	Meaning	Example
Pneumatophore	वात-मूल / श्वसन मूल	Breathing root	Mangrove roots that stick up out of mud

N.5 Protected Areas — Framework

UNESCO MAB and India's Biosphere Reserve Programme

Category	Fact
National Park	Strictest; no private rights in core idea; habitat + species
Wildlife Sanctuary	Wildlife first; some regulated rights possible
Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र)	**Core** (strict) + **buffer** + **transition** (settlements/use) — UNESCO MAB idea; India list is national programme too
Conservation / Community Reserve (सामुदायिक आरक्षित क्षेत्र)	WLPA later additions — community/landscape gaps
Tiger Reserve	NTCA unit (core+buffer); may contain NP + WLS

High-frequency NP / species

Name	State	Extra
Dachigam	J&K	**Hangul** / Kashmir stag
Keibul Lamjao	Manipur	Only floating NP; **Loktak**; Sangai
Dudhwa	UP	Only UP NP; Lakhimpur Kheri
Simlipal	**Odisha** (ओडिशा)	Not MP
Bandipur (बांदीपुर)	Karnataka	
Manas	Assam	WLS/BR/tiger landscape
Periyar (पेरियार)	Kerala	
Keoladeo	Rajasthan	Birds
Kanha	MP	Tiger
Indravati	Chhattisgarh	2020 match
Mollem	Goa	2020
Kalesar	Haryana	2020
Betla	Jharkhand	2020 (paper OCR "Betva")

Biosphere reserves (match gold)

Biosphere	State(s)	Trap
Nilgiri (नीलगिरि)	TN, KL, KA	**First** Indian BR (1986)
Nanda Devi	**Uttarakhand** (उत्तराखंड)	Not UP
Nokrek	**Meghalaya** (मेघालय)	**Not Manipur**
Great Nicobar	A&N	
Gulf of Mannar	Tamil Nadu	Also reefs
Manas	Assam	Correct in 2025
Sundarbans	West Bengal	
Simlipal	Odisha	
Pachmarhi	MP	2021 match
Agasthyamalai	**Kerala and TN**	"Kerala" alone treated correct (2025)
Achanakmar–Amarkantak	MP–CG	
Panna	MP	
Kachchh	Gujarat	
Cold Desert	HP	
Seshachalam	AP	
Khangchendzonga	Sikkim	
Dibru-Saikhowa	Assam	
Dehang-Debang	Arunachal	

N.6 Uttar Pradesh — Forests, Parks, Sanctuaries, Ramsar

Name	Type	Belt	Note
Dudhwa	NP	Lakhimpur Kheri	Only UP NP

Name	Type	Belt	Note
Dudhwa TR	Tiger Reserve	Kheri + Bahraich complex	**NP + Kishanpur + Katarniaghat**
Pilibhit TR	Tiger Reserve	Pilibhit–Shahjahanpur (शाहजहाँपुर)	Separate Terai TR — not Dudhwa NP
Amangarh TR	Tiger Reserve	**Bijnor**	Corbett buffer/extension
Ranipur TR	Tiger Reserve	**Chitrakoot** (चित्रकूट)	Bundelkhand
Sanctuary	District	Extra	
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Katarniaghat	Bahraich	Est. **1975**	
Kishanpur	Lakhimpur Kheri	Dudhwa TR	
Chandraprabha	**Chandauli**	not Sonbhadra	
National Chambal (चंबल)	Etawah (इटावा)–Agra	Gharial; **UP–MP–RJ**	
Hastinapur (हस्तिनापुर)	Meerut (मेरठ)–Bijnor–Amroha	Floodplain / swamp deer	
Kaimoor	Sonbhadra–Chandauli	Vindhyan	
Suhelwa	Balrampur–Shravasti–Gonda	Nepal fringe	
Sohagibarwa	Maharajganj	Terai	
Mahavir Swami (स्वामी)	Lalitpur	Bundelkhand	
Ranipur	Chitrakoot	TR base	
Okhla	GB Nagar fringe	Birds	

UP Ramsar (learn IN vs NOT)

#	Ramsar site (IN UP)	District / belt
1	Upper Ganga	Brijghat–Narora
2	Nawabganj	Unnao
3	Sarsai Nawar	Etawah
4	Sandi	Hardoi
5	Samaspur	Raebareli
6	Parvati Arga	Gonda
7	Saman	Mainpuri
8	Sur Sarovar (Keetham)	Agra
9	Haiderpur	Muzaffarnagar–Bijnor
10	Bakhira	Sant Kabir (कबीर) Nagar
11	Patna Bird Sanctuary	Etah
12	Sheikha / Shekha Jheel	Aligarh (अलीगढ़)
13	Surha Tal / JP Narayan Bird Sanctuary	Ballia
Name that sounds like UP	Actual state	Trap
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Rudrasagar	**Tripura**	Not UP
Sultanpur	**Haryana**	Not UP
Surinsar–Mansar	**J&K**	Not UP

Common Traps — Don't Fall For These

1. **Simlipal = Madhya Pradesh** — FALSE. **Odisha**.
2. **Nokrek = Manipur** — FALSE. **Meghalaya**.
3. **Nanda Devi = Uttar Pradesh** — FALSE. **Uttarakhand**.
4. **Botanical garden = in-situ** — FALSE. **Ex-situ**.
5. **Dudhwa = Bahraich / Pilibhit** — FALSE. **Lakhimpur Kheri** (2022).
6. **Pilibhit / Amangarh = National Parks** — FALSE. They are **Tiger Reserves**; only NP = Dudhwa.
7. **Chandraprabha = Sonbhadra** — FALSE. **Chandauli** (2023).
8. **Diversity increases toward poles** — FALSE. It **decreases**.
9. **MISHTI started 2019** — FALSE. Budget **2023–24**; aim = restore mangroves.
10. **Olivia in August** — FALSE. Nesting **~Nov–May**; Coast Guard, Odisha.
11. **Rudrasagar / Sultanpur in UP** — FALSE. Tripura / Haryana.
12. **Ballia has no ecology to ask** — FALSE. Near-nil *forest*; **Surha Tal** is Ramsar.
13. **Evergreen = largest Indian forest** — FALSE. **Moist deciduous**.
14. **Coral death = ocean cooling** — FALSE. **~2°C warming**.
15. **Project Tiger launched 1972 as open fact** — Open recall = **1 April 1973**; 2018 MCQ omitted 1973.
16. **Hotspots = 18** — FALSE. **Four** hotspots; **18** biosphere reserves.
17. **Nanda Devi = first BR** — FALSE. **Nilgiri (1986)** is first.
18. **Forest cover % = Recorded Forest Area %** — FALSE. ISFR cover ≠ legal RFA.
19. **Four hotspots = Eastern Himalaya + Western Himalaya + Western Ghats + Andaman** — FALSE. The four are Himalaya, Western Ghats–Sri Lanka, Indo-Burma, and Sundaland (**Nicobar**).
20. **Dry evergreen = Western Ghat rain forest** — FALSE. It is the **Tamil Nadu / Coromandel** NE-monsoon belt.
21. **Chir pine is an alpine juniper scrub** — FALSE. Chir is **sub-tropical pine** at about 1000–2000 m in the western Himalaya.
22. **Hotspots exist only in the tropics** — FALSE. Mediterranean, SW Australia and Chilean temperate forest are hotspots too.
23. **Northern Himalayan slopes denser than southern** — FALSE. **Southern** slopes denser (more rain + sun).
24. **Tree-line higher in Western Himalaya** — FALSE. Tree-line is **lower** in the west.
25. **Vembanad = Karnataka** — FALSE. **Kerala**.
26. **Flame of the forest = Teak / Bauhinia** — FALSE. **Butea monosperma (Palash / Dhak)**; UP state flower.
27. **Katha from Sal / Babool** — FALSE. From **khair**.
28. **Mahogany = Himalayan species** — FALSE. Tropical evergreen; Himalayan set = juniper, fir, spruce, deodar, birch.
29. **Cinchona grows in Chhattisgarh** — FALSE. Assam / Kerala / West Bengal evergreen belts.
30. **Birch = Nilgiris / Aravallis / Vindhyas** — FALSE. **Himalayas**.

Topic 8 — Minerals, Energy & Industry

Current Affairs (this topic)

Year	Fact	Why asked	Source
2023	Li inferred resources **Reasi, J&K**; India import-dependent for **Li, Ni, Co**	Frozen 2023 Q77	MoM / PYQ
2023	Union list (संघ सूची) of **24 critical minerals** (Li, Ni, Co, REE, graphite, etc.)	Scheme/list trap	Ministry of Mines
2023–26	**Critical Minerals Mission** + auction of critical-mineral blocks	Living CA	PIB / MoM
2015 / 2025	**ISA**: India+France, HQ **Gurugram**; **not 38 members**; **OSOWOG** under ISA	2019 Q63 + 2025 Q109	ISA / MEA
Jan 2026	US announced withdrawal from ISA; ISA continues with **120+** members	Member-count trap	MEA / ET
2018	Major solar plant **Mirzapur (मिर्ज़ापुर) (UP)** with **France**	UP energy PYQ	2020 Q46
DFC	**Western** Dadri (ददरी)–JNPT; **Eastern** Ludhiana–Dankuni — phased commissioning	Corridor syllabus	DFCCIL
Nuclear	Kakrapar / Kudankulam expansion; **Narora = only UP** plant	Plant–state	NPCIL

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q100

Which is the largest centre of the petrochemical industry in India?

A. Trombay B. Jamnagar C. Digboi D. Vadodara

Confused Pairs

Pair	Correct	Trap	Hindi
Iron leaders	**OD + CG** (2021–22 report)	JH+KA always lead	ओडिशा+छत्तीसगढ़
Bailadila	**Chhattisgarh**	Odisha	बैलाडीला = CG
Noamundi	**Jharkhand**	Odisha	नोआमुंडी = JH
Kudremukh	**Karnataka** (magnetite)	Odisha	कुद्रेमुख = KA
Mount Newman	**Iron**	Manganese	लौह अयस्क
Postmasburg	**Manganese (SA)**	U/mica	मैंगनीज
MALCO	**Mettur, TN**	Chennai	मेट्टूर
Khetri	**Copper, RJ**	Gold	खेतड़ी = तांबा
Panna	**Diamond, MP**	Gold	पन्ना = हीरा
Sukinda	**Chromite, OD**	Bauxite	सुकिंदा = क्रोमाइट
Zawar	**Zn–Pb, RJ**	Copper	जावर = जस्ता
Jaduguda	**Uranium, JH**	Thorium sands	जादूगुड़ा = यूरेनियम
Maolong	**Meghalaya**	Arunachal	मेघालय कोयला
Singrauli	**MP–UP**; oldest NTPC	Northern CG	सिंगरौली ≠ CG
Ruhr	**Germany**	Britain/France	रुहर = जर्मनी
Kashagan	**Kazakhstan**	Kuwait	कशागन = कजाख

Pair	Correct	Trap	Hindi
CNG	**Methane**	Butane	मीथेन
Black Revolution	**Petroleum**	Oilseeds (Yellow)	काला = पेट्रोलियम
Shivanasamudra	**Hydel**	Nuclear	शिवणसमुद्र = जल
Kalpakkam	**Nuclear**	Hydel	कल्पक्कम = परमाणु
Ahmedabad mills	In cotton **belt**; not traditionally **largest** centre	Largest = Mumbai	मुंबई कपड़ा केंद्र
Magnesite vs Mn	Magnesite = $MgCO_3$ refractory	Manganese metal	मैग्नेसाइट ≠ मैंगनीज
Neyveli	**Lignite, TN**	Gondwana bituminous	नेयवेली = लिग्नाइट
ISA HQ	**Gurugram**	New Delhi / Paris	गुरुग्राम
LPG	Propane/butane mix	CNG = methane	एलपीजी ≠ सीएनजी
Bhilwara	Mica + Rampura-Agucha **Zn**	Manganese city (2020)	भीलवाड़ा ≠ Mn
Bhilai steel	**Chhattisgarh**	Odisha	भिलाई = CG
Firozabad	**Glass**	Brass (that is Moradabad)	फिरोजाबाद = कांच
Pegu Yoma	**Mineral oil** (Myanmar)	Tin (that is Kinta / Bangka)	पेगू योमा = तेल
Mangalore–Kanpur line	Most minerals **east** of line	Think west-rich	मंगलुरु–कानपुर रेखा
Northern plains minerals	Almost **devoid** of economic minerals	Treat Ganga plain as ore belt	मैदान ≈ खाली
Non-ferrous endowment	**Poor except bauxite**	India rich in all non-ferrous	बॉक्साइट अपवाद
Magnetite vs hematite	Magnetite ~**70%** Fe; hematite most **used**	Swap grades / uses	मैग्नेटाइट ≠ हेमेटाइट मात्रा
Chota Nagpur vs Aravalli	Chota Nagpur = richest multi-mineral	Call Aravalli richest	छोटानागपुर = समृद्ध
Mica belt	**Koderma–Gaya–Hazaribagh** + Nellore + Ajmer	Only “Kodarma town” as if exclusive	कोडरमा–गया–हजारीबाग
Chitradurga	**Iron** (Ballari–Chitradurga belt, KA)	Copper city (2020 trap)	चित्रदुर्ग = लौह
Guntur	**Not** an iron-ore mining centre	List with Ballari / Bailadila as iron	गुंटूर ≠ लौह
Coal rank %	Peat **<40** → lignite **40–55** → bituminous **60–80** → anthracite **80–95**	Swap % or skip homes	कार्बन % सीढ़ी
Anthracite India	Scarce; **J&K** pockets	Call it India’s bulk coal	एन्थ्रेसाइट ≠ मुख्य
Lignite home	**Neyveli (TN)** (+ RJ / GJ)	Call Neyveli bituminous Gondwana	नेयवेली = लिग्नाइट
Bituminous share	~**80%** deposits; mostly **non-coking**	Call most Indian coal anthracite / coking	अधिकतर नॉन-कोकिंग

N.0 Framework (map first)

Rock systems (Ghatnachakra / NCERT map)

System	Character	Main minerals
Dharwar	Oldest metamorphosed belts; first studied in Karnataka	**Storehouse of metallic minerals** — iron, manganese, gold, copper; most mineralised system in teaching keys

System	Character	Main minerals
Gondwana	Palaeozoic–Mesozoic sediments in eastern/central valleys	About **95–99%** of India's **coal** (Damodar, Son, Mahanadi, Godavari, etc.)
Vindhyan	Sedimentary (limestone, sandstone, shale)	**Limestone**, sand, china clay — cement base; **not** the metallic king
Cuddapah / Tertiary	Younger / basin rocks	Tertiary coal NE; petroleum in sedimentary basins (Gujarat, Assam, offshore)

How minerals occur

Mode	How it forms	Examples
Vein / lode	Molten / gaseous minerals rise into cracks of **igneous and metamorphic** rock, then cool	Tin, copper, zinc, lead
Bed / layer	Deposition in **sedimentary** strata under heat and pressure	Coal, iron ore, gypsum, potash, common salt (लवणाध्यक्ष)
Residual	Surface rock weathers; soluble parts wash away; a concentrated cap remains	**Bauxite** on laterite
Placer / alluvial	Hard grains collect in valley-floor / hill-base sands (not corroded by water)	Gold, silver, tin, platinum, **monazite**
Ocean water / ocean bed	Dissolved salts, and nodules on the sea floor	Common salt, magnesium, bromine; **manganese nodules**

Classification

Class	Meaning	Examples
Ferrous metallic	Contain iron / feed steel	Iron ore, manganese, chromite, nickel, cobalt
Non-ferrous metallic	Metals without iron as the main metal	Bauxite (aluminium), copper, zinc, lead, tin
Precious metals	High-value non-ferrous subset	Gold, silver, platinum
Non-metallic (inorganic)	Not used mainly as metal	Mica, limestone, gypsum, graphite, diamond, salt, magnesite
Energy / mineral fuels	Organic fossil fuels + nuclear fuels	Coal, petroleum, natural gas, uranium, thorium

Mineral belts of India

Belt	Cover	Character
North-Eastern Plateau (Chota Nagpur)	Jharkhand, Odisha, West Bengal, parts of Chhattisgarh	**Richest** — iron, coal, manganese, bauxite, mica, copper, chromite; often called **India's Ruhr**
Central belt	Chhattisgarh, Madhya Pradesh, Telangana, Andhra Pradesh, Maharashtra	Manganese, bauxite, limestone, coal, iron, mica, graphite, gems
South-Western Plateau	Karnataka, Goa, adjoining Tamil Nadu–Kerala uplands	Ferrous metals, bauxite, limestone; **almost no Gondwana coal**; **Neyveli lignite** in Tamil Nadu; Kerala **monazite** sands; Goa iron
North-Western (Aravalli)	Rajasthan–Gujarat	Copper, zinc–lead, mica, building stone, gypsum, **Fuller's earth**, salt; Gujarat petroleum
Himalayan / Assam–Arakan	North-East and Himalayan foothills	Copper–lead–zinc pockets; **Tertiary coal**; Assam oil

Master map — mineral by state / centre (revise daily)

Mineral	Main state / centre	One-line note
Iron ore	**Odisha**, Chhattisgarh, Karnataka, Jharkhand	Bailadila (CG); Noamundi (JH); Kudremukh / Ballari (KA)
Manganese	**Odisha**, Maharashtra, Madhya Pradesh	Balaghat (MP); Nagpur–Bhandara (MH)
Chromite	**Odisha (Sukinda)**	Stainless steel / ferrochrome
Bauxite	**Odisha (Panchpatmali)**	Residual laterite ore of aluminium
Copper	**Khetri (RJ)**; Malanjkhand (MP); Singhbhum (JH)	India still imports copper
Zinc–lead	**Zawar**; **Rampura-Agucha (Bhilwara)** — RJ	Rajasthan near-monopoly
Gold	**Hutti** (working); **Kolar** (historic (ऐतिहासिक)) — KA	Tiny mine output vs jewellery
Mica	**Koderma–Gaya–Hazaribagh**; Nellore; Ajmer (अजमेर)–Bhilwara	Dielectric sheets
Diamond	**Panna (MP)**	India's diamond centre
Coal (Gondwana)	Damodar–Son–Mahanadi–Godavari valleys	~**97%** of Indian coal
Lignite	**Neyveli (TN)**	Brown coal; pit-head power
Petroleum	Assam; Mumbai High; Gujarat; Barmer; KG / Cauvery	Sedimentary basins only
Uranium	**Jaduguda (JH)**	Also Tummalapalle (AP); Domiasiat (ML)
Thorium / monazite	**Kerala–Tamil Nadu** beach sands	Also rare earths

N.1 Metallic Minerals

Iron

Ore	Iron content	Trait
Magnetite (Fe_3O_4)	About **70%**	Finest / richest ore; magnetic; electrical industry
Hematite (Fe_2O_3)	About **50–60%**	Most used industrial ore by quantity
Limonite	About **40–60%** (lower / variable)	Hydrated iron oxide; lesser industrial weight
Siderite	About **40–50%** (lower)	Iron carbonate ore; lesser industrial weight

Belt	Note
Odisha–Jharkhand	Mayurbhanj (मयूरभंज)–Keonjhar–Sundargarh hematite; Gua–Noamundi in Singhbhum
Durg–Bastar–Chandrapur	Bailadila high-grade hematite in Chhattisgarh; Dalli–Rajhara; export via Visakhapatnam
Ballari–Chitradurga–Chikkamagaluru–Tumakuru	Karnataka magnetite–hematite; Kudremukh slurry to Mangaluru
Maharashtra–Goa	Lower grade; export via **Mormugao**
Mine / belt	State
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Gurumahisani / Badampahar / Sulaipat	**Odisha**
Bonai / Kendujhar / Sundargarh	Odisha
Bailadila	**Chhattisgarh**
Dalli–Rajhara	Chhattisgarh

Belt	Note
Noamundi / Gua / Kiriburu	**Jharkhand**
Kudremukh	**Karnataka**
Donimalai / Ballari–Hospet	Karnataka
Ratnagiri belt	Maharashtra

Manganese

Point	Teaching line
Steel need	About **10 kg of manganese** per tonne of steel
Rock link	Mainly associated with **Dharwar** rocks
Leading states	**Odisha, Maharashtra, Madhya Pradesh** (Karnataka is another classic belt)
Madhya Pradesh belt	**Balaghat**
Maharashtra belt	**Nagpur–Bhandara**
Odisha belt	With the iron belt of **Keonjhar–Sundargarh**
World centre	**Postmasburg** (South Africa)

Chromite

Point	Teaching line
Main Indian belt	**Sukinda in Odisha** (India's leading chromite belt)
Use	**Stainless steel** and **ferrochrome**
Belt family	Eastern plateau ferrous belt with Odisha iron–manganese country

Bauxite and aluminium

Point	Teaching line
Leading state	**Odisha**
Leading mine	**Panchpatmali** (Koraput, Odisha)
Other belts	Gujarat, Jharkhand, Chhattisgarh, Maharashtra
Older city tag	**Katni** (Madhya Pradesh)
Smelting logic	Aluminium smelting is **power-hungry**, so plants sit near cheap electricity as well as near ore
Company (कंपनी)	Plant
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BALCO	**Korba**
HINDALCO	**Renukoot**
MALCO	**Mettur**
INDAL	**Hirakud** (हीराकुंड)
NALCO	Damanjodi (alumina) / Angul (smelter)

Copper

Centre	State	Note
Khetri (Khetri–Singhana belt)	**Rajasthan**	Classic textbook copper centre
Malanjkhand	**Madhya Pradesh**	Large **open-pit** copper mine
Singhbhum (Mosabani–Ghatsila)	**Jharkhand**	Eastern copper belt

Centre	State	Note
World lead	**Chile** (northern **Andes porphyry**)	Leading producer in the usual paper set

Gold

Centre	State	Note
Kolar	Karnataka	Historic goldfield; now largely closed
Hutti	Karnataka	Main working goldfield
Ramagiri	Andhra Pradesh	Smaller goldfield

Tin

Point	Teaching line
Ore	**Cassiterite**
Only producing state (recent IMYB)	**Chhattisgarh** (Dantewada belt)
Reserve tables	Haryana also appears in resource share
World centres	**Kinta Valley (Malaysia)**; **Bangka (Indonesia)**

Zinc, lead and silver

Centre	State	Metal
Zawar (near Udaipur)	Rajasthan	Historic **zinc–lead** (silver often a by-product)
Rampura-Agucha (Bhilwara)	Rajasthan	Asia's richest zinc–lead reserve belt; Hindustan Zinc link

N.2 Non-metallic and atomic minerals

Mica

Belt	State / region	Note
Koderma–Gaya–Hazaribagh–Munger	Jharkhand–Bihar	Classic belt; **Kodarma** often called the "capital of mica"
Nellore	Andhra Pradesh	Southern mica centre; AP leads **resource** share in IMYB
Ajmer–Bhilwara	Rajasthan	Rajasthan leads recent **production**; Bhilwara is also zinc country

Limestone, dolomite, gypsum, cement rocks

Rock / mineral	Use	Where (teaching centres)
Limestone	Cement; bulky → plant near quarry	UP: **Chunar–Churk–Dalla**; wide Vindhyan / peninsula belts
Dolomite	Steel **flux**; refractory	UP: **Sonbhadra–Mirzapur**; Odisha and other plateau belts
Gypsum	Cement; reclaim **usar / alkaline** soils	Rajasthan; western Uttar Pradesh

Graphite and diamond

Mineral	Where	Use / note
Graphite	Odisha, Jharkhand	Crucibles, pencils, electrodes; carbon mineral
Diamond	**Panna (Madhya Pradesh)**	India's main diamond centre

Salt, magnesite, silica, Fuller's earth

Mineral	Chemistry / use	Where
Common salt	Food; chemicals; from sea / lakes	**Sambhar Lake (RJ)** inland; Gujarat coastal solar salt
Magnesite	$MgCO_3$; **refractory** bricks	Tamil Nadu; Uttarakhand (उत्तराखंड)
Silica sand	Glass sand	**Shankargarh (Prayagraj (प्रयागराज))** → **Firozabad** glass
Fuller's earth	Clay used to clean oils and grease	North-western **Aravalli** (अरावली) belt, Rajasthan

Other industrial minerals

Mineral	Use	Where
Kyanite / sillimanite	Refractories	Eastern plateau (Jharkhand–Odisha)
Asbestos	Older industrial fibre	Rajasthan; Andhra belts
Phosphorite	Fertilizer phosphate	**Jhamarkotra (Rajasthan)**; Bundelkhand / Lalitpur

Atomic minerals and critical minerals

Mineral	Main centre	Note
Uranium	**Jaduguda (Jharkhand)**	Singhbhum shear; also Bhatin–Narwapahar–Turamdih
Uranium (other)	**Tummalapalle (AP)**; **Domiasiat / Mawthabah (Meghalaya)**	Extra centres
Thorium / monazite	**Kerala–Tamil Nadu** beach sands	Also Andhra–Odisha coasts; monazite carries **rare earths**
Lithium	**Reasi, Jammu & Kashmir**	2023 inferred-resource find; India still imports Li
Nickel / cobalt	Import-dependent	Laterite / ultramafic traces; no Khetri-class mines

N.3 Coal

Two geological ages

Age	Share / quality	Where found
Gondwana coal	About **97–98%** of India's coal; mostly **bituminous**; usually low sulphur, high ash	River valleys of the peninsula: **Damodar, Son, Mahanadi, Godavari** (and Wardha)
Tertiary coal	Small share; younger; often **high sulphur**	**Assam, Meghalaya, Nagaland, Arunachal Pradesh**; also Kalakot (J&K) in older NCERT lists

Types of coal (carbon % and Indian homes)

Type	Carbon (approx.)	Moisture / quality	Where found in India	Main use
Peat	Under about **40%**	Highest moisture; first stage of coal; burns like wet wood	Limited swamp / hill pockets (not a major commercial Indian coal)	Local fuel only
Lignite (brown coal)	About **40–55%**	High moisture; low grade; can ignite spontaneously	**Neyveli (Tamil Nadu)**; **Palana / Barsingsar (Rajasthan)**; **Panandhro (Gujarat)**; also J&K / coastal Tertiary notes	Pit-head **thermal power**
Bituminous	About **60–80%**	Most common Indian coal; dense black; high heat	Gondwana valleys — Jharkhand, West Bengal, Odisha, Chhattisgarh, Madhya Pradesh, Telangana, Maharashtra	Power + coke (if coking grade)

Type	Carbon (approx.)	Moisture / quality	Where found in India	Main use
Anthracite	About **80–95%**	Hardest; lowest moisture; best quality; burns with little smoke	**Very little** in India — small pockets in **Jammu & Kashmir**	High-efficiency fuel

Major Gondwana coalfields

Coalfield	State	Valley / note
Raniganj	**West Bengal**	Damodar; oldest Indian field; first commercial mining **1774**
Jharia	**Jharkhand**	Damodar; **coking**; known for mine fires; largest field in many notes
Bokaro / Ramgarh / Karanpura / Giridih	Jharkhand	Damodar family
Rajmahal	Jharkhand	Eastern Jharkhand
Talcher	**Odisha**	**Mahanadi** (महानदी) valley
Ib Valley	Odisha	Mahanadi family
Korba / Mand–Raigarh	Chhattisgarh	Thermal towns
Singrauli	**Madhya Pradesh–Uttar Pradesh**	**Son** valley; **not** northern Chhattisgarh
Sohagpur	Madhya Pradesh	Son family
Singareni	Telangana	**Godavari** (गोदावरी) valley
Wardha / Kamptee / Umrer	Maharashtra	Godavari–Wardha
Coalfield	Valley	
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Talcher	**Mahanadi**	
Karanpura	**Damodar** (दामोदर)	
Singrauli	**Son**	
Singareni	**Godavari**	

Tertiary and lignite centres

Centre	State	Type
Makum / Naharkatiya belt	Assam	Tertiary coal
Maolong	**Meghalaya**	Tertiary (**not Arunachal**)
Namchik–Namphuk	**Arunachal Pradesh** (Changlang)	Tertiary
Neyveli	Tamil Nadu	**Lignite** (main Indian lignite)
Palana / Barsingsar	Rajasthan	Lignite
Panandhro	Gujarat	Lignite

World coalfields (by country)

Field	Country
Appalachian	**USA**
Lancashire	**England**
Ruhr	**Germany**
Kuzbass	**Russia**

Field	Country
Donetsk	**Ukraine**

N.4 Petroleum, gas, refineries

Fuel	Main composition	Teaching line
Natural gas / **CNG**	Mainly **methane**	Piped / compressed methane fuel
LPG	**Propane–butane** mix	Not the same as CNG
Petroleum	Mixture of hydrocarbons	Black Revolution = petroleum; Yellow = oilseeds

Indian fields and basins

Field / basin	Note
Digboi	Assam — oldest Indian refinery town / onshore oil
Naharkatiya / Moran / Rudrasagar	Assam–Arakan onshore
Mumbai High	Offshore Arabian Sea; India's giant offshore field
Bassein / Tapti (ताप्ती)	Western offshore gas–oil
Ankleshwar / Cambay / Kalol / Mehsana	Gujarat onshore
KG basin (including KG-D6)	East-coast gas and oil
Ravva	Krishna–Godavari offshore field
Cauvery basin	Tamil Nadu–Puducherry coast
Barmer (**Mangala** and related fields)	Rajasthan onshore

Refineries

Refinery	State
Nayara (Vadinar / Jamnagar belt)	Gujarat
Jamnagar	Gujarat
Koyali (Vadodara)	Gujarat
Manali	Tamil Nadu
Kochi	Kerala
Mangaluru (MRPL)	Karnataka
Numaligarh	Assam
Digboi / Guwahati / Bongaigaon	Assam
Tatipaka	Andhra Pradesh
Visakhapatnam	Andhra Pradesh
Mathura	**Uttar Pradesh**
Barauni	Bihar
Panipat	Haryana
Bhatinda	Punjab
Paradip	Odisha
Bina	Madhya Pradesh
Haldia	West Bengal

N.5 Power: thermal, hydro, nuclear, renewables

Type	Sources	Teaching line
Conventional	Coal, petroleum, natural gas, **nuclear**	Long-used commercial fuels; mostly exhaustible
Non-conventional	Solar, wind, tidal, biogas, geothermal	Cleaner / renewable set in school keys
Stored solar?	Wind, biomass	Nuclear and geothermal are **not** stored solar

Hydroelectricity

Hydel name	Note
Shivanasamudra	pair with Kalpakkam
Bhakra	Satluj (multipurpose)
Tehri (टिहरी)	Bhagirathi (भागीरथी)
Hirakud	Mahanadi
Koyna	Maharashtra
Sharavathi / Linganamakki	Karnataka western-ghat hydel
Nagarjuna (नागार्जुन) Sagar (नागार्जुन सागर) / Srisailam	Krishna
Idukki	Kerala

Nuclear plants

Plant (परमाणु संयंत्र)	State (राज्य)	Key Facts (मुख्य तथ्य)
Tarapur (तारापुर)	Maharashtra (महाराष्ट्र)	First commercial nuclear plant in India (1969) / भारत का पहला परमाणु संयंत्र
Kakrapar (काकरापार)	Gujarat (गुजरात)	Tapi River / तापी नदी तट
Kudankulam (कुडनकुलम)	Tamil Nadu (तमिलनाडु)	Largest capacity, Russian VVER technology / सबसे बड़ा संयंत्र
Kaiga (कैगा)	Karnataka (कर्नाटक)	Kali River basin / काली नदी बेसिन
Rawatbhata / RAPP (रावतभाटा)	Rajasthan (राजस्थान)	Near Kota, Rana Pratap Sagar / राणा प्रताप सागर
Kalpakkam / MAPS (कल्पक्कम)	Tamil Nadu (तमिलनाडु)	Madras Atomic Power Station / फास्ट ब्रीडर रिएक्टर
Narora (नरोरा)	Uttar Pradesh (उत्तर प्रदेश)	Bulandshahr / बुलंदशहर (गंगा नदी तट, UP का एकमात्र संयंत्र)

N.6 Industry, Regions, Corridors

Iron and steel

Plant	State
Bhilai	Chhattisgarh
Rourkela	Odisha
Bokaro	Jharkhand
Durgapur / Burnpur	West Bengal
Visakhapatnam	Andhra Pradesh
Salem	Tamil Nadu
Bhadravati / Vijayanagar	Karnataka
Jamshedpur	Jharkhand (Tata (तत) pioneer)

Industrial regions

Region	Core character
Mumbai–Pune	Cotton textiles, engineering, chemicals, automobiles
Hugli	Jute, engineering, port industries
Bangalore–Tamil Nadu belt	Diversified engineering, autos, textiles; Bengaluru IT + Chennai + Coimbatore
Ahmedabad–Vadodara–Surat (Gujarat)	Textiles, petrochemicals, engineering
Chotanagpur	Mineral-based heavy industry — India's Ruhr industrial extension
Visakhapatnam–Guntur	Coastal steel, shipbuilding, engineering
Gurgaon–Delhi–Meerut	NCR automobiles, electronics; western Uttar Pradesh link
Kollam–Thiruvananthapuram	Kerala coastal–plantation and light industry

Agencies, SEZs and corridors

Agency	Headquarters
ONGC	New Delhi
MECL	Nagpur
NMDC	Hyderabad
NALCO	Bhubaneswar

Corridor / zone	Meaning
DMIC	Delhi–Mumbai Industrial Corridor — the must-know industrial corridor
Western DFC	Dedicated freight rail **Dadri–JNPT**
Eastern DFC	Dedicated freight rail **Ludhiana–Dankuni**
CBIC	Chennai–Bengaluru Industrial Corridor
AKIC	Amritsar (अमृतसर)–Kolkata Industrial Corridor
VCIC	Visakhapatnam–Chennai Industrial Corridor
SEZ	Export-oriented special zone

Coalfield	State (as given in options)
1. Maolong	Arunachal Pradesh
2. Raniganj	West Bengal
3. Ramgarh	Jharkhand
4. Talcher	Odisha

N.7 Uttar Pradesh focus

City	Industry tag
Kanpur	Textile, leather
Moradabad	Brass
Aligarh	Padlocks and hardware
Firozabad	Glass
Noida–Ghaziabad	NCR engineering / electronics
Meerut	NCR belt; **sports goods**

City	Industry tag
Agra	Light industry and tourism, not a mineral capital
Renukoot	Aluminium (HINDALCO)
Chunar–Dalla belt	Cement

Common Traps — Don't Fall For These

1. **West of Mangalore–Kanpur is mineral-rich** — FALSE. Most major minerals lie **east** of the Mangaluru–Kanpur line.
2. **Aravalli = richest mineral belt** — FALSE. **Chota Nagpur / NE Peninsular** is richest ("India's Ruhr").
3. **Jharkhand + Karnataka always lead iron** — FALSE for **2021–22**. **Odisha + Chhattisgarh** lead.
4. **Bailadila = Odisha** — FALSE. **Chhattisgarh**. Gurumahisani = Odisha; Noamundi = Jharkhand; Kudremukh = Karnataka.
5. **Bauxite = sedimentary basin only** — FALSE. Commonly linked to **laterite** weathering; Odisha lead (**Panchpatmali**).
6. **MALCO = Chennai** — FALSE. **Mettur, Tamil Nadu**. HINDALCO = Renukoot (UP); BALCO = Korba.
7. **Maolong = Arunachal** — FALSE. **Meghalaya**.
8. **Anthracite = softest / lignite = hardest** — FALSE. Rank rises peat → lignite → bituminous → **anthracite** (hardest, highest carbon).
9. **Singrauli NTPC = northern Chhattisgarh** — FALSE. **MP–UP**; it **is** NTPC's oldest unit.
10. **Kashagan = Kuwait** — FALSE. **Kazakhstan**. Ruhr = **Germany**, not Britain.
11. **CNG / natural gas = butane** — FALSE. **Methane**. LPG is the propane/butane mix.
12. **Solar / wind = conventional energy** — FALSE. Conventional = coal/oil/gas/nuclear; solar–wind–tidal–biogas–geothermal = **non-conventional**.
13. **Black Revolution = oilseeds** — FALSE. Oilseeds = **Yellow**; Black = **petroleum**.
14. **Ahmedabad = largest cotton mill centre** — FALSE. In the cotton **belt** (R true); traditionally **Mumbai** is largest.
15. **Shivanasamudra = nuclear** — FALSE. **Hydel**; Kalpakkam = nuclear.
16. **ISA Secretariat = New Delhi / Paris** — FALSE. **Gurugram**. Member count is **not 38**.
17. **Nuclear / geothermal = stored solar** — FALSE. Wind and biomass are solar-linked.
18. **Copper city = Chitradurga; Mn city = Bhilwara** — FALSE. Ballari = iron; Katni = bauxite.
19. **Diu is not the first 100% solar UT** — FALSE. It **is**. First solar airport = **Cochin**, not Ahmedabad.
20. **UP = Bhilai-class steel state / no coal at all** — FALSE. Facts = **Sonbhadra** energy, **non-coking** Singrauli coal, **Mathura** refinery, **Narora** nuclear, **Shankargarh** silica. Firozabad = **glass**, Moradabad = **brass**.
21. **Pegu Yoma = tin** — FALSE. **Mineral oil**. Tin = Kinta / Bangka.
22. **Northern plains = major ore belt** — FALSE. The plains are almost **devoid** of economic minerals.
23. **India rich in all non-ferrous metals** — FALSE. **Poor except bauxite**.
24. **Hematite has more Fe than magnetite** — FALSE. Magnetite ~**70%**; hematite is the **quantity** workhorse (~50–60%).
25. **~80% Indian coal is coking** — FALSE. Most is bituminous **non-coking** (thermal).
26. **Gondwana = metallic storehouse / Dharwar = coal king** — FALSE. **Dharwar** = metallic minerals; **Gondwana ≈ 95–99%** coal.
27. **Vindhyan = iron–gold belt** — FALSE. Sedimentary **limestone / china clay**.
28. **Tin producer = Assam / J&K** — FALSE. Only **Chhattisgarh** produces tin in recent production data.
29. **Digboi = largest petrochemical centre** — FALSE. **Oldest** oilfield/refinery centre; **Jamnagar** is the giant complex.
30. **ISPRL storages include Ankleshwar** — FALSE. **Visakhapatnam, Mangaluru, Padur**.

31. **Coal reserve order = production order** — FALSE. Proved reserves often **JH > OD > CG**; production often **CG > OD > MP**.
32. **Zawar = copper** — FALSE. **Zinc–lead** (Udaipur); copper = **Khetri / Malanjkhand**.
33. **Nathra-ki-Pal = copper of Rajasthan** — FALSE. **Iron ore**.
34. **Marble = igneous rock** — FALSE. **Metamorphic** (recrystallised limestone); **Makrana** best quality.

Topic 9 — Transport & Communication

Current Affairs (this topic)

Year	Fact	Why it matters
13 Oct 2021	**PM Gati Shakti** National Master Plan launched (not 2022)	Year trap with seven engines
17 Sep 2022	**National Logistics Policy** — complements Gati Shakti	Scheme pair
Feb 2023	Amrit Bharat: about **1,275** stations all-India; **149** of them in UP for modernisation	149 = UP stations, not new lines
2024–26	**New Pamban** vertical-lift railway sea bridge (RVNL)	RVNL + first vertical-lift fact
18 Mar 2024	**FLY91** inaugural from **Manohar Intl** (Mopa, Goa (गोवा))	New airline + airport pair
2024–26	**Noida International (Jewar)** — UP's new greenfield intl hub	UP airport CA
DFC	Western Dadri (ददरी)–JNPT / Eastern Ludhiana–Dankuni — phased opening	Freight spine

Sagarmala / Bharatmala remain living umbrella programmes; still tests **name swap**, not project lists.

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q88

Which of the following ports is on the western coastline?

A. Nhava Sheva B. Visakhapatnam C. Chennai D. Tuticorin

Q. UKPCS Prelims 2025, Q92

Which country has the highest density of railway network?

A. Canada B. U.S.A. C. Russia D. India

Confused Pairs

Pair	Correct	Trap	Hindi
NER HQ	**Gorakhpur**	Hajipur	गोरखपुर
SER HQ	**Garden Reach, Kolkata**	Cuttack	कटक नहीं
ECR HQ	**Hajipur**	Gorakhpur	हाजीपुर
Amrit Bharat	Station **modernisation**; UP **149 stations** (Feb 2023)	149 new railway lines	149 स्टेशन
Bharatmala	**Highways**	Ports	सड़क
Sagarmala	**Port-led**	Highways	पत्तन
Gati Shakti year	**2021**	2022	2021
Purvanchal (built)	**Passes Mau**; not Basti	Blind 2018 "not Mau" as today's map	मऊ हाँ
Atal Tunnel	**Pir Panjal**	"World's longest highway"	पीर पंजाल
Igarka	**Russia**	China	रूस
NW-1	**Ganga**	Brahmaputra	गंगा
NW-3	**Kerala canal**	Ganga	केरल
Harbour vs port	Shelter vs shelter+facilities	Same thing	पोताश्रय ≠ पत्तन
Fibre	**TIR**	Refraction	पूर्ण आंतरिक परावर्तन
Motihari pipe	**Amlekhganj**	Kathmandu	अमलेखगंज

Pair	Correct	Trap	Hindi
Solar airport	**Cochin**	Delhi / Ahmedabad	कोच्चि
Paradip	**Odisha**	AP / WB	पारादीप = ओडिशा
Alappuzha	**Kerala**	TN	आलप्पुझा
NH vs rural roads	NH ~**2%** length but ~**40%** traffic	Think NH = most km	राष्ट्रीय राजमार्ग
Silchar spelling	EW corridor = **Silchar / Silcher** Assam	Mix with Shillong	सिलचर

N.1 Railways

Factory (फैक्टरी) / works	Place	What it builds
Rail Coach Factory	**Kapurthala**	Coaches
Integral Coach Factory	**Perambur** (Chennai)	Coaches
Chittaranjan Locomotive Works	**Chittaranjan**	Electric locomotives
Banaras Locomotive Works	**Varanasi (वाराणसी) / Manduadih**	Diesel–electric locomotives
Modern Coach Factory	**Rae Bareli**	Third major coach factory

Zone	Headquarters
Northern	New Delhi
North Eastern	**Gorakhpur**
North Central	Prayagraj
East Central	**Hajipur**
Eastern	Kolkata (Fairlie Place)
South Eastern	**Garden Reach, Kolkata**
Southern	Chennai
South Central	**Secunderabad** (not Hyderabad (हैदराबाद) city trap)
Western	Churchgate, Mumbai
West Central	Jabalpur
Central	Mumbai CSMT
East Coast	Bhubaneswar
North Western	**Jaipur** (not Jodhpur)
South Western	**Hubballi**
South East Central	Bilaspur
South Coast	Visakhapatnam
Metro Railway Kolkata	17th zone (2010)

N.2 Roads, Expressways, Bharatmala

Class	Who builds / maintains	Note
National Highways	Central government / **NHAI**	About **2%** of road length but about **40%** of road traffic
State Highways	State governments	Link the state capital to district headquarters

Class	Who builds / maintains	Note
District roads	Zila Parishad (जिला परिषद) / district	Link the district headquarters to important nodes
Rural roads	Local bodies; **PMGSY** push	About **80%** of total road length; all-season village links
Border roads	**BRO** (established May 1960)	Northern and north-eastern borders; high-altitude strategic infrastructure; executes **Project Dantak** in Bhutan, and infrastructure projects in Myanmar and Afghanistan
International highways	Cross-border links	Neighbour connectivity

Expressway	Note
Yamuna (यमुना)	Greater Noida–Agra
Agra–Lucknow	West–central Uttar Pradesh
Purvanchal	Lucknow–Ballia via Azamgarh and **Mau**
Bundelkhand	Southern Uttar Pradesh
Ganga Expressway	Meerut (मेरठ)–Prayagraj corridor (under construction / phased)
Gorakhpur Link	Connector toward Gorakhpur / NER city

N.3 Waterways

NW	River / canal	Stretch	States / note
NW-1	Ganga–Bhagirathi (भागीरथी)–Hooghly (हुगली)	**Haldia–Prayagraj** (about 1,620 km)	UP, Bihar, Jharkhand, West Bengal
NW-2	Brahmaputra	**Dhubri–Sadiya**	Assam
NW-3	West Coast Canal, Champakara and Udyogmandal canals	**Kottapuram–Kollam**	**Kerala**
NW-4	Godavari, Krishna and connected canals (including Buckingham Canal)	**Kakinada–Puducherry**	Andhra Pradesh–Tamil Nadu–Puducherry
NW-5	Brahmani–Mahanadi delta (डेल्टा) and East Coast Canal	**Talcher–Dhamra** / Geonkhali–Charbatia stretch	Odisha–West Bengal

N.4 Ports, Harbours, Maritime Trade

Coast	Major ports
West / NW	Kandla (**Deendayal**, Gujarat), Mumbai, JNPT, Mormugao (Goa), New Mangalore (Karnataka), Cochin (Kerala)
East / SE	Kolkata–Haldia, Paradip (Odisha), Visakhapatnam (Andhra Pradesh), Chennai, Ennore (**Kamarajar**), Tuticorin (**VOC**, Tamil Nadu)

Port–state match bank (next year can ask any)

Port	State	Tag
Kandla / Deendayal	**Gujarat**	Major; NW hinterland
Mumbai	**Maharashtra**	Major; natural harbour class
JNPT / JNPA	**Maharashtra**	Container flagship

Port	State	Tag
Mormugao	**Goa**	Major; iron-ore note
New Mangalore	**Karnataka**	Major
Cochin / Kochi	**Kerala**	Major; natural harbour class
Alappuzha ★	**Kerala**	Non-major; 2025 Q50
Kolkata–Haldia	**West Bengal**	Major; riverine
Paradip ★	**Odisha** (ओडिशा)	Major; 2025 Q50
Visakhapatnam	**Andhra Pradesh**	Major; natural harbour class
Kakinada ★	**Andhra Pradesh**	Non-major; 2025 Q50
Krishnapatnam	**Andhra Pradesh**	Non-major / private class
Chennai	**Tamil Nadu**	Major; often artificial harbour
Ennore / Kamarajar	**Tamil Nadu**	Major; N of Chennai
Tuticorin / VOC ★	**Tamil Nadu**	Major; 2025 Q50
Mundra / Pipavav	**Gujarat**	Non-major / private west coast
Dhamra	**Odisha**	Non-major east coast

N.5 Airports

Major Named Airports in India (Match Desk)

Airport Name	City / Location	State / UT
Chhatrapati Shivaji Maharaj International Airport	Mumbai (Santa Cruz / Sahar)	Maharashtra
Netaji Subhash Chandra Bose International Airport	Kolkata (Dum Dum)	West Bengal
Indira Gandhi International Airport	New Delhi (Palam)	Delhi NCT
Anna / Chennai International Airport	Chennai (Meenambakkam)	Tamil Nadu
Kempegowda International Airport	Bengaluru (Devanahalli)	Karnataka
Rajiv Gandhi International Airport	Hyderabad (Shamshabad)	Telangana
Sri Guru Ram Dass Jee International Airport	Amritsar (Rajasansi)	Punjab
Dr. Babasaheb Ambedkar International Airport	Nagpur (Sonegaon)	Maharashtra
Lokpriya Gopinath Bordoloi International Airport	Guwahati (Borjhar)	Assam
Veer Savarkar International Airport	Port Blair	Andaman & Nicobar
Sardar Vallabhbhai Patel International Airport	Ahmedabad	Gujarat
Chaudhary Charan Singh International Airport	Lucknow (Amausi)	Uttar Pradesh
Lal Bahadur Shastri International Airport	Varanasi (Babatpur)	Uttar Pradesh
Lok Nayak Jayaprakash Airport	Patna	Bihar
Devi Ahilyabai Holkar Airport	Indore	Madhya Pradesh
Raja Bhoj Airport	Bhopal	Madhya Pradesh

N.7 Sagarmala, Blue Economy, Gati Shakti, NLP

Blue Economy — sector teaching card

Scheme	Focus	Trap
Sagarmala	Port-led development, coastal shipping, port connectivity, coastal communities	Not a highway scheme
Bharatmala	Highway corridors and feeders	Not ports
PM Gati Shakti NMP	Multimodal planning map of projects	Year trap
National Logistics Policy	Lower logistics cost and time	Complements Gati Shakti; does not replace it
Blue Economy	Sustainable ocean-resource use — shipping, fisheries, coastal tourism, offshore energy	Not "any coastal factory"
DFC	Dedicated rail freight hardware	Neighbour of Sagarmala, not the same scheme

N.9 Uttar Pradesh Focus

Theme	Fact
Railway HQ	**North Eastern Railway — Gorakhpur**
Expressway	Built Purvanchal **passes Mau**; Basti is off the main alignment
Airport	Kushinagar international from **24 Jun 2020**; Jewar = Noida International
Stations	Amrit Bharat: **149 UP stations** identified for **modernisation** (Feb 2023), out of about 1,275 all-India
Waterway	**NW-1** on the Ganga (Haldia–Prayagraj)
DFC pull	Western UP / NCR toward the Dadri–JNPT freight spine
Pipeline nodes	HBJ gas toward Jagdishpur / Auraiya; Mathura refinery product / crude complex

N.10 Tourism geography

Resort / hill-station match bank

Place	State / UT	Note
Chakrata	**Uttarakhand** (उत्तराखंड) (Dehradun (देहरादून))	Cantonment; Tiger Fall
Haflong	**Assam**	Only hill station of Assam (Dima (डिम) Hasao)
Kalimpong	**West Bengal**	Darjeeling hills
Kufri	**Himachal Pradesh**	Near Shimla
Kasauli	**Himachal Pradesh**	Solan
Keylong	**Himachal Pradesh**	Lahaul
Auli (औली)	**Uttarakhand**	Ski / hill station
Ooty (Udhagamandalam)	**Tamil Nadu**	Nilgiris
Gulmarg / Pahalgam	**J&K**	Valley hill stations
Chikmagalur	**Karnataka**	Coffee / hills
Khajjiar	**Himachal Pradesh**	Meadow / lake note
Banni grassland	**Gujarat**	Kachchh

Place	State / UT	Note
Bugyal	**Uttarakhand**	Alpine meadow

City nicknames and pilgrimage geography

Fact	Correct	Trap
Pink City	**Jaipur**	Not Mysore
City of Lakes	**Udaipur**	**Not Jaisalmer**
City of Joy	Kolkata	Common nickname pair
City of Mahakal	Ujjain (उज्जैन)	Correct with Jaipur pair
Incredible India	MoT brand	Not “Beautiful India”
Sabarimala	**Kerala** (Ayyappa)	Not TN / KA / AP
Srisailam	**Nallamala Hills**, Andhra Pradesh	Mallikarjuna + Bhramaramba; Jyotirlinga
Omkareshwar	Mandhata island, **Narmada** (नर्मदा), Madhya Pradesh	**Not** Satmala Hills
Pushkar	Ajmer (अजमेर), **Aravalli** (अरावली) / Brahma temple	**Not** Mahadeo Hills
Biligiriranga Hills	SE Karnataka	Standing **Ranganatha / Venkatesha** temple
Tirupati Venkateswara	**Seshachalam** (Andhra Pradesh)	Not Shevaroy / Jawadhu / Biligiri as the main Tirupati hill
Udvada	Gujarat	Zoroastrian fire temple
Point Calimere	Nagapattinam, Tamil Nadu	Apex of **Coromandel Coast**
Konark Sun Temple	Odisha coast; UNESCO **1984**; Narasimha Deva I	World Heritage
Champaner–Pavagarh	Panchmahal, Gujarat; UNESCO **2004**	World Heritage
Mamallapuram	Pallava (पल्लव) stone town, Tamil Nadu; UNESCO	World Heritage
CST Mumbai	Victorian Gothic terminus; UNESCO	World Heritage
Ghost towns	Kuldhara, Dhanushkodi, Lakhpat	**Charkhari** is **not** a ghost town (Mahoba, UP)

N.12 World transport · ports · isopleths

Note	Fact
Busiest ocean trade route	**North Atlantic** (links two industrial zones)
Trans-Siberian	Long Eurasian rail; **Sochi** is **not** on the main route
Eurotunnel / Channel Tunnel	**England–France**
Canadian Pacific	Montreal–Vancouver
Bullet / HSR network length	**China** leader
Rotterdam	Rhine mouth; Europe’s great seaport
Shanghai	Often #1 container harbour in recent frames
Santos	**Coffee port** (Brazil)
Maracaibo	Venezuela **oil** port
Port Nolloth	South Africa **diamond** port
Gwadar	**Pakistan** (Balochistan), Arabian Sea
Igarka	**Russia** (Yenisei) — not China
Alexandria	**Egypt** (मिस्र)

Line	What it joins (equal value)
Isobar	Pressure
Isohyet	Rainfall
Isotherm	Temperature
Isohaline	Salinity
Isobath	Depth
Isogonic	Magnetic declination
Contours	Elevation / relief

Common Traps — Don't Fall For These

1. **SER HQ = Cuttack** — FALSE. **Garden Reach, Kolkata**.
2. **NER HQ = Hajipur** — FALSE. **Gorakhpur**. Hajipur = **East Central**.
3. **Amrit Bharat = 149 new railway lines** — FALSE. It modernises **stations**. As of Feb 2023 about **1,275** stations all-India, of which **149** were in **Uttar Pradesh**.
4. **Gati Shakti launched 2022** — FALSE. **13 Oct 2021**; seven engines **true**.
5. **Bharatmala = ports / Sagarmala = highways** — SWAP. Bharatmala = **roads**; Sagarmala = **port-led**.
6. **Mau is off today's Purvanchal Expressway** — FALSE for the **built** road (Lucknow–Ballia via Azamgarh–Mau). **Basti** is not on the main alignment. The 2018 stem "will not pass" had a different official key — see that full question in the PYQ bank.
7. **Atal = world's longest highway tunnel** — FALSE as a bare claim. **Pir Panjal** is the safe fact.
8. **Igarka = China** — FALSE. **Russia**.
9. **NW-1 = Brahmaputra** — FALSE. **Ganga, Haldia–Prayagraj**. NW-2 = Brahmaputra. NW-3 = **Kerala** canal. NW-4 = Kakinada–Puducherry. NW-5 = Talcher–Dhamra.
10. **Kushinagar not international** — FALSE. Intl on **24 Jun 2020**.
11. **First solar airport = Delhi / Ahmedabad** — FALSE. **Cochin**.
12. **Optical fibre = refraction** — FALSE. **Total internal reflection**.
13. **Motihari–Kathmandu pipeline** — FALSE. **Amlekhganj, Nepal**.
14. **Paradip = Andhra / Alappuzha = Tamil Nadu** — FALSE. Paradip **Odisha**; Alappuzha **Kerala**.
15. **UP has a major seaport** — FALSE. Identity = Gorakhpur, NW-1, expressways, Kushinagar/Jewar, 149 Amrit stations.
16. **National Highways = most of India's road kilometres** — FALSE. NH are ~**2%** of length but carry ~**40%** of traffic; rural roads are ~**80%** of length.
17. **Inland waterways already move most Indian cargo** — FALSE. Share is tiny; sea still moves ~**95%** of trade by volume.
18. **IWAI = NHAI** — FALSE. IWAI (1986) = inland waterways; NHAI (1995) = national highways.
19. **Only three National Waterways exist** — FALSE. The 2016 Act declared **111**. The first five are the map you must know even if a paper has not yet asked NW-4 or NW-5.
20. **Kakinada is a major-port-state fact for Odisha** — FALSE. Kakinada = **Andhra Pradesh**; Paradip = **Odisha**.
21. **East–West Corridor = Dibrugarh–Surat / Dwarka** — FALSE. **Porbandar–Silchar**. Cross with NS at **Jhansi**.
22. **Konkan Railway = Kerala main line** — FALSE. **Roha–Mangaluru** through MH–Goa–KA.
23. **SER HQ = Bhubaneswar / SCR HQ = Hyderabad** — FALSE. SER = **Garden Reach, Kolkata**; East Coast = Bhubaneswar; SCR = **Secunderabad**.
24. **North Western Railway HQ = Jodhpur** — FALSE. **Jaipur**.
25. **Chennai = deepest natural port** — FALSE. Often **artificial**; Vizag/Paradip depth notes varies by stem.
26. **Kandla = LNG terminal** — FALSE. LNG terminals are Dahej, Hazira, Kochi, Dabhol, Ennore.

27. **Old NH-7 is still always the longest NH** — FALSE for current MoRTH data — often **NH-27 EW**; also note the **NH-44** corridor.
28. **NS–EW meet at Nagpur / Gwalior** — FALSE. **Jhansi**.
29. **First CNG train = Delhi–Agra** — FALSE. **Rewari–Rohtak**.
30. **Sethusamudram = Cape Comorin only** — FALSE. Links **Gulf of Mannar** with **Palk Bay**.
31. **City of Lakes = Jaisalmer** — FALSE. **Udaipur**. Jaipur = **Pink City**.
32. **Sabarimala = Tamil Nadu / Karnataka** — FALSE. **Kerala**.
33. **Haflong = Himachal / Kalimpong = Assam** — FALSE. Haflong = **Assam**; Kalimpong = **West Bengal**; Chakrata = **Uttarakhand**; Kufri = **HP**.
34. **Omkareshwar = Satmala / Pushkar = Mahadeo** — FALSE. Only **Srisailam–Nallamala** was correct in IAS 2015 among those three.
35. **Charkhari = ghost town** — FALSE. Ghost-town names are Kuldhara, Dhanushkodi, Lakhpat.
36. **BARC = Karnataka** — FALSE. **Maharashtra**. Thumba = **geomagnetic equator** site in Kerala.
37. **Moradabad sarees** — FALSE. Brassware; saree map = Chanderi / Banarasi / Kanjivaram.

Topic 10 — Tribes & Institutions

Current Affairs (this topic)

Year	Fact	Why it matters
15 Nov	**Janjatiya Gaurav Diwas** — Birsa Munda (मुंडा) birth anniversary	Tribal (आदिवासी) identity day
2023	**PM-JANMAN** — PVTG mission (housing, health, education, connectivity)	Living PVTG scheme
1875–2025	**IMD** 150 years; headquarters still **New Delhi**	Institute anniversary
Jun 2025	**Adi Karmayogi** launched by **Ministry of Tribal Affairs**	Ministry trap vs Social Justice / AYUSH
Static	Uttar Pradesh (उत्तर प्रदेश) ST list is still the **15-entry** notification set, many of them district-limited	Match traps

Masai habitat (वास स्थान) remains **East Africa**. Do not "update" it to West Africa.

Confused Pairs

Pair	Correct	Trap	Hindi
Khasi	**Meghalaya**	Arunachal	मेघालय
Rengma	**Nagaland**	Andaman	नागालैंड
Kuki	**North-East**	UP	पूर्वोत्तर
Yanadi	**Andhra Pradesh**	Rajasthan	आंध्र
Toda / Paliyan	**Tamil Nadu**	Nagaland	तमिलनाडु
Birhor	**Jharkhand**	Islands / TN	झारखंड
Keria/Kharia	**Odisha** (also JH)	Random west	ओडिशा
Masai	**East Africa**	West Africa	पूर्वी अफ्रीका
Khirghiz	**Central Asia**	Sudan	मध्य एशिया
Koryak	**NE Siberia**	Alaska	साइबेरिया
Semang	**Malaysia**	Indonesia	सेमांग
Maasai	**East Africa**	West Africa / Zambia	मसाई
Bantu	E / C / S Africa	Sahara	बांटू
Transhumance	Seasonal herd move	Permanent emigration	ट्रांसह्यूमेंस
Tharu	**Terai**	Sonbhadra only	तराई
Agariya	Iron smelting	Farming-only tag	लोहा गलाना
Saharya	**Lalitpur**	Terai	ललितपुर
NDRI	**Karnal**	Lucknow / Izatnagar	करनाल
IVRI	**Izatnagar (Bareilly)**	Karnal	इज्जतनगर
FRI	**Dehradun**	Delhi / Lucknow	देहरादून
NDRI vs NDDB	Karnal **lab** vs Anand **co-op board**	Same body	एनडीआरआई ≠ एनडीडीबी
PVTG count	**75** groups	95	75 पीवीटीजी

Pair	Correct	Trap	Hindi
Fifth vs Sixth Schedule	Fifth = many Scheduled Areas; Sixth = **four** NE states' councils	Swap	पाँचवीं ≠ छठी

N.1 Tribes of India

State / UT	High-yield tribes
Andhra Pradesh / Telangana	**Chenchu**, **Yanadi**, Koya, Lambadi, Gond
Arunachal Pradesh	Nyishi, **Apatani**, Adi, Monpa — **not Khasi**
Assam	**Bodo**, Mishing, Karbi, Dimasa
Bihar	Santhal, Oraon, Munda, Tharu (north)
Chhattisgarh	Gond, Halba, Kawar, Maria and Muria of Bastar
Gujarat	Bhil, Siddi, Rathwa
Himachal (हिमाचल) / J&K–Ladakh (लद्दाख)	Gaddi, Gujjar, Kinnaura, **Gujjar–Bakarwal**
Jharkhand	**Santhal, Munda, Oraon, Ho, Kharia, Birhor**, Asur, Birjia
Karnataka	Siddi, Soliga, Koraga, Jenu Kuruba
Kerala	Irular, Kurumba, Kadar, Paniyan, Kattunayakan
Madhya Pradesh	Bhil, Gond, Baiga, Kol, Korku, Sahariya, Bharia
Maharashtra	Bhil, Gond, **Warli** (वारली), Katkari
Manipur	Naga (नागा) groups, **Kuki**
Meghalaya	**Khasi, Jaintia (जयंतिया), Garo**
Mizoram	Mizo (मिज़ो), Chakma, Bru / Reang
Nagaland	**Angami**, Ao, Sumi, Lotha, **Rengma**, Konyak
Odisha	Santhal, Munda, Kondh, Saora, Juang, Bonda, **Kharia**, Ho, Gond
Rajasthan	Bhil, **Meena**, Garasia, Sahariya
Sikkim	**Lepcha**, Bhutia, Limboo
Tamil Nadu	**Toda**, Kota, Irular, **Paliyan**, Kadar
Tripura	Tripuri, Reang, Jamatia
Uttarakhand (उत्तराखंड)	Tharu, Bhotia, Buksa, Jaunsari, Raji
Uttar Pradesh	**Tharu, Buksa, Raji, Bhotia, Jaunsari**, Gond, Agariya, Saharya (15 notified entries)
West Bengal	Santhal, Munda, Oraon, **Toto**
Andaman & Nicobar	**Great Andamanese, Onge, Jarawa, Sentinelese, Nicobarese, Shompen**

N.2 Tribes of Uttar Pradesh

#	Tribe	Note
1	Bhotia	Himalayan trading and herding
2	Buksa (Bhoksa)	Terai–Bhabar (भाबर) on the Bijnor fringe; **PVTG**
3	Jaunsari	Himalayan / Uttarakhand–UP set
4	Raji	Tiny northern community; **PVTG**
5	Tharu	Terai — Kheri, Bahraich, Balrampur, Shravasti, Pilibhit

#	Tribe	Note
6	Gond (Dhuria, Nayak, Ojha, Pathari, Raj Gond)	District-limited in Maharajganj, Siddharthnagar, Basti, Gorakhpur (गोरखपुर), Deoria, Mau, Azamgarh, Jaunpur (जौनपुर), Ballia, Ghazipur, Varanasi (वाराणसी), Mirzapur, **Sonbhadra** (सोनभद्र)
7	Kharwar	Deoria–Ballia–Ghazipur–Varanasi–Sonbhadra
8	Saharya	**Lalitpur**
9	Parahiya	Sonbhadra
10	Baiga	Sonbhadra
11	Pankha / Panika	Sonbhadra, Mirzapur
12	Agariya	Sonbhadra / Mirzapur — traditional **iron smelting**
13	Patari	Sonbhadra
14	Chero	Sonbhadra, Varanasi
15	Bhuiya	Sonbhadra
Zone	Districts	Key tribes
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Terai / north	Kheri, Bahraich, Balrampur, Shravasti, Pilibhit, Bijnor fringe	Tharu, Buksa
Vindhyan / south	Sonbhadra, Mirzapur, Lalitpur and notified east pockets	Gond, Kharwar, Chero, Baiga, Agariya, Saharya

N.6 World tribes

People	Habitat / note
Maasai	E Africa plateau — Kenya, N Tanzania, E Uganda; cattle; **kraal** huts
Bushman / San	**Kalahari** (Botswana, Namibia, parts of SA) — hunting–gathering; termites
Zulu	**South Africa** — agriculture + cattle; Isandlwana 1879
Semang	**Malay Peninsula / Malaysia** (Negrito hunter–gatherers) — **not** Indonesia
Sakai	Malay Peninsula / Indonesia forests — blowpipe hunting
Papuan	Papua New Guinea
Bedouin	Arabian deserts (esp. N Saudi) — pastoral nomads
Kirghiz / Kyrgyz	Central Asia (Pamir–Tianshan) — seasonal herding; **yurt**
Yupik	E Russia / W Alaska — dog **sledge**
Yukaghir	NE Siberia (Verkhoyansk / Stanovoy)
Maya	Central America — Mexico, Guatemala, Honduras
Maori	**New Zealand** (Polynesian)
Fulani	West Africa
Bantu	E / C / S Africa — **not** Sahara
Nuba	Sudan
Vedda	Sri Lanka
Berber	Morocco / N Africa
Ainu	Japan
Aeta	Philippines

People	Habitat / note
Bora (बोरा)	W Amazon (Brazil–Peru–Colombia)
Bindibu / Aborigines	Western Australia desert
Chukchi / Koryaks / Samoyeds / Yakuts	Siberia / NE Asia tundra (Russia)
Aleuts	Alaska / Aleutian Islands
Tuareg (Turregs)	Sahara Desert (West & Central)
Hausa	Northern Nigeria
Hottentots (Khoikhoi)	South / SW Africa
Orang Asli	Peninsular Malaysia (indigenous forest peoples)
Ibans	Sarawak / Borneo equatorial forests
Tapiro	Papua New Guinea (pygmy group)
Kalmuk / Buryat	Central Asia / Siberia
Kazakhs	Kazakhstan steppes
Jews / Teda / Beja / Lur	Israel / Libya / Egypt (मिस्र) / Iran (ethnic match sets)

Wrong claim	Correct teaching
Semang live in Indonesia	The **Semang** belong to the **Malay Peninsula / Malaysia**
Maasai live in West Africa or Zambia	The **Maasai** live in **East Africa** (Kenya–Tanzania)
Bantu live in the Sahara	The **Bantu** belt is **East / Central / Southern Africa**, not the Sahara
Pygmy peoples are pastoral	They are **hunters–gatherers** of the Congo rainforest
Gypsies originally came from Egypt	Their original home in standard notes is **India**

Tribe / people	Habitat
1. Bushman	Kalahari Desert
2. Eskimo	Northern Canada
3. Pygmy	Congo Basin
4. Masai	West Africa

N.7 IMD and ICAR

Body	Headquarters	Role
IMD	**New Delhi**	Weather, climate, cyclone warnings
ICAR	**New Delhi**	Apex agricultural research and education
IARI	**Pusa, New Delhi**	Flagship ICAR institute

Common Traps — Don't Fall For These

1. **Khasi = Arunachal** — FALSE. **Meghalaya**.
2. **Rengma = Andaman native** — FALSE. **Nagaland**.
3. **Kuki = Uttar Pradesh** — FALSE. **North-East**.
4. **Yanadi = Rajasthan** — FALSE. **Andhra Pradesh**. Paliyan = **Tamil Nadu**. Keria = **Odisha**.
5. **Masai = West Africa** — FALSE. **East Africa**.
6. **Khirghiz = Sudan** — FALSE. **Central Asia**.

7. **Alaska = Koryak** — FALSE. Koryak = **north-east Siberia**.
8. **Tharu = Sonbhadra only** — FALSE. **Terai**.
9. **UP PVTG = Tharu + Gond** — FALSE. **Buksa + Raji**.
10. **Agariya = only farmers** — FALSE. Traditional **iron smelting**.
11. **Birjia / Santhal = island tribes** — FALSE. **Jharkhand / mainland**.
12. **NDRI = Izatnagar / IVRI = Karnal** — SWAP. NDRI **Karnal**; IVRI **Bareilly**.
13. **FRI = Delhi or Lucknow** — FALSE. **Dehradun**.
14. **Toda = Nagaland** — FALSE. **Tamil Nadu** (Nilgiris).
15. **IMD / ICAR HQ = a regional campus** — FALSE. Both **New Delhi**. NDRI ≠ NDDB; FRI ≠ WII.
16. **PVTG count = 95** — FALSE. There are **75** groups. Odisha has the most communities.
17. **Fifth Schedule = only four NE states** — FALSE. Those four are the **Sixth Schedule** set.
18. **PTG rename year = 1973** — FALSE. Dhebar identified PTGs in 1973. The rename to PVTG is **2006**.
19. **One all-India ST roster** — FALSE. Lists are **state/UT-wise** under Article 342.
20. **Any forest dweller = PVTG** — FALSE. PVTG is a tighter vulnerability subset with set criteria.

Topic 11 — Population Geography

Current Affairs (this topic)

Year	Fact	Why asked	Source
Census 2011	Official number set until **Census 2027**	Every density/SR/literacy MCQ	ORGI
NFHS-5	TFR ~**2.0** nationally	NFHS-4 (2015–16) figure used in 2018 was **2.2**	MoHFW
UNFPA	World Population Report	2021 Q79	UNFPA
~2023	UN estimate: India most populous	CA only — 2011 still 2nd in Census table	UN / UNFPA
NPP 2000	Stable population **2045**	Policy year fact	MoHFW
11 July	World Population Day	2018 Q114	UN

Confused Pairs

Pair	Correct	Trap	Hindi
First census	**1872** non-sync; paper **1871** (2023)	1881 as "first ever"	प्रथम जनगणना
First synchronous	**1881**	1872	समकालिक
2011 rank	**15th** / **7th** after Independence	14th / 8th	15वीं
Density India	**382**	324 (2001)	382
Highest density **state**	**Bihar**	WB / UP / Delhi	बिहार
Lowest density	**Arunachal**	HP / Mizoram	अरुणाचल
Sex ratio	**943**	CSR 919	943
Lowest CSR (rural+urban)	**Haryana**	UP / J&K	हरियाणा
Literacy	**74.04%**, age **7+**	0–6 / 18+	74.04
TFR 2.1	Children **per woman**	"per thousand" (that is CBR)	कुल प्रजनन दर
Natural growth	**CBR − CDR**	Includes migration	प्राकृतिक वृद्धि
Dividend	**15–59**	60+ or 0–6	जनसांख्यिकीय लाभांश
Acceleration (urban curve)	**2nd stage**	3rd / 4th	द्वितीय
Most urbanised religion	**Jains**	Christians	जैन
Kanpur / Lucknow million	**1971 / 1981**	1961/1971	कानपुर 1971
Shrawasti	Lowest **female literacy** district (UP)	Balrampur / Bahraich	श्रावस्ती
Lorenz curve	**Income inequality**	Illiteracy	लोरेंज
World Population Report	**UNFPA**	IMF / WHO / UNDP	यूएनएफपीए
Arithmetic density	**Population / total area**	Physiological density	अंकगणितीय घनत्व
Physiological density	**Population / net sown area**	Agricultural density	शारीरिक घनत्व
Natural vs induced growth	Natural = CBR−CDR; induced = migration	Same thing	प्राकृतिक ≠ प्रेरित
Immigration / emigration	In / out	Swap	आप्रवास / उत्प्रवास

Pair	Correct	Trap	Hindi
SC / ST share (2011)	SC ~**16.6%** · ST ~**8.6%**	Swap or invent	अनुसूचित जाति / जनजाति
Optimum Population	**Edwin Cannan**	Robbins / Dalton / Carr-Saunders as the first name	अनुकूलतम जनसंख्या
Social mal-adjustment	**Henry George**	Dalton's "maladjustment" formula, or Malthus	सामाजिक कुसमायोजन
Demographic Transition	**Thompson** (Notestein also)	Cannan / Malthus	जनांकिकीय संक्रमण
Population–food	**Malthus**	Marx (he criticises Malthus)	जनसंख्या–खाद्य
Geometric vs arithmetic	Population **geometric**; food **arithmetic**	Swap the two progressions	गुणोत्तर / समानांतर
Positive vs preventive checks	Positive = **death** up; preventive = **birth** down	Swap	धनात्मक / निवारक नियंत्रण
Social capillarity	**Arsène Dumont**	Henry George	सामाजिक केशिका

N.1 Census of India

Phase	Years	Character
I	1901–1921	Stagnant / low growth (ends at the Great Divide)
II	1921–1951	Steady growth
III	1951–1981	Rapid growth / population explosion
IV	1981–2011	High growth with a **declining rate**

N.3 Population Density

Type	Formula	What it shows
Arithmetic density	Total population ÷ total area	Everyday "persons per km²" fact
Physiological density	Total population ÷ **net sown** area	Pressure of people on cultivated land
Agricultural density	Agricultural population ÷ **net sown** area	Pressure of farm workers on cultivated land

N.13 Demographic Transition & Theories

Theory	Propounder
Optimum Population	**Edwin Cannan**
Social mal-adjustment	**Henry George**
Demographic Transition	**Thompson**
Population–food	**Malthus**

Demographic Transition — Thompson (Notestein)

Stage	Birth rate	Death rate	Growth
1 High stationary	High	High	Slow
2 Early expanding	High	Falls first	Rapid (explosion)
3 Late expanding	Falls	Low	Slows

Stage	Birth rate	Death rate	Growth
4 Low stationary	Low	Low	Stable / very slow
5 Declining (Blacker)	Very low	Low, but can exceed births	Shrinks / ages

Other scholars who appear as options

Idea	Person
Surplus population under capitalism	**Karl Marx** (with Engels)
Population growth can intensify farming	**Ester Boserup**
Social capillarity (rise in status → fewer children)	**Arsène Dumont**
Five-stage transition including decline	**C.P. Blacker**
Fertility falls as density rises	**Sadler**
Diet theory (better fed → lower fertility)	**Doubleday**
Energy to mind versus reproduction	**Herbert Spencer**
Optimum restated (not the first name)	Robbins / Dalton / Carr-Saunders
DTT restated (not the first name)	**Notestein**

N.14 Uttar Pradesh Focus

Item	Uttar Pradesh 2011	India 2011
Rank	Most populous state; largest rural population	—
Population	~**19.98 crore** (~**16.5%** of India)	**121.09 crore**
Density	**829**	**382**
Sex ratio	**912**	**943**
Literacy	~**67.7%**	**74.04%**
Urban share	~**22.3%**	**31.16%**
Rural share	~**77.7%**	**68.84%**
Decadal growth 2001–11	~**20.23%**	**17.64%**
Female literacy floor (district)	**Shrawasti**	—
Million years	Kanpur **1971**, Lucknow **1981**	—
UA order	Kanpur > Lucknow > Ghaziabad > Agra	—

Common Traps — Don't Fall For These

1. **First census = 1881** — FALSE as "first ever." **1881 = first synchronous**; 2023 key **1871**.
2. **Density 324** — that is **2001**. **2011 = 382**.
3. **Highest density = West Bengal / Delhi as a "state"** — state fact is **Bihar**.
4. **Lowest density = Himachal / Mizoram** — **Arunachal**.
5. **CSR = 943** — 943 is **overall** SR; CSR = **919** (0–6).
6. **Lowest CSR = UP** — FALSE. **Haryana** (rural and urban).
7. **TFR 2.1 = 2.1 per thousand** — FALSE. That wording is **CBR**. TFR = per **woman**.
8. **Natural growth includes migration** — FALSE. **CBR − CDR** only.
9. **Dividend = 60+ or 0–6** — FALSE. **15–59**.
10. **Urbanisation acceleration = 3rd/4th stage** — FALSE. **2nd stage**.

11. **Most urbanised religion = Christians** — FALSE. **Jains**.
12. **Kanpur million in 1961 / Lucknow 1971** — FALSE. **1971 and 1981**.
13. **Largest rural population = Maharashtra** — FALSE. **UP**.
14. **Shrawasti vs Balrampur/Bahraich** — female literacy floor = **Shrawasti**.
15. **Replace 2011 with UN 2023 headcount in Prelims tables** — FALSE until **Census 2027**.
16. **Lorenz curve = illiteracy** — FALSE. It measures **income inequality**.
17. **World Population Report = WHO / UNDP** — FALSE. **UNFPA**.
18. **Highest density state = Delhi** — FALSE as a state answer. Fact **Bihar**.
19. **Nagaland = least populous** — FALSE. Negative growth ≠ least populous; least populous = **Sikkim**.
20. **Literacy age = 0–6 or 18+** — FALSE. Census literacy = age **7+**.
21. **Arithmetic density = population ÷ net sown area** — FALSE. That is **physiological** density.
22. **Natural growth includes migration** as "induced natural" — FALSE. Migration = **induced**; natural = CBR−CDR.
23. **Immigration = movement out** — FALSE. Immigration = **in**; emigration = **out**.
24. **ST share = 16.6%** — FALSE. ST ~**8.6%**; SC ~**16.6%**.
25. **Skip factors of distribution because UPPCS never asked them** — FALSE. Next year can ask terrain/soil/industry pulls from the same topic.
26. **Optimum Population = Robbins / Dalton / Carr-Saunders as the first name** — FALSE. Propounder = **Edwin Cannan**. The others restated it.
27. **Social mal-adjustment = Dalton or Malthus** — FALSE. **Henry George**. Dalton's M = (A − O) / O is the optimum **gap**, not George's land-distribution argument.
28. **Demographic Transition = Cannan** — FALSE. **Thompson** (Notestein also appears). Cannan is optimum.
29. **Population geometric / food arithmetic swapped** — FALSE. Population is **geometric**. Food is **arithmetic**.
30. **Malthusian Theory = poverty / economy / unemployment** — FALSE. It is a theory of **population**. Poverty-from-distribution is **Henry George**.
31. **Positive checks = delayed marriage** — FALSE. Positive checks **raise deaths** (famine, disease, war). **Preventive** checks lower births.
32. **Marx = geometric vs arithmetic** — FALSE. Marx criticised Malthus and treated surplus population as a product of **capitalism**.
33. **Dumont = Social Mal-adjustment** — FALSE. Dumont is **social capillarity**. George is social mal-adjustment.

Topic 12 — Human Geography (Settlements)

Current Affairs (this topic)

Year	Fact	Why asked	Source
25 Jun 2015	Smart Cities Mission + PMAY-U same day	Year pair vs JNNURM 2005	MoHUA
Jun 2015	AMRUT (~500 cities)	2025 chronology	MoHUA
1 Oct 2021	**AMRUT 2.0** — water-secure towns	Successor of AMRUT	PIB / MoHUA
21 Feb 2016	**SPMRM** launched (not "2015 launch")	2023 Q143	MoRD
ISAC-2020	Indore/Surat best cities; **UP best state**; category winners above	2022 Q26	PIB / MoHUA
100 / 10	100 Smart Cities nationally; **10 Central in UP**	2018 + 2020	MoHUA
2019	UP **State** Smart Cities (7 Nagar Nigams)	Does **not** change 2018/2020 keys of 10	UP DoUD
Mission window	Smart Cities Mission timeline extended to **31 Mar 2025**	Completing works	MoHUA

Million-city **population (जनसंख्या) ranks**. Missions and settlement types

Confused Pairs

Pair	Correct	Trap	Hindi
Census town	5,000 + 400/km² + 75% non-agri male main	Any big village / always a municipality	जनगणना नगर
Clustered village	Plains + **Rajasthan water** + Bundelkhand/Nagaland defence	"Thar = dispersed"	संकुल
Dispersed village	**Meghalaya, UK, HP, Kerala**, NE forests/hills	Ganga plain / all Rajasthan	प्रकीर्ण
Hamleted local names	**panna, para, palli, nagla, dhani**	A separate census town	पुरवा/ढाणी
Semi-clustered	Dominant caste **centre**; lower strata **flanks** (GJ/RJ)	Same as hamleted	अर्ध-संकुल
Class I town	≥**1 lakh** (468 in 2011; ~**60%** urban pop)	≥10 lakh	श्रेणी I
Metropolitan (NCERT)	**10 lakh–50 lakh**	10 million	महानगर
Mega city (NCERT)	**> 50 lakh**; **six** in 2011	Only 1 crore / only Mumbai	मेगा
UN megacity	≥ **1 crore**	Mixing with NCERT 50-lakh mega	UN मेगा
2011 size rank	**Mumbai > Delhi > Kolkata > Chennai**	Mumbai > Kolkata > Delhi (that is **2001**)	2011 क्रम
Bhopal	**Not** on a major river bank	Agra/Patna/Kolkata are	भोपाल ≠ नदी
UP Smart Cities (papers)	**10 Central** listed	12/15/18; Ghaziabad	10 शहर
Ghaziabad	Not in **Mission-100 / Central-10**	"Is a Smart City" (only **State** 2019)	गाजियाबाद नहीं
SPMRM year	**2016 launch**	2015	2016
HRIDAY in UP	**Varanasi + Mathura**	Prayagraj / Ayodhya	काशी+मथुरा
AMRUT	**Jun 2015**	2005 (that is JNNURM)	2015

Pair	Correct	Trap	Hindi
JNNURM	**2005**	2015	2005
Smart Cities vs SPMRM	MoHUA vs **MoRD**	Same ministry	आवास vs ग्राम
Global Village	Transport + communication	UN / politics only	परिवहन+संचार
Rurbanization (options)	**G.S. Ghurye**	M.N. Srinivas	घुर्ये
Determinism	Nature controls man (**Ratzel / Semple**)	Possibilism	नियतिवाद
Possibilism	Man chooses within nature (**Vidal de la Blache**)	Determinism	संभावनावाद
Neo-determinism	Stop-and-go determinism (**Griffith Taylor**)	Pure free will	नव-नियतिवाद
Site vs situation	Local ground vs wider regional location	Same thing	स्थल ≠ स्थिति

N.0 Framework — Nature & Scope of Human Geography

Approach	Core fact	Name pair
Environmental determinism	Nature controls human life and culture	**Friedrich Ratzel**; **Ellen C. Semple**
Possibilism	Nature offers possibilities; humans choose	**Vidal de la Blache**
Neo-determinism / stop-and-go	Nature sets limits; man can proceed carefully	**Griffith Taylor**

N.1 Settlements — Rural vs Urban

Feature	Rural	Urban
Economy	Mainly primary (farm, forest, fish, village-scale mine)	Secondary / tertiary / quaternary
Density / size	Lower; hamlet to village	Compact; town to mega city
Link	Supply food and raw (रॉ) material	Supply goods, services and jobs
Census status	Village / rural	**Statutory town** or **census town**
Social life	Face-to-face primary groups	Secondary groups and specialised jobs

N.2 Types of Villages

Type	Form	Where (India) — match this
Clustered / nucleated / agglomerated	Compact houses with living area **separate** from fields; streets may be rectangular, radial or linear	Fertile **alluvial (जलोढ़) plains**; also **NE plains**; **Rajasthan** (थार) (scarce water → cluster at well/oasis); **Bundelkhand + Nagaland** (defence)
Semi-clustered / fragmented	One main cluster plus groups pushed a little away	**Gujarat** and **Rajasthan** fragmentation. The dominant / land-owning community sits at the **centre**; lower strata / menial workers sit on the **outer flanks**
Hamleted	Several physically separated units, **one village name**	**Middle & lower Ganga**, **Chhattisgarh**, lower **Himalayan valleys**
Dispersed / isolated	Isolated huts / tiny hamlets on hills, jungle, pasture slopes	**Meghalaya, Uttarakhand, Himachal, Kerala**; forests & hills of **North-East**

Name	Region
Panna / para / palli	East / Ganga–Chhattisgarh belt

Name	Region
Nagla	Western UP / Ganga plain
Dhani	Rajasthan

Shape	Note
Linear	Road, river, canal, coast, levee
Rectangular	Planned / northern plains field grid
Circular / radial	Around tank, temple, well
Star / T / Y	Where routes meet

Belt	Note
Heavy rain (W. coast, NE)	**Sloping** roofs; bamboo / wood in NE
Arid Rajasthan	**Flat** roofs; compact around water
Himalaya (हिमालय)	Stone / slate
Ganga plain	Mud / brick; courtyard houses

N.3 Functional Classification of Towns

Class	Population	2011 fact
I (city)	**≥ 1,00,000**	**468** towns; ~**60%** of India's urban population
II	50,000–99,999	
III	20,000–49,999	
IV	10,000–19,999	
V	5,000–9,999	
VI	< 5,000	

Type	Example
Administrative	Chandigarh, New Delhi, Bhopal, Shillong, Guwahati, Imphal, Srinagar, Gandhinagar, Jaipur, Chennai, **Lucknow** (लखनऊ)
Industrial	Jamshedpur, Bhilai, Durgapur, Modinagar, Barauni, Sindri, Mumbai, Madurai, **Kanpur** (कानपुर)
Transport / port	Kandla, Kochi, Visakhapatnam, **Mughalsarai / Pt. Deen Dayal Upadhyaya Jn.**, Itarsi, Katni
Commercial	Kolkata, **Saharanpur**, Satna
Mining	Raniganj, Jharia / Dhanbad, Digboi, Ankleshwar, **Singrauli**
Garrison / cantonment	Ambala, Jalandhar, Mhow, Babina, Udhampur, **Meerut** (मेरठ)
Educational	Roorkee, **Aligarh**, Pilani, Varanasi (BHU), Prayagraj
Religious / cultural	**Varanasi** (वाराणसी), Mathura, Amritsar (अमृतसर), Madurai, Puri (पुरी), Ajmer (अजमेर), Pushkar, Tirupati, Kurukshetra, Haridwar (हरिद्वार), Ujjain (उज्जैन)
Tourist	Nainital (नैनीताल), Mussoorie (मसूरी), Shimla, Pachmarhi, Jodhpur, Jaisalmer, Udhagamandalam (Ooty), Mount Abu, **Agra** (आगरा)

Layer	Fact
Ancient (~2000+ yrs)	**Varanasi, Prayag (Prayagraj), Pataliputra (पाटलिपुत्र) (Patna), Madurai** — religion / river / administration

Layer	Fact
Medieval (~100 towns)	Fort / court HQ — **Delhi, Hyderabad, Jaipur, Lucknow, Agra, Nagpur**
Modern — European foothold	First **ports** at Surat, Daman, Goa (गोवा), Puducherry; later **Mumbai, Chennai, Kolkata**; hill stations as summer resorts; cantonments
After Independence	New capitals such as **Chandigarh, Bhubaneswar, Gandhinagar, Dispur**; industrial towns such as **Durgapur, Bhilai, Sindri, Barauni**
Satellite towns	Old towns pulled into a metro's orbit — **Ghaziabad, Gurugram, Rohtak** around Delhi; Navi Mumbai / Thane around Mumbai

City	Note
New Delhi	Lutyens / Baker
Chandigarh	**Le Corbusier**
Bhubaneswar	Otto Koenigsberger
Jamshedpur	Company (कंपनी) / steel town (Tata (तत))

N.4 Metropolitan Cities & Urban Hierarchy

Term	Fact (Census / NCERT)
Town	Urban centre below Class I, or loosely any urban place
City / Class I	≥ **1 lakh**
Million city	≥ **10 lakh** · Census 2011 = **53** UAs · largest **Greater Mumbai** · smallest **Kota**
Metropolitan (NCERT)	**1–5 million** (10 lakh–50 lakh)
Mega city (NCERT table)	**> 5 million** · **six** in 2011: Greater Mumbai, Delhi, Kolkata, Chennai, Bengaluru, Hyderabad · ~**21%** of urban pop
Megacity (UN)	≥ **10 million** · 2011 India: Mumbai, Delhi, Kolkata
Conurbation	Once-separate cities fused (Mumbai–Thane)
Megalopolis	Chain of conurbations (Boswash type)
Primate city	One city dwarfs the national urban system — **India does not** have a single primate city (Mumbai vs Delhi)
NCR	Delhi + **Ghaziabad, Noida, Greater Noida, Meerut** (UP) + Gurugram, Faridabad (HR)

N.5 Smart Cities & Heritage Cities

Round	When	Cities
Fast-track / first UP entry	**May 2016**	**Lucknow** (first UP city on the Mission list)
Round 2	**Sep 2016**	**Kanpur, Agra, Varanasi**
Round 3	**Jun 2017**	**Prayagraj (Allahabad (इलाहाबाद)), Aligarh, Jhansi**
Round 4	**2018**	**Moradabad, Bareilly, Saharanpur**

Category	Winner in the paper's list
Culture	**Indore**
Governance	**Vadodara**
Social Aspects	**Tirupati**
Urban Environment	**Bhopal**

N.6 Smart Villages & Scheme Years

Scheme	Year / fact	Trap
SPMRM (Shyama Prasad Mukherji Rurban Mission)	Launch **21 Feb 2016** (Cabinet **16 Sep 2015**); **MoRD**; about **300** clusters	**2015** is **not** the launch year (2023 NOT-matched)
Sansad Adarsh Gram Yojana (SAGY)	**11 Oct 2014**	
DDU-GKY	**2014**	
Rural Housing Interest Subsidy Scheme	**2017** (that paper)	
JNNURM	**2005**	Confusing with AMRUT
National Urban Housing and Habitat (वास स्थान) Policy	**2007**	
AMRUT	**June 2015**; about **500** cities (all 1-lakh+ ULBs, capitals, HRIDAY cities and selected others)	2005
AMRUT 2.0	**1 Oct 2021** — water security; all statutory towns	Calling it 2015
Jal Jeevan Mission	**2019** (Har Ghar Jal)	Putting it before AMRUT
PMAY-U	**25 Jun 2015** (same day as Smart Cities) — housing neighbour, year fact only	
SBM-Urban	**2 Oct 2014** (with rural SBM) — sanitation neighbour	Calling it 2015 Smart Cities
DAY-NULM	Urban livelihoods (MoHUA)	Mixing with SPMRM / DDU-GKY

Scheme	Year
JNNURM	**2005**
National Urban Housing and Habitat Policy	**2007**
AMRUT	**2015**
Jal Jeevan Mission	**2019**

N.7 Uttar Pradesh Focus — Urban Centres

City	Functional / tag
Lucknow	State capital — administrative; first UP Smart City (May 2016)
Kanpur	Industrial / leather–textile; first UP million city (**1971**)
Prayagraj (प्रयागराज)	Ancient Prayag; educational; Smart City round 3; **not** HRIDAY
Varanasi	Ancient; religious–educational (BHU); HRIDAY; Smart City round 2
Agra	Medieval; tourist (Taj); Smart City round 2
Aligarh	Educational (AMU); industrial neighbour of the western UP belt; Smart City round 3
Meerut	Garrison / NCR satellite — **not** Central Smart-10
Ghaziabad	NCR satellite / large UA — **not** Central Smart-10 (State 2019 only)
Noida / Greater Noida	Planned satellite of Delhi — **not** Mission-100
Saharanpur	NCERT **commercial** town; Smart City round 4
Bareilly / Moradabad / Jhansi	Smart City round 4 / 3; Moradabad brass
Mathura	Religious; **HRIDAY**; State Smart City 2019 (not Central-10)
Ayodhya (अयोध्या)	Religious; State Smart City 2019; **not** HRIDAY

City	Functional / tag
Singrauli belt (Sonbhadra (सोनभद्र))	Mining / energy belt — mining-town type
Mission fact	Fact
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Central Smart Cities	**10** — Ghaziabad is **out** of Mission-100
State Smart Cities (2019)	7 Nagar Nigams — do **not** change 2018/2020 keys
HRIDAY	**Varanasi** and **Mathura** only
ISAC-2020	UP is the **best state**
NCR (UP slice)	Ghaziabad, Noida, Greater Noida, Meerut

Common Traps — Don't Fall For These

1. **Ghaziabad = Central Smart City** — FALSE for Mission-100 / 2018–2020 keys. State-2019 list is a **different** tag.
2. **UP Smart Cities = 12/15/18** — FALSE. Papers key **10**.
3. **HRIDAY = Prayagraj / Ayodhya** — FALSE. **Varanasi + Mathura**. National total **12**, not 100.
4. **SPMRM launched 2015** — FALSE as launch year. **21 Feb 2016**.
5. **Class I town = 10 lakh** — FALSE. **1 lakh**. 10 lakh = million UA.
6. **Census town = any big village / always a municipality** — FALSE. Triple test **or** statutory town; 75% is **male main** workers.
7. **AMRUT = 2005** — FALSE. **2015**. **JNNURM = 2005**.
8. **Rurbanization = M.N. Srinivas** — FALSE in that 2022 set. **G.S. Ghurye**.
9. **Global Village = UN created it** — FALSE. **Transport + communication**.
10. **Dispersed villages = Ganga plain / all of Thar** — FALSE. Plains = clustered. **Rajasthan = clustered (water)**. Dispersed = **Meghalaya, UK, HP, Kerala, NE forests**.
11. **Smart village scheme = MoHUA Smart Cities** — FALSE. SPMRM = **MoRD**.
12. **NCERT mega city = 10 lakh or only 1 crore** — FALSE. NCERT mega = **> 50 lakh (six)**; metro = **10–50 lakh**; UN megacity = **1 crore**.
13. **ISAC Culture = Vadodara** — FALSE. Culture **Indore**; Governance **Vadodara**.
14. **Determinism = Vidal / Possibilism = Ratzel** — SWAP. Determinism = **Ratzel/Semple**; Possibilism = **Vidal**.
15. **Neo-determinism = free will without limits** — FALSE. **Griffith Taylor** stop-and-go.
16. **Site = regional location** — FALSE. Site = **local ground**; situation = region.
17. **Skip nature/scope because recent papers asked Smart Cities** — FALSE. Next year can ask approaches from the same topic.
18. **Varanasi = major seaport town** — FALSE. Religious–cultural; HRIDAY yes.
19. **2011 UA rank = Mumbai > Kolkata > Delhi** — that is **2001**. **2011 = Mumbai > Delhi > Kolkata > Chennai**.
20. **Bhopal = river-bank town like Agra/Patna** — FALSE.
21. **Hamleted = a census town** — FALSE. Local **panna/para/palli/nagla/dhani**, one village name.
22. **75% workers in primary sector = census town** — FALSE. Need **non-agriculture** male main workers.

🌐 Human Geography Core Theories & Concepts (Ghatnachakra Complete)

- **Neo-Determinism ("Stop-and-Go Determinism" / Scientific Determinism)**: Advanced by **Griffith Taylor**. Holds that humans can accelerate, slow down, or stop nature's programme, but cannot depart from the broad physical conditions nature sets.
- **Cultural Worlds (Introducing Cultural Geography)**: **J.E. Spencer and W.L. Thomas** divided the world into **11 cultural worlds**.

- **Migration Theory**: Advanced by **Everett S. Lee** (Push and Pull factors; permanent or semi-permanent change of residence).

Topic 13 — Disaster Geography

Current Affairs (this topic)

Year	Fact	Why asked	Source
Mar 2026	BIS **IS 1893:2025 / Zone VI withdrawn**	Do **not** replace II–V in the hall	The Hindu / BIS
2015–30	Sendai Framework	Successor of Hyogo	UNDRR
2019	**CDRI** launched by India	Infrastructure + DRR	MEA / NDMA
2016 / 2019	NDMP issued / revised	First Sendai-aligned national plan	NDMA
2023	Joshimath subsidence; Sikkim GLOF	Landslide / flood CA	NDMA / news
2021	Tauktae (Arabian Sea → Gujarat)	"Arabian Sea never hits west coast" trap	IMD
2020	Amphan (BoB → WB/Odisha)	Super cyclone east coast	IMD

Cyclone **formation** (SST ~27°C, Coriolis, low shear) and IMD wind ladder are taught in N.10. Ring-of-Fire volcano list remains a Kilimanjaro trap pair.

Confused Pairs

Pair	Correct	Trap	Hindi
Focus vs epicentre	Focus = **inside**; epicentre = **surface**	Swapped	उद्गम / अधिकेन्द्र
Body vs surface waves	Body = through Earth (**P, S**); surface = along ground (**Love, Rayleigh**)	Treating all as one family	शरीर तरंग / पृष्ठ तरंग
P-wave vs S-wave	**P** = fastest, solids+liquids+gases; **S** = slower, **solids only**	S through outer core	प्राथमिक / द्वितीयक
P shadow vs S shadow	P miss ~**103–142°**; S absent beyond ~**103°**	Same belt for both / both recorded past 105°	छाया क्षेत्र
Richter vs Mercalli	Magnitude vs **intensity/damage**	Same scale	रिक्टर / मर्कली
Zone V vs II	V = **highest** (NE, Himalaya, Kutch, A&N)	V = safest	क्षेत्र V
Latur 1993	**Peninsular** / Killari (Zone III)	"Only Himalaya shakes"	लातूर
Koyna 1967	**Reservoir-induced**	Pure Himalayan thrust	कोयना
Rajasthan / Deccan	Mostly **II–III**	All India Zone V	
Tsunami	**Seafloor slip** / landslide / volcano — Japanese harbour wave	Wind-raised waves / "tidal wave"	सुनामी
INCOIS	Tsunami warning — **Hyderabad** (MoES)	NDMA HQ / IMD Pune as the tsunami hub	INCOIS
Kilimanjaro	**E. African Rift** — not Ring of Fire	Fuji/Pinatubo/St Helens are	किलिमंजारो
Cloudburst	**≥100 mm in one hour**	Any heavy rain	बादल फटना
Heat wave plains	Gate **≥40°C** then +4.5°C	Any hot afternoon	लू / ऊष्मा तरंग
BoB vs Arabian	**BoB more** cyclones	Equal / Arabian more	बंगाल की खाड़ी
Baguio	**Philippines**	Japan / USA	बागिओ
Willy-willies	**Australia**	USA	विली-विली
NDMA chair	**Prime Minister**	Home Minister	प्रधानमंत्री
SDMA chair	**Chief Minister**	Governor	मुख्यमंत्री

Pair	Correct	Trap	Hindi
Sendai vs Hyogo	**2015–30** vs 2005–15	Yokohama 2015	सेंडाई
DPAP year	**1973–74**	1952 (that is CDP)	DPAP
Hazard vs disaster	Hazard = potential; disaster = when capacity fails	Same word	आपदा ≠ संकट बीज
Mitigation vs response	Before impact vs during/after	Swap	न्यूनीकरण ≠ अनुक्रिया
NPDM year	**2009**	2005 (that is DM Act)	2009 नीति
Chemical / industrial	**Bhopal 1984** man-made	"Natural only India"	भोपाल
GLOF vs ordinary flood	Glacial-lake outburst (e.g. **Sikkim 2023**)	Any river monsoon rise	GLOF
Joshimath vs Kedarnath	Joshimath **2023** = subsidence; Kedarnath **2013** = rain–debris–flood	Same mechanism	जोशीमठ / केदारनाथ
Heat gate vs hot day	Plains need **Tmax ≥40°C** then departure/actual cut	Any afternoon above 35°C	ऊष्मा तरंग
BoB peak months	**May–Jun** and **Oct–Dec**	Mid-monsoon as the only peak	चक्रवात ऋतु
NDRF year	Raised **2006** (MHA)	2005 = DM Act only	NDRF
NEC chair	**Union Home Secretary**	Home Minister / PM	NEC

N.1 Earthquakes — Mechanics

Seismic waves — body first, then surface

Wave	Family	Nature	Travels through	Arrival / damage
P (primary)	Body	Longitudinal / sound-like; **fastest**	Solids, liquids, gases	First
S (secondary)	Body	Transverse / shear; slower than P	**Solids only**	Second; stopped by liquid outer core
Love / Rayleigh	Surface	Along the free surface	Near-surface rock / soil	Last; usually most destructive at ground

Shadow zones

Term	Recall
Focus / hypocentre	Inside the crust
Epicentre	Surface above the focus
Richter / Mw	Magnitude (energy)
Mercalli	Intensity (damage)
Body waves	Through Earth — **P** and **S**
Surface waves	Along ground — Love and Rayleigh
P then S then surface	Arrival order
S shadow	Beyond ~**103°** (liquid outer core)
P shadow	Gap ~**103–142°** (core refraction)

N.2 Earthquakes in India

Hall earthquakes — Cause, Course, Result

Year	Event	Trap
1819	Allah Bund / Kutch	
1897	Shillong / Assam	
1905	**Kangra**	
1934	**Bihar–Nepal** (east UP felt)	
1950	Assam–Tibet (~8.6)	
1967	**Koyna** (MH) — reservoir-induced	"Only Himalaya"
1991	Uttarkashi	
1993	**Latur / Killari** — stable peninsula	Himalayan
1999	Chamoli (चमोली)	
2001	**Bhuj** (26 Jan)	
2004	Sumatra–Andaman (tsunami)	
2005	Kashmir / Muzaffarabad	
2011	Sikkim	
2015	Nepal Gorkha (Bihar/east UP)	

N.3 Seismic Zones of India

Zone	Risk	Where (list)
V	Very high	NE India, Andaman & Nicobar, Himalayan pockets, Kutch, N Bihar
IV	High	Rest of Himalayan belt, Delhi, N/E UP Terai
III	Moderate	Broad peninsula, most of UP, many metros, Latur–Koyna
II	Low	Large stable Deccan / western Rajasthan interiors

N.4 Tsunami

26 December 2004 — Indian Ocean mega-event

Item	Fact
Word origin	Japanese **harbour wave**
Trigger	Seafloor slip — quake / landslide / volcano
Not the same as	Wind waves / cyclone swell
Speed behaviour	Fast in deep water; piles up near coast
Warning hub	**INCOIS, Hyderabad** (MoES)
India high-risk coasts	A&N, east coast, Kerala; west coast lower but not zero
Hall event	**26 Dec 2004** — Sumatra megathrust
Soft shield	**Mangroves** / coastal forests cut surge and wave force

N.5 Landslides

Types (coaching map)

Type	What moves
Fall	Rocks detach and freefall from a cliff
Slide	A mass moves on a distinct slip surface
Flow	Wet debris / mud moves like a fluid (**debris flow**)
Creep	Very slow downhill soil movement over years

N.7 Drought

Types

Type	Meaning
Meteorological	Rainfall well below normal
Hydrological	Rivers / tanks / groundwater fail
Agricultural	Soil moisture / crop stress
Socio-economic	Livelihood failure

Schemes chronology (asked order)

Scheme	Year
Community Development Programme	**1952**
DPAP (Drought Prone Areas Programme)	**1973–74**
Food for Work	**1977**
DDP (Desert Development Programme)	**1977–78**
TRYSEM	**1979**

N.9 Heat Waves & Cold Waves

Heat wave (IMD)

Region	Gate (Tmax)
Plains	≥ **40°C**
Coast	≥ **37°C**
Hills	≥ **30°C**
Rule	Heat wave
------	----------
Departure from normal	**4.5–6.4°C**
Actual Tmax (plains)	≥ **45°C**

Confused pair

Pair	Correct
Heat wave vs any hot day	Need **gate** + departure / actual cut

Pair	Correct
Cold wave vs ordinary winter night	Need abnormal **Tmin** drop / departure
Loo	Hot dry plains wind of north India in summer

N.10 Cyclone Basins of India

IMD wind ladder (increasing intensity)

Stage	Approx. wind (km/h)
Low pressure area	< 31
Depression	31–49
Deep Depression	50–61
Cyclonic Storm	62–88
Severe Cyclonic Storm	89–117
Very Severe Cyclonic Storm	118–165
Extremely Severe Cyclonic Storm	166–220
Super Cyclonic Storm	≥ 221

Basins and seasons

Basin	Disaster fact
Bay of Bengal	**Higher frequency** (~4:1 vs Arabian in many decades); Odisha–AP–TN–WB–Sundarbans (सुंदरबन) surge
Arabian Sea	Fewer storms, but they **do** hit **Gujarat / Maharashtra**
Peaks	**May–Jun** and **Oct–Dec** (retreating monsoon (लौटता मानसून)). Mid-monsoon shear suppresses BoB

Local names (match bank)

Name	Country / basin	Note
Baguio / Baguios ★	**Philippines**	High-yield match
Hurricane ★	**USA** / Atlantic–Caribbean	High-yield match
Typhoon ★	NW Pacific — **China** class	High-yield match
Taifu ★	**Japan**-facing NW Pacific name	Often paired with Japan
Willy-willies ★	**Australia**	Cyclone ≠ Brickfielder (ब्रिकफील्डर)
Cyclone	North Indian Ocean (**India**/BoB/Arabian)	IMD names
Cordonazo	West Mexico Pacific	Coaching distractor

2019 named-storm match (core four) + nearby names

Storm	Most affected (fact)
Dorian ★	**Bahamas**
Hagibis ★	**Japan**
Lekima ★	**China**
Mitag ★	**Jeju, South Korea**
Amphan (2020)	West Bengal / Odisha
Fani (2019)	Odisha

Storm	Most affected (fact)
Tauktae (2021)	Gujarat / west coast
Biparjoy (2023)	Gujarat / Arabian Sea
Odisha Super Cyclone (1999) / Phailin / Hudhud / Yaas	Classic India east-coast set

N.11 Disaster Management Framework

International and CA pairs

Body	Role
DM Act	**23 Dec 2005**
NDMA	Apex body; the **PM** is ex-officio Chair; up to 9 members
SDMA	**CM** chairs
DDMA	District Magistrate chairs (elected head co-chairs)
NEC	National Executive Committee — **Union Home Secretary**
NDRF	**2006**, specialised response under **MHA**; started at 8 Bn and later rose to **16 Bn**
NIDM	Training / research in **New Delhi**; grew from **NCDM (1995, under IIPA)**
Nodal ministry	**MHA** (Disaster Management Division)
NDMP	**1 Jun 2016** (revised 2019) — aligned to Sendai
Sendai	**2015–2030**, Sendai (Japan), after **Hyogo 2005–2015** and Yokohama 1994; **7** global targets + **4** priorities; "Build Back Better" in Priority 4
NPDM	National Policy on Disaster Management — **2009**
CDRI	Coalition for Disaster Resilient Infrastructure — India, **2019**
IDDR	International Day for DRR — **13 October**
NCS	National Centre for Seismology — **MoES**
INCOIS	Tsunami — **Hyderabad**
IMD	Cyclone, rainfall, heat/cold

N.12 Uttar Pradesh Focus

Institutions

Hazard	Fact
Seismic	**No Zone V.** Terai / north–east is **IV**; Lucknow–Kanpur–Prayagraj–Varanasi–Agra belt is **III**
Flood	East UP — **Ghaghara, Rapti, Gandak, Ganga**; trans-Ghaghara belt
Drought	**Bundelkhand** + Vindhyan / south-west pockets
Heat	May–Jun plains (**Loo**)
Cold / fog	Western UP + Terai, Dec–Jan
Cloudburst	Not a plains speciality; hills of neighbouring Uttarakhand
Biological-disaster label (2018)	**Gomati** (pollution) — not Yamuna/Sai/Tamsa in that paper
Framework	SDMA under **CM**; DDMA under DM

Common Traps — Don’t Fall For These

1. **NDMA chair = Home Minister** — FALSE. **Prime Minister**.
2. **Zone V = safest / Zone I still exists** — FALSE. **V = highest**; I abolished.
3. **Entire UP = Zone V** — FALSE. **III–IV**.
4. **Only Himalaya shakes** — FALSE. **Latur 1993, Koyna 1967**.
5. **Richter = damage scale** — FALSE. **Mercalli** = intensity.
6. **S-waves travel through the liquid outer core / both P and S recorded past 105°** — FALSE. **S** stop beyond ~**103°**; **P** miss ~**103–142°**.
7. **Surface waves arrive first / P waves do most surface damage** — FALSE. Arrival is **P** → **S** → **surface**; surface waves usually do worst ground damage.
8. **Tsunami = wind waves / cyclone swell / tidal wave** — FALSE. **Seafloor displacement**; Japanese **harbour wave**.
9. **INCOIS = NDMA HQ / IMD Pune** — FALSE. Tsunami warning = **INCOIS, Hyderabad**.
10. **Kilimanjaro = Ring of Fire** — FALSE. **East African Rift**.
11. **Cloudburst = any hill rain** — FALSE. **≥100 mm/hour**.
12. **Heat wave = any day above 35°C** — FALSE. Plains **gate 40°C** + departure/actual cuts.
13. **Arabian Sea never hits India** — FALSE. **Gujarat/MH** (Tauktae).
14. **Baguio = Japan; Willy-willy = USA** — FALSE. **PH and Australia**.
15. **Sendai = 2005–15** — FALSE. That is **Hyogo**. Sendai **2015–30**.
16. **NPDM = 2005** — FALSE. DM Act **2005**; NPDM **2009**.
17. **NDMA chair = Home Minister** — FALSE. **Prime Minister**.
18. **Hazard = disaster always** — FALSE. Disaster needs exposure + weak capacity.
19. **Skip HVRC / Sendai priorities because recent papers asked zones** — FALSE. Next year can ask the framework from the same topic.
20. **Arabian Sea never hits India** — FALSE. Gujarat–Maharashtra landfalls exist.
21. **India is disaster-free** — FALSE (2018 key).
22. **DPAP = 1952** — FALSE. **1973–74**; 1952 = CDP.
23. **Gomati vs Yamuna as 2018 biological-disaster river** — paper key **Gomati**.
24. **2004 deaths = 50 lakh / Bangladesh worst-hit by that wave path** — FALSE. Order **2–3 lakh**; Bangladesh relatively few for wave-direction reasons.
25. **Joshimath 2023 = same as Kedarnath 2013 rain wall** — FALSE. Joshimath is mainly **subsidence / slope distress**; Kedarnath is a rain–debris–flood cascade.
26. **GLOF = ordinary Ganga bank flood** — FALSE. **Glacial-lake outburst** (Sikkim Teesta / South Lhonak 2023).
27. **Cyclones need the equator / form only in mid-monsoon BoB** — FALSE. Need **Coriolis** (usually outside ~5°) and peak in **May–Jun / Oct–Dec**.
28. **Bhopal = natural earthquake disaster** — FALSE. **Man-made** chemical / industrial (1984).

Topic 14 — Earth & Universe

Current Affairs (this topic)

Year	Fact	Why asked	Source
2013–14	Mangalyaan / MOM	First Indian Mars orbiter (2024)	ISRO
2023	Aditya-L1	Solar mission — not Mars	ISRO
1987 / 1997	Montreal vs Kyoto (क्योटो)	Ozone vs climate	UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) / UNFCCC (यूएनएफसीसीसी)
2015	Paris Agreement (पेरिस समझौता)	NDC successor to Kyoto path	UNFCCC
2018	Science — Mars subglacial lake	2018 Q85	Science
2026	El Niño tendency vs Indian monsoon	IMD	

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q95

Which planet takes the longest time completing one revolution around the sun?

A. Saturn B. Uranus C. Neptune D. Jupiter

Confused Pairs

Pair	Correct	Trap	Hindi
Ozone layer	**Stratosphere**	Troposphere	समतापमंडल
Montreal vs Kyoto	Ozone vs **climate/GHG**	Kyoto “save water”	मॉन्ट्रियल / क्योटो
El Niño fish	**Less** plankton	Warm water → more plankton	अल नीनो
Coriolis SH	**Left**	Right / magnetic	कोरिऑलिस
Jet streams	Mainly **westerly**	“All easterly” (2024 A)	जेट धारा
TEJ	Summer **easterly** ~14°N	Same as STWJ	उष्णकटिबंधीय पूर्वी जेट
ITCZ	**Low** / rising air	High-pressure belt	आईटीसीजेड
Atmosphere heat	**Long-wave terrestrial**	Direct short-wave	दीर्घ तरंग
Fujita vs Saffir–Simpson	Tornado vs **Hurricane**	Swapped	फुजिता / सफिर–सिंपसन
Willy-Willy	**Australia**	USA hurricane	विली-विली
Horse latitudes	Subtropical **high** ~30°	Low pressure	हॉर्स लैटिट्यूड
Cumulonimbus	Intense rain / thunder	Fair-weather cumulus	कपासी वर्षा मेघ
Albedo	**Reflecting** power	Absorbing	अल्बीडो
GHG	Not **argon**	Argon as GHG	आर्गन नहीं
Saturn (that paper)	**Methane**	SO2 / CO	शनि
Mars	Life-candidate + 2018 lake	Jupiter lake	मंगल
Kilimanjaro	**E. African Rift**	Ring of Fire	किलिमंजारो
Fold vs fault	Compressional folds vs **block/rift**	All Himalaya = fault	वलित / भ्रंश

Pair	Correct	Trap	Hindi
Perihelion	**January** (NH winter)	"Closest = summer"	उपसौर
IDL	~**180°**; west is ahead	Straight 180° only	अंतरराष्ट्रीय तिथि रेखा
IST	**82°30′ E, +5:30**, whole India	Only east of Mirzapur	भारतीय मानक समय
Nuclear / geothermal	**Not** stored solar	All energy = solar	नाभिकीय / भूतापीय
Hottest planet	**Venus** (greenhouse)	Mercury	शुक्र
Largest moon	**Ganymede** (Jupiter)	Titan / Moon	गेनीमीड
Asteroid belt	**Mars–Jupiter**	Earth–Mars	क्षुद्रग्रह
Solar vs lunar eclipse	**New Moon** vs **Full Moon**	Swapped	सूर्य / चंद्र ग्रहण
Roaring Forties	~**40°S westerlies**	40°N trades	गर्जन-चालीसा
Horse latitudes	**~30°** subtropical high	ITCZ	अश्व अक्षांश
Stromboli	**Mediterranean / Italy**	Ring of Fire Pacific	स्ट्राम्बोली
Sial vs Sima	Continents vs **ocean crust**	Swapped	सियाल / सीमा
Heat budget	Earth absorbs ~**51**/100; planet albedo ~**35** reflected	"All 100 heat the air directly"	ऊष्मा बजट
Seasons cause	**23½° tilt**	Perihelion distance	ऋतु
Divergent boundary	Plates **apart** / MOR	Convergent trench	अपसारी
Chandrasekhar limit	~**1.44 M⊙** white dwarf	Eddington / Fowler alone	चंद्रशेखर
Milky Way shape	**Barred spiral**	Plain spiral only	आकाशगंगा
Earth density	**Highest** mean among 8	Jupiter densest	घनत्व
Albedo Mercury vs Earth	Earth **higher** (~0.3)	Mercury brighter albedo	अल्बीडो
Brittle star	**Sea** echinoderm	Pulsar / quasar class	ब्रिटल स्टार
0°–0° meet	**Atlantic** / Guinea	On land / India	शून्य
Cancer / Capricorn traps	Not **Iran** / not **Bolivia**	Wrong country lists	कर्क / मकर
Granite	**Igneous**	Metamorphic	ग्रेनाइट
Limestone →	**Marble**	Quartzite	संगमरमर
Sandstone →	**Quartzite**	Marble	क्वार्टजाइट
Shale →	**Slate**	Gneiss first	स्लेट
Magma vs lava	Below vs **surface**	Swapped	मैग्मा / लावा
Volcanic gas	Mostly **H_2O vapour**	CO_2 / SO_2 first	जल वाष्प
Stromboli	**Active** Mediterranean lighthouse	Extinct / Etna only	स्ट्राम्बोली
Kilimanjaro	**Dormant** (E. African Rift)	Ring of Fire active	किलिमंजारो
Aconcagua	Peak / **not** active volcano in GC keys	Same class as Etna	एकोंकागुआ
Focus vs epicentre	Rupture inside vs **surface above**	Swapped	फोकस / उपरिकेंद्र
Richter vs Mercalli	Magnitude vs **intensity**	Swapped / both "intensity"	रिक्टर / मर्कली
Richter +1 step	**10×** amplitude; ~**32×** energy	100× energy	रिक्टर ऊर्जा
S-waves	**Solids only**	Through liquid core	S-तरंग
Ring of Fire	Quakes + volcanoes + Pacific	Forest fires	रिंग ऑफ फायर
NW India quakes	**Plate convergence**	Volcano / coral	प्लेट
Tsunami word	**Japanese** harbour wave	Arabic / Latin	सुनामी

N.1 Origin of Universe

Universe-scale theories

Theory	Propounder	Teaching line
Nebular	**Kant – Laplace**	Rotating gas cloud flattened into Sun + planets
Planetesimal	Chamberlin – Moulton	Passing star pulled matter into planetesimals
Tidal	Jeans – Jeffreys	Tidal filament from a passing star broke into planets
Binary star	Russell – Lyttleton	Companion star disrupted into planetary debris
Interstellar dust	**Otto Schmidt**	Sun captured a gas–dust cloud that formed planets
Big Bang	**Lemaître** (often + **Gamow**)	Hot dense start ~**13.7–13.8 bn yr**; expansion
Steady State	**Hoyle – Bondi – Gold**	Universe looks constant; continuous creation of matter
Pulsating	Oscillating model	Repeated expansion and contraction

Galaxy, stars and distances

Note	Fact
Expansion	**Hubble** — galaxies recede; redshift evidence
Evidence	Expanding space and the **CMB**
Galaxy	The **Milky Way** is a **barred spiral** with about **100 billion** stars
Galactic year	The Sun orbits the galactic centre in about **22.5–25 crore** years
Distance	One **AU** is about **149.6 million km**. One **light-year** is about **9.46×10^{12} km**
Star colour	Blue stars are hottest. Red stars are cooler
Star life	Nebula to main sequence to red giant to nova/supernova to white dwarf / neutron star / black hole
Chandrasekhar limit	About **1.44 M⊙** is the white-dwarf ceiling
Black hole	Massive-star remnant from which light cannot escape
Constellations	The IAU lists **88**. **Saptarishi** points to the Pole Star
Closest star (after Sun)	**Proxima Centauri**
Not space	A **brittle star** is a marine animal. A **byte** is a computing unit

N.2 Solar System

Planet desk (tilt · spin sense · moons · periods)

Planet	Axial tilt (approx.)	Rotation sense	Rotation (day)	Revolution (year)	Moons (coaching)	Must-score note
Mercury	**~0°**	Prograde	**~59** Earth days	**~88** Earth days	**0**	Smallest planet, fastest orbit, no moon
Venus	**~177°**	**Retrograde**	**~243** Earth days	**~225** Earth days	**0**	Hottest (~464°C); Morning/Evening Star
Earth	**23½°**	Prograde	**~24 h**	**365¼** days	**1** (Moon)	Densest mean; habitable zone
Mars	**~25°**	Prograde	**~24.6 h**	**~687** Earth days	**2** (Phobos, Deimos)	Red planet; day/tilt close to Earth

Planet	Axial tilt (approx.)	Rotation sense	Rotation (day)	Revolution (year)	Moons (coaching)	Must-score note
Jupiter	**~3°**	Prograde	**~10 h**	**~12** Earth years	Many; Galilean **Io, Europa, Ganymede, Callisto**	Largest / heaviest; fastest spin
Saturn	**~27°**	Prograde	**~10.7 h**	**~29.5** Earth years	Many; **Titan** largest of Saturn	Clearest rings
Uranus	**~98°**	**Retrograde** (sideways)	**~17 h**	**~84** Earth years	Many	Sideways rotator; Sun rises in the west framing
Neptune	**~28°**	Prograde	**~16 h**	**~165** Earth years	Many; **Triton** famous	Coldest / farthest in 8-planet set
Pluto (dwarf)	High / retrograde framing	Retrograde	Long	**~248** Earth years	Charon, Nix, Hydra and others	IAU dwarf since **2006**

N.3 Motions of Earth

Motion	Period	Effect
Rotation	**~24 h** (west to east)	Day and night, apparent Sun motion, and **Coriolis**
Revolution	**365¼** days	Year; leap year every 4 years (century ÷400)
Orbit shape	Ellipse	**Perihelion ~3 Jan** (closest); **Aphelion ~4 Jul** (farthest)

Date	Event	Sun overhead / polar day
21 Mar / 23 Sep	Equinox	Sun over Equator; day equals night
21 Jun	NH summer solstice	Tropic of **Cancer**; Arctic 24 h day
22 Dec	NH winter solstice	Tropic of **Capricorn**; Arctic 24 h night

Note	Fact
Axis to orbital plane	**66½°** (complement of 23½°)
Circle of illumination	Great circle separating day/night; tilts with seasons
Sidereal day	**~23 h 56 min** (true 360° spin) vs mean solar day **24 h**
Leap year	Extra day because revolution ≈ **365¼ d**; century years leap only if ÷ **400**

N.4 Latitude

Parallel	Value	Note
Tropic of Cancer	**23½° N**	Sun overhead at NH summer solstice
Tropic of Capricorn	**23½° S**	Sun overhead at NH winter solstice
Arctic / Antarctic circles	**66½°**	90 – 23½
Heat zones	See bullets below	

N.5 Longitude, Time, Time Zones, IDL

Note	Note
Earth 360° / 24 h	Roughly **15°** of longitude equals **1 hour**; **1°** equals **4 minutes**

Note	Note
East	Local time is **ahead** of Greenwich
IST	Based on **82°30′ E** near **Mirzapur, UP**
Time zones	Roughly 15° wide; countries zigzag for politics
IDL	Follows about **180°** (bent around islands / Aleutians)
Prime Meridian land	Full land list below — **not Niger** / Nigeria as the classic wrong option
GMT places	London, Lisbon, Accra, Dublin — **not Addis Ababa**
Cairo	About **30° E**, so about **2 h ahead** of GMT

Country pair	Time zones
Russia	Most (11)
USA	6 (contiguous 4 + AK + HI)
China	**1** despite width
India	**1** (IST)

N.6 Atmosphere & Its Structure

Composition of dry air (by volume)

Gas	Approx. share	Teaching note
Nitrogen (N_2)	**~78.08%**	Most abundant gas in dry air
Oxygen (O_2)	**~20.95%**	Needed for respiration and combustion
Argon (Ar)	**~0.93%**	Main noble gas; **not** a greenhouse gas
Carbon dioxide (CO_2) (कार्बन डाइऑक्साइड)	**~0.04%** (~400 ppm class)	Trace greenhouse gas that varies with season and human activity
Neon, helium, methane, krypton, hydrogen, etc.	Trace	Very small permanent or near-permanent fractions

Permanent vs variable constituents

Type	Gases / constituents	Teaching note
Permanent	N_2, O_2, Ar (+ other rare noble gases)	Shares stay nearly constant day to day in the lower atmosphere
Variable	Water vapour, CO_2, O_3, aerosols / dust	Change with place, season, height and human activity

Vertical structure (bottom to top)

Layer	Approx. height	Temp. with height	Must-score note
Troposphere	~8 km at poles to ~18 km at the equator (avg ~13 km)	Usually **falls** (~**6.5°C/km**)	Almost all weather and most water vapour
Tropopause	Top of troposphere	Transition	Higher over the equator; lower over the poles
Stratosphere	~18–50 km	**Rises** because ozone absorbs UV	Ozone UV shield; dry and largely weather-free
Stratopause	~50 km	Transition	Boundary above the stratosphere
Mesosphere	~50–80 km	**Falls**	Coldest broad layer; meteors burn here

Layer	Approx. height	Temp. with height	Must-score note
Mesopause	~80 km	Transition	Coldest region of the atmosphere
Thermosphere / Ionosphere	above ~80 km (often framed ~80–640 km)	**Rises** strongly	Radio-wave reflection and aurora
Exosphere	outermost fringe	Merges into space	Very thin outer fringe

Homosphere (सममंडल) and heterosphere (विषममंडल)

Zone	Rough extent	Mixing
Homosphere	Surface to ~80–100 km	Turbulent mixing keeps major gas ratios roughly uniform
Heterosphere	Above ~80–100 km	Gases sort more by molecular weight

Heating, ozone and inversion

Idea	Correct
Main atmospheric heating	**Long-wave terrestrial** radiation from Earth's surface
Ozone maximum / UV shield	**Stratosphere**
Ozone hole (ओजोन छिद्र) focus	**Antarctica** in spring
Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल)	Cut **CFCs** on the ozone path (not the Kyoto climate path)
Normal lapse (troposphere)	Temperature falls about **6.5°C** per km
Temperature inversion	Temperature rises with height in a short layer

N.6A Insolation · Albedo · Heat budget

Surface	Typical albedo
Fresh snow	**75–95%**
Deserts	**20–30%**
Prairie (प्रेयरी) / grassland	**10–20%**
Paddy / dark wet fields	**3–15%**
Earth mean (planetary)	**~30%**

Heat budget (NCERT 100-unit picture)

Fate	Units
Reflected / scattered back to space (**planetary albedo**)	**~35**
Absorbed by the atmosphere	**~14**
Absorbed by Earth's surface	**~51**

N.6B Humidity · Dew

Humidity terms — Teaching Card

Term	Quick recall
Absolute humidity	Mass of vapour per volume of air
Relative humidity	% of saturation at that temperature
Specific humidity	Mass of vapour per mass of moist air

N.6C Clouds

Height families

Family	Typical height (mid-latitudes)	Make-up	Teaching note
High	About **5–13 km**	Mostly **ice crystals**	Names often begin with **cirro-**
Middle	About **2–7 km**	Water drops and/or ice	Names often begin with **alto-**
Low	Surface to about **2 km**	Mostly water drops	Includes stratus sheets and many rain clouds
Vertical / towering	Base low; top may reach the tropopause	Strong convection	**Cumulus** and **Cumulonimbus** grow upward through layers

Form families

Form idea	Look	Classic members
Cirriform	Thin, wispy, fibrous	Cirrus, cirrostratus, cirrocumulus
Stratiform	Sheet or layer	Stratus, altostratus, nimbostratus
Cumuliform	Heap / cauliflower	Cumulus, cumulonimbus, altocumulus
Nimbus	Rain-bearing	**Nimbostratus**, **Cumulonimbus**

Main cloud types

Cloud	Family	Teaching note
Cirrus	High	Thin, wispy ice trails. Often fair-weather or approaching weather aloft
Cirrostratus	High	Thin icy sheet. May make a Sun or Moon **halo**
Cirrocumulus	High	Small icy ripples or “mackerel sky” patches
Altocumulus	Middle	White or grey patches and rolls in the middle troposphere
Altostratus	Middle	Grey sheet that can dull the Sun. May thicken before steadier rain
Stratus	Low	Grey layer hugging the weather layer. Light drizzle or dull sky
Nimbostratus	Low	Thick rain sheet. Steady, often widespread rain or snow
Cumulus	Vertical	Puffy cauliflower heaps. Isolated fair-weather cumulus means unsettled tall storm not yet formed
Cumulonimbus	Vertical	Towering storm cloud. Intense rain, hail, thunder and lightning

Fog, halo and thunder links

Idea	Correct
Fog	Cloud with its base at or very near the **ground**
Halo around Sun / Moon	Often **cirrostratus**
Steady widespread rain	Often **nimbostratus**
Intense convective rain / hail / thunder	**Cumulonimbus**
Fair-weather afternoon heaps	Isolated **cumulus**

N.6D Tropical cyclones · Tornado · Anticyclone

Tropical cyclone

Local name	Region
Hurricane	USA / Atlantic–Caribbean
Typhoon	China Sea / western Pacific
Willy-Willy	North-west Australia
Cyclone	North Indian Ocean / India
Baguio	Philippines
Taifu	Japan

Anticyclone

System	Centre	Surface spin (NH)	Surface spin (SH)	Typical weather
Cyclone / low	Low	Anticlockwise	Clockwise	Cloud, rain, strong wind
Anticyclone / high	High	Clockwise	Anticlockwise	Clear, settled, light wind

N.7 Weather vs Climate

Important Geographical Isopleths (Lines of Equal Value)

Line (Isopleth)	Parameter / Value
Isobars	Equal atmospheric pressure
Isotherms	Equal temperature
Isohyets	Equal rainfall / precipitation
Isohalines	Equal ocean / water salinity
Isobaths	Equal depth below water/sea surface
Isohypse / Contours	Equal elevation above mean sea level
Isohels	Equal amount of sunshine / sunshine duration
Isonif	Equal amount of snowfall / snow
Isorymes	Equal frost
Isobronts	Places experiencing a thunderstorm at the same time
Isochrones	Places located at equal travel time from a common centre
Isogeotherms	Equal subterranean / underground temperature
Isoseismals	Equal earthquake shaking intensity
Isogonic lines	Equal magnetic declination (Agonic line = zero declination)

N.8 Global Warming

Is a greenhouse gas	Not a greenhouse gas
Water vapour, **CO_2**, **CH_4**, **N_2O**, **O_3**, **CFCs / HCFCs**	**Argon**
Natural coaching set often lists CO_2, CH_4, O_3 and N_2O	"Nitrogen oxide" (NO / NO_2) in the 2020 key is **not** treated as that natural GHG set

Climate and ozone treaties (do not swap)

Instrument	Year	Focus	What it does
Montreal Protocol	**1987**	**Ozone layer**	Cuts ozone-depleting substances, especially **CFCs** and related chemicals
Kyoto Protocol	**1997**	**Climate / GHG**	Sets binding emission cuts for listed greenhouse gases and frames **carbon credits**
Paris Agreement (पेरिस समझौता)	**2015**	**Climate / NDCs**	Each country files Nationally Determined Contributions under the UNFCCC path

Wrong pairing	Correct
Kyoto → save water	Kyoto → climate / GHG / carbon credit
Montreal → climate mitigation (जलवायु शमन)	Montreal → ozone / CFCs
Ramsar (रामसर) → climate mitigation	Ramsar → wetlands
CITES (साइट्स) → climate mitigation	CITES → wildlife trade
Paris → ozone CFCs	Paris → climate NDCs

N.12 Pressure Belts, Planetary Winds, ITCZ

Latitude	Belt	Wind
0°	Equatorial low / **ITCZ**	Doldrums with rising air and rain
30°	Subtropical **high**	Horse latitudes; **Trade winds** (NE/SE) blow toward ITCZ
60°	Subpolar **low**	**Westerlies**
90°	Polar **high**	Polar easterlies

Named Southern Hemisphere westerlies

Named belt	Fact
Roaring Forties	About **40°S**
Furious Fifties	About **50°S**
Shrieking / Screaming Sixties	About **60°S**

N.13 Interior of the Earth

Boundary	Depth pair	Fact
Moho	Base of crust (~8–40 km)	Crust / mantle
Gutenberg	**~2900 km**	Mantle / **outer core** — **S-waves stop** (liquid)
Lehmann	~5100 km	Outer / **inner core** (solid iron–nickel)
Layer chemistry (names)	Fact	
----------	------	
Sial	Continental crust (~2.7–3.0 g/cm³) — silica + aluminium; granite; less dense than sima	
Sima	Oceanic crust — silica + magnesium; basalt; denser than sial	

Boundary	Depth pair	Fact
Nife	Core (~13+ g/cm³ at centre) — nickel + iron (thickness ~3,471 km, core temp ~5000°C)	

Shadow	Note
S-wave	None beyond ~**103°** from epicentre (liquid outer core)
P-wave	Weak/absent ~**103–142°** (refraction in the core)

N.14 Continental Drift

How the idea grew after Wegener

Later piece	Fact
Mantle convection	**Holmes** (~1928–29)
Sea-floor spreading	**Harry Hess** (1960s); paleomagnetic stripes on MOR
Plates	Lithosphere pieces; India was in **Gondwana**

N.15 Plate Tectonic Theory

Boundary	Motion	Result
Divergent	Apart	MOR, rift (East African), new crust
Convergent	Together	Trenches, fold mountains, volcanoes, quakes
Transform	Slide	San Andreas; quakes, little crust created

N.16A Rock Types & Rock Cycle

Family	How it forms	Key traits
Igneous (primary)	Cooling of **magma / lava**	No fossils; not layered like sediment
Sedimentary	Weathering → transport → deposition → **lithification**	Fossils common; occur in **layers (strata)**
Metamorphic	Any rock remade by heat + pressure + chemistry (**metamorphism**)	New fabric / minerals; forms deeper / hotter than surface conditions

Kind	Setting	Examples
Plutonic / intrusive	Magma cools **slowly deep** inside the crust	**Granite**, gabbro, pegmatite
Volcanic / extrusive	Lava cools at / near the surface	**Basalt**, volcanic breccia

Type	Examples
Mechanical (clastic)	Sandstone, shale, conglomerate; limestone often listed here in older books
Organic / carbonic	Coal; limestone / geyserite note
Chemical	Halite, potassium salts

Parent	Metamorphic product
Granite	**Gneiss**
Basalt	Schist / amphibolite

Parent	Metamorphic product
Shale	**Slate**, then phyllite, then schist at higher grade
Limestone / chalk / dolomite	**Marble**
Sandstone	**Quartzite**
Conglomerate	Gneiss (listed in some GC sheets)
Gabbro	Serpentine (GC sheet)

N.17 Volcano Types

Note	Fact
Magma	Molten rock **below** the surface
Lava	Magma that reaches the surface (~600–1200°C); cools to igneous rock
Most abundant volcanic gas	**Water vapour (H_2O)** (जलवाष्प) — then CO_2, SO_2 and others
Acidic lava	High silica makes it sticky / viscous
Basic lava	Low silica makes it runny / fluid
Caldera	Large collapsed vent / crater
Geyser	Intermittent hot water + steam (e.g. Old Faithful, Yellowstone) — near magmatic heat
Fumarole / solfatara	Late-stage gas vents (a sulfurous vent is a solfatara); Katmai's "Valley of Ten Thousand Smokes" is the classic note
Type	Fact / example
------	----------
Shield	Low-silica, runny lava; classic Hawaii pair is **Mauna Loa / Mauna Kea**
Composite / strato	Sticky, explosive; **Fuji, Vesuvius, Etna, Pinatubo, St Helens, Rainier, Cotopaxi**
Cinder cone	Small, steep; **Paricutin** (Mexico)
Caldera	Collapse after blast; **Krakatoa / Krakatau** (Indonesia), Yellowstone
Fissure / flood basalt	**Deccan Trap** (दक्कन ट्रैप)

Status	Fact
Active	Open vent / recent eruptions — **Stromboli** ("lighthouse of the Mediterranean"), **Etna** (Sicily), **Kilauea** (often ranked among most active), **Mauna Loa**, **Barren Island** (India's only confirmed active — Andaman Sea), **Fuji**, **Vesuvius**, **Erebus** (Ross Island, Antarctica), **Cotopaxi** (Ecuador)
Highest active (world)	Coaching often cites **Ojos del Salado** (Andes, Chile–Argentina, ~6893 m). Among common options **Cotopaxi** may be the tallest listed
Dormant	Quiet but may revive — **Kilimanjaro** (Kibo dormant; Mawenzi / Shira extinct) and **Damavand** (Iran)
Extinct	Unlikely to erupt again — Kulal (Kenya), Chimborazo (Ecuador), Popa (Myanmar), Koh-e-Sultan (Pakistan) note

Volcano / peak	Country	Teaching note
Sabancaya	**Peru**	Andean volcano
Colima	**Mexico**	Mexican volcano match
Merapi	**Indonesia**	Classic Indonesian volcano
Kinabalu	**Malaysia**	Peak often listed with volcano/country sets
Elburz (Alborz)	**Iran**	Iranian range / peak match

Volcano / peak	Country	Teaching note
Aconcagua	**Argentina**	High peak; **not** treated as an active volcano in standard keys
Krakatoa / Krakatau	**Indonesia**	Famous caldera volcano

N.17B Earthquakes · Seismic Waves · Tsunami · Scales

Earthquake terms — Teaching Card

Term	Quick recall
Focus / hypocentre	Underground rupture point
Epicentre	Surface point above the focus
Seismograph / seismometer	Instrument that records waves
Seismogram	The wave record itself

Wave	Nature	Travels through	Note
P (primary)	Longitudinal / sound-like; **fastest**; arrives first	Solids, liquids, gases	Body wave
S (secondary)	Transverse / light-like; slower than P	**Solids only**	Body wave
Surface	Slowest; Love + Rayleigh; **most damaging** at surface	Near surface	Love waves are the faster surface type

Scale	Measures	Note
Richter (C.F. Richter, **1935**)	**Magnitude** ≈ energy released ($\log_{10}$ amplitude)	+1 magnitude ≈ **10×** amplitude; ≈ **~32×** energy (not 100×)
Mercalli	**Intensity** (felt damage / effects)	Different from magnitude

Event pair	Fact
26 Dec 2004 Indian Ocean	Sumatra subduction quake. Waves reached as far as Africa. Deaths were about **2–3 lakh**, not "50 lakh". Bangladesh saw relatively few casualties because the fault trended roughly N–S while main wave energy ran roughly E–W
11 Mar 2011 Japan	A great quake and tsunami damaged **Fukushima** Daiichi and caused a nuclear disaster

N.18 Major Earthquake & Volcano Belts

Belt	Note
Circum-Pacific / Ring of Fire	Pacific rim arc (Chile–California–Alaska–Japan–Philippines–New Zealand)
Mid-Atlantic	Iceland, Azores — divergent; quakes + volcanoes
Alpine–Himalayan / Mid-Continental	Cape Verde–Atlantic–Mediterranean–Alps–Caucasus–**Himalaya**; India's main seismic framing
East African Rift	**Kilimanjaro, Kenya** — **not** Ring of Fire ()
Cocos plate	Between **Central America** and the **Pacific** plate

Common Traps — Don't Fall For These

1. **Ozone = troposphere** — FALSE. **Stratosphere**.
2. **El Niño → more Peruvian fish** — FALSE. **Less** upwelling / plankton.
3. **SH deflection = right / magnetic** — FALSE. **Coriolis, left**.
4. **All jets are easterly** — FALSE. **Westerly**; TEJ is the easterly exception.

5. **ITCZ = high pressure** — FALSE. **Low**.
6. **Atmosphere heated by incoming short-wave** — FALSE. **Long-wave from Earth**.
7. **Argon is a GHG** — FALSE.
8. **Kyoto = ozone / save water** — FALSE. **Climate / carbon credit**. Montreal = ozone.
9. **Kilimanjaro = Ring of Fire** — FALSE. **East African Rift**.
10. **Perihelion = NH summer** — FALSE. **Early January**.
11. **Seasons = changing distance from Sun** — FALSE. **Tilt**.
12. **IST only east of Mirzapur** — FALSE. **Whole India**.
13. **India = 6th largest / wholly tropical** — FALSE. **7th**; **2.4%**; Tropic through the middle.
14. **Nuclear/geothermal = stored solar** — FALSE.
15. **Fold = Vosges; fault = Himalaya** — swapped. Himalaya = **fold**; Vosges = **block**.
16. **Hottest planet = Mercury** — FALSE. **Venus**.
17. **Largest moon = Titan / our Moon** — FALSE. **Ganymede**.
18. **Solar eclipse at Full Moon** — FALSE. **New Moon**. Lunar = Full.
19. **Roaring Forties = 40°N trades** — FALSE. **~40°S westerlies**.
20. **Nebular theory = Wegener / Hoyle** — FALSE. **Kant–Laplace**. Wegener = drift; Hoyle = Steady State.
21. **Sial = ocean floor** — FALSE. **Continents**. Sima = ocean.
22. **Kilimanjaro / Stromboli = same belt** — FALSE. Stromboli = **Mediterranean**; Kilimanjaro = **E. African Rift**.
23. **All 100 insolation units heat the air directly** — FALSE. ~**35** reflected; surface ~**51**; air then heated by **long-wave**.
24. **Seasons caused by perihelion** — FALSE. Seasons = **axial tilt**.
25. **Main jet streams = easterly** — FALSE. Main jets are **westerly**; TEJ is the exception.
26. **Skip spheres / rock cycle / plate types because recent papers asked ozone** — FALSE. Next year can ask any standard NCERT angle from this topic.
27. **Kilimanjaro = Ring of Fire** — FALSE. **East African Rift**.
28. **Hottest planet = Mercury / densest = Jupiter** — FALSE. Hottest = **Venus**; densest mean = **Earth**.
29. **Stratosphere = weather layer** — FALSE. Weather = **troposphere**. Stratosphere = ozone + jet flight (little weather).
30. **Aurora = mesosphere** — FALSE. **Ionosphere / thermosphere**.
31. **Horse latitudes = low pressure** — FALSE. Subtropical **high** ~30°.
32. **Eye of cyclone is 10°C colder** — FALSE. Eye is **warmer**.
33. **Jet streams = NH only** — FALSE. Both hemispheres.
34. **Fresh snow = lowest albedo** — FALSE. Fresh snow has **highest** albedo among common options.
35. **Ganymede = Saturn's moon / largest moon = Titan** — FALSE. **Ganymede = Jupiter**; Titan = Saturn's largest.
36. **Chandrasekhar limit ≈ Eddington / Fowler name alone** — FALSE. ~**1.44 M⊙** white-dwarf limit; black-hole path above collapse residue note.
37. **Nebula = solar-system member** — FALSE. Nebulae are star-forming clouds, not planets/asteroids/comets of the Sun's family.
38. **Brittle star = pulsar class** — FALSE. Marine animal; pulsars / quasars / black holes are cosmic.
39. **Mercury albedo > Earth** — FALSE. Earth albedo higher (~0.3 vs ~0.1).
40. **Equator through Mexico / Capricorn through Bolivia** — FALSE. Check country lists carefully.
41. **0°–0° on land / in India** — FALSE. Intersection in **Atlantic** (Gulf of Guinea).
42. **Solar eclipse at Full Moon / lunar at New Moon** — SWAP. Solar = **New**; lunar = **Full**.
43. **Granite = metamorphic** — FALSE. **Igneous**; gneiss is its metamorphic child.
44. **Fossils common in granite** — FALSE. Fossils favour **sedimentary** rocks.
45. **Most volcanic gas = CO_2 / SO_2** — FALSE. **Water vapour** leads.

46. **Magma = surface melt** — FALSE. Surface = **lava**; magma is underground.
47. **Aconcagua = active like Etna** — FALSE. Peak / not active in GC keys; Etna–Vesuvius–Fuji are active.
48. **Kilimanjaro = Ring of Fire / always extinct** — FALSE. **East African Rift**; mainly **dormant** framing.
49. **No volcanoes possible in Circum-Pacific** — FALSE. That belt is the **maximum** volcanic belt.
50. **Baltic Sea = volcanic hotspot** — FALSE. Classic "eruptions do **not** occur" option among seas.
51. **Epicentre = underground rupture** — FALSE. That is the **focus/hypocentre**; epicentre is the surface point above it.
52. **Richter step = 100× energy** — FALSE. ~**10×** amplitude and ~**32×** energy; Mercalli measures **intensity**.
53. **S-waves travel through liquid outer core** — FALSE. **Solids only**; that makes the S shadow.
54. **Ring of Fire includes forest fires** — FALSE. Quakes + volcanoes around the **Pacific**.
55. **NW India quakes = coral / volcano only** — FALSE. Mainly **Indian–Eurasian plate** convergence.
56. **2004 tsunami killed 50 lakh** — FALSE. Order of ~**2–3 lakh**; Bangladesh had few deaths for wave-direction reasons.
57. **Tsunami is Latin / Arabic** — FALSE. **Japanese** harbour wave.

🛰️ Satellite Orbits Master Fact (Ghatnachakra Complete)

- **Geostationary Satellite Orbit**: Revolves at a **fixed altitude of approximately 35,786 km** directly above Earth's equator. Orbital period matches Earth's rotation (23 hours 56 minutes 4 seconds), appearing stationary from ground.

Topic 15 — Geomorphology & Landform Processes

Current Affairs (this topic)

Year	Fact	Why it matters	Source
Static	Trellis / bird's-foot / Imphal lacustrine surface	Process stems stay map-static	NCERT / coaching
Static	Mekong delta = southern Vietnam (not Cambodia)	Country trap in delta stems	Atlas
Static	Moribund wording: Bengal subdivision vs Cauvery lobe	Two different stem shapes	Coaching
Static	Himalaya fossil belts (Himadri / Lesser / Shiwalik)	Mixed landform–fossil stems	NCERT

Confused Pairs

Pair	Correct	Trap	Hindi
Weathering vs erosion	In situ vs **transport**	Same thing	अपक्षय / अपरदन
Granite vs basalt	Intrusive acidic vs **extrusive basic**	Swapped	ग्रेनाइट / बेसाल्ट
Marble	From **limestone**	From sandstone	संगमरमर
Quartzite	From **sandstone**	From limestone	क्वार्टजाइट
Gneiss	From **granite**	From shale	नाइस
Trellis	**Folded** structures	Dendritic on folds	जाल / जाली
Bird's-foot delta	**Mississippi only**	Ganga / Nile / a second river	पक्षी-पाद
Arcuate delta	**Nile, Ganga–Brahmaputra**	Mississippi	चापाकार
Moribund wording	**Subdivision → Bengal**; **lobe → Cauvery**	One answer for both	मृतप्राय
Mekong delta	**Southern Vietnam**	Cambodia	मेकांग
Nile delta	**Northern Egypt**	Sudan / Ethiopia	नील
Indus delta	**Sindh, Pakistan**	Indian west coast	सिंधु
Irrawaddy delta	**Southern Myanmar**	Thailand	इरावदी
Danube delta	**Black Sea (Romania–Ukraine)**	Hungary	डेन्यूब
Volga delta	**Caspian Sea**	Black Sea	वोल्गा
Niger deltas	**Mali inland; Nigeria at the sea**	One mouth only	नाइजर
Imphal basin	**Lacustrine**	Loess / glacial	झीलकृत
Barchan horns	Point **downwind**	Upwind	बरखान
Stalactite	**Ceiling**	Floor	स्टैलेक्टाइट
Stalagmite	**Floor**	Ceiling	स्टैलेग्माइट
U-valley vs V-valley	Glacier vs **river youth**	Swapped	U / V
Fjord	Glacial trough drowned	Any estuary	फियोर्ड
East vs west India mouth	**Delta** vs **estuary**	All India deltas	डेल्टा / ज्वारनदमुख
Fault-valley river	**Damodar** (2019)	Chambal (ravines)	भ्रंश घाटी
Antecedent vs superimposed	Cuts rising land vs **inherited** on cover	Swapped	पूर्ववर्ती / अध्यारोपित
Consequent vs subsequent	Follows slope vs **follows structure**	Same	अनुगामी / परवर्ती
Sill vs dyke	**Parallel** to beds vs **cuts** beds	Swapped	सिल / डाइक

Pair	Correct	Trap	Hindi
Conglomerate vs breccia	Rounded pebbles vs **angular**	Swapped	समूहिका / ब्रेशिया
Peneplain vs pediplain	Davis humid old-age vs **King arid**	Same word	समप्राय मैदान / पदस्थली
Mesa vs butte	Broad flat cap vs **narrow remnant**	Swapped	मेसा / ब्यूट
Cuesta vs hogback	Gentle dip + steep scarp vs **steep both sides**	Same landform	क्वेस्टा / हॉगबैक
Atoll shape	**Horseshoe / ring** around a lagoon	Triangular / rectangular / rod	एटॉल आकार
Ria vs fjord	Drowned **river** vs drowned **glacier**	Same estuary	रिया / फियोर्ड
Gorge vs canyon	Steep mountain cut vs **arid stair-step**	Always identical	गॉर्ज / कैनियन
Endogenic vs exogenic	Interior build vs surface wear	Same	अंतर्जात / बहिर्जात
Weathering vs denudation	In situ breakdown vs weathering+erosion	Same	अपक्षय ≠ अनाच्छादन

N.1 Rocks — Igneous, Sedimentary, Metamorphic

Class	How formed	rocks
Igneous (primary)	Cooling of magma/lava	**Granite** (intrusive, acidic, coarse); **basalt** (extrusive, basic, Deccan Trap (दक्कन ट्रैप)); gabbro, pegmatite, pumice, obsidian
Sedimentary (secondary)	Compaction of debris / chemical / organic	Sandstone, shale, **limestone**, conglomerate, coal, loess beds — **fossils possible**
Metamorphic	Heat + pressure on existing rock	See pair table

Parent Rock	Metamorphic Product	Type / Notes
Limestone / dolomite	**Marble**	Sedimentary → Metamorphic (calcite recrystallised)
Sandstone	**Quartzite**	Sedimentary → Metamorphic (quartz grains welded)
Granite	**Gneiss**	Igneous → Metamorphic (banded crystalline)
Gabbro	**Serpentine**	Igneous → Metamorphic
Basalt	**Hornblende / Amphibolite / Schist**	Igneous → Metamorphic
Shale / Clay	**Slate → Phyllite → Schist**	Progressive foliation series
Coal	**Graphite / Diamond**	Organic Sedimentary → Metamorphic

Body	Role
Batholith	Huge granite mass; floor not seen; mountain cores
Laccolith	Lens, **convex up**; arches overlying beds
Lopolith	Saucer, **concave up**
Phacolith	Lens in fold crest/trough
Sill	Sheet **parallel** to bedding
Dyke	Wall that **cuts** bedding (discordant (असंगत))
Volcanic neck / plug	Filled pipe left standing

Family	Rocks
Clastic	Conglomerate (**rounded**), **breccia** (angular), sandstone, shale/mudstone
Chemical	Limestone, dolomite, gypsum, rock salt (लवणाध्यक्ष), chert
Organic / biogenic	Coal, peat, chalk, some limestone

N.2 Rock Cycle

From	Process	To
Magma	Cooling / crystallisation	Igneous
Any exposed rock	Weathering → erosion → deposition → lithification	Sedimentary
Igneous or sedimentary	Heat + pressure (no full melt)	Metamorphic
Any rock	Melting	Magma

N.3 Weathering, Erosion, Deposition

Term	Meaning
Weathering	In situ breakdown
Erosion	Pickup + transport
Deposition	Drop of load
Denudation	Weathering + erosion
Mass wasting	Gravity movement

Process	Quick recall
Frost / ice wedging	Water freezes and pries cracks
Exfoliation / unloading	Onion sheets on granite
Thermal / insolation	Desert day–night flake
Salt wedging	Crystal growth in arid pores
Block vs granular	Joint slabs vs grain-by-grain
Spheroidal	Rounded corestones

Process	Quick recall
Carbonation	Carbonic acid → limestone karst
Solution	Direct dissolve (salt, gypsum)
Oxidation	Iron rusts; red soils
Hydration	Water uptake → swell and crack
Hydrolysis	Feldspar → clay
Chelation	Organic acids leach metals

N.4 Agents of Erosion

Agent	Work words	Signature
River	Hydraulic, abrasion/corrasion, attrition, solution; load = traction · saltation · suspension · solution	V-valley → meander → delta
Wind	Deflation, abrasion, attrition	Barchan, yardang, loess
Glacier	Plucking, abrasion	U-valley, cirque, moraine
Sea	Hydraulic, abrasion, corrosion, attrition	Cliff, stack, spit
Groundwater	Solution	**Karst** in limestone

N.5 Fluvial Landforms (Davis cycle)

Stage	Landforms
Youth	Steep **V-valley**, gorge/canyon, waterfall, rapids, **potholes**, interlocking spurs
Mature	**Meander**, slip-off slope + river cliff, **floodplain**, **natural levee**, bluff, widening valley
Old	Broad floodplain, **ox-bow** / mortlake / meander scar, yazoo, **delta**, deferred tributary
Rejuvenation	Uplift / base-level fall → **knickpoint**, river **terrace**, **incised meander**

Fluvial forms — Teaching Card

Form	Quick recall
Meander	Outer cut cliff; inner point bar
Ox-bow	Cut-off meander lake
Floodplain	Point-bar + overbank silt
Levee	Raised silt ribbon along channel
Yazoo	Deferred tributary behind levee
Fan / bajada	Mountain-foot dump; coalescing fans
Braided	Many threads (Brahmaputra type)
Gorge vs canyon (गॉर्ज)	Narrow steep cut vs wider arid stair-step

Theory	Quick recall
Davis	Structure–process–time → **peneplain** + **monadnock**
Penck	Uplift and denudation **together** (no neat old-age wait)
L.C. King	Pediment retreat → **pediplain**

Stream origin names — Teaching Card

Stream	Follows
Consequent	Initial slope
Subsequent	Weaker beds / **faults** (trellis valleys)
Obsequent	Opposite to consequent (scarp face)
Resequent	Same direction as consequent, later
Insequent	No obvious control (dendritic on uniform rock)

N.6 Types of Deltas

Moribund — two different questions

Wording in the stem	What it asks	Answer
Subdivision of a named delta	Which delta is split into moribund–mature–active belts	**Bengal Delta**
Which **river's** delta is the moribund / abandoned **lobe**	Which delta is old and no longer growing	**Cauvery**
Either wording, with Krishna–Godavari or Mahanadi as the pick	Distractor on both stems	**Neither**

If the stem swaps the river — where the delta lies

River	Rises in	Delta lies in	Usual wrong place
Mekong (मेकांग)	Tibetan Plateau	**Southern Vietnam**	Southern Cambodia
Red River	Yunnan, China	**Northern Vietnam**	Southern Vietnam (that is the Mekong)
Irrawaddy (इरावदी)	Northern Myanmar	**Southern Myanmar**	Thailand
Chao Phraya	Northern Thailand	**Central Thailand**	Cambodia
Nile (नील)	White Nile: Lake Victoria; Blue Nile: Lake Tana	**Northern Egypt**	Sudan or Ethiopia
Indus (सिंधु)	Tibet, near Mansarovar	**Sindh, Pakistan**	Indian west coast
Ganga–Brahmaputra	Himalaya	**Bengal** (India and Bangladesh)	Assam alone
Godavari (गोदावरी)	Trimbakeshwar	**Andhra Pradesh**	Odisha or Tamil Nadu
Krishna (कृष्णा)	Mahabaleshwar	**Andhra Pradesh**	Karnataka at the sea
Mahanadi (महानदी)	Sihawa, Chhattisgarh	**Odisha**	Andhra Pradesh
Cauvery (कावेरी)	Talakaveri, Karnataka	**Tamil Nadu**	Karnataka at the sea
Narmada / Tapi	Amarkantak / Multai	**Estuary**, not a large delta	An arcuate fan
Danube (डेन्यूब)	Black Forest, Germany	**Black Sea**, Romania and Ukraine	Hungary or Austria
Rhine (राइन)	Swiss Alps	**Netherlands**	Germany only
Volga (वोल्गा)	Valdai Hills, Russia	**Caspian Sea**, Russia	Black Sea
Niger (नाइजर)	Guinea highlands	Inland delta in **Mali**; sea delta in **Nigeria**	Treating Mali as the sea mouth
Amazon	Andes	Broad **estuary in Brazil**	A fan delta in Peru
Hwang Ho	Tibetan Plateau	**Bohai**, China	A bird's-foot mouth
Mississippi	Lake Itasca, Minnesota	**Louisiana**, Gulf of Mexico	Any second bird's-foot river

Type	Shape / fact	Classic
Arcuate	Bow / fan	Nile, Ganga–Brahmaputra (Sundarbans), Rhine, Niger, Hwang Ho
Bird's-foot	Distributaries like toes	**Mississippi only**
Cuspate	Tooth / pointed	Tiber (Italy), Ebro (Spain)
Estuarine	Fills a drowned mouth	Narmada, Tapi
Moribund (subdivision wording)	Dying belt inside a three-belt scheme	**Bengal Delta**
Moribund (lobe wording)	Abandoned / ageing delta	**Cauvery**

N.7 Structural Basins & Fault Valley Drainage

Example	Type
Imphal basin	**Lacustrine** plain (old lake) — **not** loess / glacial / ordinary alluvial trap
Kashmir vale	Lacustrine / structural (Pir (पीर) Panjal (पीर पंजाल) wall)
Chhattisgarh / Michigan type	Structural/sedimentary basin

Plain	Agent
Alluvial	River (Indo-Gangetic)
Lacustrine	Old lake floor — **Imphal**, Kashmir
Glacial / outwash	Ice
Loess	Wind silt — **China**; not Imphal
Lava / volcanic	Flood basalt (Deccan surface)

Structural landforms — Teaching Card

Form	Dip of beds	Side view
Cuesta	Gentle	One gentle slope + one steep scarp
Hogback	Steep	Steep on both sides

Form	Size feel	Cap
Mesa	Broad flat top	Hard cap still wide
Butte	Smaller / pillar-like	Same idea, much less area

Drainage patterns — Teaching Card

Pattern	Looks like	Control (why)	River / ground
Dendritic	Tree branches	Homogeneous rock	**Ganga** and **Indus** plains; **Godavari**; **Mahanadi**; **Krishna**; **Kaveri**
Trellis	Lattice / garden trellis	Folded hard–soft belts	**Singhbhum** (Chotanagpur)
Radial	Spokes outward	Dome / volcano / highland	**Narmada** west and **Son** toward the Ganga from **Amarkantak**; Girnar; **Mahaweli** (Sri Lanka)
Centripetal	Spokes inward	Closed basin	**Loktak** (Imphal basin); **Mendha** and **Rupangarh** into **Sambhar**
Rectangular	Right-angle bends	Joints / faults	Jointed **Vindhyan** streams
Parallel	Side-by-side lines	Steep uniform slope	**Sharavati** (Western Ghats)
Annular	Rings	Concentric weak beds	**Black Hills** — no single Indian river
Deranged	Chaotic / unfinished	New glacial dump	**Canadian Shield** — no master river

N.8 Glaciers & Glacial Landforms

Glacier types — Teaching Card

Type	Quick recall
Cirque glacier	Armchair hollow
Valley / alpine	Tongue in a trough
Piedmont	Coalesced on a plain
Ice sheet	Greenland, Antarctica
Ice cap	Smaller dome
Ice shelf	Floating fringe

Erosional glacial landforms — Teaching Card

Form	Quick recall
Cirque / tarn	Armchair hollow; lake inside
Arête	Knife ridge between cirques
Horn	Pyramidal peak (Matterhorn type)
U-trough	Glacier (≠ river V)
Hanging valley	Higher tributary floor → waterfall
Fjord	Drowned glacial trough
Roche moutonnée	Abraded + plucked bedrock
Crag-and-tail	Hard rock + lee debris

Depositional glacial landforms — Teaching Card

Form	Quick recall
Moraine	Lateral, medial, terminal, ground
Drumlin	Streamlined till; blunt up-ice
Esker	Sinuous subglacial sand–gravel
Kame	Ice-contact mound
Outwash / sandur	Meltwater plain beyond snout
Erratic	Far-travelled boulder
Kettle	Melt pit → lake

N.9 Aeolian Landforms

Dunes and loess — Teaching Card

Form	Quick recall
Barchan	Crescent; horns **downwind**
Parabolic	Crescent; horns **upwind** (opposite barchan)
Seif / longitudinal	Ridge parallel to wind
Transverse	Crest across wind
Star / pyramidal	Multi-directional winds
Loess	Wind silt; China — **not** Imphal
Erg	Sand sea

N.10 Coastal Landforms (+ karst)

Depositional coastal forms — Teaching Card

Form	Quick recall
Beach	Wave-built sand / shingle strip
Spit	Attached at one end; free tip
Bar	Closes a bay; lagoon behind

Form	Quick recall
Tombolo	Island tied to mainland
Hook	Curved spit tip
Barrier island	Offshore, parallel to coast

Reef type	Where it sits	Lagoon?
Fringing	Against the shore	Little / none
Barrier	Offshore, parallel to land	Yes — between reef and land
Atoll	Ring / horseshoe in open sea	Yes — **central** lagoon; island mostly gone
Coast type	Meaning	
---	---	
Fjord	Drowned **glacial** trough	
Ria	Drowned **river** valley (non-glacial)	
Dalmatian	Concordant drowned ranges // coast (Adriatic type)	
Emergent	Raised beach / wave-cut platform now inland	
Submergent	Ria / fjord / drowned mouth	

Common Traps — Don't Fall For These

1. **Ganga delta = bird's-foot** — FALSE. Bird's-foot has **one** example: the **Mississippi**. Ganga = **arcuate**.
2. **Every moribund stem = Cauvery** — FALSE. **Subdivision** wording → **Bengal**; **abandoned lobe** wording → **Cauvery**. Krishna–Godavari and Mahanadi fit neither.
3. **Folds → dendritic** — FALSE. **Trellis**.
4. **Imphal = loess / glacial** — FALSE. **Lacustrine**.
5. **Marble from sandstone** — FALSE. **Limestone**. Quartzite = sandstone.
6. **Weathering = erosion** — FALSE. Weathering is **in situ**.
7. **Barchan horns upwind** — FALSE. **Downwind**.
8. **Stalactite on the floor** — FALSE. **Ceiling**.
9. **U-valley = youthful river** — FALSE. **Glacier**. River youth = **V**.
10. **Mekong delta = Cambodia** — FALSE. **Southern Vietnam**. The Red River delta is the **northern** Vietnam one.
11. **West Indian rivers = big arcuate deltas** — FALSE. **Estuaries** (Narmada/Tapi).
12. **Nile delta = Sudan / Ethiopia** — FALSE. **Northern Egypt**.
13. **Indus delta = Indian west coast** — FALSE. **Sindh, Pakistan**.
14. **Irrawaddy delta = Thailand** — FALSE. **Southern Myanmar**.
15. **Danube delta = Hungary** — FALSE. **Black Sea** (Romania and Ukraine).
16. **Volga delta = Black Sea** — FALSE. **Caspian Sea**.
17. **Niger has one delta** — FALSE. Inland delta in **Mali**; sea delta in **Nigeria**.
18. **Greater Himalaya full of marine fossils** — FALSE. That's **Lesser**; Greater is fossil-poor crystalline.
19. **Fjord = any estuary** — FALSE. **Drowned glacial trough**.
20. **Basalt = intrusive granite twin** — FALSE. Basalt = **extrusive**.
21. **Loess = river silt of Imphal** — FALSE. **Wind silt**; Imphal = lake plain.
22. **Fault-valley = Chambal** — FALSE. **Damodar**.
23. **Endogenic = only rivers** — FALSE. Endogenic = interior build (fold/fault/volcano).
24. **Weathering always transports** — FALSE. Weathering is **in situ**.

25. **Skip process framework because recent papers asked trellis/Damodar** — FALSE. Next year can ask any standard NCERT agent or pair.
26. **Sill cuts the beds** — FALSE. **Dyke** cuts; sill is **parallel**.
27. **Point bar on the concave bank** — FALSE. Concave **erodes**; convex **deposits**.
28. **Ria = fjord** — FALSE. Ria = drowned **river**; fjord = drowned **glacier**.
29. **Peneplain = King's arid pediment** — FALSE. Peneplain = **Davis**; pediplain = **King**.
30. **Marble is foliated** — FALSE. Marble/quartzite = **non-foliated**.
31. **Intertrappean = sea fossils** — FALSE. **Freshwater / land** fossils.

🏞️ Fluvial Landform Concept — Peneplain (Ghatnachakra Complete)

- **Peneplain**: An almost featureless, gently undulating plain produced by end-stage **fluvial (river) erosion and deposition**, reducing land close to base level. The term was coined by **William Morris Davis**.

Topic 16 — Oceans

Current Affairs (this topic)

Year	Fact	Why it matters	Source
Feb 2026–ongoing	After US–Israel strikes on Iran (from **28 Feb 2026**), Iran moved to control / restrict the **Strait of Hormuz**; tanker traffic collapsed; mines, boardings and attacks on merchant ships (नावाध्यक्ष) were reported	World's top oil–LNG chokepoint in the news	Reuters / IEA / CRS
2025 baseline	About **20 mb/d** of oil products (~**25%** of seaborne oil) and ~**20%** of world **LNG** (mainly Qatar + UAE) normally transit Hormuz	Scale of any closure	IEA
Static geography	Hormuz joins **Persian (फ़ारसी) Gulf** ↔ **Gulf of Oman**; shores are **Iran (north)** and **Oman (Musandam, south)**; UAE sits on the approaches	Country / strait MCQs	Atlas
Bypass note	Only **Saudi Arabia** and the **UAE** have meaningful pipeline routes that can skip Hormuz; Iraq, Kuwait, Qatar, Bahrain and Iran lean almost entirely on the strait	Who is trapped if closed	IEA
Demand side	Most Hormuz oil goes to **Asia** (China, India, Japan, Korea); India is a major importer through this gate	India linkage	EIA / visualisers
Other 2026 themes	Dual blockade talk (Iran on the strait / US on Iranian ports), insurance spikes, and later escort / corridor politics kept oil prices elevated	Chokepoint = price risk	Reuters / Brookings
Static	**BBNJ 2023** high-seas biodiversity treaty — India has signed; UNCLOS **12 / 24 / 200 nm** unchanged	Law of the sea	UN
Static	India Arctic Policy **2022**; Arctic Council **observer** since **2013**; Deep Ocean Mission / Samudrayaan / Blue Economy (नीली अर्थव्यवस्था)	Polar + deep ocean	MEA / MoES

Confused Pairs

Pair	Correct	Trap	Hindi
Benguela	**Atlantic** cold (SW Africa)	Pacific	बेंगुएला
Humboldt / Peru	**Pacific** cold	Atlantic	हम्बोल्ट
Agulhas	**Warm**, Indian Ocean	Cold / Atlantic	अगुलहास
Brazil current	**Warm**	Cold	ब्राज़ील
California / Canary	**Cold**	Warm	कैलिफोर्निया / कैनरी
Kuroshio vs Oyashio	Warm Japan vs **cold Kurile**	Swapped	कुरोशियो / ओयाशियो
Spring vs neap	Syzygy **large** vs quadrature **small**	Straight line = small tide	वृहत् / लघु
Salinity max	**Tropics** belt	Equator	लवणता
Mariana	**Pacific**	Atlantic	मारियाना
Sunda/Java trench	**Indian**	Pacific	सुंडा
Molloy Deep	**Arctic**	Atlantic	मोलॉय
Datum line	Mean **sea level** reference	Date Line / 0° meridian	डेटा म रेखा
Fresh after ice	Largest remaining = **groundwater**	Lakes/rivers larger than GW	भूजल
Sargasso	No **coastline**	Has coasts like Black Sea	सारगैसो
Red Sea form	**Axial trough**	Ordinary eroded valley	लाल सागर
NAD	**Blanket of Europe** / warm	Cold Labrador clone	उत्तरी अटलांटिक ड्रिफ्ट
Halocline	**Salinity** jump	Thermocline (temp)	हैलोक्लाइन

Pair	Correct	Trap	Hindi
Lake Van	Very high lake salinity (~330‰)	Always pick Dead Sea	वैन झील
Diamantina	**Indian Ocean**	Pacific	डायमंटिना
Ninetyeast Ridge	**Indian Ocean**	Mid-Atlantic	नाइंटी ईस्ट
Suez	**Med ↔ Red**, **sea-level** (no climb)	Directly Atlantic–Indian	स्वेज
Panama	Atlantic ↔ **Pacific** + **stepped chambers** / Gatun	Same as Suez / no chambers	पनामा
EEZ vs territorial	**200 nm** vs **12 nm**	Swapped	EEZ / क्षेत्रीय जल
Maitri + Bharati	Operational Antarctica	Dakshin Gangotri still open	मैत्री / भारती
Himadri	**Arctic** (Svalbard)	Antarctica	हिमाद्री
Guyot vs seamount	**Flat** drowned top vs peaked	Same	गायट / सीमाउंट
El Niño fish	**Less** plankton	Warm → more fish	अल नीनो
Red clay vs ooze	Deepest **no carbonate** vs **biogenic** mud	Same deposit	लाल मृत्तिका
Globigerina vs diatom	**Calcareous** vs **siliceous** (cold)	Swapped	ग्लोबिजेरिना / डायटम
Ninetyeast Ridge	Volcanic / hotspot trail	Mid-ocean spreading ridge	नाइंटीईस्ट
Carlsberg Ridge	**NW Indian** spreading	Atlantic MAR	कार्ल्सबर्ग
Drake vs Magellan	Open south of Cape Horn vs **Chilean channels**	Same strait	ड्रेक / मैगलन
Tidal bore	Funnel river **wave**	Ordinary spring tide	ज्वार-भित्ति
ISA / Area	Seabed **beyond** national zones	EEZ mining	ISA
Arctic Council	India = **observer** (2013)	Full member	आर्कटिक परिषद
Neritic vs hadal	Over **shelf** vs inside **trench**	Same depth word	नेरिटिक / हैडल
Territorial vs EEZ	**12 nm** vs **200 nm**	Swapped	12 ≠ 200
India west vs east shelf	**West wider** (overall)	East always wider	पश्चिमी शेल्फ
Constructive vs destructive wave	Builds beach vs **erodes**	Same	रचनात्मक / विनाशकारी
Upwelling coast	**West** coast of continent	East coast rule	उत्प्रवाह
Türkiye north	**Black Sea**	Caspian / Med	काला सागर
W→E seas ladder	Med → Black → Caspian → Aral	Black first / Aral before Caspian	मेड→ब्लैक→कैस्पियन→अराल
Baltic coast	Norway **out**	Norway in	बाल्टिक
Jordan sea	**Aqaba / Red** only	Mediterranean	जॉर्डन
Hormuz shores	**Iran + Oman**	UAE as both shores	होरमुज़
Persian Gulf coast	**Not Oman**	Oman inside Gulf	फारस की खाड़ी
Syria sea	**Mediterranean**	Black Sea	सीरिया
Gaza water	**Mediterranean**	Red / Dead Sea as coast	गाजा

N.1 Oceans & Seas

Ocean	Area & Depth	Salinity / Features	Key Historical & Geographic Facts
Pacific	~166.24 million sq. km; Avg depth ~**4,200 m** (deepest ocean)	~35‰	Named *Pacific* (meaning calm or peaceful) by **Ferdinand Magellan** during world circumnavigation. Mariana Trench is deepest point (11,033 m). Contains >20,000 islands, mostly volcanic or coral.

Ocean	Area & Depth	Salinity / Features	Key Historical & Geographic Facts
Atlantic	~86.56 million sq. km; Avg depth ~3,646 m	High salinity in subtropics (~37‰)	Derived from **Atlas** (Titan in Greek mythology). Formed when Gondwanaland rifted (South America and Africa separated) and is still widening. Has the **longest coastline** and is the **busiest ocean for global commerce/trade** (connects industrialised W. Europe & NE North America).
Indian	~73.43 million sq. km; Avg depth ~**3,741 m** (deeper on average than the Atlantic)	Variable (~35–37‰ in Arabian Sea, lower in Bay of Bengal)	**Only ocean named after a country** (India). Greeks called it Erythraean Sea. Current circulation completely reverses seasonally with Monsoon winds. Contains major continental islands (Madagascar, Sri Lanka) and volcanic/coral islands (Mauritius, Seychelles, Maldives, Lakshadweep, Andaman & Nicobar).
Southern	~20.3 million sq. km	Low	Encircles Antarctica; carries the continuous Antarctic Circumpolar Current (West Wind Drift).
Arctic	~13.23 million sq. km; Avg depth ~**1,500 m** (shallowest ocean)	**Least salinity (~20‰)**	Smallest ocean, centered on the North Pole within the Arctic Circle. Largely ice-covered year-round with widest continental shelves.

South China Sea

- The **South China Sea** is a **western Pacific** marginal sea in Southeast Asia.
- Picture the frame: **China** on top, **Vietnam** on the left, the **Philippines** on the right, and **Borneo / Malaysia** below.

- Core coastal set: China, Taiwan, Vietnam, Philippines, Malaysia, Brunei, and parts of Indonesia.
- It is often the **deepest** option among common seas (~**5377 m**).
- Island traps inside the basin include the **Paracel** and **Spratly** groups.

Red Sea

Sea	Just remember this
South China Sea	China – Vietnam – Philippines – Borneo
Bering Sea	**Russia ↔ Alaska**
Mediterranean	**Europe ↔ Africa**, Asia east
Black Sea	Ukraine – Russia – Georgia – Türkiye – Bulgaria – Romania
Sea of Azov	**North of Black Sea**, via Kerch Strait
Tyrrhenian	**West of Italy**, Italy–Sardinia–Corsica–Sicily
Sargasso	**North Atlantic; no coastline; currents define it**
Aral	**Kazakhstan + Uzbekistan; shrinking lake**
Red Sea	**Africa ↔ Arabia; rift**

Direction	Sea	Easy cue
North (12)	**Black Sea**	Cold wall **above** Türkiye
South (6)	**Mediterranean**	Warm "middle sea" **below** Türkiye
West (9)	**Aegean Sea**	Greek-islands side
North-west pocket	**Sea of Marmara**	Tiny inland sea between Black and Aegean

Letter	Sea
M	**Mediterranean**
B	**Black**
C	**Caspian**
A	**Aral**

Sea / water body	Coastal countries (core list)	Classic "does NOT" trap
Mediterranean	Spain, France, Monaco, Italy, Slovenia, Croatia, Bosnia & Herzegovina, Montenegro, Albania, Greece, **Türkiye**, **Syria**, **Lebanon**, **Israel**, Egypt, Libya, Tunisia, Algeria, Morocco (+ island states)	**Jordan**, Chad, Iraq, Saudi Arabia
Black Sea	Russia, Ukraine, Romania, Bulgaria, **Türkiye**, Georgia	**Syria** (Med only)
Caspian (lake)	Russia, Kazakhstan, Turkmenistan, Iran, Azerbaijan	**Türkiye**, Uzbekistan
Aral (lake)	Kazakhstan, Uzbekistan	Azerbaijan / Russia as pair
Red Sea	Egypt, Sudan, Eritrea, Djibouti, Saudi Arabia, Yemen, Israel, **Jordan** (Aqaba tip)	Jordan–Mediterranean
Persian / Arab Gulf	Iran, Iraq, Kuwait, Saudi Arabia, Bahrain, Qatar, **UAE**	**Oman** (faces Gulf of Oman / Arabian Sea)
Gulf of Oman	Oman, Iran, UAE (approaches)	Calling it "inside" the Persian Gulf
Baltic Sea	Denmark, Germany, Poland, Lithuania, Latvia, Estonia, Russia, Finland, Sweden	**Norway**
North Sea	UK, Norway, Denmark, Germany, Netherlands, Belgium, France	Mixing with Baltic list

Sea / water body	Coastal countries (core list)	Classic "does NOT" trap
Arabian Sea	India, Pakistan, Iran, Oman, Yemen, Somalia, Maldives approaches	Treating Persian Gulf as identical
Bay of Bengal	India, Bangladesh, Myanmar, Sri Lanka approaches	—
South China Sea	China, Taiwan, Vietnam, Philippines, Malaysia, Brunei, Indonesia (parts)	—

Name	Where	Opens into	Note
Persian / Arab Gulf	Between Iran and Arabia	Gulf of Oman → Arabian Sea	Mouth at the **Strait of Hormuz** (oil chokepoint)
Gulf of Oman	SE of Hormuz	Arabian Sea	Outer side of Hormuz — **not** the same as the Persian Gulf
Gulf of Aden	Between Yemen and Somalia	Arabian Sea	Inner side of **Bab-el-Mandeb**, leading into the Red Sea
Gulf of Suez	NW arm of **Red Sea** (Egypt)	Red Sea	**Not** the Suez Canal lakes. The canal starts from this gulf toward the Mediterranean
Gulf of Aqaba	NE arm of Red Sea	Red Sea	Shared by Israel, Jordan, Egypt and Saudi Arabia, parallel to the Gulf of Suez
Gulf of Mexico	S of USA, E of Mexico	Atlantic (via Florida / Yucatan straits)	Caribbean neighbour with the Loop Current and major oil fields
Gulf of Guinea	West Africa (Nigeria–Ghana (घन) coast)	Atlantic	Warm **Guinea Current** and the Niger delta
Gulf of California (Sea of Cortez)	Between Baja and mainland Mexico	Pacific	Pacific arm, not Atlantic
Gulf of Alaska	S of Alaska	Pacific	Warm **Alaska Current**
Gulf of Thailand	Cambodia / Thailand / Vietnam	South China Sea	Shallow gulf, not part of the Indian Ocean
Gulf of Tonkin	N Vietnam / S China	South China Sea	Northern South China Sea arm
Gulf of Khambhat / Cambay	Gujarat, between Kathiawar and mainland	Arabian Sea	India's classic **tidal-energy** coast
Gulf of Kutch	N of Kathiawar (Gujarat)	Arabian Sea	Second tidal site, with reefs
Gulf of Mannar	Between TN and Sri Lanka	Indian Ocean	Reefs and biosphere reserve (जैवमंडल आरक्षित क्षेत्र). **Palk Strait** is the narrower passage to the north
Hudson Bay	NE Canada	Atlantic via Hudson Strait	Huge inland sea, not ranked as a true ocean
Bay of Fundy	Nova Scotia–New Brunswick, Canada	Atlantic	**Highest tidal range in the world** (not Khambhat)
Gulf of Carpentaria	N Australia	Arafura Sea	Shallow tropical gulf
Bay of Biscay	W France / N Spain	Atlantic	Rough seas on Europe's continental shelf
Gulf of Bothnia	Between Sweden and Finland	Baltic	Low-salinity Baltic arm
Gulf of Finland	Finland / Estonia / Russia (St Petersburg)	Baltic	Eastern Baltic arm
Bay of Bengal	E of India	Indian Ocean	Named a **bay** but treated as a sea; cyclone belt

N.2 Ocean Floor, Ridge, Shelf, Slope, Guyot

Step	Depth / slope	What it is	Key fact
Continental shelf	Shore to **shelf break ~100–200 m**; slope ≤1°; ~**7.5%** of ocean area; avg width ~**80 km**	Drowned edge of the continent	Rich in **fisheries and oil/gas**. World's widest shelves are **Siberian / Arctic**. India's **west shelf is wider than the east** (Gujarat–Mumbai / Khambhat up to ~300 km; Konkan–Kerala narrower). The east shelf is narrower overall but carries big **delta / Bengal Fan** sediment. Oil pair: **Bombay High** (west) and KG basin (east)
Shelf break	~130–200 m	Sudden steepening	Marks the end of the "neritic" zone
Continental slope	~200 m down to **~3000 m**; 2–5°	True edge of the continental block	Cut by **submarine canyons**. **Turbidity currents** race down them
Deep-sea fan / cone	Foot of a canyon	Turbidite dump	Same idea as an alluvial (जलोढ़) fan, underwater
Continental rise	~3000–4000 m; gentler than slope	**Sediment apron** from shelf mud and canyon fans	Present on **passive** Atlantic-type margins. Often **missing** on **active** Pacific margins where a **trench** swallows the sediment
Abyssal plain	**~3000–6000 m**; almost flat	Most of the ocean floor (~**76%** of ocean area in GC framing)	Hosts red clay and Mn nodules. "Abyssal hills" are small volcanic bumps on the plain
Trench (hadal)	**>6000 m** (Mariana ~11 km)	Convergent plate where oceanic slab bends down	Island arcs or the Andes rise on the land side. Deepest points are in **N.3**
Mid-ocean ridge	Crest ~2000–3000 m (shallower than plains)	**Divergent** zone of new crust	Central **rift valley**, **transform faults**, **fracture zones**, and black smokers. Hess linked ridges to seafloor spreading

Seamount, guyot, atoll and related forms — Teaching Card

Form	Quick recall
Seamount	Peaked undersea volcano (>1000 m)
Guyot	Flat-topped drowned seamount
Volcanic island	Seamount that breaks the surface
Atoll	Ring / horseshoe reef around a lagoon
Submarine canyon	Gorge across shelf–slope
Bank	Flat shallow elevation (Dogger, Grand Banks)

Ridge	Fact
Mid-Atlantic	S-shaped; **Telegraphic Plateau** on the **North** arm
East Pacific Rise	Fast spreading
Carlsberg	NW **Indian** Ocean (Arabian Sea)
Mid-Indian / SW + SE Indian	Indian Ocean spreading
Ninetyeast	N–S volcanic trail in E Indian Ocean — **not** a classic spreading MAR

Word	Where
Neritic	Over the **shelf**
Bathyal	Over the **slope**
Abyssal	Over the **plain**
Hadal	Inside the **trench**

Deposit	Fact
Terrigenous	Land mud and sand on the shelf and slope
Red clay	Deepest abyssal deposit from volcanic and meteoric dust, with **carbonate dissolved away**
Globigerina ooze	Calcareous foraminifera mud at mid-depth
Pteropod ooze	Calcareous ooze, shallower than globigerina
Diatom ooze	**Siliceous** ooze of cold high-latitude seas
Radiolarian ooze	Siliceous ooze of tropical deep water
Manganese nodules	Potato-shaped ores on abyssal plains — classic in the **Pacific**; India's claim is the **Central Indian Ocean Basin**

N.3 Trenches

Ocean	Deepest Point (Trench)	Exact Max Depth	Geographical Location
Pacific	**Mariana Trench** (Challenger Deep)	**11,033 m** (~36,201 ft)	Near Guam / Mariana Islands (W. Pacific)
Atlantic	**Puerto Rico Trench** (Milwaukee Deep)	**8,605 m** (up to ~9,460 m in classical texts)	North of Puerto Rico (Caribbean margin)
Indian	**Java / Sunda Trench**	**7,450–7,542 m**	South of Java & Sumatra (NE Indian Ocean)
Arctic	**Molloy Deep** (Fram Strait)	**~5,550 m**	Between Greenland and Svalbard

N.4 Ocean Currents (warm & cold)

Current	Type
Kuroshio (Japan)	**Warm**
North Pacific	Warm drift
Alaska	Warm
California	**Cold**
Humboldt / Peru	**Cold** — upwelling, rich fish. El Niño shuts this belt
East Australian	Warm
Oyashio / **Kurile**	**Cold**
Equatorial (N/S) + Equatorial Counter	

Current	Type
Gulf Stream / Florida	**Warm**
North Atlantic Drift	Warm drift that keeps NW Europe mild — the **Blanket of Europe**, helping keep Norway harbours milder and freer of ice
Canary	**Cold**
Labrador	**Cold** (meets Gulf Stream → Grand Banks fog/fish)
Brazil	**Warm**
Benguela	**Cold** — **SE Atlantic**, SW Africa — **not** Pacific
Falkland / Malvinas	Cold
Guinea	Warm

Current	Type
Agulhas	**Warm** current off SE Africa — the classic **Indian Ocean** option (not Florida, Canary or Kurile)
Mozambique	Warm
Somali	Seasonal (SW monsoon (दक्षिण-पश्चिम) upwelling off Somalia)
West Australian	**Cold**
Monsoon current	**Reverses** with monsoon — why N Indian Ocean current direction changes seasonally

N.5 Salinity

Belt	Salinity
Tropics of Cancer / Capricorn	**Maximum** in the open ocean — high evaporation, less rain
Equator	Lower than the tropics (heavy rain)
Poles	Lower (ice melt, low evaporation)

Hypersaline lake	Approx. salinity	Place note
Lake Assal	**~348‰**	Often cited as among the world's highest lake salinities
Lake Van	**~330‰**	**Türkiye**
Dead Sea	**~238‰**	Jordan Rift; classic high-salinity lake
Great Salt Lake	**~220‰**	**Utah, USA**

N.6 Tides

Tide types — Teaching Card

Type	Quick recall
Spring	Syzygy (new/full); **large** range
Neap	Quadrature (quarters); **small** range
Semi-diurnal	2 high + 2 low
Diurnal	1 high + 1 low
Mixed	Unequal highs / lows

N.7 Straits

Strait of Hormuz — must-score chokepoint

Item	Fact
Shores	**Iran** on the north; **Oman** (Musandam Peninsula) on the south
Nearby	**UAE** maritime zones sit on the Gulf and Arabian Sea **approaches**, but the classic shore pair is Iran–Oman
Width	About **21–29 nm** at the narrowest teaching band; shipping lanes are only a few miles wide each
Oil	Roughly **20 mb/d** / about **25%** of world **seaborne oil** in recent IEA figures
Gas	About **20%** of world **LNG** (mainly **Qatar** and **UAE**)
Gulf states behind the gate	Iran, Iraq, Kuwait, Saudi Arabia, Bahrain, Qatar, UAE — **Oman is outside** the Persian Gulf coast list

Item	Fact
Bypass	Only Saudi Arabia and the UAE have useful pipelines that can skip Hormuz; the others are almost fully dependent on the strait
Buyers	Most cargo goes to **Asia** (China, India, Japan, Korea)

Other high-yield straits — seas joined + shores

Strait	Joins	Shore / country cue
Hormuz	Persian Gulf ↔ Gulf of Oman	**Iran–Oman**
Bab-el-Mandeb	Red Sea ↔ Gulf of Aden	**Djibouti / Eritrea** side ↔ **Yemen**; gate to the Indian Ocean for Suez traffic
Malacca	Andaman / South China Sea approaches ↔ Indian Ocean	**Malaysia–Indonesia** (Singapore at the tip)
Gibraltar	Mediterranean ↔ Atlantic	**Spain–Morocco** (UK: Gibraltar Rock)
Bosphorus	Black Sea ↔ Marmara	Entirely **Türkiye**
Dardanelles	Marmara ↔ Aegean	Entirely **Türkiye**
Bering	Arctic ↔ Pacific	**Russia–USA** (Alaska)
Palk	Bay of Bengal side ↔ Gulf of Mannar / Laccadive approaches	**India–Sri Lanka**
10° Channel	Andaman ↔ Nicobar (निकोबार)	Inside India
9° Channel (नौ डिग्री)	Minicoy ↔ Lakshadweep (लक्षद्वीप) proper	Inside India
8° Channel	Maldives ↔ Minicoy	Maldives–India
Duncan Passage	Little Andaman ↔ South Andaman	Inside India
Sunda	Indian Ocean ↔ Java Sea approaches	**Sumatra–Java** (Indonesia)
Torres	Coral / Arafura approaches	**Australia–Papua New Guinea**
Bass	Tasman Sea approaches	**Australia–Tasmania**
Magellan	Atlantic ↔ Pacific via Chilean channels	**Chile**
Drake Passage	Atlantic ↔ Pacific (open water)	South of Cape Horn — **not** Magellan
Dover	North Sea ↔ English Channel	**UK–France**
Mozambique	Indian Ocean corridor	**Mozambique–Madagascar**
Davis	Baffin Bay approaches	**Greenland–Canada**
Denmark Strait	Greenland Sea / Atlantic	**Greenland–Iceland**
Cook	Pacific between NZ islands	North Island ↔ South Island
Florida / Yucatan	Gulf of Mexico gates	USA–Cuba / Mexico–Cuba approaches
Formosa / Taiwan	East China Sea ↔ South China Sea	Taiwan Strait
Korea / Tsushima	Japan Sea approaches	**Japan–Korea**
Lombok / Makassar	Indonesian passages	Tanker alternatives when Malacca is tight
Kerch	Black Sea ↔ Sea of Azov	**Russia–Ukraine** zone
Otranto	Adriatic ↔ Ionian	Italy–Albania approaches
Skagerrak / Kattegat	North Sea ↔ Baltic approaches	Scandinavia–Denmark gate

N.8 Canals & Waterways

Other canals (match-list desk)

Canal	Joins	Country / note
Kiel	**North Sea ↔ Baltic**	**Germany** (Elbe estuary side ↔ Baltic). Not Mediterranean
Corinth	Ionian ↔ Aegean	Greece
Soo (Sault Ste. Marie)	Lake **Superior ↔ Huron**	USA–Canada. **Not** Superior–Michigan
Welland	Erie ↔ Ontario	Bypasses Niagara Falls
Erie Canal	Lake Erie ↔ Hudson River	Historic (ऐतिहासिक) US inland route
St Lawrence (लॉरेंस) Seaway	Great Lakes ↔ Atlantic	USA–Canada ocean door
Rhine–Main–Danube	North Sea inland system ↔ **Black Sea**	Europe cross-continent inland
White Sea–Baltic	Baltic ↔ White / Arctic approaches	Russia
Ludwig's	S Germany inland waterway	Match-list filler
Volga–Don	Caspian approaches ↔ Black Sea / Azov system	Russia

N.9 UNCLOS, Territorial Waters, EEZ, Shelf

Zone	Breadth
Territorial sea	**12 nautical miles** — full sovereignty, with innocent passage
Contiguous zone	**24 nm** — customs, fiscal, immigration and sanitary control
EEZ	**200 nm** — sovereign rights to resources (fish, oil, wind)
Continental shelf	At least **200 nm**, and may extend to **350 nm** if geology allows
High seas	Beyond national zones
The Area	Seabed beyond national jurisdiction — **common heritage**, managed by the **ISA** (Kingston, Jamaica)

N.10 Important Fisheries

Ground	Why
Grand Banks (Newfoundland)	Labrador cold + Gulf Stream
Dogger Bank	North Sea shelf
Peru–Chile	**Humboldt upwelling** (fails in El Niño)
Japan / Kurile	Kuroshio + Oyashio mix
Iceland / Norway	Arctic/Atlantic mix
India	West coast and Wadge Bank, under EEZ rights

Common Traps — Don't Fall For These

1. **Benguela = Pacific** — FALSE. **Atlantic**.
2. **Agulhas / Brazil = cold** — FALSE. **Warm** (2025).
3. **Syzygy = small tide** — FALSE. **Spring / large**.

4. **Salinity max at equator** — FALSE. **Tropics**.
5. **Tide every exactly 12 h 30 min** — FALSE. **~12 h 25 min**.
6. **El Niño → more Peruvian plankton** — FALSE. **Less**.
7. **Suez = Atlantic–Indian directly in 2025 codes** — FALSE. **Red + Mediterranean**.
8. **Suez lakes start at Timsah in the north** — FALSE. **Manzala** first.
9. **Mariana = Atlantic** — FALSE. **Pacific**. Puerto Rico = Atlantic.
10. **Molloy = Atlantic** — FALSE. **Arctic**.
11. **Dakshin Gangotri still operational** — FALSE. **Maitri + Bharati**.
12. **Himadri = Antarctica** — FALSE. **Arctic**.
13. **EEZ = 12 nm** — FALSE. Territorial **12**; EEZ **200**.
14. **Panama is sea-level like Suez** — FALSE. **Suez** is sea-level; Panama uses **stepped chambers** + Gatun.
15. **Kurile/Oyashio = Indian Ocean** — FALSE. **Pacific cold**. Indian MCQ = **Agulhas**.
16. **Telegraphic Plateau = Indian Ocean Ridge** — FALSE. **North Atlantic Ridge**.
17. **Guyot = peaked seamount** — FALSE. **Flat top**.
18. **Caspian = true ocean sea** — FALSE. **Lake**.
19. **Ninetyeast Ridge = Mid-Atlantic spreading** — FALSE. Volcanic trail in the **Indian** Ocean.
20. **Red clay = shallow coral mud** — FALSE. **Deepest** abyssal, carbonate gone.
21. **Diatom ooze = calcareous tropics** — FALSE. **Siliceous**, cold.
22. **Fundy = India's tidal-energy MCQ** — FALSE. World range king; India option = **Khambhat**.
23. **Drake = Magellan** — FALSE. Drake is open south of the Horn.
24. **India is an Arctic Council member** — FALSE. **Observer**.
25. **Himadri = Maitri's other name** — FALSE. Himadri = **Arctic**.
26. **ISA mines the EEZ** — FALSE. ISA = **the Area** beyond national zones.
27. **Skip ocean floor / deposits because recent papers asked currents** — FALSE. Next year can ask any standard NCERT trench, deposit or UNCLOS zone.
28. **India east shelf always wider than west** — FALSE. Coaching fact = **west wider** overall (Gujarat–Mumbai).
29. **Density ignores salinity** — FALSE. Cold + salt (+ pressure) set density and deep flow.
30. **Waves = tides** — FALSE. Waves need wind/fetch; tides need Moon/Sun gravity.
31. **Upwelling on east coasts of continents** — FALSE. Classic upwelling = **west** coasts.
32. **OMT depth is always 129 m** — FALSE. IAS key uses ~**59 m** mean for SW Indian Ocean Jan–Mar; statement 1 in that stem is wrong.
33. **Fresh usable = 95% in ice of all Earth water** — FALSE. Ice is most of *fresh* water; usable total is **<1%** of all water.
34. **Rivers+lakes > groundwater** — FALSE among remaining fresh after ice. **Groundwater** is larger.
35. **Sargasso has a coastline** — FALSE.
36. **Telegraph Plateau = Indian Ocean** — FALSE. **North Atlantic Ridge**.
37. **Neap tide = new/full moon** — FALSE. New/full = **spring**. Neap = quadrature (~7–8th day).
38. **Benguela = Pacific / Indian** — FALSE. **South Atlantic**.
39. **Agulhas = cold** — FALSE. **Warm** Indian Ocean.
40. **Suez uses Panama-style stepped chambers** — FALSE. Suez is **sea-level**; Panama climbs chambers + **Gatun Lake**.
41. **Persian Gulf borders Oman** — FALSE. Oman faces **Gulf of Oman**.
42. **Soo Canal = Superior–Michigan** — FALSE. Superior–**Huron**.
43. **Great Barrier Reef = Atlantic / New Zealand** — FALSE. **Pacific**, NE **Queensland / Australia**.
44. **Coral reefs = cold coasts only** — FALSE. Tropical warm clear water.

45. **Türkiye north = Mediterranean / Caspian** — FALSE. North = **Black Sea**; south = Mediterranean.
46. **W→E = Black → Med → Caspian → Aral** — FALSE. **Mediterranean → Black → Caspian → Aral** ("My Black Cat Ate").
47. **Norway on the Baltic** — FALSE. Baltic ring excludes Norway.
48. **Jordan opens to the Mediterranean** — FALSE. Only **Aqaba / Red Sea**.
49. **Gaza faces the Red Sea** — FALSE. Gaza faces the **Mediterranean**.
50. **Hormuz shores = Iran + UAE only** — FALSE. Classic pair = **Iran + Oman** (Musandam).
51. **Syria borders the Black Sea** — FALSE. Syria is **Mediterranean**.
52. **Hormuz joins Red Sea and Mediterranean** — FALSE. That is **Suez**. Hormuz = Persian Gulf ↔ Gulf of Oman.

🌊 Bermuda Triangle & Fishing Grounds Master Facts (Ghatnachakra Complete)

- **Major World Fishing Grounds**: Formed where **cold and warm ocean currents meet** in shallow continental shelves (e.g. Labrador Cold + Gulf Stream Warm). Plankton thrives abundantly, creating rich marine ecosystems.
- **Grand Banks**: Off the coast of Newfoundland (Canada), in the North Atlantic Ocean. Formed by convergence of cold Labrador Current and warm Gulf Stream.
- **Bermuda Triangle ("Devil's Triangle")**: Oceanic area in the **North Atlantic Ocean** bounded roughly by points in **Miami (Florida)**, **San Juan (Puerto Rico)**, and **Bermuda**.

Topic 17 — World Rivers & Lakes

Current Affairs (this topic)

Year	Fact	Why it matters	Source
Static	GERD on Blue Nile (Ethiopia)	Dam–river pair	News / atlas
Static	Aral / Chad shrinkage (Amu–Syr diversion for Aral)	Endorheic lake stems	Atlas
UKPCS 2025	Grand Canyon — Colorado River	City/river style stem	UKPCS
Static	Nile length vs Amazon discharge	Superlative swap	Coaching

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q94

The Grand Canyon of the U.S.A. is formed by which river?

A. St. Lawrence (लॉरेंस) B. Missouri C. Ohio D. Colorado

Confused Pairs

Read as: **this** is correct · **that** is the usual wrong option.

	Correct	Usual wrong option	Hindi
New Orleans is on the	**Mississippi**	Missouri (Missouri joins Mississippi at **St Louis**)	न्यू ऑर्लीन्स
Hyderabad (India) is on the	**Musi**	Godavari / Paleru	मुसी
Mekong flows	**South / south-east**	South-west	मेकांग
Mekong delta is in	**Southern Vietnam**	Cambodia	मेकांग डेल्टा
Limpopo Capricorn	Crosses **twice**	Once only	लिम्पोपो
Mahaweli	**Sri Lanka**	India / Bangladesh	महावेली
Inselberg	**Wind** residual	Glacier	इंसेलबर्ग
Grand Canyon river	**Colorado**	Missouri / St Lawrence	ग्रैंड कैन्यन
Berlin river	**Spree**	Rhine	बर्लिन
Vienna river	**Danube**	Volga	वियना
USA–Mexico border river	**Rio Grande**	Colorado or Mississippi	रियो ग्रांडे
Lake Onega is in	**Russia**	Canada	ओनेगा
Lake Michigan lies	**Wholly in the USA**	Shared USA–Canada (Superior / Huron / Erie / Ontario are shared)	मिशिगन
Largest lake on Earth	**Caspian** (saline)	Superior	कैस्पियन
Largest **fresh** lake by **area**	**Superior**	Caspian / Baikal	सुपीरियर
Deepest lake + most fresh **volume**	**Baikal**	"Largest lake"	बैकाल
Longest river (tradition)	**Nile**	Amazon	नील
Largest **discharge** / basin	**Amazon**	Nile	अमेज़न
Highest waterfall	**Angel** (Venezuela)	Victoria	एंजेल

	Correct	Usual wrong option	Hindi
Victoria Falls is on the	**Zambezi**	Nile / Niagara	विक्टोरिया
Most countries on one river	**Danube** (~10)	Rhine	डेन्यूब
Busiest European inland navigation	**Rhine**	Danube	राइन
Budapest is on the	**Danube**	Rhine	बुडापेस्ट
Cologne is on the	**Rhine**	Danube	कोलोन
Khartoum stands at	**White Nile + Blue Nile**	Only one of them	खार्तूम
Congo uniqueness	**Deepest** + crosses Equator **twice**	Amazon does both	कांगो
Tanganyika vs Baikal	Tanganyika **2nd** deepest / longest fresh	Tanganyika deepest	टंगाईका
Titicaca	Highest **navigable**	Highest of any puddle	टिटिकाका
Dead Sea	Lowest land + hypersaline	Ordinary sea	मृत सागर
Niger	**Paradox** (inland arc first)	Straight to Gulf	नाइजर
Volga mouth	**Caspian**	Black Sea	वोल्गा

N.1 World Rivers

Continent	Rivers to place first
North America	Mississippi–Missouri, Mackenzie, St Lawrence, Colorado, Rio Grande
South America	Amazon, Paraná, Paraguay, Uruguay, Orinoco
Europe	Danube, Rhine, Volga, Dnieper, Don
Africa	Nile, Congo, Niger, Zambezi, Orange
Asia	Ob, Yenisei, Lena, Amur, Huang He, Yangtze, Mekong, Ganga, Brahmaputra, Indus, Tigris–Euphrates
Australia	Murray–Darling

River	Continent	Mouth	What the means by "famous"
Nile	Africa	Mediterranean	**Longest** river in the usual Prelims fact. White Nile from Lake Victoria + Blue Nile from Lake **Tana** (Ethiopia) meet at **Khartoum**. Cairo and Aswan sit on it.
Amazon	S America	Atlantic	**Not** the longest in most keys — but **largest discharge** and largest basin. Rises in the **Andes**.
Yangtze (Chang Jiang)	Asia (China)	East China Sea	**Longest river in Asia**. Three Gorges dam. Chongqing, Wuhan, Shanghai.
Mississippi–Missouri	N America (USA)	Gulf of Mexico	Longest **system** in North America. Missouri is a tributary that joins at **St Louis**. The mouth city is **New Orleans** on the Mississippi, **not** on the Missouri.
Ob, Yenisei, Lena	Siberia	Arctic Ocean	Three great north-flowing Siberian rivers. **Angara** comes **north** out of Lake Baikal and joins the Yenisei.
Volga	Europe (Russia)	**Caspian** (inland, not the ocean)	**Longest river in Europe**. Linked to the Black/Azov side by the Volga–Don canal.
Murray–Darling	Australia	Southern Ocean (via Murray mouth)	Australia's main river system. **Darling Range** in Western Australia is a **hill range**, not this river.

River	Source	Mouth	Direction / countries	
Mekong	**Tibetan Plateau**	South China Sea at **S Vietnam**	China → Myanmar → Laos → Thailand → Cambodia → Vietnam. Flows **south / south-east**	**Not SW**. Delta is **Vietnam**, not Cambodia
Yangtze	Tibet	East China Sea	China, west → east	Asia's longest river. Three Gorges Dam. Shanghai near the mouth
Hwang Ho / Yellow	Tibet	Bo Hai (Yellow Sea)	China	Called the "Sorrow of China". Heavy **loess** silt. Arcuate delta
Amur	Mongolia–China border	Tatar Strait (Pacific)	Russia–China border river	Flows **north-east / east**
Ob (with Irtysh)	Altai	Arctic	Russia / Kazakhstan	North to the Arctic
Yenisei	Mongolia / Tuva	Arctic	Russia	The **Angara** leaves Baikal northward and joins the Yenisei
Lena	Highlands near Baikal	Arctic	Russia	North to the Arctic
Amu Darya	Pamir	Aral Sea	Central Asia	Aral is shrinking
Syr Darya	Tien Shan	**Aral Sea**	Kazakhstan / Uzbekistan	Flows **north-west**. With the Amu Darya it fed (and starved) the Aral
Tigris	Turkey	Shatt al-Arab → Persian (फ़ारसी) Gulf	Turkey–Iraq	**Baghdad** sits on the Tigris
Euphrates	Turkey	Shatt al-Arab → Persian Gulf	Turkey–Syria–Iraq	Twin of the Tigris
Jordan	Hermon / Anti-Lebanon	**Dead Sea**	Israel–Jordan	Ends in the Dead Sea with no ocean mouth
Irrawaddy	N Myanmar	Andaman (अंडमान) Sea	Myanmar lifeline	
Mahaweli Ganga	Sri Lanka highlands	Bay of Bengal	Sri Lanka's longest river (~335 km). The island shows **radial** drainage	
Salween	Tibet	Andaman Sea	China–Myanmar–Thailand	

River	Source	Mouth	Cities / dams / falls	Note
Nile	White: Victoria · Blue: **Tana**	Mediterranean	**Khartoum** (confluence), Cairo, **Aswan High Dam** → **Lake Nasser**	Arcuate delta. Traditional **longest** river. **GERD** sits on the **Blue Nile** in Ethiopia (Egypt (मिस्र)–Sudan dispute)
Congo / Zaire	E African highlands	Atlantic	Congo Basin rainforest	**Deepest** major river. Only major river that **crosses the Equator twice**. Second-largest discharge after the Amazon
Niger	Guinea highlands	Gulf of Guinea (Nigeria)	Inland delta in Mali, then a coastal delta	**Paradox river** — arcs inland / away from the nearby Atlantic before turning south-east to the Gulf
Zambezi	Zambia / Angola uplands	Mozambique Channel	**Victoria Falls** · Kariba · Cahora Bassa	Zambia–Zimbabwe border at the falls
Orange	Drakensberg	Atlantic	South Africa / Namibia	
Limpopo	S Africa	Indian Ocean (Mozambique)	Crosses **Tropic of Capricorn twice**	

River	Mouth	Cities	
Volga	Caspian	Source in the **Valdai Hills**. Linked by the Volga–Don canal	Longest river in Europe. Mouth is the **Caspian**, not the Black Sea
Danube	Black Sea	Vienna · **Budapest** · Belgrade	Crosses the **most countries** (~10)
Rhine	North Sea at Rotterdam	Basel · Strasbourg · **Cologne** · Bonn	**Busiest** European inland navigation. Flows in a **rift valley**
Thames	North Sea	**London**	
Seine	English Channel	**Paris**	
Po	Adriatic	N Italy	
Dnieper	Black Sea	Ukraine	
Vistula	Baltic	**Warsaw**	
Elbe	North Sea	**Hamburg**	
Rhône	Mediterranean	Lyon / S France	
Tagus	Atlantic	**Lisbon**	

River	Source → mouth	Cities / dams	
Amazon	Andes → Atlantic	—	Largest discharge
Paraná–Paraguay	Interior S America → Río de la Plata	**Itaipu** dam (Brazil–Paraguay)	
Orinoco	Venezuela / Colombia → Atlantic	—	
Mississippi	Minnesota → Gulf of Mexico	**New Orleans** at the mouth	Bird's-foot delta
Missouri	Rockies → joins Mississippi at **St Louis**	St Louis	**Not** the New Orleans river
Rio Grande (Río Bravo)	Rockies → Gulf of Mexico	—	**USA–Mexico border** (2023)
Colorado	Rockies → Gulf of California	Hoover Dam and the Grand Canyon	**Not** the USA–Mexico border river
Columbia	Rockies → Pacific	Grand Coulee Dam	NW USA
St Lawrence	Great Lakes → Atlantic	Montreal · Quebec	Seaway
Mackenzie (मैकेंज़ी)	Great Slave Lake → Arctic	—	Canada
Yukon	Canada → Bering Sea (Alaska)	—	
Murray–Darling	SE Australia → Southern Ocean	—	Main Australian system

N.2 World deltas

Shape	Meaning	Classic river
Arcuate	Bow / fan	**Nile**, Hwang Ho, Niger, Rhine, Ganga–Brahmaputra
Bird's-foot	Distributaries like toes	**Mississippi** — Nile is **not** this
Cuspate	Pointed tooth	Tiber (Italy)
Moribund	Abandoned / dying lobe	Cauvery (India)

Continent	Delta	River	Opens into
Asia	**Ganga–Brahmaputra–Meghna**	Ganga / Brahmaputra	**Bay of Bengal**
Asia	Mekong Delta	Mekong	**South China Sea**

Continent	Delta	River	Opens into
Asia	Yangtze Delta	Yangtze	**East China Sea**
Asia	Pearl River Delta	Pearl	South China Sea approaches
Asia	Indus Delta	Indus	**Arabian Sea**
Africa	Nile Delta	Nile	**Mediterranean**
Africa	Niger Delta	Niger	**Gulf of Guinea** (Atlantic)
Europe	Danube Delta	Danube	**Black Sea**
North America	Mississippi Delta	Mississippi	**Gulf of Mexico**
South America	Amazon mouth	Amazon	**Atlantic**
South America	Orinoco Delta	Orinoco	**Atlantic**

N.2A River landforms (world desk)

Form	Note
Gorge	Deep narrow valley with steep walls (e.g. **Chisapani** on Karnali, Nepal)
Canyon	Broader/deeper eroded gorge — **Grand Canyon** cut by **Colorado** (Arizona)
V-shaped valley	Youthful vertical + lateral erosion
Meander	Looping course on gentle plains
Alluvial (जलोढ़) fan / cone	Sediment cone where a stream leaves mountains into a plain
Delta	Mouth deposit. The Ganga–Brahmaputra–Meghna is the world's largest. The Mississippi is **bird's-foot**. The Nile is arcuate
Watershed / catchment	Land unit that collects, stores and releases water to a common outlet

N.3 Cities on Rivers

City	Country	River	Trap
Budapest	Hungary	**Danube**	Not Rhine
Vienna	Austria	Danube	
Belgrade	Serbia	Danube (+ Sava)	
Cologne (Köln)	Germany	**Rhine**	Not Danube
Rotterdam	Netherlands	Rhine mouth	
Basel	Switzerland	Rhine	
Strasbourg	France	Rhine	
New Orleans	USA	**Mississippi**	**Not Missouri**
St Louis	USA	Mississippi **meets Missouri**	
Cairo	Egypt	Nile	
Khartoum	Sudan	White Nile + Blue Nile	Not a single-arm city
Alexandria	Egypt	Nile delta fringe	
London	UK	Thames	
Paris	France	Seine	
Rome	Italy	Tiber	

City	Country	River	Trap
Hamburg	Germany	Elbe	
Warsaw	Poland	Vistula	
Moscow	Russia	Moskva	
Berlin	Germany	Spree	
Lisbon	Portugal	Tagus	
Baghdad	Iraq	Tigris	Not Euphrates
Shanghai	China	Yangtze	
Wuhan	China	Yangtze	
Chongqing	China	Yangtze	
Montreal	Canada	St Lawrence	
Quebec	Canada	St Lawrence	
Washington DC	USA	Potomac	
New York	USA	Hudson	
Hyderabad	India	**Musi** (joins Krishna (कृष्णा))	Not Godavari; not Paleru
Phnom Penh	Cambodia	Mekong	
Hanoi	Vietnam	Red River	
Yangon	Myanmar	Irrawaddy	Former capital
Madrid	Spain	Manzanares	
Asunción	Paraguay	Paraguay	
Niamey	Niger	Niger	
Basra	Iraq	Shatt al-Arab	
Kinshasa / Brazzaville	DRC / Congo	Congo (Zaire)	
Bangkok	Thailand	Chao Phraya (Menam)	
Lahore	Pakistan	Ravi (रावी)	
Perth	Australia	Swan	
Caracas	Venezuela	Orinoco basin note	
Kabul	Afghanistan	Kabul River	
Ankara	Turkey	Kizil River	
Glasgow	Scotland (UK)	Clyde	
Bristol	UK	Avon	
Newcastle	UK	Tyne	
Philadelphia	USA	Delaware	
Canton (Guangzhou)	China	Si-Kiang (Pearl River)	
Nanking	China	Yangtze (Yang-tse-Kiang)	
Karachi	Pakistan	Indus	
Chittagong	Bangladesh	Karnaphuli / Maiyani	

N.4 International / Border Rivers

River	Countries / border stretch	Trap
Rio Grande	**USA–Mexico** for a long stretch	Not Colorado, not Mississippi
Colorado	Only a **short** US–Mexico bit near Yuma	Do not pick for "the" border river
Amur	**Russia–China**	—
Mekong	Parts of Myanmar–Thailand, Laos–Thailand, Cambodia–Vietnam region	Do not treat the whole course as one border
Salween	Myanmar–Thailand stretches	—
Orange	South Africa–Namibia	—
Limpopo	South Africa with Botswana / Zimbabwe stretches	—
Zambezi	Zambia–Zimbabwe (Victoria Falls belt)	—
Danube	Portions of several European borders; crosses ~**10** countries overall	—
Senegal	Senegal–Mauritania	—
Congo	Parts of DRC–Republic of Congo	—
Kunene	Angola–Namibia	—
Jordan	Israel–Jordan	Into Dead Sea
Rhine	Switzerland–France–Germany–Netherlands corridor	—
St Lawrence	USA–Canada	—
Niagara River	USA–Canada, between Lakes Erie and Ontario	—

N.5 River Navigation

River navigation — how inland waterways work

Waterway	From / to	
Rhine	Swiss Alps → Rotterdam (North Sea)	Europe's **busiest** inland waterway
Danube	Germany → Black Sea	Linked to the Rhine by the Rhine–Main–Danube canal
Volga	Russia → Caspian	Volga–Don canal opens a way toward Azov / Black Sea
St Lawrence Seaway	Great Lakes → Atlantic	Ocean ships into the lakes
Yangtze	Interior China → East China Sea	China's main inland trunk
Mississippi	Interior USA → Gulf of Mexico	USA inland trunk

N.6 Dams

Dam	Country	River
Three Gorges	China	**Yangtze**
Itaipu	Brazil–Paraguay	**Paraná**
Guri	Venezuela	**Caroní**
Aswan High	Egypt	**Nile** (creates **Lake Nasser**)
Grand Coulee	USA	**Columbia**
Tucuruí	Brazil	**Tocantins**
Robert-Bourassa	Canada	**La Grande**

Dam	Country	River
Sayano-Shushenskaya	Russia	**Yenisei**
Tarbela	Pakistan	**Indus**
Nurek	Tajikistan	**Vakhsh**
GERD	Ethiopia	**Blue Nile**
Kariba	Zambia–Zimbabwe	**Zambezi**
Hoover	USA	**Colorado**
Akosombo	Ghana (घन)	Volta
Merowe	Sudan	Nile
Cahora Bassa	Mozambique	Zambezi
Xiaowan	China	Lancang / Mekong
Zangmu	Tibet (China)	Yarlung Tsangpo / Brahmaputra
Oroville	USA	Feather

Chain	Path
Yangtze	Tibetan Plateau → China → **Three Gorges** → Shanghai belt → **East China Sea**
Nile	Victoria / Tana → Sudan–Egypt → **Aswan High Dam** → **Nile Delta** → **Mediterranean**
Paraná	Interior South America → **Itaipu** (Brazil–Paraguay) → Río de la Plata → **Atlantic**

N.7 Lakes

Type	Fact / example
Tectonic / rift	Baikal, Tanganyika, Dead Sea, Malawi
Glacial	Many Canadian and Alpine lakes. The Great Lakes have glacial history (इतिहास)
Crater / caldera	Crater Lake (USA). Lonar in India is impact, not volcanic
Lagoon / coastal	Chilika-type coastal idea. Barred coastal lakes
Oxbow	Cut-off meander lakes on floodplains
Endorheic	Caspian, Aral, Chad, Dead Sea, Eyre — inland, no sea outlet
Lake	Country
------	---------
Caspian	RU / IR / KZ / TM / AZ
Superior	USA–Canada
Baikal	**Russia**
Tanganyika	TZ / DRC / BI / ZM
Victoria	E Africa (TZ / UG / KE)
Michigan	**USA only**
Huron	USA–Canada
Erie	USA–Canada
Ontario	USA–Canada
Onega	**Russia** (northwest)
Ladoga	Russia

Type	Fact / example
Maracaibo	**Venezuela**
Dead Sea	Jordan–Israel
Titicaca	Peru–Bolivia
Aral	Kazakhstan–Uzbekistan
Chad	Africa (multi-state)
Eyre	Australia
Great Bear / Great Slave	Canada
Malawi / Nyasa	Africa rift

Lake / pair	Fact
Faguibine	Mali (W Africa) — dried / desertified
Balkhash	**Kazakhstan**
Finger Lakes	New York, **USA**
Boiling Lake	Dominica
Five Flower Lake	Sichuan, **China** (not Mongolia)
Red Lagoon	Bolivia
Urmia	Iran
Reindeer / Athabasca	**Canada** (Uranium City near Athabasca)
Plitvice	Croatia — UNESCO lake park
Great Lakes E→W	Ontario → Erie → Huron → Michigan → Superior
Cities	Duluth on Superior. Chicago on Michigan. Buffalo on Erie. Detroit on the Detroit River
Victoria	Shared by Tanzania, Uganda and Kenya (**three** countries). Rift-associated. White Nile outflow near Jinja
Volume ladder (fresh note)	Caspian (saline) > Baikal > Tanganyika > Superior > Malawi
Area ladder (fresh, approximate coaching order)	**Superior → Victoria → Huron → Michigan → Tanganyika → Baikal → Great Bear → Malawi → Great Slave → Erie → Winnipeg → Ontario**

N.8 Waterfalls

Fall	River / basin	Countries	
Angel	Churún (Caroní / Orinoco)	**Venezuela**	**Highest** on Earth
Tugela	Tugela	South Africa	Usual 2nd-height fact
Victoria	**Zambezi**	Zambia–Zimbabwe	Not the highest
Niagara	Niagara (Erie → Ontario)	USA–Canada	
Iguazu / Iguaçu	Iguazu (Paraná system)	Brazil–Argentina	
Ribbon	Ribbon Creek	California (Yosemite)	
Browne	—	New Zealand	high fall note
Yosemite	Yosemite Creek	USA (California)	

Common Traps — Don’t Fall For These

1. **New Orleans = Missouri** — FALSE. **Mississippi**. Missouri joins at **St Louis**.
2. **Hyderabad = Godavari / Paleru** — FALSE. **Musi**.
3. **Mekong = south-west** — FALSE. **South / SE**.
4. **Mekong delta = Cambodia** — FALSE. **Vietnam**.
5. **USA–Mexico = Colorado** — FALSE. **Rio Grande**.
6. **Onega = Canada** — FALSE. **Russia**.
7. **Michigan is shared like Superior** — FALSE. Michigan is **wholly USA**.
8. **Baikal = largest lake** — FALSE. **Caspian** largest; Baikal = **deepest / fresh volume**.
9. **Amazon = longest in every book** — FALSE. **Nile** is the traditional longest fact.
10. **Budapest = Rhine** — FALSE. **Danube**. Cologne = Rhine.
11. **Angel = Victoria** — FALSE. Angel = **highest**; Victoria = **Zambezi**.
12. **Three Gorges = Hwang Ho** — FALSE. **Yangtze**.
13. **Volga → Black Sea** — FALSE. **Caspian**.
14. **Congo = largest discharge** — FALSE. **Amazon**. Congo = deepest + Equator twice.
15. **Tanganyika = deepest lake** — FALSE. **Baikal**; Tanganyika is 2nd.
16. **Titicaca = highest of any lake always** — FALSE. Fact = highest **navigable**.
17. **Skip lake types because papers asked Onega** — FALSE. Next year can ask rift / endorheic / Great Lakes order.
18. **GERD = Aswan** — FALSE. GERD = Blue Nile Ethiopia; Aswan = Egypt Nile / Lake Nasser.
19. **Darling Range = Murray–Darling river** — FALSE. Darling Range = hills of **Western Australia**.
20. **Hyderabad = Godavari** — FALSE. **Musi**.
21. **Inselberg = glacier landform** — FALSE. Wind residual (with zeugen). Moraine = glacier.
22. **Grand Canyon = Missouri** — FALSE. **Colorado**.
23. **Berlin = Rhine** — FALSE. **Spree**. Vienna = Danube (not Volga).
24. **Lisbon = Tiber** — FALSE. Tiber = Rome; Lisbon = **Tagus**.
25. **Mahaweli = India** — FALSE. **Sri Lanka**.
26. **Congo Basin excludes Cameroon** — FALSE. Cameroon is in the Congo Basin set.
27. **Baikal = Ukraine** — FALSE. **Russia**.
28. **Five Flower Lake = Mongolia** — FALSE. **Sichuan, China**.
29. **Angel = Tugela as world’s tallest** — FALSE. **Angel (Venezuela)** is tallest.
30. **Hoover = Tennessee** — FALSE. **Colorado**.
31. **Lake Victoria bordered by four countries** — FALSE. **Three** (TZ, UG, KE).
32. **Onega = Canada** — FALSE. **Russia**.

Topic 18 — World Landforms

Current Affairs (this topic)

Year	Fact	Why it matters	Source
Static	Andes longest vs Himalaya highest	Fold-chain swap	Atlas
Static	Kilimanjaro = East African Rift (not Ring of Fire)	Volcano belt trap	Coaching
Static	Hot-desert west-coast + cold-current logic	Desert location stems	NCERT
Static	Grassland name set (Pampas / Campos / Llanos / Prairie / Steppe)	Match-list diet	Lucent

Confused Pairs

Pair	Correct	Trap	Hindi
Andes vs Himalaya	Andes = **longest** fold chain; Himalaya = **highest**	Swapped	एंडीज़ / हिमालय
Kilimanjaro vs Fuji	Kilimanjaro = **E African Rift**; Fuji = Ring of Fire	Kilimanjaro in Pacific	किलिमंजारो
Pyrenees vs Alps	Pyrenees = **Spain–France**; Alps = Switzerland / C Europe	Alps on Spain–France MCQ	पिरेनीज़
Apennine vs Balkan	Apennine = **Italy**; Balkan = **Bulgaria**	Swapped	एपिनाइन / बाल्कन
Atlas vs Hoggar	Atlas high peak **Toubkal = Morocco**; Hoggar = **Algeria**	Both "North Africa" dump	अटलास / होगार
Gobi vs Sahara	Gobi = **cold**, Mongolia+China; Sahara = **hot**, Africa	Gobi in Kazakhstan	गोबी
Gibson vs Sonoran	Gibson = **Australia**; Sonoran = **USA**	Gibson–Brazil	गिब्सन
Pampas vs Campos	Pampas = **Argentina**; Campos = **Brazil**	Pampas–Brazil/Chile	पम्पास
Prairie vs Steppe	Prairie = N America; Steppe = Eurasia	Names swapped	प्रेयरी / स्टेपी
Savanna vs hot desert	"Land of Big Games" = **Savanna**, not Sahara	Desert = safari trap	सवाना
Maquis vs Chaparral	Maquis = **Med basin**; Chaparral = **California**	Same shrub, wrong coast	माकी / चैपरैल
Fynbos vs Matorral	Fynbos = **South Africa**; Matorral = **Chile**	Swapped SH west coasts	फिनबॉस
Taiga vs Tundra	Taiga = **conifer forest**; Tundra = **treeless** cold	Both "cold vegetation"	टैगा / टुंड्रा
Borneo vs Java	Borneo = **Sunda shelf**, not volcanic dump; Java = volcanic arc	"All Indonesia = volcano"	बोर्नियो
Greenland vs New Guinea	Greenland = **largest island**; New Guinea = 2nd	Australia is a continent	ग्रीनलैंड
Med vs monsoon	Med rain = **winter**; monsoon rain = **summer**	Med summer rain	भूमध्यसागरीय
Pamir vs Tibet	Pamir = phrase **Roof of the world**; Tibet = **highest large** plateau	Swapped	पामीर / तिब्बत
Hot desert coasts	**West** coasts + cold currents + subtropical high	East-coast deserts	पश्चिमी तट
Chernozem	Temperate grassland / **steppe–prairie** black soil	Equatorial laterite	चेरनोज़ेम
Residual mountains	Worn leftovers of old folds (Appalachians type)	Young Himalaya	अवशिष्ट
Continent area	Asia largest; **Australia** smallest	Africa largest	महाद्वीप
Most countries	**Africa (~54)**	Europe / Asia	अफ्रीका
Guyana	**South America**	Africa (Gabon/Guinea set)	गुयाना
Highest mean elevation	**Antarctica**	Asia	अंटार्कटिका

Pair	Correct	Trap	Hindi
Longest range	**Andes**	Himalaya / Rockies	एंडीज़
Sierra Nevada	**Block** mountain	Young fold with Alps	सिएरा नेवादा
Death Valley	**Rift** + extreme heat (California)	Cold Siberia only	डेथ वैली
Telegraphic Plateau	**N Atlantic Ridge**	Indian Ocean	टेलीग्राफिक
Thar	**Most populated** desert	Sahara densest people	थार
Downs vs tropical	Downs = **temperate** (Australia)	Called tropical	डाउन्स
Selvas vs grassland	Selvas = Amazon **rainforest**	Treated as grassland	सेल्वास
Veld vs Downs	Veld = **South Africa**; Downs = **Australia**	Veld–Australia	वेल्ड / डाउन्स
Faroe	**Sheep Islands** (Denmark)	Canary / Madeira	फेरो
Honshu	Japan's **largest**; Tokyo	Hokkaido largest	होन्शू
Falklands	**S Atlantic**	Indian Ocean	फॉकलैंड
Galápagos	**Ecuador**	Peru	गैलापागोस
Socotra	**Yemen**	Oman	सोकोत्रा
Majuli	**Brahmaputra** river island	Amazon	माजुली

N.0A Continents

Rank	Continent	Area pair (km²)	% of Earth's land (Philip's-type)	Countries (GC sheet)
1	**Asia**	~44.6 million	~29.8%	~48
2	**Africa**	~30.1 million	~20.3%	~**54** (most of any continent)
3	**North America**	~24.5 million	~16.2%	~23
4	**South America**	~17.8 million	~11.8%	~12 (Brazil largest)
5	**Antarctica**	~14.2 million	~9.2%	—
6	**Europe**	~9.9 million	~6.8%	~49
7	**Australia**	~7.7 million	~5.9%	Oceania ~14 in wider lists

N.1 Mountains

Family	Force / story	World examples
Fold	Compression (orogeny)	Himalaya, Alps, Andes, Rockies, Atlas, Pyrenees, Zagros. The **Appalachians** are an old worn fold
Block / fault / rift	Tension / uplifted blocks	Vosges, Black Forest, Sierra Nevada; African rift shoulders (**Kilimanjaro, Kenya, Rwenzori**)
Volcanic cone	Magma	Fuji, Etna, Cotopaxi, Mauna Loa — location table in **N.4**
Residual / dome	Old highland left after long denudation; or magma push (dome)	Residual leftovers of old folds (Appalachian-type worn highlands); **Black Hills** / laccolith **dome**

Continent	Peak	Range	Country fact
Asia	**Everest** (8848 m)	Himalaya	Nepal–China
Asia (2nd)	**K2 / Godwin Austen**	Karakoram (काराकोरम)	Pakistan–China
South America	**Aconcagua**	Andes	**Argentina** — highest outside Asia

Continent	Peak	Range	Country fact
North America	**Denali / McKinley**	Alaska Range	USA
Africa	**Kilimanjaro**	—	**Tanzania** (not Kenya)
Europe	**Elbrus**	Caucasus	Russia
Alps (W Europe)	**Mont Blanc**	Alps	France–Italy
Australia (mainland)	**Kosciuszko**	Snowy Mts	Australia
Oceania (often asked)	**Puncak Jaya / Carstensz**	Sudirman	Indonesia (New Guinea)
Antarctica	**Vinson Massif** (4897 m)	Ellsworth Mts	Deepest point: Bentley Subglacial Trench (-2555 m)

Continental Extremes: Highest and Lowest Points

Continent	Highest Point (Peak & Elevation)	Lowest Point (Depression & Elevation)
Asia	**Mount Everest** (8,848 m)	**Dead Sea** (-397 m / -400 m below sea level)
Africa	**Mount Kilimanjaro** (5,895 m)	**Lake Assal** (-156 m, Djibouti)
North America	**Denali / Mt. McKinley** (6,194 m)	**Death Valley** (-86 m, California)
South America	**Mount Aconcagua** (6,960 m)	**Valdes Peninsula** (-40 m, Argentina)
Europe	**Mount Elbrus** (5,642 m)	**Caspian Sea shore** (-28 m)
Australia	**Mount Kosciuszko** (2,228 m) / Puncak Jaya (4,884 m, Oceania)	**Lake Eyre** (-16 m)
Antarctica	**Vinson Massif** (4,897 m)	**Bentley Subglacial Trench** (-2,555 m)

Range	Where	Highest / note
Andes	W South America, ~7,000 km	**Longest** fold chain. Chimborazo (**Ecuador**). Copper Chile–Peru. Rain-shadow → Atacama + Patagonia
Himalaya	India–Nepal–Bhutan–China	**Highest**. Young Tertiary. Everest / Kanchenjunga (कंचनजंगा)
Karakoram (काराकोरम)	PK–India–China	**K2**. Not a Himalaya "range name" swap
Rockies	W Canada–USA–Mexico fringe	Young cordillera. **Not** the Appalachians
Appalachians	**Eastern** USA	**Old** fold; worn; coal fields
Alps	France–Switzerland–Austria–Italy	**Mont Blanc**. Switzerland is the classic Alps country. The foehn is a warm downslope wind on the **north** side
Pyrenees	**Spain–France** (Andorra)	**Not** Alps
Apennines	**Italy**	Spine of the peninsula
Balkans	**Bulgaria** / SE Europe	Not Carpathians
Carpathians	Romania / Slovakia / Ukraine	Transylvania arc — do not place on Bulgaria
Caucasus	Russia–Georgia–Azerbaijan	**Elbrus** is Europe's high point
Urals	Russia	Conventional **Europe–Asia** divide
Atlas	Morocco–Algeria–Tunisia	Highest peak **Toubkal** is in **Morocco** (not Algeria)
Drakensberg	South Africa / Lesotho	SE African escarpment
Ethiopian Highlands	Ethiopia	Blue Nile / Lake Tana source side
Rwenzori	Uganda–DRC	**Mt Stanley** — country fact **Uganda**
Hoggar / Ahaggar	**Algeria**	Sahara highland. **Not** Atlas
Tibesti	Chad	Sahara highland

Range	Where	Highest / note
Zagros / Elburz	Iran	Zagros is the west fold. Elburz is the north range, with **Damavand**
Tien Shan / Altai	C Asia	Syr Darya / Ob headwaters
Kunlun	N of Tibet	Tibet's northern wall
Great Dividing Range	Eastern Australia	Old; not Himalaya-young; ~3700 km; Mt Kosciuszko
Southern Alps	South Island, **New Zealand**	Aoraki / **Mt Cook**
Darling Range	**SW coast** of Australia (Western Australia)	Not NE Australia
Arakan Yoma	**Myanmar**	India–Myanmar hills: Naga, Chin, Patkai / Lushai; peak Natma Taung / Mt Victoria — **Khasi is Meghalaya (मेघालय)**, not this border trio
White Mountains	New Hampshire / Maine, **USA**	
Black Mountains	North Carolina, **USA** (Appalachian)	Not Black Forest
Black Forest	**Germany** (Feldberg); east of **Rhine**; Vosges west in France	Block mountain
Salt Range	Pakistan	Block mountain note
Allegheny	USA (Appalachian)	
Cantabrian	**Spain**	
Mackenzie (मैकेंज़ी)	**Canada**	
Jura	France–Switzerland	
Dinaric Alps	Balkans coast belt	
Pennines	Europe (UK)	Residual / old with Appalachians & Aravallis

Peak	Country	Do not confuse with
Kilimanjaro	**Tanzania**	Kenya (Mt Kenya is a different peak); **not** Ring of Fire
Toubkal	**Morocco**	Hoggar
Stanley	**Uganda**	Kilimanjaro
Hoggar	**Algeria**	Atlas / Toubkal
Chimborazo	**Ecuador**	Peru/Chile Andes dump
Aconcagua	**Argentina**	Chile
Cotopaxi	Ecuador	Volcanic Andes

N.2 Plateaus

Kind	Meaning	World example
Intermontane	High block **between** fold ranges	**Tibet** (Himalaya–Kunlun); **Altiplano** (Andes)
Piedmont	At the **foot** of mountains	Appalachian piedmont; Patagonian edge
Continental / shield	Old worn-down block	**Deccan** (दक्कन), Brazilian, African, Western Australian
Lava / volcanic	Flood basalt	**Columbia** (USA), **Deccan Trap** (दक्कन ट्रैप) cover
Plateau	Where	Note
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Kind	Meaning	World example
Tibetan Plateau	China / Tibet	**Highest large** plateau. Headwaters: Indus (सिंधु), Sutlej (सतलुज), Brahmaputra (ब्रह्मपुत्र), Mekong, Yangtze, Hwang Ho
Pamir	Tajikistan knot	Phrase **"Roof of the world"**. Knot of Himalaya–Karakoram–Hindu Kush–Tien Shan
Deccan	Peninsular India	Old gneiss + **Deccan Trap** lava in the west. **Not** Brazil
Malwa / Chotanagpur (छोटानागपुर)	India	Chotanagpur = Jharkhand coal–iron table
Brazilian	Brazil	Coffee on **Terra Roxa**; iron. Separate from Andes
Guiana	Venezuela–Guyana–N Brazil	
Patagonian	S Argentina	Rain-shadow plateau east of Andes
East African / Ethiopian	Africa	Rift lakes; Blue Nile from Ethiopia
African (south)	SA / Botswana / Namibia	High veld sits on it
Colorado	SW USA	**Grand Canyon** (Colorado River) cut into it
Columbia	Washington–Oregon–Idaho	**Lava** plateau
Colorado Plateau ≠ Columbia	—	Canyon vs lava — do not swap
Mexican (Mesa Central)	Mexico	
Anatolian	Türkiye	Interior of Asia Minor
Iranian	Iran	Dasht-e-Kavir / Lut on it
Mongolian	Mongolia	**Gobi** on the dry side
Altiplano	**Bolivia–Peru**	Highest inhabited Andean plateau; Lake Titicaca
Meseta	**Spain**	Interior tableland of Iberia
Loess Plateau	N China	Wind-silt; Hwang Ho yellow colour
Kimberley / Western Plateau	Australia	Old shield
Ozark	USA	Domed plateau note
Shan	Myanmar	Fluvial / tableland note
Telegraphic Plateau	**North Atlantic Ridge** (Newfoundland–Ireland note)	**Not** Indian Ocean

N.2A Valleys

Valley types and named valleys — Teaching Card

Place	Quick recall
Death Valley	California rift; heat; salt pan; ~−86 m
Panjshir	NE Afghanistan (Hindu Kush)
Taylor Valley	Antarctica — "Valley of the Dead"
Silicon Valley	California chip / tech belt
Valley of the Kings	Egypt (Luxor / Nile west bank)
Great Artesian Basin	Australia groundwater basin
Great Basin	USA endorheic interior

Place	Quick recall
Blind valley / doline	Karst limestone

N.3 Plains

How plains form — Teaching Card

Kind	How formed	Example
Alluvial	River dump	Indo-Gangetic, Nile, Mesopotamia, Hwang Ho, Mississippi
Glacial / till	Ice-sheet	North European Plain, Prairie of Canada–USA
Coastal	Sea + river	Atlantic coastal USA; eastern India
Lacustrine	Old lake floor	Imphal-type locally; some interior basins
Plain	Where	Note
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Indo-Gangetic	India–Pakistan–Bangladesh	Alluvium; wheat–rice–cane
North / Great European	France → Poland → Russia	Wheat–dairy; glacial + river
Great Plains / Prairie	USA–Canada	Temperate grassland overlay
Pampas	**Argentina** (into Uruguay)	Temperate grassland **and** a plain. **Not Chile, not Brazil**
Llanos	Orinoco (Venezuela–Colombia)	Tropical grassland plain
Manchurian	NE China	Soy / wheat
West Siberian	Russia	Ob–Yenisei lowland
Mesopotamian	Iraq	Tigris–Euphrates
Nile valley / delta (डेल्टा)	Egypt	Alluvium in a desert
Po valley	N Italy	Alpine alluvium
Hungarian (Alföld)	Hungary	**Puszta** grassland sits here
Canterbury	New Zealand	Downs / sheep
Nullarbor	S Australia	Limestone plain, not a grassland name

N.4 Volcanoes (world locations)

Belt	Share of great volcanoes	names
Circum-Pacific / Ring of Fire	**~80%**	Fuji, Pinatubo, Mayon, Apo, St Helens, Rainier, Popocatépetl, Andes cones, Krakatoa, Tambora
Mid-Atlantic	Divergent	**Iceland** (Hekla), Azores
Alpine–Himalayan / Med	Collision	**Etna, Vesuvius, Stromboli** (Italy)
East African Rift	Tension	**Kilimanjaro, Kenya** — **not** Ring of Fire
Volcano	Country	Belt / extra
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Fuji	Japan	Pacific. Classic cone
Pinatubo	Philippines	Pacific. 1991 eruption tag
Mayon	Philippines	Pacific. Perfect cone

Belt	Share of great volcanoes	names
Apo	Philippines	Highest PH; 2021 list
St Helens	USA (Washington)	Cascades / Pacific. 1980
Rainier	USA	Cascades / Pacific. 2021 list
Popocatépetl	Mexico	Pacific
Paricutin (paper: **Pericutine**)	**Mexico**	Cinder cone; born 1943. 2021 list
Cotopaxi	Ecuador	Andes / Pacific rim
Krakatoa / Krakatau	Indonesia	Sunda arc / Pacific
Tambora	Indonesia	1815; "year without summer"
Mauna Loa / Mauna Kea	Hawaii (USA)	**Hot-spot shield**, not a collision arc
Etna	**Italy** (Sicily)	Med. 2021 list. **Not** Pacific
Vesuvius	Italy	Pompeii AD 79
Stromboli	Italy	"Lighthouse of the Mediterranean"
Kilimanjaro	Tanzania	**Rift**. 2024 "which is NOT Ring of Fire"
Mt Kenya	Kenya	Rift. Lower than Kilimanjaro
Erebus	Antarctica	Ross Island; southern active
Hekla	Iceland	Mid-Atlantic

N.5 Islands

Rank	Island	Note
1	**Greenland**	Largest island; Denmark politically
2	**New Guinea**	Indonesia (Papua) + Papua New Guinea
3	**Borneo**	Indonesia, Malaysia, Brunei
4	**Madagascar**	Africa's large island
5	**Baffin**	Canada (Nunavut)
6	**Sumatra**	Indonesia
7	**Honshu**	Japan's main island
8	**Victoria**	Canada (Arctic)
9	**Great Britain**	England–Scotland–Wales
10	**Ellesmere**	Canada (Arctic)

Kind	How made	Examples
Continental	Piece of a continent, shallow shelf	**Greenland**, **Borneo**, **New Guinea**, **Madagascar**, Britain, Sri Lanka
Volcanic arc / hot-spot	Magma	**Java**, **Iceland**, **Hawaii**, Japan (Honshu mixed arc), Sicily (Etna)
Coral / atoll	Reef on a sinking volcanic peak or shelf	Maldives, Lakshadweep (लक्षद्वीप), many Pacific atolls
Coral reef (not an island rank)	—	Great Barrier Reef (Australia) is a **reef**, not a "largest island" option

Kind	How made	Examples
Island	Fact	
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Greenland	**Largest island**. Denmark. Ice sheet. Inuit	
New Guinea	2nd. Split **Indonesia (Papua) + Papua New Guinea**	
Borneo (Kalimantan)	3rd. **Three countries: Indonesia, Malaysia (Sabah/Sarawak), Brunei**. **Sunda shelf** — **not** volcanic deposition (2022 **Only 1**)	
Madagascar	4th. Africa's big island; unique fauna (प्राणीजात); **not** three-country split	
Baffin	Canada / Arctic	
Sumatra	Indonesia; Sunda arc **is** volcanic — contrast with Borneo	
Java	Indonesia; dense, volcanic	
Sulawesi	Indonesia; K-shaped	
Honshu	Japan's main island. Hokkaido / Kyushu / Shikoku = the other three	
Victoria	Canada / Arctic; after Honshu in the area ladder	
Great Britain	England–Scotland–Wales (Ireland is separate)	
Ellesmere	Canada / Arctic; after Great Britain in the top-ten set	
Iceland	Mid-Atlantic Ridge; volcanic	
Hawaii	Pacific hot-spot chain	
Sri Lanka	Continental; Palk Strait / Gulf of Mannar vs India	
Tasmania	S of Australian mainland	
Sicily / Sardinia / Corsica / Crete / Cyprus	Med islands. Sicily has **Etna**	
New Zealand	North Island + South Island; Cook Strait	

Island / group	Note
Nuuk	Capital of Greenland; elevation ~**1–3 m**
Diego Garcia	Largest of **Chagos** (Indian Ocean); UK/US base
Socotra	**Yemen**; UNESCO natural heritage **2008**; Guardafui / Arabian Sea mouth — **not Oman**
Seychelles / Mauritius / Madagascar	SW Indian Ocean set
Falklands	**South Atlantic** (UK), ~650 km E of S America — **not** Indian Ocean
Azores / Madeira / Cape Verde	Atlantic (Azores/Madeira = Portugal; Cape Verde = sovereign)
Bahamas	Atlantic / Lucayan
Fiji	**Pacific** Melanesia; Viti Levu + Vanua Levu hold most people
Melanesia	New Guinea, Bismarck, Solomon, Vanuatu, New Caledonia, **Fiji** (not Gilbert/Marshall (मार्शल)/Society = Micronesia/Polynesia)
Galápagos	Pacific volcanic; **Ecuador** (not Peru)
Grenada	Caribbean; "Island of Spices"
Balearic	Spain; **W Mediterranean**; Palma de Mallorca
Faroe	Denmark; **Sheep Islands**; N Atlantic
Macau	Handed to China **1999**

Island / group	Note
Majuli	Largest river island; **Brahmaputra** (ब्रह्मपुत्र), Assam
Japan N→S	**Hokkaido → Honshu → Shikoku → Kyushu**; Tokyo on **Honshu** (largest)
Indonesia W→E	**Sumatra → Java → Bali (बलि) → Lombok**
NZ South Island	Larger than NZ North / Java / Newfoundland among common options
Australia	**Island continent** (smallest continent)
Polynesia	Tonga, Tuvalu, Tahiti — **Tinian** (Marianas) is **not** Polynesia
Saint Martin	Caribbean salt-pan / lagoon coast pair
Indonesia island count	~**17,500** islands (highest among common MCQ options; Sweden note for absolute (निरपेक्ष) world count)

N.6 Deserts

Kind	Why dry	Examples
Hot subtropical	Descending air ~**30°** (horse latitudes)	Sahara, Arabian, Australian, Kalahari, Thar
Coastal / fog	Cold current + stable air	**Atacama** (Peru/Humboldt), **Namib** (Benguela)
Rain-shadow	High range steals moisture	**Patagonia** (Andes), **Atacama** (Andes), Thar (Aravalli leeward)
Cold interior	Far from sea, winter freeze	**Gobi**, **Taklamakan**, Great Basin, Colorado Plateau dry side
Desert	Country / belt	
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Sahara	N Africa (many states)	**Largest hot** desert. Hoggar / Tibesti are highland islands in it
Arabian	Arabian Peninsula	2nd large hot desert. **Rubʿ al-Khali** = Empty Quarter
Gobi	**Mongolia + China** (Inner Mongolia, Gansu, Xinjiang fringe)	**Cold**. 2025: **Only 1 and 2**. **Not** Russia, **not** Kazakhstan
Great Sandy	**NW Australia**	Smallest of the 2023 four
Great Victoria	S/W Australia	Largest **Australian** desert by many lists
Gibson	**Australia** (between Victoria and Sandy)	**Not Brazil** (2018)
Simpson	Central Australia	Red dunes
Sonoran	**USA** + Mexico	2018 correct pair. Cactus desert
Mojave	USA	Death Valley
Chihuahuan	USA–Mexico	
Taklamakan	**China** (Xinjiang)	Cold; Tarim basin. 2018 correct pair
Karakum	**Turkmenistan**	2018 correct pair. **Not** Kazakhstan's default dump
Kyzylkum	Uzbekistan–Kazakhstan	Red sand; separate from Karakum
Kalahari	Botswana / South Africa / Namibia	Bushman / San. Semi-desert
Namib	Namibia coast	Fog; among the oldest
Atacama	N **Chile** (into Peru)	Among the **driest** on Earth
Patagonia	**Argentina**	Cold rain-shadow. Not Chile's Atacama
Thar	India–Pakistan	Pleistocene–recent sands
Dasht-e-Lut / Kavir	Iran	Hot interior; Lut = extreme surface-temperature renown (NASA Aqua note)
Great Basin	USA	Cold interior

Kind	Why dry	Examples
Rubʿ al-Khali	Arabian Peninsula	Largest contiguous sand sea
An Nafud	Saudi Arabia	

N.7 Grasslands

Local name	Where	Climate	Economy tag
Prairie	USA–Canada (Great Plains)	Temperate	Wheat / maize / cattle; chernozem-type soils
Pampas	**Argentina** (+ Uruguay)	Temperate	Cattle / wheat
Veld (High / Middle / Low)	**South Africa**	Temperate–subtropical	Maize, gold belt overlay
Downs	**Australia**	Temperate	Sheep / wheat
Canterbury Downs	New Zealand	Temperate	Sheep
Steppe	Ukraine–S Russia–Kazakhstan–Mongolia	Semi-arid **BS**	Wheat / nomadism; classic **chernozem**
Puszta	**Hungary**	Temperate	Hungarian Alföld fact
Campos	**Brazil** (plateau)	Tropical / temperate mix	Cattle. **Not** Pampas
Llanos	**Venezuela–Colombia** (Orinoco)	Tropical	Cattle
Savanna / Sudan type	Africa (also N Australia, Brazilian edges)	Tropical **Aw**	**Land of Big Games**
Sahel	S of Sahara	Semi-arid fringe	Not the same as savanna core
Gran Chaco	Paraguay–N Argentina–Bolivia	Dry woodland/grass	Not Pampas

N.8 Vegetation (world belts)

Belt	Where	Plant picture
Equatorial rainforest (Af)	Amazon, Congo, Malaysia–Indonesia	Multilayer, **epiphytes**, hardwood (mahogany, ebony). Brazil name = **Selva**
Tropical monsoon	India, mainland SE Asia, N Australia	Teak, sal; sheds in dry season
Savanna	Sudan belt, Campos, Llanos, N Australia	Grass + scattered trees; **baobab**
Hot desert / thorn	Sahara, Thar, Australia	**Acacia**, cactus, xerophytes
Mediterranean sclerophyll	5 west-coast patches (N.9)	Olive, vine, cork, **cedar**; shrubs below
China type / laurel	E coasts ~25–40°	Camphor, tea in hills; summer rain forest
Temperate deciduous	W Europe, E USA, N China	Oak, beech, maple
Taiga / boreal	Canada, Scandinavia, Siberia	**Conifer** — spruce, fir, pine, larch. **Not** treeless
Tundra	Arctic fringe, high Andes/Tibet alpine	**Treeless** — moss, lichen, dwarf shrub. **Cryophyte steppe** = Arctic tundra (2024)
Ice	Greenland, Antarctica	None

Name	Region
Maquis	**Mediterranean** Sea coast
Fynbos (paper: **Fynbus**)	**South Africa** (Cape)
Chaparral	**California**
Matorral	**Chile**

Plant	Region
Epiphytes	Equatorial
Acacia	Sahara / thorn
Baobab	**Savanna**
Cedars	**Mediterranean**

Brazil pair	Meaning
Selva	Equatorial **rainforest**
Terra Roxa	Fertile dark **basaltic coffee soil** — **not** a forest name. Parallel to Deccan fertility, **not** the same as Indian regur (रेगुर) as a name

N.9 World Climatic Regions

Type (Köppen (कोपेन))	Where	Rain timing	Vegetation / crop
Equatorial (Af)	Amazon, Congo, Malaysia	**Year-round** convection	Selva; **not** a citrus belt
Savanna / Sudan (Aw)	Africa belt, Campos, Llanos, N Australia	**Summer** wet, winter dry	Grass + trees; baobab; **Land of Big Games**
Tropical monsoon	India, mainland SE Asia	**Summer** monsoon	Deciduous teak/sal
Hot desert (BWh)	Sahara, Arabia, Thar, Australia	Rare	Xerophytes
Cold desert (BWk)	Gobi, Taklamakan	Rare / snow	Sparse
Steppe (BS)	Ukraine to Mongolia; rain-shadow edges	Scanty	Short grass
Mediterranean (Cs)	**30–45° west coasts** — five patches below	**Winter** cyclonic	Olive, citrus, vine, cork, wheat, cedar, maquis
China / warm-temperate east coast	E China, SE USA, NSW, Natal	**Summer** rain	Laurel / tea hills
British / W European (Cfb)	Britain, NW France, W Norway	**Year-round** cyclonic	Deciduous oak–beech
Taiga / boreal	50–70°N	Modest, summer max	Conifers
Tundra (ET)	Arctic coast	Low	Moss/lichen; permafrost (स्थायी हिम मृदा)
Ice cap (EF)	Greenland interior, Antarctica	—	None

Common Traps — Don't Fall For These

1. **Kilimanjaro = Ring of Fire** — FALSE. **East African Rift**.
2. **Pampas = Brazil** — FALSE. **Argentina**. Brazil = **Campos**.
3. **Gibson = Brazil** — FALSE. **Australia**.

4. **Gobi in Kazakhstan / Russia** — FALSE for the 2025 code. **Mongolia + China**.
5. **Sahara smaller than Gobi** — FALSE. Sahara is the giant.
6. **Borneo = volcanic island** — FALSE. **Sunda shelf**; three countries.
7. **Maquis = California** — FALSE. **Chaparral**. Maquis = Med basin.
8. **Fynbos = Chile** — FALSE. **South Africa**. Chile = **Matorral**.
9. **Land of Big Games = Sahara** — FALSE. **Tropical savanna**.
10. **Mediterranean rain in summer** — FALSE. **Winter**.
11. **Taiga = treeless** — FALSE. That's **tundra**.
12. **Selva = Brazilian soil** — FALSE. Selva = **forest**; soil = **Terra Roxa**.
13. **Pyrenees = Italy** — FALSE. **Spain–France**. Italy = **Apennine**.
14. **Toubkal = Algeria** — FALSE. **Morocco**. Hoggar = Algeria.
15. **Etna = Pacific** — FALSE. **Italy / Mediterranean**.
16. **Australia = largest island** — FALSE. Australia is a **continent**; largest island = **Greenland**.
17. **Hot deserts only on east coasts** — FALSE. Classic set = **west** coasts + cold currents + Horse Latitudes.
18. **Pamir = highest large plateau** — FALSE. That is **Tibet**; Pamir = **Roof of the world** phrase.
19. **Chernozem = equatorial laterite** — FALSE. Chernozem = temperate grassland / steppe–prairie.
20. **Skip mountain families because papers asked country matches** — FALSE. Fold / block / residual can all appear.
21. **Atlas = France/Spain** — FALSE. That is **Pyrenees**. Atlas peak Toubkal = **Morocco**.
22. **Pamir = highest large plateau** — FALSE. Phrase is Pamir; **Tibet** is the highest large plateau.
23. **Colorado Plateau = lava** — FALSE. **Columbia** is lava; Colorado holds the **Grand Canyon**.
24. **Aconcagua = Chile** — FALSE. **Argentina**. Chimborazo = Ecuador.
25. **Africa = fewest countries / Australia = largest continent** — FALSE. Africa has the **most countries**; Asia is largest by area; Australia is **smallest**.
26. **Guyana = Africa** — FALSE. **South America**.
27. **Highest mean elevation = Asia** — FALSE. **Antarctica**.
28. **Sierra Nevada = young fold like Alps** — FALSE. **Block** mountain.
29. **Telegraphic Plateau = Indian Ocean** — FALSE. **North Atlantic Ridge**.
30. **Death Valley = cold Siberia** — FALSE. Hot **rift** in **California**.
31. **Thar = emptiest desert** — FALSE. **Most densely populated** desert.
32. **Europe has the worst desertification** — FALSE. **Least** among continents in GC framing.
33. **Downs = tropical grassland** — FALSE. Downs are **temperate** (Australia).
34. **Veld = Australia** — FALSE. Veld = **South Africa**; Australia = Downs.
35. **Selvas = grassland** — FALSE. Selvas = Amazon **tropical rainforest**.
36. **Savanna forest check = burrowers + soil only** — FALSE. Lead trio = **fire + grazers + seasonal rain**.
37. **Falklands = Indian Ocean** — FALSE. **South Atlantic**.
38. **Galápagos = Peru** — FALSE. **Ecuador** (volcanic Pacific).
39. **Socotra = Oman** — FALSE. **Yemen** (UNESCO 2008).
40. **Japan N→S = Hokkaido–Kyushu–Honshu** — FALSE. **Hokkaido → Honshu → Shikoku → Kyushu**.
41. **Borneo = volcanic dump like Hawaii** — FALSE for standard/UPPCS. **Sunda shelf**; three countries.
42. **Tinian = Polynesia** — FALSE. Marianas / Micronesia set.

Topic 19 — World Regional Geography

Current Affairs (this topic)

Year	Fact	Why it matters	Source
Static	Korea **38th** parallel / USA–Canada **49th** parallel	Boundary-line stems	Atlas
Static	Central Asian capital set (Tashkent–Dushanbe–Bishkek–Ashgabat–Astana)	Stan capital swaps	Atlas
2023–26	West Asia / Palestine map units (Gaza, West Bank) remain high-visibility	Map CA without rewriting physical facts	News / UN
Static	India Antarctic operational pair **Maitri + Bharati**	Polar station stem	MoES / MEA

Confused Pairs

Pair	Correct	Trap	Hindi
Nobi / Kanto	**Japan**	Korea	नोबी / कांतो
Esperanto	Artificial world language	Mountain / seaport	एस्पेरान्तो
Bahasa	**Indonesia**	Thailand	बहासा
Tamil abroad	**Singapore** official set	Myanmar major	तमिल
Sunni Triangle	Baghdad, Tikrit, **Ramadi**	**Basra** (Shia south)	सुन्नी त्रिभुज
Pontic vs Zagros	Pontic = **N Turkey**; Zagros = **W Iran**	Swapped on W→E list	पोंटिक / ज़ाग्रोस
Kara Kum vs Kyzylkum	Kara Kum = **Turkmenistan**	Kazakhstan / Tajikistan	काराकुम
Tashkent vs Ashgabat	Tashkent = **Uzbekistan**; Ashgabat = **Turkmenistan**	Swapped stans	ताशकंद
Bishkek vs Dushanbe	Bishkek = **Kyrgyzstan**; Dushanbe = **Tajikistan**	Swapped	बिश्केक
Darling vs Great Dividing	Darling = **SW** Australia; GDR = **east**	Darling on east coast	डार्लिंग रेंज
N Australia climate	**Tropical**	Temperate (2025 R)	उत्तरी ऑस्ट्रेलिया
Cape Verde capital	**Praia**	Bamako (**Mali**)	केप वर्डे
Madeira	NE **Atlantic** (Portugal)	Caribbean	मदेरा
Maitri / Bharati	Both **operational** Antarctica	Dakshin Gangotri still open	मैत्री / भारती
W Europe rain	**All months** (westerlies)	Mediterranean winter-only	पश्चिमी यूरोप
Philippines cane/coconut	**Spanish and Americans**	British / Dutch	फिलीपींस
Palestine vs Israel map	West Bank + Gaza as units	"No such region on map"	फिलिस्तीन
Seoul vs Pyongyang	Seoul = **South**; Pyongyang = **North**	Swapped	सियोल / प्योंगयांग
Sumatra vs Java	Sumatra is **west** of Java	Java first on W→E list	सुमात्रा / जावा
Bali vs Lombok	Bali **west** of Lombok (Wallace Line between them)	Lombok before Bali	बाली / लॉम्बोक
Mekong delta	**S Vietnam**	Cambodia	मेकांग डेल्टा
Bolivia vs Uruguay	Bolivia = **landlocked**; Uruguay has Atlantic	Uruguay landlocked	बोलीविया
Igarka	**Russia** (Yenisei)	China	इगारका
Casablanca vs Rabat	Rabat = **capital**	Casablanca	रबात
UK vs Great Britain	UK = GB + **N Ireland**; GB = Eng+Scot+Wales	Ireland in UK	यूके

Pair	Correct	Trap	Hindi
Ethiopia vs Eritrea	Ethiopia **landlocked**; Eritrea has Red Sea	Ethiopia still coastal	इथियोपिया
Bolivia vs Paraguay	**Both** landlocked S America	Only one of them	पराग्वे
NZ vs Tasmania	NZ = separate country, capital **Wellington**	NZ is an Australian state	न्यूज़ीलैंड
Continent size	Asia largest; Australia / Oceania smallest inhabited	Europe largest	महाद्वीप
Double landlocked	**Uzbekistan** / Liechtenstein	Any coastal state	दोहरा स्थलरुद्ध
38th vs 49th parallel	Korea DMZ vs Canada–USA	Swapped	38° / 49°
Capital ≠ mega-city	Ankara / Canberra / Brasília / Abu Dhabi	Istanbul / Sydney / Rio / Dubai	राजधानी

N.1 Asia (overview + East + South)

East Asia	Capital	Physical fact
China	Beijing	Tibet; Hwang Ho / Yangtze; Manchurian plain NE; Tropic through south; Shanghai on Yangtze mouth
Japan	Tokyo	Islands **Hokkaido–Honshu–Kyushu–Shikoku** (N→S). **Kanto** is the Tokyo plain. **Nobi** is the Nagoya plain. These are **not** in Korea (2021)
Mongolia	Ulaanbaatar	Landlocked; Gobi to the south
Taiwan	Taipei	Off Fujian; not Caribbean
North / South Korea	See **N.5**	Peninsula — not the Japanese plains

South Asia (India's neighbourhood as a *region*)

Country	Capital	Fact
India	New Delhi	Peninsula; Tropic
Pakistan	Islamabad	Indus (सिंधु); Karachi ≠ capital
Nepal	Kathmandu	**Landlocked**; Himalaya (हिमालय)
Bhutan	Thimphu	**Landlocked**
Bangladesh	Dhaka	Ganga (गंगा)–Brahmaputra (ब्रह्मपुत्र) delta
Sri Lanka	Sri Jayawardenepura Kotte / Colombo	Island; Palk Strait
Maldives	Mal (माल)é	Atolls; equator-near
Afghanistan	Kabul	Landlocked hinge to West/Central Asia; Hindu Kush

N.2 West Asia

Order	Range	Where
1	**Pontic**	North **Turkey**, Black Sea coast
2	**Zagros**	Western **Iran** (Iraq border)
3	**Hindu Kush**	**Afghanistan**
4	**Karakoram** (काराकोरम)	Pakistan–China

Country	Capital	Extra
Turkey	Ankara	Pontic + Taurus; Istanbul is **not** the capital
Iran	Tehran	Zagros; Elburz / Damavand

Country	Capital	Extra
Iraq	Baghdad	Tigris; Basra on the Shatt
Saudi Arabia	Riyadh	Rubʿ al-Khali; Mecca / Jeddah west
Syria	Damascus	
Lebanon	Beirut	
Jordan	Amman	East of the Jordan R. / Dead Sea
Yemen	Sanaa	Bab el-Mandeb
Oman	Muscat	
UAE	**Abu Dhabi**	Dubai is not the capital
Qatar	Doha	
Kuwait	Kuwait City	
Bahrain	Manama	
Israel	Jerusalem (claimed)	Mediterranean west

N.3 Central Asia

Country	Capital	Desert / extra
Uzbekistan	**Tashkent**	Aral / Amu–Syr irrigation
Tajikistan	**Dushanbe**	Pamir knot
Kyrgyzstan	**Bishkek**	Tien Shan
Turkmenistan	**Ashgabat**	**Kara Kum / Karakum** desert (2022)
Kazakhstan	**Astana** (Nur-Sultan was the 2019–22 name)	Steppe (स्टेपी); **not** Kara Kum

N.4 South-East Asia

Country	Capital	Fact
Indonesia	**Jakarta**	Thousands of islands; equator. Port–country pair **Jakarta–Indonesia is correct** (2023)
Malaysia	Kuala Lumpur	Peninsula + **Sabah/Sarawak on Borneo**
Singapore	Singapore	City-state; Malacca approach
Brunei	Bandar Seri Begawan	North Borneo
Philippines	Manila	Luzon / Visayas / Mindanao; Spanish then US
Thailand	Bangkok	Never colonised; Chao Phraya
Myanmar	Naypyidaw (Yangon = old/port)	Pegu Yoma; Bay of Bengal
Laos	Vientiane	**Landlocked**; Mekong
Cambodia	Phnom Penh	Mekong **passes through** — **delta is not here**
Vietnam	Hanoi (Ho Chi Minh City = south)	**Mekong delta = southern Vietnam** (2021)
Timor-Leste	Dili	East of Indonesia's Timor

N.6 Palestine

Piece	Fact
Gaza Strip	Mediterranean coast; borders **Egypt** + Israel
West Bank	West of Jordan River; borders **Jordan** + Israel
Jordan River	Sea of Galilee → **Dead Sea** (no ocean outlet)
Dead Sea	Israel–Jordan; lowest land, saline (लवणीय)
Jerusalem	Claimed capital; Old City holy sites
Ramallah	West Bank administrative centre often in news

Extra map piece	Fact
Golan Heights	NE; borders **Syria**
Sinai	Egyptian peninsula; Suez Canal on its west; Gaza at its NE corner
Occupied / disputed labels	Learn **where** Gaza, West Bank, Golan sit — do not write a politics essay

N.7 Europe

Belt	Who	Note
Western Europe climate	Britain, NW France, Low Countries	**Westerlies** + **rain in all months** (2024 Both)
Mediterranean Europe	Spain, S France, Italy, Greece	Winter rain — **not** the "all months" fact
British Isles	UK + Ireland	The **UK** is Great Britain plus **Northern Ireland**. **Great Britain** is England, Scotland and Wales. The Republic of Ireland is **not** in the UK. Capital **London**
Iberia	Spain, Portugal	Iberian Peninsula
Benelux	Belgium, Netherlands, Luxembourg	Brussels, **Amsterdam** (government also The Hague), Luxembourg. **Rotterdam** is the Netherlands port (2023 correct pair)
Nordic / Scandinavia	Norway, Sweden, Denmark; Finland + Iceland often bundled	Oslo, Stockholm, Copenhagen, Helsinki, Reykjavik
Baltic states	Estonia, Latvia, Lithuania	Tallinn, Riga, Vilnius
Alps	Switzerland, Austria, parts of FR/IT/DE	Bern, Vienna
Balkans	Greece, Bulgaria, Serbia, Croatia, Albania, N Macedonia, Bosnia, Montenegro, Kosovo, Romania/Slovenia fringe	Bulgaria holds the Balkan range name
East-Central	Poland, Hungary, Czechia, Slovakia, Ukraine	Warsaw, Budapest, Prague, Kyiv
Russia	Moscow	Europe + Asia; **Igarka** (Yenisei) is in **Russia**, not China

Port–country	Right?
Rotterdam – Netherlands	Yes
Montevideo – Uruguay	Yes
Jakarta – Indonesia	Yes
Igarka – China	**NO** — Igarka is **Russia** (Yenisei, Siberia)

N.8 Africa

Belt	Countries / fact
Maghreb	Morocco, Algeria, Tunisia, Libya (+ Western Sahara disputed)
Sahel	Senegal–Mauritania–Mali–Burkina–Niger–Chad–Sudan fringe **south of Sahara**
Horn	**Ethiopia, Somalia, Eritrea, Djibouti**. Ethiopia became **landlocked** after **Eritrea** (1993) took the Red Sea
West Africa	Nigeria (Abuja, not Lagos as capital), Ghana (घन) (Accra), Senegal (Dakar), Mali (**Bamako**)
Central	DRC (Kinshasa), Congo-Brazzaville, Cameroon, CAR
East	Kenya (Nairobi), Tanzania (Dodoma; Dar es Salaam = port), Uganda (Kampala, landlocked)
Southern	South Africa (Cape Town / Pretoria / Bloemfontein — **three capitals**), Namibia, Botswana (**landlocked**), Zimbabwe (**landlocked**), Zambia (**landlocked**), Mozambique (coast)
Islands	Madagascar (4th largest island), Cape Verde (**Praia**), Mauritius, Seychelles, Comoros. **Madeira / Canary = Atlantic, not "African Caribbean"**

Country	Capital
Morocco	**Rabat** (not Casablanca)
Tunisia	**Tunis**
Algeria	**Algiers**
Libya	Tripoli
Cape Verde (Cabo Verde)	**Praia**
Mali	**Bamako**

N.9 North America

Country	Capital	Physical spine
Canada	**Ottawa** (not Toronto/Vancouver)	Shield, Prairies, Rockies, Arctic islands; **Baffin**; 49th parallel with USA
USA	**Washington DC** (not New York)	Rockies west, Appalachians east, Great Plains, Mississippi, Great Lakes. **Alaska** (Arctic/Pacific) + **Hawaii** (Pacific islands) are states
Mexico	Mexico City	Plateau; Yucatan; Gulf of Mexico
Greenland	Nuuk	**Denmark**; largest **island**, not a US state

N.10 South America

Country	Capital	Coast?	Regional fact
Brazil	**Brasília** (not Rio / São Paulo)	Atlantic	**Selva** + **Terra Roxa** (2025 both). Amazon
Argentina	Buenos Aires	Atlantic	Pampas; Aconcagua
Chile	Santiago	Pacific	Atacama; **not** landlocked
Peru	Lima	Pacific	Andes; Titicaca with Bolivia. **Not** landlocked (2022 trap)
Ecuador	Quito	Pacific	Equator; Galápagos
Colombia	Bogotá	Caribbean **and** Pacific	
Venezuela	Caracas	Caribbean	Llanos / Orinoco
Bolivia	Sucre (const.) / La Paz (seat)	**Landlocked**	2022 answer. Lost Pacific in 19th c.

Country	Capital	Coast?	Regional fact
Paraguay	Asunción	**Landlocked**	The other S American landlocked state
Uruguay	Montevideo	Atlantic	Port pair 2023 correct. **Not** landlocked
Guyana	Georgetown	Atlantic	Former British Guiana
Suriname	Paramaribo	Atlantic	**Not** landlocked (2022 trap)
French Guiana	Cayenne	Atlantic	France overseas — often in "Guianas"

N.11 Australia

Coast	What sits there
South-west	**Darling Range** (Perth / WA) — 2020
East	**Great Dividing Range**; Sydney, Brisbane
North-east	Great Barrier Reef; tropical Queensland
North	**Tropical** (monsoon / savanna (सवाना)) — **not temperate**
South	Melbourne, Adelaide; more temperate
Tasmania	South of Bass Strait

Extra	Fact
Outback	Dry interior
Great Artesian Basin	Groundwater under the east-central plains
Murray–Darling	SE basin; not the Darling **Range**
Uluru / Ayers Rock	NT
Tropic of Capricorn	Cuts WA–NT–QLD

New Zealand (Oceania pair — always asked with Australia)

Note	Note
Capital	**Wellington** (not Auckland)
Islands	**North Island + South Island**; **Cook Strait**
Alps	Southern Alps on South Island
Not	An Australian state. Tasmania is the Australian island

N.12 Antarctica

Station	Fact
Dakshin Gangotri	First Indian (1983/84). **Not operational** now (supply base / abandoned as permanent station)
Maitri	1988/89, Schirmacher Oasis — **operational**
Bharati	2012, Larsemann Hills — **operational**

N.13 Caribbean

Group	Islands
Greater Antilles	Cuba, Jamaica, Hispaniola (**Haiti + Dominican Republic**), Puerto Rico

Group	Islands
Lesser Antilles (names)	**Grenada, Montserrat, Anguilla**, Barbados, Trinidad (S edge), Antigua
NOT Caribbean	**Madeira** (Portugal, NE Atlantic off Morocco), Canary Islands (Spain, Atlantic), **Azores**, **Cape Verde**, Bermuda (N Atlantic)
Island	Capital / extra
--------	----------
Cuba	Havana; largest Caribbean island
Jamaica	Kingston
Haiti / Dominican Rep.	Port-au-Prince / Santo Domingo — **one island, two countries**
Puerto Rico	US territory
Trinidad and Tobago	Port of Spain; near Venezuela
Bahamas	Atlantic archipelago — often **not** the "Caribbean **Sea**" MCQ

N.14 Landlocked · Equator · Capitals · Boundaries (map diet)

Line	Note
38th parallel	Approx. Korea DMZ / North–South Korea
49th parallel	Much of **Canada–USA** border
Durand Line	Afghanistan–Pakistan
Radcliffe Line	India–Pakistan (1947)
McMahon Line	India–China (eastern sector)
Maginot / Siegfried	Historic France–Germany fortification names

Country	Capital	Famous non-capital
Australia	**Canberra**	Sydney
Brazil	**Brasília**	Rio / São Paulo
Turkey	**Ankara**	Istanbul
Nigeria	**Abuja**	Lagos
Tanzania	**Dodoma**	Dar es Salaam
South Africa	Pretoria / Cape Town / Bloemfontein	Johannesburg
USA	**Washington DC**	New York
Canada	**Ottawa**	Toronto
Morocco	**Rabat**	Casablanca
UAE	**Abu Dhabi**	Dubai
New Zealand	**Wellington**	Auckland
Pakistan	**Islamabad**	Karachi
Myanmar	**Naypyidaw**	Yangon
Bolivia	Sucre / La Paz	—
Netherlands	Amsterdam	Rotterdam (port); The Hague = government/ICJ

N.15 World languages

Note	Fact
Largest by **L-1**	Often **Mandarin Chinese**
Largest by **total** (L-1+L-2) in recent Ethnologue frames	Often **English**
Hindi	Huge L-1 + L-2; classic "top world languages" set
Spanish	Official across much of Latin America

Country / place	Language
Singapore	Official set includes **Malay** (national), **English**, **Mandarin**, **Tamil**
Chile / Colombia / Cuba	**Spanish** official
Republic of Congo	**Not** Spanish (French / local)
Myanmar	**Burmese** (not Tamil as major)
Indonesia	**Bahasa Indonesia** (Bahasa ≠ Thailand)
Denmark / Netherlands / China	Danish / Dutch / Mandarin
Pakistan	**Urdu + English** official; Punjabi widely spoken but not the sole "official = Punjabi" story
Bangladesh	Stronger **ethnic/linguistic homogeneity** than Pakistan in classic A/R stems

Common Traps — Don't Fall For These

1. **Nobi/Kanto = Korea** — FALSE. **Japan**.
2. **Basra in the Sunni Triangle** — FALSE. **South Iraq**.
3. **Northern Australia = temperate** — FALSE. **Tropical**. Interior dry is still true.
4. **Darling Range = east coast** — FALSE. **South-west**. East = Great Dividing Range.
5. **Kara Kum = Kazakhstan** — FALSE. **Turkmenistan**.
6. **Bishkek = Kazakhstan** — FALSE. **Kyrgyzstan**.
7. **Cape Verde = Bamako** — FALSE. **Praia**. Bamako = Mali.
8. **Madeira = Caribbean** — FALSE. **Atlantic / Portugal**.
9. **Dakshin Gangotri still operational** — FALSE. **Maitri + Bharati**.
10. **W Europe = Mediterranean winter rain only** — FALSE. Rain in **all months**.
11. **Philippines plantations = Dutch** — FALSE. **Spanish and Americans**.
12. **Istanbul = capital of Turkey** — FALSE. **Ankara**.
13. **Borneo = volcanic + two countries** — FALSE. **Three countries**, continental shelf.
14. **Iraq borders Israel** — FALSE.
15. **Greenland = US state** — FALSE. **Denmark**.
16. **Java is west of Sumatra** — FALSE. **Sumatra → Java → Bali → Lombok**.
17. **Mekong delta = Cambodia** — FALSE. **Southern Vietnam**.
18. **Uruguay / Peru / Suriname = landlocked** — FALSE. **Bolivia** (and Paraguay).
19. **Igarka = China** — FALSE. **Russia**.
20. **Ethiopia has a Red Sea coast** — FALSE since Eritrea. **Landlocked**.
21. **Wellington is in Australia** — FALSE. **New Zealand**. Canberra = Australia.
22. **Equator through India** — FALSE.
23. **Paraguay has a seacoast** — FALSE. Landlocked with Bolivia.
24. **Casablanca / Istanbul / Sydney / Rio / Auckland = capitals** — FALSE. Rabat / Ankara / Canberra / Brasília / Wellington.

25. **Europe is the largest continent** — FALSE. **Asia** is largest; Australia/Oceania is smallest inhabited.
26. **Double-landlocked = Bolivia** — FALSE. Bolivia is landlocked; **Uzbekistan / Liechtenstein** are the double-landlocked pair.
27. **49th parallel = Korea** — FALSE. Korea ≈ **38th**; 49th ≈ Canada–USA.
28. **Skip capitals map because recent papers asked Australia A/R** — FALSE. Next year can ask any Lucent capital or landlocked trap.

Topic 20 — World Agriculture

Current Affairs (this topic)

None mandatory. If a paper quotes a **year** (e.g. 2016 coffee), freeze **that** ranking. Do not chase every FAO revision in the teaching home.

Confused Pairs

Pair	Correct	Trap	Hindi
Plantation vs food grain	Tea (also coffee, rubber, cocoa, cane, oil palm, banana)	Wheat / rice / maize as "plantation"	बागान
Producer vs exporter (rice/wheat)	China often **#1 volume**; **not** the classic exporter	China = wheat/rice export king	उत्पादक / निर्यातक
Brazil vs Vietnam coffee	Brazil **#1**; Vietnam **#2** (2016 paper)	Vietnam first	ब्राज़ील / वियतनाम
Arabica vs Robusta	Brazil/Colombia highland = arabica; Vietnam = robusta bulk	Swapped	अरेबिका / रोबस्टा
Cocoa vs coffee Africa	Cocoa = **Côte d'Ivoire + Ghana**; coffee East Africa too (Ethiopia/Uganda)	Latvia as cocoa	कोकोआ
Kerala vs Assam	Rubber **Kerala**; tea volume **Assam**	Assam rubber / Gujarat tea	केरल / असम
Kinta Valley	**Tin** (Malaysia)	Rubber / tea / coffee	किंटा घाटी
Amazon origin vs SE Asia crop	Rubber **and** cocoa **originated** in Amazon; bulk crop now elsewhere	"Origin = Malaysia / Ghana"	अमेज़न
Jhum vs plantation	Jhum = shifting subsistence	Jhum = tea estate	झूम
Ladang vs milpa	Ladang = **Malaysia**; milpa = **Mexico**	Swapped	लदांग / मिल्पा
Roca vs chena	Roca = **Brazil**; chena = **Sri Lanka**	Swapped	रोका / चेना
Cane vs beet	Cane = tropics (**Brazil**); beet = **Europe**	Beet in Kerala	गन्ना / चुकंदर
Citrus belt	**Mediterranean**	Equatorial / Himalayan valleys as *world* fact	नींबू वर्गीय
Oil palm vs rubber	Oil palm volume = **Indonesia + Malaysia**	Oil palm = Kerala only	तेल ताड़
Black tea vs green tea	Black = **fermented**; green = **not**	Swapped	काली / हरी चाय
Philippines cane	**Spanish and Americans**	Dutch (Indonesia) / British	फिलीपींस
Intensive vs extensive	Intensive = small + labour + monsoon rice; extensive = huge + machines + Prairie wheat	Swapped	सघन / विस्तृत
Producer vs exporter (coffee/tea)	Brazil coffee #1 volume; Kenya often tea **exporter** story	Same word	
Soybean triangle	**USA–Brazil–Argentina**	Only India	सोया
Milk volume vs dairy export	India = milk **volume**; NZ/Netherlands = **export** dairy	India = cheese ship king	दूध
Golden Crescent	AFG–Iran–Pakistan	Includes Iraq	गोल्डन क्रिसेंट
Fazenda vs Chena	Brazil plantation vs Sri Lanka shifting	Fazenda = jhum	फाजेंडा
Mocha vs Kona	Yemen vs Hawaii	Swapped	मोचा / कोना

Pair	Correct	Trap	Hindi
Olericulture	Vegetables	Call it horticulture-only	शाककृषि
Von Thünen vs Whittlesey	Thünen = **distance rings** from market; Whittlesey = **world type map**	Same theory	

N.1 Agriculture (world systems)

Split	Meaning	picture
Subsistence	Family food; little surplus	Wet-rice monsoon Asia; shifting tropics
Commercial	Surplus for **market / export**	Prairie wheat, plantation, dairy NZ
Intensive	High labour (or capital) **per hectare**; small farms	Monsoon rice; Dutch dairy; Japanese rice
Extensive	Low labour per hectare; **huge** farms, machines	Canadian Prairie, Australian Downs, Pampas
Sedentary	Same fields year after year	Most of Eurasia
Nomadic / shifting	People or plots **move**	Sahara herders; Amazon/Congo jhum

System	Climate / where	What is grown / raised	Note
Mixed farming	W Europe, US Midwest	Crops **+** livestock on the **same** farm	Fodder rotation; manure; not "only wheat"
Dairy farming	NW Europe, Great Lakes, NZ, Denmark, Netherlands	Milk, butter, cheese	Mild climate + fodder + urban (नगरीय) market; New Zealand is classic grass-export dairy
Mediterranean	30–45° **west coasts** (five patches)	**Citrus, vine, olive**, winter wheat, vegetables	Winter rain + dry bright summer (2020 citrus answer **C**)
Livestock ranching	Pampas, W USA Prairie, Australian Downs, S African Veld	Cattle / sheep, **commercial**	Not nomadic herding (that is subsistence)
Nomadic herding	Sahara–Arabia–C Asia–tundra (टुंड्रा)	Camel, sheep, yak, reindeer	Moves with pasture; low density
Truck / market gardening	Urban fringes	Vegetables, fruit, flowers	High value, perishable, near city
Horticulture	Med + hill + greenhouse	Fruit, veg, flowers	Viticulture means **grape / wine**
Plantation	Humid tropics	Tea, coffee, cocoa, rubber, cane, oil palm, banana, coconut, spices	Humid-tropic estate monoculture
Shifting / slash-and-burn	Humid tropics	Millets, maize, cassava, upland rice	See shifting-cultivation card below
Collective / cooperative	Former USSR kolkhoz; Israel kibbutz; Denmark dairy co-ops	Shared marketing / land	Secondary example

Term	Recall
Viticulture	Grapes (wine) — France, Italy, Spain, California, Chile, Australia, Cape
Sericulture	Silk — **China** volume; India mulberry + tussar/eri/muga
Pisciculture	Fish rearing
Apiculture	Bees / honey
Floriculture	Flowers (Netherlands auction)

N.2 Whittlesey agricultural regions

Type	Core belt
Nomadic herding	Sahara, Arabia, Gobi, tundra, C Asian drylands
Livestock ranching	Pampas, W USA, Australia, Veld
Shifting cultivation	Amazon, Congo, SE Asia hills, NE India, C America
Rudimentary sedentary tillage	Parts of tropics where plots are fixed but tools simple
Intensive subsistence — **rice dominant**	E / SE / S Asia deltas and coasts
Intensive subsistence — **without paddy rice**	N China, interior India wheat–millet, Nile
Commercial plantation	Caribbean, NE India–SL tea, Malaysia–Indonesia rubber/palm, Brazil coffee, W Africa cocoa
Mediterranean agriculture	Five west-coast patches (Med proper, California, C Chile, Cape, SW Australia)
Commercial grain farming	Prairies, Pampas, Downs, Ukrainian–Russian steppe
Mixed crop-and-livestock	W Europe, US Midwest Corn Belt edge
Subsistence crop-and-livestock	Parts of highlands / poorer temperate belts
Commercial dairy	NW Europe, Great Lakes, NZ
Specialised horticulture / market gardening	W Europe urban belt, California, Mediterranean coasts

Patch	Where
1	Mediterranean Sea basin (Spain, Italy, Greece, Levant (लेवेंट), Maghreb)
2	California
3	Central Chile
4	Cape of South Africa
5	SW Australia (Perth side)

N.3 World cereals

Wheat

Belt	Where
Prairies	Canada + USA
Pampas	Argentina
Downs	Australia
Steppe	Ukraine / S Russia / Kazakhstan
N European plain	France (export), Germany
Indo-Gangetic rabi	India (volume, mostly domestic)

N.4 Fibres, sugar, oilseeds

Sugarcane vs sugar beet

	Sugarcane	Sugar beet
Climate	Tropical; long season **10–18 months**; frost-free	Temperate; **4–7 months**

	Sugarcane	Sugar beet
Rain / water	**~75–150 cm** or irrigation	Moderate (नरम दल) rain; Europe plains
World centre	**Brazil** (also ethanol from cane)	**France, Germany, Russia, Ukraine, USA**
Asia names	India, China, Thailand, Pakistan	Not a Kerala crop

Oilseeds and tree oils

Crop	Climate / belt	Usual leaders
Soybean	Warm temperate	**USA, Brazil, Argentina** (export triangle). China = big user
Groundnut	Tropical / subtropical	China, India, Nigeria, USA
Rapeseed / canola	Cool temperate	Canada, China, India, EU
Sunflower	Temperate	Russia, Ukraine, Argentina
Oil palm	Equatorial, high rain	**Indonesia + Malaysia** (volume). Nigeria origin story, not today's tonnage king
Olive	Mediterranean winter-rain	**Spain, Italy, Greece**, Tunisia

N.5 Plantation crops (the system)

Feature	Fact
Climate	Tropical / subtropical; **no severe frost**
Organisation	**Estate** (large), often colonial origin
Capital + labour	High capital; **hired** labour (not only family)
Crop	**Monoculture** for **export**
Processing	Factory (फैक्टरी) on or near the estate (tea factory, rubber smoke-house, sugar mill)
Classic list	**Tea, coffee, cocoa, rubber, sugarcane, banana, oil palm, coconut, spices**
Not plantation	Wheat, rice, maize as the *predominant* system

Crop	India	Do not write
Tea	Assam (volume), WB Darjeeling, Nilgiri (नीलगिरि)	Gujarat — Tea
Coffee	Karnataka > Kerala > Tamil Nadu (shade, Western Ghats (पश्चिमी घाट))	Assam as coffee king
Rubber	**Kerala** (Kottayam–Ernakulam)	Assam — Wheat; UP — Jute
Spices	Kerala pepper (काली मिर्च) + cardamom	
Coconut	Kerala–TN–Karnataka coast	Philippines is the *world* Spanish–US story
Banana	India = huge **producer**; **Ecuador** = classic **exporter**	India = world's banana-ship king

Place	Crop	Credit / trap
Philippines	Coconut + sugarcane	**Spanish and Americans** (2023) — not Dutch
Indonesia / Java	Cane, coffee, rubber	**Dutch** (Hollanders)
Malaya	Rubber estates	**British**
NE India / Ceylon	Tea	British
West Africa	Cocoa	Shifted from Amazon after colonial planting
Brazil	Coffee	Portuguese / highland Terra Roxa

N.6 Tea

Type	Processing	Where the puts it
Black	Fully **fermented**	India, Sri Lanka, Kenya
Green	**Not** fermented	China, Japan
Oolong	Semi-fermented	Taiwan / SE China
CTC	Crush–tear–curl (granular, tea-bag)	Indian plains/Assam bulk
Orthodox	Whole-leaf	Darjeeling, Ceylon high-grown
Region	Fact	
--------	------	
China	Usual **largest producer**; original home; green tea	
India	**Assam** (Brahmaputra valley — **volume**), **Darjeeling** (GI, hill, orthodox), **Dooars/Terai (तराई)**, **Nilgiri** (नीलगिरि), Kangra, Munnar	
Sri Lanka	"**Ceylon** tea"; central highlands; high / mid / low grown	
Kenya	Leading **African** name; often among top **exporters**; highland	
Indonesia, Vietnam, Bangladesh, Turkey	Also revise	

N.7 Coffee

Type	Character	Where
Arabica (*C. arabica*)	Milder aroma; higher hills (~600–1600 m)	Brazilian highlands, **Colombia**, Ethiopia, India's Ghats
Robusta (*C. canephora*)	Stronger, more caffeine, hardier, lower altitude	**Vietnam**, Uganda, lowland tropics
Liberica	Secondary example	W Africa pockets

Country	Note
Brazil	Usual **#1**. São Paulo–Minas–Paraná highlands. **Terra Roxa** soil. Mostly **arabica**
Vietnam	**#2** in the 2016 fact. **Robusta** bulk. Central Highlands. Trap = putting it first
Colombia	Mild **arabica**; **third** in 2016 four (चातुर्याम)
Indonesia	Sumatra / Java / Sulawesi; **fourth** in 2016 four
Ethiopia	**Origin** story of arabica — **not** the 2016 tonnage king
India	Karnataka (**Kodagu, Chikmagalur, Hassan**) > Kerala (Wayanad) > Tamil Nadu (Nilgiri). Shade often **silver (रूपक) oak**

N.8 Cocoa

Country	Fact
Côte d'Ivoire (Ivory Coast)	World's usual **#1**
Ghana (घन)	#2 belt; quality name
Cameroon	Major (2021 list)
Nigeria	West African belt
Indonesia, Ecuador, Brazil	New World / SE Asia

Country	Fact
Latvia	**NOT** a cocoa country (2021) — Baltic **temperate forest**, not equator

N.9 Rubber

Country	Fact
Thailand	Usual world leader in recent textbook years
Indonesia	Large
Vietnam	Fast growth
Malaysia	Classic **estate** region; **not** Kinta Valley
India	**Kerala** (Kottayam–Ernakulam); also TN, KA, Andamans, NE pockets
China	Growing consumer-producer
Brazil	Origin; wild/native story, not the estate king now

N.10 Other plantation / tropical cash crops

Crop	Climate	World centre	Trap
Oil palm	Equatorial, high rain	**Indonesia + Malaysia**	Latvia / Kerala-only
Banana	Hot, wet, frost-free	India = huge **producer**; **Ecuador** (also Philippines, Costa Rica) = **export**	Equating producer with exporter
Coconut	Coastal tropics	Indonesia, Philippines, India	Philippines development = **Spanish and Americans**
Pepper (काली मिर्च)	Hot, wet vine	Vietnam / India / Indonesia (ranks move)	"Only Kerala in the world"
Cardamom	Humid hills	Guatemala + India (Kerala–Karnataka)	
Clove / nutmeg	Equatorial islands	Indonesia (Maluku / "Spice Islands")	
Tobacco	Warm, frost-careful	China, Brazil, India, USA	Not a slope-tea crop
Citrus	Med winter-rain + some subtropics	Spain, Italy, California, Brazil (orange juice)	**World belt name = Mediterranean** (2020)

N.11 Livestock, dairy, wool, silk

Item	Note
Cattle numbers	**India** often #1 inventory; then Brazil, China, USA
Beef export	Brazil, Australia, USA — **not** "India exports most beef because it has most cattle" as a blind fact
Dairy	**India** leads **milk** volume (domestic). **New Zealand**, the Netherlands and Denmark lead commercial **export** dairy. Do not treat India as the cheese-ship king
Sheep / wool	**Australia** (Merino), plus China and New Zealand. Wool is not cotton
Pigs	**China**
Ranching map	Pampas cattle, Australian Downs sheep, Veld, W USA
Silk	**China** volume; India second-story (mulberry + wild silks)

N.12 Shifting cultivation

Local name	Where
Jhum / jhoom	North-East India
Podu	Andhra / Odisha hills
Bewar / dahia	Madhya Pradesh
Penda	Chhattisgarh / Bastar
Kuruwa	Jharkhand
Kumari	Western Ghats / Kerala hills
Khil	Himalaya (हिमालय)
Valre / waltre	SE Rajasthan
Bringa / koman	Odisha pockets

Local name	Where
Ladang	**Malaysia**
Caingin	Philippines
Tamrai	Thailand
Chena	**Sri Lanka**
Humah	Java
Milpa	**Mexico / Central America**
Roca	**Brazil**
Masole	Congo basin
Ray	Vietnam
Fang	Equatorial Africa

N.13 World Green Revolution and FAO (short)

Note	Note
HYV package	Seeds + water + fertiliser + pesticide — same logic worldwide
Wheat	**Norman Borlaug**; CIMMYT **Mexico**; Nobel **Peace** (1970) — **not** "Nobel in Agriculture"
Rice	**IRRI**, **Los Baños, Philippines**
Spread	Mexico wheat into India/Pakistan; Philippines IR8 rice across Asia
FAO	Food and Agriculture Organization; HQ **Rome**; **World Food Day 16 October**
Golden rice	**Vitamin A** (correct 2018 pair)

N.14 GC extras — cash, livestock, terms

Crop / item	Usual coaching leader / note
Rice + wheat volume	**China** often leads both
Rice area	**India** often has the largest harvested area
Hybrid rice	Especially associated with **China**

Crop / item	Usual coaching leader / note
IRRI / Rice Gene Bank	Los Baños, **Philippines**
Sugarcane	**Brazil** leads
Sugar beet	**Russia** and **France** lead beet sugar
Ukraine sugar	Almost entirely from **beet**
Natural rubber	**Thailand** (with **Indonesia** close behind)
Coffee	**Brazil** and **Vietnam** lead; **Robusta** is an African species also grown in India
Tea leaves	**China** leads
Tobacco	**China** leads
Silk	**China** leads
Bananas	**India** leads
Milk volume	**India** leads
Castor	**India** often leads
Coconut	**Indonesia** leads
Maize	**USA** leads
Livestock inventory	**China** often leads total headcount

Note	Fact
Mocha coffee	**Yemen** (port (बंदरगाह) name)
Kona coffee	**Hawaii** (USA)
Sacramento–San Joaquin	**California** — grapes / citrus
Sugar Bowl	Classic tag for **Cuba**
Fazenda	Brazil plantation (coffee) — **not** shifting cultivation
Taungya	Agroforestry (कृषि वानिकी) / forest–farm system in **Myanmar**
Alley cropping	Crops in alleys between tree rows
Golden Crescent	Illicit opium — **Afghanistan, Iran, Pakistan** (not Iraq)
Shahtoosh	Tibetan antelope / **Chiru** hair (Tibet / Ladakh (लद्दाख) story)
Catechu (**kattha**)	From **Khair** (*Acacia catechu*) wood
Groundnut native	**Brazil** / Mato Grosso; Gambia's classic peanut economy
Alfalfa / Lucerne	Fodder legume (Fabaceae)
Heroin	From **opium**

Term	Recall
Apiculture	Bees / honey
Viticulture	Grapes (viniculture = wine side)
Pomology	Fruits
Olericulture	Vegetables only
Horticulture	Fruits + vegetables + flowers
Green Box (AoA)	Allowed support — research, pest control, training, marketing

Common Traps — Don’t Fall For These

1. **Wheat/rice/maize = plantation** — FALSE. **Tea**.
2. **Vietnam #1 coffee 2016** — FALSE. **Brazil**, then Vietnam.
3. **Latvia = cocoa** — FALSE. **Côte d’Ivoire / Ghana / Cameroon**.
4. **Kinta Valley = rubber** — FALSE. **Tin**.
5. **Gujarat = tea / Assam = wheat / UP = jute** — FALSE. **Kerala = rubber**.
6. **Jhum = tea estate** — FALSE. **Shifting subsistence**.
7. **Ladang = Mexico** — FALSE. **Malaysia**. Milpa = Mexico.
8. **Citrus = equatorial** — FALSE. **Mediterranean**.
9. **Philippines cane = Dutch** — FALSE. **Spanish and Americans**.
10. **Rubber originated in Kerala** — FALSE. **Amazon**; Kerala is India’s producer.
11. **Cocoa originated in West Africa** — FALSE. **Amazon**; Africa is production.
12. **Colombia first in 2016 coffee four** — FALSE. **Third**.
13. **Oil palm = Latvia** — FALSE. **Indonesia + Malaysia**.
14. **Shifting = intensive Ganga rice** — FALSE.
15. **Brazil lost all coffee** — FALSE. Still the 2016 **#1**.
16. **China = rice/wheat export king** — FALSE. Volume **producer**; exporters often others.
17. **Borlaug Nobel = Agriculture** — FALSE. **Peace**.
18. **Green tea = fermented** — FALSE. **Black** is fermented.
19. **India = banana export king because it produces most** — FALSE. **Ecuador** ships; India eats.
20. **Ranching = nomadic herding** — FALSE. Ranching is **commercial**.
21. **Von Thünen = Whittlesey** — FALSE. Rings vs world-type map.
22. **India = dairy export king because milk volume #1** — FALSE. Volume ≠ NZ/Netherlands export story.
23. **Soy = only Asian crop belt** — FALSE. USA–Brazil–Argentina triangle.
24. **Skip systems because papers asked tea/coffee** — FALSE. Next year can ask any Lucent system or crop belt.

Topic 21 — World Minerals & Energy

Current Affairs (this topic)

None mandatory. Producer ranks and "leading country" freeze to the **paper's year**. Do not chase every USGS revision in the teaching home.

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q83

Which country is the leading producer of gold in the world in the year 2023?

A. China B. U.S.A. C. Canada D. South Africa

Confused Pairs

Pair	Correct	Trap	Hindi
Ruhr	**Germany**	Britain / France / Russia	रुहर
Kuzbass	**Russia**	Ukraine / Germany	कुजबास
Donetsk	**Ukraine** coal	Kyiv / Kharkiv as the coal answer	दोनेत्स्क
Appalachian	**USA**	England	अप्लेशियन
Lancashire	**England**	USA / Germany	लंकाशायर
Mount Newman	**Iron**, Australia	Mn / Cu / bauxite	माउंट न्यूमैन
Krivoy Rog	**Ukraine** iron	Russia	क्रिवॉय रोग
Normandy iron	**France**	Germany (2020 wrong pair)	नॉर्मंडी
Karaganda	Kazakhstan (coal renown; 2020 iron list accepted)	Kuwait oil	कारागांदा
Chile copper	**Andes porphyry**, N Chile	"Chile has no copper"	चिली तांबा
Kinta Valley	**Tin**	Rubber / tea / coffee	किंटा
Pegu Yoma	**Mineral oil**	Tin (that is Kinta / Bangka / Tenasserim)	पेगू योमा
Kashagan	**Kazakhstan**	Kuwait	कशागन
Burgan	**Kuwait**	Kashagan dump	बुर्गान
Haft Kel	**Iran**	Iraq	हफ्त केल
Zubair	**Iraq**	Iran	ज़ुबैर
Dhahran	**Saudi Arabia**	Kuwait	धहरान
Methane	Natural gas **and** CNG	LPG (propane/butane)	मीथेन
Nuclear vs solar	Nuclear **not** stored solar	Marking only wind as non-solar	नाभिकीय
Iodine	**Chile**	Japan / USA / China as the 2018 fact	आयोडीन
Postmasburg	**Manganese**, South Africa	Uranium / mica / bauxite	पोस्टमासबर्ग
Weipa vs Newman	Weipa = **bauxite**; Newman = **iron**	Swap	वीपा / न्यूमैन
Witwatersrand	**Gold**, South Africa	Postmasburg Mn dump	विटवाटरसैंड
Sudbury	Canada **Ni + Cu**	Tin / oil	सडबरी

Pair	Correct	Trap	Hindi
North Field / South Pars	Qatar / Iran — **same** Gulf gas giant	Two unrelated fields	नॉर्थ फील्ड
OPEC vs IAEA	Both **Vienna**; OPEC = oil, IAEA = nuclear	Swap jobs	वियना
Brass vs bronze	Brass = Cu+Zn; bronze = Cu+Sn	Swap	पीतल / कांसा
Morocco	**Phosphate** reserves	Oil / copper	मोरक्को
Broken Hill	Australia **Pb–Zn**	Iron / tin	ब्रोकन हिल
Lithium Triangle	**Chile–Argentina–Bolivia** (brines)	Brazil in the triangle	लिथियम
Rare earths	**China** dominates processing	"India has zero REE"	दुर्लभ मृदा
German silver	Cu–Ni–Zn (**no Ag**)	Contains silver	जर्मन सिल्वर
Peace Pipeline	Iran–Pakistan gas	Russia–Europe only	पीस पाइपलाइन
Uranium City	**Canada**	Australia / USA	यूरेनियम सिटी
Kinta Valley	**Tin** (Malaysia)	Rubber	किंटा घाटी
Carajás vs Newman	Both **iron**; Carajás = Brazil, Newman = Australia	Swap countries	काराजास
Producer vs exporter (coal/iron)	China often huge **volume**; Australia/Indonesia **ship**	China = export king	

N.1 Coal

Type	Carbon	Character	Region
Peat	Lowest	First stage; high water	Ireland, Russia, Canada bogs
Lignite (brown coal)	Low	Soft; thermal only	Germany (Lusatia), Australia, Neyveli-type
Bituminous	Medium–high	Steam **and** coking	Ruhr, Appalachian, Damodar (दामोदर), Kuzbass
Anthracite	Highest	Hardest; cleanest burn	**Pennsylvania** (USA); South Wales

Role	Usual names
Volume miners	**China** (eats most of it), India, Indonesia, USA, Australia, Russia, South Africa
Exporters	**Indonesia, Australia, Russia, USA, South Africa**
Trap	China is **not** the classic export king — it is the volume **consumer**

Coalfield	Country	Note
Appalachian	**USA**	Eastern USA (not the Rockies as the name)
Pennsylvania anthracite	USA	Highest-rank US coal
Interior / Illinois	USA	Bituminous
Lancashire	**England**	Historic English coal–textile belt
Yorkshire / Northumberland / Midlands	England	Same island; do not place on the USA
South Wales	UK	anthracite
Ruhr	**Germany**	**Not** Britain / France / Russia
Saar	Germany–France border	Also revise
Lusatia / Rhineland	Germany	**Lignite**
Silesia (Upper)	**Poland**	
Kuzbass / Kuznetsk	**Russia**	Siberia

Coalfield	Country	Note
Pechora	Russia	Arctic
Kansk–Achinsk	Russia	Lignite / brown coal
Donets (Russian side)	Russia	Do not swap with Ukraine Donetsk in a "country" MCQ
Donetsk / Donbas	**Ukraine**	2023 largest among Kyiv / Kharkiv / Zaporizhzhia
Karaganda	Kazakhstan	Coal city; 2020 iron list also tagged it
Shanxi / Shaanxi / Inner Mongolia	**China**	Volume belt
Hunter Valley (NSW)	Australia	Export
Bowen (Queensland)	Australia	Export
Witbank / Highveld	South Africa	
Damodar — Jharia, Raniganj	India	Jharia = coking; Raniganj = West Bengal

N.2 Iron ore

Ore	Colour	Fe	Fact
Hematite	**Red**	~70%	**Bulk** ore of world trade (पण्याध्यक्ष)
Magnetite	**Black**	~72%	**Highest grade**; magnetic
Limonite	Yellow-brown	~60%	Hydrated; lower grade
Siderite	Grey-brown	~40%	Iron carbonate; poorest of the four (चातुर्याम)

Role	Usual names
Volume miners	Australia, Brazil, **China**, India, Russia
Exporters	**Australia** (Pilbara) and **Brazil** (Minas / Carajás)
Trap	China mines a lot and still **imports**. India is a miner, not the Pilbara-style ship king

Area	Country	Note
Mount Newman	**Australia**	2023. Pilbara / Hamersley
Hamersley Range	Australia	Same belt
Itabira	**Brazil**	Minas Gerais; old "iron quadrangle"
Carajás	Brazil	Giant Amazon-side mine
Cerro Bolívar	Venezuela	Also revise
Mesabi Range	**USA**	Lake Superior / Minnesota
Labrador Trough	Canada	Quebec–Labrador
Lorraine	**France**	Minette ore; **not** Germany
Normandy	**France**	**Not Germany** (2020 wrong pair)
Pyrenees	France	2020 accepted pair
Kiruna / Gällivare	**Sweden**	Magnetite; Arctic rail to Narvik
Bilbao / Basque	Spain	
Salzgitter / Siegerland	**Germany**	Real German iron — do **not** write Normandy
Cleveland / Scunthorpe	UK	Historical
Krivoy Rog / Kryvyi Rih	**Ukraine**	2020 correct pair. **Not** Russia
Kursk Magnetic Anomaly	Russia	

Area	Country	Note
Magnitogorsk / Urals	Russia	Steel town on the ore
Karaganda	Kazakhstan	Coal city; 2020 iron list treated as matched
Anshan / Wuhan	China	
Sishen	South Africa	
Liberia / Mauritania	W Africa	Classic export names
Odisha–Chhattisgarh–Jharkhand–Karnataka	India	Bailadila, Noamundi, Kudremukh — India-map detail

N.3 Copper

Place	Country	Note
Chuquicamata	**Chile**	Open-pit giant; Atacama / N Chile
El Teniente	Chile	One of the world's largest underground Cu mines
Escondida	Chile	Modern volume name
Andes (Peru)	Peru	2nd Andes producer
Katanga (Shaba)	**DRC**	African copper-cobalt
Copperbelt	**Zambia**	Ndola–Kitwe belt
Bingham Canyon	**USA**	Utah
Butte	USA	Montana; historical
Sudbury	Canada	**Nickel + copper** together
Norilsk	Russia	Ni–Cu–PGE
Grasberg	Indonesia	Cu + gold (Papua)
China	China	Huge **user**; also mines
Khetri / Malanjkhand / Singhbhum	India	India-map fact

N.4 Tin

Place	Country	Note
Kinta Valley	**Malaysia**	2020 PYQ. Rubber grows in Malaysia; Kinta still = **tin**
Bangka / Belitung	Indonesia	
Yunnan / Guangxi	China	Usual **volume** leader in modern lists
Potosí / Andes lodes	**Bolivia**	Highland tin
Andes	Peru	
Jos Plateau	Nigeria	Older textbook name
Cornwall	UK	Historical; worked out
Rondônia (राष्ट्रीय अन्वेषण एजेंसी)	Brazil	Also revise

N.5 Petroleum

Field / area	Country	Trap
Ghawar	Saudi Arabia	Often tagged world's largest conventional field

Field / area	Country	Trap
Dhahran	**Saudi Arabia**	2024 correct pair
Abqaiq / Safaniya	Saudi Arabia	Safaniya = offshore
Ras Tanura	Saudi Arabia	Export **terminal**, not a field name swap
Burgan	**Kuwait**	Kuwait's giant. **Not** Kashagan
Kirkuk / Mosul	**Iraq**	North Iraq
Rumaila / **Zubair** / Basra	Iraq	South Iraq. Zubair = 2024 correct
Haft Kel	**Iran**	2024 correct. **Not** Iraq
Masjid-i-Suleiman	Iran	First Middle-East commercial well (1908 story)
Agha Jari / Gachsaran	Iran	
Kashagan	**Kazakhstan**	Caspian offshore. **Kuwait–Kashagan = WRONG** (2024)
Tengiz	Kazakhstan	Onshore Caspian
Baku	Azerbaijan	Caspian; Nobel/old Russian oil story
West Siberia / **Samotlor**	Russia	Volume belt
Volga–Urals	Russia	Older Russian belt
North Sea	**UK + Norway**	Ekofisk (NO), Forties/Brent (UK)
Prudhoe Bay	USA (Alaska)	
East Texas / Permian / Gulf of Mexico	USA	
Maracaibo / Orinoco	Venezuela	Heavy oil / sands story
Cantarell	Mexico	
Campos / pre-salt	Brazil	
Hassi Messaoud	Algeria	
Bonny / Niger Delta (डेल्टा)	Nigeria	
Dukhan	Qatar	
Murban	UAE	
Daqing	China	
Minas	Indonesia	
Seria	Brunei	
Yenangyaung / Chauk	**Myanmar**	With **Pegu Yoma** oil (2022)
Mumbai High / Digboi / Ankleshwar	India	India-map fact

N.6 Natural gas

Role	Usual names
Reserves	**Russia, Iran, Qatar**, USA, Turkmenistan
Production	**USA, Russia**, Iran, China, Qatar, Australia, Norway
LNG export	**Qatar, Australia**, USA

Field / area	Country	Note
Urengoy / Yamburg / Medvezhye	Russia	West Siberia
North Field	**Qatar**	Same giant reservoir as Iran's South Pars

Field / area	Country	Note
South Pars	**Iran**	Persian (फ़ारसी) Gulf; world's largest gas accumulation with Qatar
Groningen	Netherlands	Historic European giant
Troll / Ekofisk gas	Norway	North Sea
Hugoton / Marcellus shale	USA	
Hassi R'Mel	Algeria	
Galkynysh	Turkmenistan	
Shah Deniz	Azerbaijan	Caspian; pipelines to Turkey/EU
NW Shelf / Gorgon	Australia	LNG
North Sea	UK + Norway	Oil **and** gas
KG / Mumbai High gas	India	India-map fact

N.7 Energy resources

Project	River	Country
Three Gorges	Yangtze	**China**
Itaipu	Paraná	**Brazil–Paraguay**
Grand Coulee	Columbia	USA
Hoover	Colorado	USA
Aswan High	Nile	Egypt (मिस्र)
Guri	Caroní	Venezuela
Churchill Falls	Churchill	Canada
Sayano–Shushenskaya	Yenisei	Russia

Country	Fact
Iceland	Power + heating
New Zealand	Rotorua / Taupo
Philippines	Ring of Fire
USA	The Geysers, California
Italy	Larderello (first plant)

Body	HQ
OPEC	Vienna
IAEA	Vienna
IEA	Paris
IRENA	Abu Dhabi

N.8 Other world minerals

Place	Country
Postmasburg / Kalahari	**South Africa** (2020 — not U / mica / bauxite)
Moanda	Gabon

Place	Country
Groote Eylandt	Australia
Nikopol	Ukraine
Odisha / Maharashtra	India

Place	Country
Weipa / Gove / Darling Range	**Australia**
—	**Guinea** (reserves renown)
Jamaica / Suriname	Caribbean–Guiana
Trombetas	Brazil
Les Baux	France (name origin)
Odisha–Gujarat–Jharkhand	India

Place	Country
Witwatersrand	South Africa
Kalgoorlie	Australia
Yukon / Klondike	Canada
Homestake	USA (historical)
Grasberg	Indonesia (with copper)
—	China (modern volume)

Mineral	World centre
Sheet **mica**	**India** (Jharkhand–Andhra–Rajasthan) classic
Graphite	China; India; Sri Lanka (vein)
Sulphur	USA, Mexico, Poland; volcanic Italy/Japan
Platinum	Bushveld (South Africa); Norilsk
Cobalt	**DRC** (with Katanga copper)
Lithium	Australia hard-rock (spodumene); **Lithium Triangle** brines = **Chile–Argentina–Bolivia** (Brazil is **not** in the triangle)
Rare earths (REE)	**China** dominates mining **and** especially **processing**; India has monazite sands but not China's refining fact
Tungsten	**China**; Bolivia
Mercury	Almadén (Spain) historical
Asbestos	Russia, Kazakhstan; Canada historical

N.9 GC extras — silver German · oil · uranium city

Note	Fact
German silver	Cu–Ni–Zn alloy — **no silver** (name from appearance)
Silver volume	Often **Mexico** #1 in recent frames
Aluminium ore	**Bauxite** (≠ haematite / magnetite / limonite)
Primary Al volume	Often **China** first in recent frames
Tin	Mostly **placer**; **Kinta Valley** (Malaysia) = tin
Brent crude	Major crude class; sourced from **North Sea**; low sulphur — not "zero sulphur"

Note	Fact
Gasohol	~90% gasoline + ~10% ethanol; USA classic producer/consumer
Peace Pipeline	**Iran–Pakistan** gas pipeline name
First commercial oil (1857 register)	**Romania** (Drake 1859 Titusville is the other classic "first well" story)
Crude reserves	Often **Venezuela** #1 proved in IMYB frames; Saudi #2
SE Asia oil	**Indonesia**
Venezuela	Oil-dominated export economy
Ras Tanura	Refinery — **Saudi Arabia**
Pegu Yoma (Myanmar)	**Mineral oil** pair
Uranium City	**Canada** (Athabasca)
U production	Often **Kazakhstan** #1; reserves **Australia** (Olympic Dam)
Radium	From **pitchblende** (uranium ore)
Rare earths	**China** dominates; export curbs + electronics demand
Lithium Afghanistan find	Compared with oil discovery in Iran
Chile	**Nitrate / iodine** (Atacama caliche)
Diamond volume	Often **Russia**
Cobalt (~¾ of batteries)	**DRC**
Camphor	Tree native to **China/Japan** (*Cinnamomum camphora*)

Common Traps — Don't Fall For These

1. **Ruhr = Britain / France / Russia** — FALSE. **Germany**.
2. **Kuzbass = Ukraine** — FALSE. **Russia**. Donetsk = Ukraine.
3. **Appalachian = England** — FALSE. **USA**. Lancashire = England.
4. **Mount Newman = manganese** — FALSE. **Iron**, Australia. Mn = **Postmasburg**.
5. **Germany–Normandy iron** — FALSE. Normandy = **France**.
6. **Chile has no copper** — FALSE. 2022 **leading** + Andes **porphyry**.
7. **Kinta = rubber** — FALSE. **Tin**.
8. **Pegu Yoma = tin** — FALSE. **Mineral oil**.
9. **Kuwait = Kashagan** — FALSE. Kashagan = **Kazakhstan**. Kuwait = **Burgan**.
10. **LPG = methane / CNG** — FALSE. CNG = methane; LPG = propane+butane.
11. **Iodine = Japan 2018** — FALSE. **Chile**.
12. **Kyiv = Ukraine's coal centre** — FALSE. **Donetsk**.
13. **Nuclear / geothermal = stored solar** — FALSE.
14. **2022 steel #1 = India** — FALSE. **China**.
15. **Haft Kel = Iraq** — FALSE. **Iran**. Zubair = Iraq. Dhahran = Saudi.
16. **Mount Newman = bauxite** — FALSE. **Iron**. Weipa = bauxite.
17. **Postmasburg = gold** — FALSE. **Manganese**. Witwatersrand = gold.
18. **Pegu Yoma = Kinta tin** — FALSE. Pegu = **oil**; Kinta = **tin**.
19. **North Field = Russian Urengoy** — FALSE. **Qatar** (pair = Iran South Pars).
20. **OPEC HQ = Riyadh / Geneva** — FALSE. **Vienna**.
21. **Lithium Triangle includes Brazil** — FALSE. Chile–Argentina–Bolivia.
22. **Rare earths = Chile only** — FALSE. Processing fact = **China**.

23. **Carajás = Mount Newman** — FALSE. Carajás = Brazil iron; Newman = Australia iron.
24. **Skip non-PYQ fields because papers asked Ruhr/Kinta** — FALSE. Next year can ask any Lucent field.

Topic 22 — World Industries

Current Affairs (this topic)

None mandatory.

Confused Pairs

Pair	Correct	Trap	Hindi
Osaka	**Cotton textile**	Auto / ship	ओसाका
Detroit	**Automobile**	Cotton	डेट्रॉइट
Cuba	**Cigar**	Ship	क्यूबा
St. Petersburg	**Shipbuilding**	Cotton	सेंट पीटर्सबर्ग
Igarka	**Russia** (Yenisei timber)	China	इगारका
Rotterdam	**Netherlands**	Germany / Belgium	रॉटरडैम
Montevideo	**Uruguay**	Argentina	मोंटेवीडियो
Suez waters	**Mediterranean + Red Sea**	Atlantic as a canal bank	स्वेज
Suez cut	**~7,000 km** India–Europe	5,000 / 10,000	7,000 किमी
Suez lakes N→S	Manzala → Timsah → Great Bitter → Little Bitter	Reverse / skip Timsah	मंज़ाला पहले
Panama	Atlantic (Caribbean) ↔ **Pacific**	Med ↔ Red (that is Suez)	पनामा
Kiel	North Sea ↔ **Baltic**	Med	कील
Chinook	Warm **dry**, **Rockies**	Cold wind	चिनूक
Foehn	**Alps**, warm dry	Australia	फॉन
Mistral	Southern **France** (Rhône)	**Australia**	मिस्ट्रल
Shamal	**Arabia / Persian Gulf**	**Austria**	शमाल
Brickfielder	**Australia**	France	ब्रिकफील्डर
Santa Ana	**California**	Spain	सांता आना
Harmattan	West Africa	Alps	हरमट्टन
Bora	Adriatic (cold)	Warm Chinook dump	बोरा
Suez vs Panama	Suez = no facts, Med–Red; Panama = **facts**, Atlantic–Pacific	Swap	स्वेज / पनामा
Duisburg	Inland Rhine port, **Germany**	Netherlands / sea entrepôt	ड्यूसबर्ग
Willy-willy	Australian **cyclone**	Local wind like Brickfielder	विली-विली
Akron	**Tyres / rubber** (USA)	Steel	एक्रॉन
Toulouse	**Airbus**	Boeing (Seattle)	टूलूज़
Khamsin	**Egypt**	Australia	खामसिन
Osaka vs Detroit (Japan)	Osaka = **Manchester** (cotton); Nagoya = **Detroit** (auto)	Swap nicknames	
Footloose vs steel	Footloose = electronics / R&D; steel = ore + coal	Electronics on coalfield only	
Silicon Valley	California Bay Area **electronics / IT**	Detroit auto dump	सिलिकॉन वैली

N.1 Industrial location factors

Factor	Industry
Ore + coking coal	Iron and steel (Ruhr, Pittsburgh, Damodar (दामोदर))
Sugarcane / timber	Sugar mill, sawmill — bulky, perishable
Limestone	Cement — bulky
Cheap **hydel**	**Aluminium**, electro-chemicals
Cheap / skilled labour	Cotton textiles, garments, watches
Market	Bread, printing, consumer goods
Break-of-bulk (port, lake, rail)	Oil refining, flour, lake-steel
Agglomeration	Auto parts around Detroit / Nagoya / Turin
Government / SEZ	Planned estates
Capital + R&D	Pharma, electronics

Industry	Typical location
Iron–steel	Coalfield **or** ore **or** lake/port mixing both (Mesabi ore + Appalachian coal → Pittsburgh–Great Lakes)
Cotton textile	Humid labour districts (Lancashire, Osaka, Mumbai–Ahmedabad (अहमदाबाद))
Wool	Sheep hinterland + mill town (Yorkshire)
Shipbuilding	Deep estuary (ज्वारनदमुख) (Glasgow, St. Petersburg, Yokohama, Busan, Belfast)
Automobiles	Market + parts cluster (Detroit, Nagoya, Wolfsburg, Turin)
Oil refining	Port / pipeline end (Rotterdam, Houston, Abadan, Jamnagar)
Aircraft	Tech + large airfield (Seattle Boeing; **Toulouse** Airbus)
Films	Hollywood / Los Angeles
Watches	Skilled Alps–Jura (Switzerland)
Electronics	**Footloose** — Silicon Valley, East Asia coasts

N.2 Industrial regions

Region	Country	Fact
Ruhr	Germany	Coal, steel, heavy engineering; Essen–Dortmund–Duisburg
Saar	Germany–France border	Coal–steel
Lorraine	France	Iron ore + steel
Sambre–Meuse	Belgium	Coal–steel
Randstad	Netherlands	Rotterdam port + manufacturing
Upper Silesia	Poland	Coal–steel
Catalonia	Spain	Barcelona textiles / diversified
Lancashire	England	**Cotton** — Manchester–Liverpool
Yorkshire	England	**Wool** — Leeds–Bradford
Midlands / Black Country	England	Birmingham engineering
South Wales	UK	Coal–steel (historical)
Clydeside / Scottish Lowlands	UK	Glasgow **shipbuilding**
Po Basin	Italy	Milan–Turin–Genoa

Region	Country	Fact
Moscow–Nizhny	Russia	Diversified
Donbas	Ukraine	Coal–steel
Kuzbass	Russia	Coal–steel, Siberia
Urals / Magnitogorsk	Russia	Ore + steel

Region	Fact
New England	Early cotton; now diversified
New York–Philadelphia	Port, chemicals, publishing
Pittsburgh–Lake Erie	Steel — Appalachian coal + Mesabi ore via lakes
Chicago–Gary	Steel, machinery, meat-packing
Detroit	Automobiles
Birmingham (Alabama)	Southern steel
Gulf Coast	Petrochemicals (Houston)
Southern California	Aerospace, films, LA–Long Beach port
Silicon Valley	Electronics
Windsor–Ontario	Canada auto (next to Detroit)

Region	Fact
Keihin	Tokyo–Yokohama — Japan's largest
Hanshin	Osaka–Kobe — cotton, ship, port
Chukyo	Nagoya — automobiles
Kitakyushu / N Kyushu	Japan — steel
Setouchi	Inland Sea, Japan — heavy industry
Shanghai–Nanjing / Yangtze	China — port + diversified
Pearl River Delta	Guangzhou–Shenzhen–Hong Kong
Beijing–Tianjin	China — north coast
Manchuria / Anshan–Shenyang	China — steel
Wuhan	China — steel on Yangtze
Seoul–Incheon	South Korea — auto, port
Busan	South Korea — ship + port

Region	Fact
São Paulo–Santos	Brazil
Buenos Aires	Argentina
Witwatersrand	South Africa — gold + industry
SE Australia	Sydney–Melbourne
Mumbai–Pune (पुणे)	India — cotton, engineering
Hugli	India — jute, engineering
Chotanagpur (छोटानागपुर)	India — minerals, steel

N.3 Industrial cities

Centre	Industry
Osaka	**Cotton textile**
Detroit	**Automobile**
Cuba (Havana)	**Cigar**
St. Petersburg	**Shipbuilding**

Nickname	City	Fact
Manchester of Japan	**Osaka**	Cotton
Detroit of Japan	**Nagoya** (Chukyo)	Automobiles
Pittsburgh of Japan	**Kawasaki**	Steel / heavy
Russian Manchester	**Ivanovo**	Cotton textiles

City	Industry
Manchester	Cotton
Ahmedabad	Cotton (“Manchester of India”)
Leeds / Bradford	Wool
Lyon	Silk
Milan	Textiles / fashion
Dundee	Jute (historical)
Sheffield	Steel / cutlery
Solingen	Cutlery (Germany)
Essen	Krupp steel / engineering
Birmingham (UK)	Engineering
Pittsburgh	Steel
Chicago	Meat-packing, steel, machinery
Minneapolis	Flour
Milwaukee	Brewing
Akron	Rubber / tyres
Hollywood / Los Angeles	Films
Glasgow / Belfast / Gdansk	Shipbuilding
Yokohama / Busan	Shipbuilding
Turin / Wolfsburg / Nagoya / Stuttgart	Automobiles
Magnitogorsk / Anshan / Jamshedpur	Steel
Zurich / Jura	Watches
Dresden / Limoges	Porcelain
Toulouse	Airbus aircraft
Seattle	Boeing aircraft
Bournville	Chocolate
Havana	Cigar (Cuba row in 2022)
Abadan / Houston	Oil refining

City	Industry
Johannesburg	Gold
Baku (Azerbaijan)	Petroleum / oil refinery
Cadiz (Spain)	Cork
Venice (Italy)	Shipbuilding / glass
Nepanagar (MP, India)	Newsprint paper
Firozabad (UP, India)	Glass bangles
Moradabad (UP, India)	Brassware / cutlery
Kolar (Karnataka, India)	Gold mines
Khetri (Rajasthan, India)	Copper mines
Chittaranjan (WB, India)	Railway locomotives
Perambur (TN, India)	Integral Coach Factory (rail coaches)
Titagarh (WB, India)	Paper and jute
Sialkot (Pakistan)	Sports goods
Sindri (Jharkhand, India)	Chemical fertilizers
Dhariwal / Ludhiana (Punjab, India)	Woolen goods / hosiery
Coimbatore	Cotton / engineering (India)

N.4 Ports and seaports

Type	Recall	Example
Outport	Deep gate for up-river city	Cuxhaven, Tilbury, Le Havre, Bremerhaven
Entrepôt	Re-export hub	Singapore, Rotterdam, Hong Kong
Oil port	Crude / product	Ras Tanura, Houston, Rotterdam, Abadan
Container	Box trade	**Shanghai**, Singapore, Ningbo, Shenzhen, Busan, Rotterdam
Naval	Fleet base	Portsmouth, Pearl Harbor, Vladivostok
Packet	Short ferry	Dover–Calais
Inland / river	Sea-going on river/lakes	**Igarka**, Duisburg, Chicago, Kolkata

Port	Country
Rotterdam	**Netherlands** — Rhine mouth; Europort; 2023 correct
Antwerp	Belgium
Hamburg	Germany
Duisburg	Germany — largest **inland** port
London / Southampton	UK
Marseille	France
Genoa / Trieste	Italy
Piraeus	Greece
Gdansk	Poland

Port	Country
Shanghai / Ningbo / Shenzhen / Guangzhou	China — container volume

Port	Country
Hong Kong	China
Singapore	Singapore — Malacca entrepôt
Yokohama / Tokyo / Kobe / Nagoya	Japan
Busan / Incheon	South Korea
Jakarta / Tanjung Priok	**Indonesia** — 2023 correct
Manila	Philippines
Bangkok / Laem Chabang	Thailand
Ho Chi Minh	Vietnam
Colombo	Sri Lanka
Karachi / Gwadar	Pakistan
Chittagong	Bangladesh
Yangon	Myanmar
Jebel Ali (Dubai)	UAE
Bandar Abbas	Iran
Basra / Umm Qasr	Iraq
Aden / Djibouti	Red Sea–Aden gate
Mumbai / JNPT / Chennai / Kolkata / Kandla / Paradip / Vizag	India

Port	Country
New York / Houston / LA–Long Beach / New Orleans	USA
Vancouver / Montreal / Halifax	Canada
Santos	Brazil (São Paulo's gate)
Buenos Aires	Argentina
Montevideo	**Uruguay** — 2023 correct; **not** Buenos Aires
Callao	Peru
Valparaíso	Chile
Durban / Cape Town	South Africa
Lagos	Nigeria
Mombasa	Kenya
Alexandria / **Port Said** / Suez	Egypt
Sydney / Melbourne / Fremantle	Australia
Auckland	New Zealand
Igarka	**Russia** — Yenisei timber; **not China**
Vladivostok / Novorossiysk / Murmansk	Russia

N.5 Trade routes

Canal / seaway	Links	Extra
Kiel	North Sea ↔ **Baltic**	Germany; not Med
Corinth	Ionian ↔ Aegean	Greece; ship canal

Canal / seaway	Links	Extra
Soo (Sault Ste. Marie)	Lake Superior ↔ Huron	USA–Canada
Welland	Erie ↔ Ontario	Bypasses Niagara
St Lawrence (लॉरेंस) Seaway	Great Lakes ↔ Atlantic	USA–Canada
Rhine–Main–Danube	North Sea ↔ Black Sea inland	Europe
Volga–Don	Caspian system ↔ Black Sea	Russia

Passage	Gates
Malacca	Indian Ocean ↔ South China Sea (Singapore)
Sunda / Lombok	Alternate Indonesia routes
Hormuz	Persian (फ़ारसी) Gulf oil
Bab el-Mandeb	Red Sea ↔ Gulf of Aden
Gibraltar	Med ↔ Atlantic
Bosporus / Dardanelles	Black Sea ↔ Med
Dover	English Channel
Bering	Arctic ↔ Pacific
Magellan	S America (inshore of Cape Horn)
Cape of Good Hope	South of Africa — pre-Suez / ULCC tankers / Suez blockage
Cape Horn	South of America — pre-Panama

Common Traps — Don't Fall For These

1. **Osaka = automobile** — FALSE. **Cotton**. Detroit = auto.
2. **Cuba = shipbuilding** — FALSE. **Cigar**. St. Petersburg = ships.
3. **Igarka = China** — FALSE. **Russia**.
4. **Suez = Atlantic + Med** — FALSE. **Red Sea + Med**.
5. **Suez cut = 10,000 km** — FALSE. **~7,000 km**.
6. **Suez lakes start at Little Bitter** — FALSE. North door = **Manzala**.
7. **Panama = Med–Red** — FALSE. That is Suez. Panama = Atlantic–Pacific.
8. **Kiel = Med** — FALSE. **North Sea–Baltic**.
9. **Mistral = Australia** — FALSE. **France**. Brickfielder = Australia.
10. **Shamal = Austria** — FALSE. **Arabia**.
11. **Chinook is cold** — FALSE. Warm **and dry**.
12. **Foehn = Rockies** — FALSE. **Alps**. Chinook = Rockies.
13. **Rotterdam = Germany** — FALSE. **Netherlands**.
14. **Montevideo = Argentina** — FALSE. **Uruguay**.
15. **Lancashire = wool** — FALSE. **Cotton**. Yorkshire = wool.
16. **Hanshin = Tokyo** — FALSE. Osaka–Kobe. Keihin = Tokyo–Yokohama.
17. **Suez has Panama-style facts** — FALSE. Suez is **sea-level**.
18. **Willy-willy = Brickfielder** — FALSE. Willy-willy is a **cyclone**.
19. **Duisburg = Netherlands sea port** — FALSE. **German inland** Rhine port.
20. **Toulouse = Boeing** — FALSE. **Airbus**. Boeing = Seattle.
21. **Manchester of Japan = Nagoya** — FALSE. **Osaka**. Nagoya = Detroit of Japan.
22. **Footloose = steel on coal only** — FALSE. Electronics / R&D flexible location.

23. **Silicon Valley = Detroit** — FALSE. California electronics / IT.
24. **Skip non-PYQ cities/winds** — FALSE. Next year can ask any Lucent pair on this map.

Topic 23 — Political & Map-Based Geography

Current Affairs (this topic)

None mandatory. 12 / 24 / 200 nm are static.

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q89

What is the correct latitudinal extent of the mainland of India?

A. 6 degrees 45′ N to 37 degrees 6′ N B. 6 degrees 4′ N to 36 degrees 7′ N C. 8 degrees 4′ N to 36 degrees 12′ N D. 8 degrees 4′ N to 37 degrees 6′ N

Confused Pairs

Pair	Correct	Trap	Hindi
Territorial vs EEZ	**12 nm** vs **200 nm**	Swapped	क्षेत्रीय जल / EEZ
Contiguous	**24 nm**	12 or 200	सन्निहित
Shelf vs EEZ	Shelf may go to **350 nm**; EEZ stops **200**	Shelf = 12	महाद्वीपीय मग्नतट
McMahon	**India–China** (Arunachal)	Pakistan / Nepal / Myanmar	मैकमोहन
Durand	**Pakistan–Afghanistan**	India–China	डूरंड
Radcliffe	**1947** India–Pak / Bangladesh	McMahon	रैडक्लिफ
Longest coast	**Gujarat**	AP / MH / Kerala	गुजरात
Longest land border	**Bangladesh**	China / Pakistan	बांग्लादेश
Shortest land border	**Afghanistan**	Bhutan	अफ़ग़ानिस्तान
Suez waters	**Med + Red Sea**	Atlantic	स्वेज
Panama	Atlantic ↔ Pacific	Med–Red	पनामा
Kiel	North Sea ↔ Baltic	Med	कील
Hormuz	Persian Gulf ↔ Gulf of Oman	Malacca	होरमुज़
Gibraltar	Med ↔ Atlantic	Bosporus	जिब्राल्टर
Bosporus	Black Sea ↔ Marmara	Gibraltar	बोस्फोरस
Mistral / Köppen Cs	Med **winter rain**	Rain all year (that is Cfb W Europe)	भूमध्य
Koppen vs Thornthwaite	**Thornthwaite** = "vegetation is true index of climate"	Köppen letter codes	थॉर्नथ्वेट / कोपेन
Bolivia	**Landlocked**	Uruguay / Peru / Suriname	बोलीविया
Nobi / Kanto	**Japan**	Korea / China	जापान
Igarka	**Russia**	China	इगारका
Cape Verde capital	**Praia**	Bamako (Mali)	प्राइया
India area rank	**7th** largest; **2.4%** of world	6th; wholly tropical	भारत क्षेत्र
Tropic of Cancer / UP	Passes 8 states — **not UP**	UP on the Tropic	कर्क रेखा
Double landlocked	**Uzbekistan**, Liechtenstein	Nepal / Bolivia as "double"	दोहरा स्थलरुद्ध

Pair	Correct	Trap	Hindi
Kazakhstan	Largest **landlocked by area**	Has ocean coast	कज़ाखस्तान
Lesotho	**Enclave** inside South Africa	Coastal SA dump	लेसोथो
49th Parallel	**USA–Canada** (long stretch)	USA–Mexico	49वीं
38th Parallel	**North–South Korea**	India–China	38वीं
Horn of Africa	Djibouti, Eritrea, Ethiopia, Somalia	Includes Sudan	हॉर्न ऑफ अफ्रीका
Slovenia capital	**Ljubljana**	Bratislava (Slovakia)	ल्युब्लियाना
Gold Coast	Old name of **Ghana**	Dutch Guiana / Suriname	गोल्ड कोस्ट
Great Britain vs UK	GB = England+Wales+Scotland	Includes Northern Ireland	ग्रेट ब्रिटेन
Greenland	Denmark politically; N America geographically	Independent continent	ग्रीनलैंड
Caspian littoral	Five states; not Armenia/Iraq	Armenia on Caspian	कैस्पियन
Oceania	Melanesia/Micronesia/Polynesia + Australia	Indonesia included	ओशिआनिया
Laos	Only SE Asia **landlocked**	Cambodia / Thailand	लाओस
Rio Grande	**USA–Mexico**	USA–Canada	रियो ग्रांडे
Baikonur	**Kazakhstan**	Ukraine	बैकोनुर
Midnight Sun vs Rising Sun	Midnight Sun = **Norway** (Arctic); Rising Sun = **Japan**	Swapped	मध्यरात्रि सूर्य / उगता सूर्य
Morning Calm vs Rising Sun	Morning Calm = **Korea**; Rising Sun = Japan	Korea = Japan	सुबह की शांति
Thousand Lakes	**Finland**	Switzerland / Netherlands	हज़ार झीलें
Bird Continent	**South America**	Australia	पक्षी महाद्वीप
Mistress of Eastern Sea	**Sri Lanka**	India / Myanmar	पूर्वी सागर की स्वामिनी
Manchester of the East	**Osaka**	Tokyo / Nagoya	पूर्व का मैनचेस्टर
Pearl of Siberia	**Baikal**	Great Bear	साइबेरिया का मोती
Venice of the North	**St. Petersburg** (usual Russia key)	Moscow / Vladivostok	उत्तर का वेनिस
City of Smoke	**Chicago**	London / Kolkata	धुएँ का शहर
Oil Capital of Europe	**Aberdeen**	Liverpool / Belfast	यूरोप की तेल राजधानी
Land of Promise	**Mindanao**	Java / Cuba	वादा द्वीप
Ninety East Ridge	**Indian Ocean**	Pacific / Atlantic	नाइंटी ईस्ट रिज

N.3 Boundary lines

Other lines (one home)

Line	Note
49th Parallel	Long USA–**Canada** international border (not USA–Mexico)
38th Parallel	**North Korea–South Korea** armistice border
Maginot Line	Historical **France–Germany** fortification line (WWII)
Oder–Neisse Line	**Germany–Poland** post-WWII border
Hindenburg Line	WWI German boundary line between Germany and Poland/Russia
Mannerheim Line	Historic Finland / Russian front fortification line
17th Parallel	Historical North & South Vietnam boundary line (also claimed line in Sir Creek / Rann of Kutch disputes)

Line	Note
Rio Grande	Forms much of **USA–Mexico** border
Alpine / Little Maginot	France–Italy fortification line
Siegfried Line	German fortification facing the French Maginot Line

Central Asia capitals (match bank)

Country	Capital	Trap
Uzbekistan ★	**Tashkent**	—
Tajikistan ★	**Dushanbe**	Not Ashgabat
Kyrgyzstan ★	**Bishkek**	—
Turkmenistan ★	**Ashgabat**	Not Dushanbe
Kazakhstan	**Astana** (Nur-Sultan phase)	Almaty is **not** the capital
Afghanistan	Kabul	Neighbourhood
Iran	Tehran	Neighbourhood

N.5A Country ranks · regions · dependent territories · old names

Region	Members / pair
Horn of Africa	Djibouti, Eritrea, Ethiopia, **Somalia** — **not Sudan**
Balkans	Albania, Bosnia and Herzegovina, Bulgaria, Croatia, Greece, Kosovo, Montenegro, North Macedonia, Romania, Serbia, Slovenia, and European Türkiye (Eastern Thrace). **Austria is not** Balkan.
Scandinavia (geo)	Norway, Sweden, Denmark — Finland often cultural/historical set but MCQ may exclude Finland
Central America	Belize, Costa Rica, El Salvador, Guatemala, Honduras, Nicaragua, Panama — **not Sierra Leone** (W Africa)
Oceania	Melanesia / Micronesia / Polynesia + Australia — **Indonesia not** Oceania
Arabian Peninsula	Saudi Arabia, Yemen, Oman, UAE, Bahrain, Qatar, Kuwait (+ Iraq fringe) — **Syria not**
UAE emirates	Abu Dhabi, Dubai, Sharjah, Ajman, Umm Al-Quwain, Fujairah, Ras Al-Khaimah
Gaza Strip	Borders **Egypt** and **Israel** (Mediterranean strip)
Caspian shores	Russia, Kazakhstan, Turkmenistan, Iran, Azerbaijan — **not Armenia / Iraq**
Dead Sea	Israel / West Bank / Jordan — **not Lebanon**
Afghanistan neighbours	Iran, Pakistan, China, Tajikistan, Uzbekistan, Turkmenistan (+ India SE note) — **not Russia / Azerbaijan / Kyrgyzstan**
Myanmar neighbours	India, Bangladesh, China, Laos, Thailand — **not Vietnam/Cambodia/Malaysia as the trio**
Israel land borders	Lebanon, Syria, Jordan, Egypt
Chile	Longest N–S extension among common traps (W South America)
Japan	Largest among common "island without land border" options vs NZ / Philippines / Cuba
Indonesia land borders	Malaysia, Papua New Guinea, Timor-Leste — **not Brunei**

Place	Owner / note
Greenland	Denmark politically; geographically **N America**; largest island

Place	Owner / note
Christmas Island	Australia (Indian Ocean)
Martinique	France (Caribbean)
Falklands / St Helena / Gibraltar / Pitcairn / BIOT	UK
Puerto Rico / American Samoa / US Virgin / Midway	USA
Canaries / Ceuta / Melilla / Balearics	Spain
Azores / Madeira	Portugal
Cook / Niue / Tokelau / Ross Dependency	NZ association
Chechnya	Republic of **Russia** (Caucasus; capital Grozny)
Aleutian	USA; **Bear** Norway; **Franz Josef** Russia

Old Name	New Name
Siam	**Thailand**
Formosa	**Taiwan**
Mesopotamia (मेसोपोटामिया)	**Iraq**
Burma	**Myanmar**
Gold Coast	**Ghana** (घन)
Dutch Guiana	**Suriname**
Dutch East Indies	**Indonesia**
British Guiana	**Guyana**
Southern Rhodesia	**Zimbabwe** (capital Harare was Salisbury)
Northern Rhodesia	**Zambia**
Abyssinia	**Ethiopia**
Basutoland	**Lesotho**
Bechuanaland	**Botswana**
Zaire	**Republic of the Congo / DR Congo**
Nyasaland	**Malawi**
South West Africa	**Namibia**
Dahomey	**Benin**
Upper Volta	**Burkina Faso**
Ceylon	**Sri Lanka**
Batavia	**Jakarta**
Peking	**Beijing**
Angora	**Ankara**
Constantinople	**Istanbul**
Saigon	**Ho Chi Minh City**
Petrograd / Leningrad	**St. Petersburg**
Stalingrad	**Volgograd**

N.5B Geographical sobriquets

Sobriquet	Place
Land of the Midnight Sun	**Norway** (Arctic set)
Bird Continent / Continent of Birds	**South America**
Mistress of the Eastern Sea / Pearl of the Indian Ocean	**Sri Lanka**
Gateway to Asia	**Singapore**
Land of White Elephants	**Thailand**
Land of the Rising Sun / Nippon	**Japan**
Land of Thousand Lakes / Garden of Lakes	**Finland**
Land of Morning Calm	**(South) Korea**
Dark Continent	**Africa**
Land of Thunderbolts	**Bhutan**
Land of Golden Pagodas	**Myanmar**
Land of Kangaroos	**Australia**
Britain of the South	**New Zealand**
Key of the Mediterranean	**Gibraltar**
Holy Land	Palestine / Jerusalem
Sick Man of Europe	**Türkiye**
Playground of Europe	**Switzerland**
Island of Cloves	**Zanzibar**
Pearl of the Antilles / Sugar Bowl of the World	**Cuba**
Island of Pearls / Pearl of the Persian (फ़ारसी) Gulf	**Bahrain**
Gift of the Nile	**Egypt**
Land of Canals	**Pakistan** (coaching tag)
Blue Grass State	**Kentucky** (USA)
Land of Promise	**Mindanao** (Philippines)

Sobriquet	Place
Venice of the North	**St. Petersburg** (Russia) / Stockholm (Sweden)
Gateway to the West	**Istanbul**
City of Canals / Queen of the Adriatic / City of Dreams	**Venice** (Italy)
City of Seven Hills / Eternal City	**Rome** (Italy)
Manchester of the East / Nation's Kitchen	**Osaka** (Japan)
City of the Golden Gate	**San Francisco** (USA)
City of Sky Scrapers / Windy City / City of Smoke	**New York** (Sky Scrapers) / **Chicago** (Windy City / Smoke)
City of Dreaming Spires	**Oxford** (England)
Granite City	**Aberdeen** (Scotland)
Cockpit of Europe	**Belgium**
Forbidden City	**Lhasa** (Tibet)
Gate of Tears	**Bab-el-Mandeb**

Sobriquet	Place
Herring Pond	**Atlantic Ocean**
Hermit Kingdom	**Korea**
Island Continent / Land of Golden Fleece	**Australia**
Pillars of Hercules	**Strait of Gibraltar**
Pearl of the Pacific	**Guayaquil Port** (Ecuador)
Paris of South America	**Buenos Aires** (Argentina)
Empire City	**New York**
Oil Capital of Europe	**Aberdeen**
Pearl of Siberia	**Lake Baikal** (Russia)
Roof of the World	**Pamir** (Central Asia)

Sobriquet	Place
City of Golden Temple (स्वर्ण मंदिर)	Amritsar (अमृतसर)
Land of Five Rivers	Punjab
Gateway of India / City of Seven Islands	Mumbai
Queen of the Arabian Sea	Kochi (Cochin)
Spice Garden of India	Kerala
Switzerland of India	Kashmir
Twin Cities	Hyderabad (हैदराबाद)–Secunderabad
Manchester of India	Ahmedabad (अहमदाबाद)
Venice of the East	Alappuzha (Alleppey)
Sorrow of Bengal	Damodar (दामोदर)
Sorrow of Bihar	Kosi (कोसी)
Sorrow of China	Huang He (Hwang Ho)
Queen of Chotanagpur (छोटानागपुर)	Netarhat (Jharkhand)
Queen of Mountains (India)	Mussoorie (मसूरी)
World's Bread Basket	Prairies of North America

N.8 Straits

Other straits (list)

- Bab el-Mandeb — Red Sea and Gulf of Aden (Yemen–Djibouti/Eritrea).
- Palk — India and Sri Lanka, with Adam's Bridge.
- 10° Channel (दस डिग्री) — Andaman and Nicobar. 8°/9° Degree — Lakshadweep / Maldives side.
- Dover — North Sea and English Channel (UK–France).
- Magellan — Atlantic and Pacific inshore (Chile); Cape Horn is the open-ocean south.
- Torres — Australia and Papua New Guinea. Bass — Tasmania and Australia. Cook — NZ islands.
- Florida / Yucatán — doors of the Gulf of Mexico.
- Mozambique — Africa and Madagascar.
- Tsushima / Korea / Taiwan — East Asia.
- Davis / Denmark — Greenland–Canada / Iceland Arctic doors.

N.10 Mountains (map)

Range–region match bank (next year can ask any)

Range / highland	Region / country fact	Tag
Pontic ★	Northern **Turkey** (Black Sea side)	2024 W→E
Zagros ★	Western **Iran**	2024 W→E
Hindu Kush ★	**Afghanistan** (into N Pakistan)	2024 W→E
Karakoram (काराकोरम) ★	N Pakistan / Ladakh / China border	2024 W→E; K2
Alps	**Europe** (France–Switzerland–Italy–Austria)	Mont Blanc
Atlas	NW **Africa** (Morocco–Algeria–Tunisia)	
Andes	**South America** (spine)	Aconcagua in Argentina
Rockies	**North America** (W USA–Canada)	
Appalachians	Eastern **USA**	Older fold belt
Urals	**Russia** — Europe / Asia conventional divide	
Caucasus	Russia / Georgia / Azerbaijan belt	**Elbrus** (Europe peak)
Drakensberg	**South Africa** / Lesotho	
Great Dividing Range	**Australia** (east)	
Pamir	Tajikistan / "Roof of the World" knot	
Tien Shan	Central Asia / China–Kyrgyzstan	
Kunlun	North of Tibetan Plateau (**China**)	
Altai	Russia / Mongolia / China / Kazakhstan	
Ethiopian Highlands	**Ethiopia** / Horn of Africa	

Highest peak by continent (match bank)

Continent	Peak	Country fact / trap
Asia	**Everest**	Nepal–China
South America	**Aconcagua**	**Argentina**
North America	**Denali**	USA (Alaska)
Africa	**Kilimanjaro**	**Tanzania** (not Kenya)
Europe	**Elbrus**	Russia / Caucasus
Australia (mainland)	**Kosciuszko**	Australia
Antarctica	Vinson Massif	Also revise

Common Traps — Don't Fall For These

1. **EEZ = 12 nm** — FALSE. Territorial **12**; EEZ **200**.
2. **McMahon = Pakistan** — FALSE. **China**.
3. **Durand = India–China** — FALSE. **Pak–Afghan**.
4. **Longest coast = AP / Kerala** — FALSE among 2018 options. **Gujarat**.
5. **Longest land border = China** — FALSE. **Bangladesh**.
6. **Sri Lanka = land neighbour** — FALSE. **Maritime**.

7. **Suez = Atlantic** — FALSE. **Med + Red Sea**.
8. **Panama = Med–Red** — FALSE. Atlantic–Pacific.
9. **Kiel = Suez cousin on Med** — FALSE. North Sea–Baltic.
10. **Hormuz = Malacca** — FALSE. Hormuz = Gulf oil mouth.
11. **Nobi/Kanto = Korea** — FALSE. **Japan**.
12. **Igarka = China** — FALSE. **Russia**.
13. **Bolivia has a Pacific coast in that 2022 four** — FALSE. **Landlocked**.
14. **"Vegetation is true index of climate" = Köppen** — FALSE. **Thornthwaite** for that quote; Köppen = letter-code classes.
15. **Mediterranean = rain all year** — FALSE. **Winter** rain. All-year = **W Europe Cfb**.
16. **Turkmenistan = Dushanbe** — FALSE. **Ashgabat**. Dushanbe = Tajikistan.
17. **Yangon = Myanmar capital** — FALSE. **Naypyidaw**.
18. **Cape Verde = Bamako** — FALSE. **Praia**. Bamako = Mali.
19. **India = 6th largest / wholly tropical** — FALSE. **7th**; Tropic through middle; north is extra-tropical. **UP is not on the Tropic**.
20. **Uzbekistan has a sea** — FALSE. It is **double landlocked**.
21. **Kazakhstan has an ocean coast** — FALSE. Largest landlocked by area; Caspian ≠ ocean access.
22. **Lesotho = double landlocked Asia** — FALSE. Enclave in South Africa.
23. **49th Parallel = USA–Mexico** — FALSE. USA–Canada stretch.
24. **Skip non-PYQ atlas pairs** — FALSE. Next year can ask any Lucent capital, strait, or line.
25. **Horn includes Sudan** — FALSE. Horn = Djibouti, Eritrea, Ethiopia, Somalia.
26. **Slovenia = Bratislava** — FALSE. Ljubljana. Bratislava = Slovakia.
27. **Ghana = Dutch Guiana** — FALSE. Gold Coast → Ghana; Dutch Guiana → Suriname.
28. **Great Britain includes Northern Ireland** — FALSE. That makes the **United Kingdom**.
29. **Greenland = independent continent / only Europe** — FALSE. Largest island; Denmark politically; N America geographically.
30. **Armenia / Iraq on Caspian** — FALSE. Caspian five exclude both.
31. **Indonesia = Oceania** — FALSE. SE Asia.
32. **Longest world coast = Australia / Brazil** — FALSE. **Canada**.
33. **Baikonur = Ukraine** — FALSE. **Kazakhstan**.
34. **Durand = Bangladesh–India** — FALSE. Pak–Afghan. Bangladesh–India = Radcliffe.
35. **Norway = Rising Sun** — FALSE. Rising Sun = **Japan**. Norway = **Midnight Sun**.
36. **Japan = Morning Calm** — FALSE. Morning Calm = **Korea**.
37. **Bird Continent = Australia** — FALSE. **South America**.
38. **Mistress of Eastern Sea = India** — FALSE. **Sri Lanka**.
39. **Manchester of the East = Tokyo** — FALSE. **Osaka**.
40. **Pearl of Siberia = Great Bear** — FALSE. **Baikal**.
41. **Venice of the North = Moscow** — FALSE. **St. Petersburg**.
42. **City of Smoke = London** — FALSE in that GC key. **Chicago**.
43. **Ninety East Ridge = Pacific** — FALSE. **Indian Ocean**.

🗺️ World Political & Geopolitical Essentials (Ghatnachakra Complete)

- **European Union (EU)**: Headquarters in **Brussels, Belgium**; established under Maastricht Treaty (1993); currently 27 member states.

- **International Maritime Organization (IMO)**: UN specialized agency; Headquarters in **London, United Kingdom**.
- **Food and Agriculture Organization (FAO)**: UN specialized agency; Headquarters in **Rome, Italy** (NOT New York).
- **SAARC (South Asian Association for Regional Cooperation)**: Secretariat located in **Kathmandu, Nepal**; founded in Dhaka (Dec 1985); 8 member nations.
- **Organization of Turkic States**: Established 2009; founding members: Azerbaijan, Kazakhstan, Kyrgyzstan, Turkiye; Uzbekistan joined as full member in 2019 at Baku.
- **ASEAN Status**: India has the status of **"Dialogue Partner"** in ASEAN (while Vietnam, Cambodia, Singapore, etc. are full members).
- **Look East Policy**: Initiated in **1991 by PM P.V. Narasimha Rao**; aimed to cultivate extensive strategic, economic, and cultural partnerships with Southeast Asia.
- **Southeast Asia Geostrategy**: Pivot region between the Pacific and Indian Oceans with pre-eminent maritime character (Spykman's Rimland theory).
- **Kailash Mansarovar**: Located in **Chinese Tibet**; Indian pilgrims require a visa to visit.
- **Distance Units**: **Terrestrial Mile (1,609 metres / 1,760 yards) < Nautical Mile (1,852 metres)**.
- **Harmattan Wind**: Dry and dusty north-easterly trade wind blowing from Sahara across West Africa into Gulf of Guinea ("The Doctor").
- **Iberian Peninsula**: Comprises Spain, Portugal, Andorra, and Gibraltar (Greece & Albania are Balkan, NOT Iberian).
- **Denisovan**: Extinct archaic human species discovered in Denisova Cave (Altai mountains, Siberia) and Baishiya Karst Cave (Tibetan Plateau, Xiahe, Gansu, China).
- **Kyrgyzstan**: Name derived from 'Kyrk' meaning **forty** ("Country of 40 tribes", celebrated in the epic of Manas).
- **Venice**: Known as **'Queen of the Adriatic Sea'** ('La Dominante', City of Canals, City of Bridges).
- **Dendrochronology**: Method of dating and determining tree age by counting annual growth rings on trunk cross-section.
- **Largest Number of Post Offices**: **India** has the world's largest postal network (>1.59 lakh post offices).
- **Tallest Flyer Bird**: **Crane** (Ostrich is tallest bird overall but flightless).
- **Cape Canaveral / Cape Kennedy**: Major US satellite and space launch centre located on coast of **Florida, USA**.
- **RADARSAT-1**: Canadian satellite that created the first complete high-resolution map of Antarctica (1997).
- **Copacabana Beach**: Famous 4 km balneario beach in **Rio de Janeiro, Brazil**.
- **Highest Ground-based Observatory**: Indian Astronomical Observatory at **Hanle** (Ladakh, 4,500 m) & University of Tokyo Atacama Observatory (Chile, 5,640 m).
- **Christmas in Australia**: Celebrated during **Summer** (Southern Hemisphere tilt).
- **Great Leap Forward (1958–1962)**: Chinese campaign leading to rural disruption and severe nationwide famine.

Topic 24 — Geography of Uttar Pradesh

Current Affairs (this topic)

- On **5 Jun 2026**, India's **100th Ramsar** site became **Surha Tal / JP Narayan Bird Sanctuary** in **Ballia**. UP's Ramsar total is now **13**.
- In **2025**, the CIP South Asia Regional Centre was approved at **Singna, Agra**, not Aligarh (अलीगढ़). UP remains India's potato leader.
- The **Ken–Betwa (बेतवा)** link is the only National Perspective Plan link under implementation. Its foundation stone was laid in December 2024 for Bundelkhand across Madhya Pradesh and Uttar Pradesh.
- **Jewar** hosts **Noida International Airport**.
- Census district ranks stay on **2011** until Census 2027.

Confused Pairs

Pair	Correct	Trap	Hindi
Dudhwa NP district	**Lakhimpur Kheri**	Bahraich / Pilibhit	दुधवा = खीरी
NP vs TR	Only NP = Dudhwa	Pilibhit as NP	केवल एक NP
Chandraprabha	**Chandauli**	Sonbhadra	चंदप्रभा
Katarniaghat	Bahraich, **1975**	Kishanpur's year	कटरनियाघाट
Forest %	**~6%** TGA	12.8%	वन क्षेत्र
>20% forest 2023	**Chandauli + Shravasti**	Bahraich	चंदौली + श्रावस्ती
Ramsar NOT UP	Rudrasagar + Sultanpur	Sarsai Nawar / Samaspur	त्रिपुरा / हरियाणा
Mar / Parua	Mar–Kabar **black**; Parua–Rakar **red**	Swap	मार काली
Devipatan	Bahraich + Balrampur + Gonda + Shravasti	Basti	देवीपाटन
CIP potato	**Singna, Agra**	Aligarh	आगरा
Gomti	Pollution / 'biological disaster' tag (2018)	Yamuna as that paper's river	गोमती
Female literacy floor	**Shrawasti**	Balrampur / Bahraich / Rampur	श्रावस्ती
Smallest area	**Hapur**	Hathras / Hardoi	हापुड़
Lalitpur	**One** UP-district neighbour	Saharanpur / Sonbhadra	ललितपुर
Mau on Purvanchal Exp	**Built corridor passes Mau**	Blind 2018 "not Mau" as today's map	मऊ
Ghaziabad Smart City	**Not** Central-10 (2018/2020)	"Biggest UA so it must be"	गाजियाबाद
Surinsar–Mansar	**J&K**, not UP	Grouped with Sarsai Nawar	जम्मू
Betwa confluence	Yamuna at **Hamirpur**	Prayagraj sangam	हमीरपुर
Ramganga confluence	Ganga at **Kannauj**	Prayagraj	कन्नौज
Yamuna right-bank W→E	Chambal → Sind → Betwa → Ken	Reverse Ken first	चंबल→केन
Gomti origin	**Pilibhit** (Madho Tala)	Lucknow as source	गोमती स्रोत
NER HQ	**Gorakhpur**	Hajipur	गोरखपुर
Kushinagar airport	Intl **24 Jun 2020**	Agra / Prayagraj	कुशीनगर
Amrit Bharat UP	**149** stations, modernisation	57/82/123; new lines	149
HRIDAY in UP	**Varanasi + Mathura**	Prayagraj / Ayodhya	काशी+मथुरा
Nanda Devi	**Uttarakhand**	UP	नंदा देवी

Pair	Correct	Trap	Hindi
HINDALCO	**Renukoot, Sonbhadra**	Kanpur leather	रेणुकूट
UP–Jute	**Wrong**	WB is jute	जूट ≠ UP
Chambal Sanctuary	**UP–MP–RJ**	Haryana	चंबल ≠ हरियाणा
Rain high vs low	**Gorakhpur** wetter · **Mathura** drier	Swap	गोरखपुर / मथुरा
Extremes N/S/E/W	Saharanpur / Sonbhadra / Ballia / Shamli	Swap Ballia–Sonbhadra	सीमा ज़िले
Tubewell vs canal	Tubewells **dominant**; west dark-zone pressure	Only canals everywhere	नलकूप

N.2 Physiographic regions

Physiographic region — crop and soil facts (quick table)

Region	Dominant soil / terrain	Main crops / economy
Bhabar (भाबर)	Pebbly, porous	Poor tillage; forests
Terai (तराई)	Moist alluvium	Rice, wheat, cane; Dudhwa belt
Doab (दोआब)	Khadar/bangar	Wheat, sugarcane, potato
Awadh (अवध)	Deep alluvium	Rice, wheat; Lucknow–Ayodhya
Rohilkhand	Alluvial on Ramganga	Sugarcane, mentha
Bundelkhand	Mar/Kabar black; Parua red	Pulses, oilseeds; drought stress
Purvanchal (पूर्वांचल)	Gangetic alluvium	Rice; lowest forest cover belt
Vindhyan	Bhonta; limestone	Thermal power, Rihand; Sonbhadra minerals

N.10 Wetlands / Ramsar

#	Ramsar site (IN UP)	District / belt
1	Upper Ganga	Brijghat–Narora
2	Nawabganj	Unnao
3	Sarsai Nawar	Etawah
4	Sandi	Hardoi
5	Samaspur	Raebareli
6	Parvati Arga	Gonda
7	Saman	Mainpuri
8	Sur Sarovar / Keetham	Agra
9	Haiderpur	Muzaffarnagar–Bijnor
10	Bakhira	Sant Kabir Nagar
11	Patna (पटना) Bird Sanctuary	Etah
12	Shekha Jheel	Aligarh
13	Surha Tal / JP Narayan Bird Sanctuary	Ballia
Name that sounds like UP	Actual state	Trap
---	---	---
Rudrasagar	**Tripura**	Not UP
Sultanpur	**Haryana**	Not UP

#	Ramsar site (IN UP)	District / belt
Surinsar–Mansar	**J&K**	Not UP

Common Traps — Don't Fall For These

1. **Dudhwa = Bahraich / Pilibhit** — FALSE. **Lakhimpur Kheri**.
2. **Pilibhit = National Park** — FALSE. **Tiger Reserve**. Only NP = Dudhwa.
3. **Chandraprabha = Sonbhadra** — FALSE. **Chandauli**.
4. **UP forest = 12.8%** — FALSE. **~6%**.
5. **Bahraich >20% forest 2023** — FALSE. **Chandauli + Shravasti**.
6. **Sarsai Nawar / Samaspur not in UP** — FALSE. They **are**. Drop **Rudrasagar + Sultanpur**.
7. **Basti = Devipatan** — FALSE. **Bahraich–Balrampur–Gonda–Shravasti**.
8. **CIP potato = Aligarh** — FALSE. **Agra**.
9. **Shrawasti is not the literacy floor** — FALSE. It **is** (female literacy 2011).
10. **Mar = red** — FALSE. Mar/Kabar = **black**. Parua/Rakar = red.
11. **Badua dam = UP** — FALSE.
12. **Lalitpur touches many UP districts** — FALSE. **One**.
13. **Mau is off today's Purvanchal Expressway** — FALSE for the **built** road. On the **2018** paper the "will not pass" answer was **Mau**; on today's map Mau lies on that corridor.
14. **UP Day = 26 January** — FALSE. **24 January**.
15. **Ghaziabad = Central Smart City (2018/2020)** — FALSE. Central count = **10**.
16. **Surinsar–Mansar = UP Ramsar** — FALSE. **J&K**.
17. **Rakhigarhi = UP Indus site** — FALSE. **Haryana**. Mandi + Hulas are UP.
18. **Betwa joins Yamuna at Prayagraj** — FALSE. **Hamirpur**.
19. **Ramganga joins at Prayagraj** — FALSE. **Kannauj**.
20. **Nanda Devi = UP** — FALSE. **Uttarakhand** (9 Nov 2000 split).
21. **UP = jute / tea / rubber leader** — FALSE.
22. **Amrit Bharat = 149 new lines** — FALSE. **149 stations modernised**.
23. **HRIDAY = Prayagraj / Ayodhya** — FALSE. **Varanasi + Mathura**.
24. **Kushinagar intl = Agra 2020** — FALSE. **Kushinagar, 24 Jun 2020**.
25. **NER HQ = Hajipur** — FALSE. **Gorakhpur**. Hajipur = ECR.
26. **Tropic of Cancer crosses UP** — FALSE.
27. **Gomti rises at Lucknow** — FALSE. **Pilibhit**.
28. **Pilibhit / Amangarh / Ranipur = extra National Parks** — FALSE. **Tiger Reserves**.
29. **Usar remedy = lime** — FALSE. **Gypsum + drainage**. Lime is for acid soils.
30. **Rain: Mathura highest / Gorakhpur lowest** — FALSE. Swap those name facts.
31. **Easternmost = Sonbhadra** — FALSE. E = **Ballia**; S = **Sonbhadra**.
32. **Irrigation = canals only** — FALSE. Tubewells dominate; west dark-zone pressure.
33. **Skip non-PYQ UP districts** — FALSE. Next year can ask any standard district extreme, canal, or soil.

Uttarakhand — Topic 1 — Location, Relief & Structure

Current Affairs (this topic)

- Peak / glacier / Char Dham corridor news can refresh district tags (Nanda Devi (नंदा देवी)–Chamoli (चमोली), Tehri (टिहरी) dam (टिहरी बांध) landscape).
- Keep MBT–MCT zone wording stable — mismatch stems recycle every year.

Quick Revision — Spine Only

Head	Correct
Location	Central Himalaya (हिमालय); between **Himachal** (हिमाचल) (W), **Nepal/Tibet** (E/N), **UP plains** (S)
Lat–long	Roughly **28°43′–31°27′ N** · **77°34′–81°02′ E**
South → north belts	**Tarai** → **Bhabar (भाबर)** → **Shiwalik** → **Dun (दून)** → **Lesser Himalaya** → **Great Himalaya** (Trans/Tethys farther north)
Between MBT and MCT	**Lesser Himalayan zone**
South of MBT	**Shiwalik** (शिवालिक) zone
North of MCT	**Central crystalline / Higher Himalaya** approach
Duns	Longitudinal valleys in Shiwalik belt (e.g. Dehradun (देहरादून))
Highest peak	**Nanda Devi** (नंदा देवी) (~**7816–7817 m**, **Chamoli** (चमोली)); **Kamet** (कामेत) (~**7756 m**) next among major UK peaks
Divisions	**Garhwal** (गढ़वाल) (west) · **Kumaon** (कुमाऊँ) (east)

Confused Pairs

Pair	Correct
MBT–MCT belt vs Shiwalik	Lesser Himalaya ≠ Shiwalik (Shiwalik is **south** of MBT)
Tarai last in N→S list	Tarai is **southernmost** — south→north starts with Tarai
Bhabar vs Tarai	Bhabar = porous gravel, streams **sink**; Tarai = streams **reappear**, marshy, fertile
Great Himalaya vs Tethys	Tethys / Trans-Himalaya lies farther north — a different question family
UK vs HP relief vocabulary	Same thrust names; apply to **Uttarakhand** districts in state papers

Common Traps

1. **MBT–MCT = Shiwalik** → Lesser Himalaya.
2. **South→north ends with Tarai** → Tarai is first from the south.
3. **Central crystalline between MBT and MCT** → that belt is north of MCT.
4. **Bhabar = marshy fertile belt** → that is Tarai; Bhabar is porous gravel.
5. **Nanda Devi = Uttarkashi only** → Chamoli (Garhwal high Himalaya).

Uttarakhand — Topic 2 — Climate & Drainage

Quick Revision — Spine Only

Head	Key Fact
Longest River in UK	**Kali (Sharda (शारदा))** — **252 km**; source: Lipulekh (लिपुलेख) / Kalapani; international boundary with Nepal
Ganga Formation	**Bhagirathi (भागीरथी) (205 km)** + **Alaknanda (अलकनंदा) (195 km)** unite at **Devprayag** (देवप्रयाग) to form the **Ganga** (गंगा)
West \rightarrow East Rivers (2025)	**Yamuna (यमुना) \longrightarrow Bhilangana \longrightarrow Alaknanda \longrightarrow Gori Ganga** (UKPCS 2025 Q109)
Panch Prayag (प्रयाग) (N \rightarrow S)	**Vishnuprayag (विष्णुप्रयाग) \rightarrow Nandaprayag \rightarrow Karnaprayag (कर्णप्रयाग) \rightarrow Rudraprayag (रुद्रप्रयाग) \rightarrow Devprayag**
Largest Glacier in UK	**Gangotri (गंगोत्री) Glacier** (Uttarkashi (उत्तरकाशी)) — **30 km long**, 2 km wide; source of Bhagirathi
Largest Glacier in Kumaon (कुमाऊँ)	**Milam (मिलाम) Glacier** (Pithoragarh (पिथौरागढ़)) — **16 km long**; source of Gori Ganga
2nd Longest Glacier	**Pindari (पिंडारी) Glacier** (Bageshwar (बागेश्वर)) — source of Pindar (पिंडर) river; discovered by G. W. Traill
Highest Shiva Temple	**Tungnath** (Rudraprayag) — at **3,680 m**; part of Panch Kedar (arms worshipped)
Mystery / Skeleton Lake	**Roopkund** (Chamoli (चमोली), 5,029 m) — human skeletons from 9th cent. CE (*Raja (राजा) Jasdhaval*)
Deepest Lake in Kumaon	**Naukuchiatal** (Nainital (नैनीताल)) — 9-cornered lake; approx. 40 m deep
Largest Lake in Kumaon	**Bhimtal** (Nainital) — triangular lake with an island restaurant in the center

Confused Pairs & Common Traps

Confused Pair	Critical Distinction
Longest River vs Deepest River	Longest river = **Kali (252 km)**; River with greatest water volume = **Alaknanda**.
Gangotri Glacier vs Milam Glacier	Gangotri = **Largest in Uttarakhand** (Uttarkashi, 30 km); Milam = **Largest in Kumaon** (Pithoragarh, 16 km).
Vishnuprayag vs Devprayag	Vishnuprayag = Alaknanda + **Western Dhauliganga**; Devprayag = Alaknanda + **Bhagirathi** (creates the Ganga).
Roopkund vs Dodital	Roopkund = **Skeleton lake** (Chamoli, 5,029 m); Dodital = **6-cornered lake of Ganesha** (Uttarkashi).
Tungnath vs Kedarnath	Tungnath = **Highest Shiva Temple** (3,680 m, arms); Kedarnath = **Supreme Jyotirlinga Dham** (hump).

River Systems of Uttarakhand: Comprehensive Master Matrix

River System	Source Glacier / Origin	Length in UK	Major Tributaries & Confluences
Kali (Sharda)	Lipulekh / Kalapani (Pithoragarh)	**252 km** *(Longest)*	Kuti Yankti (Gunji), Eastern Dhauliganga (Tawaghat), Gori Ganga (Jauljibi (जौलजीबी)), Saryu (Pancheshwar), Ladhia (Chuka).
Bhagirathi (भागीरथी)	Gaumukh (गौमुख) (Gangotri Glacier, Uttarkashi)	**205 km**	Rudraganga (Gangotri), Jadhganga/Jahanvi (Bhaironghati), Asiganga (Uttarkashi), Bhilangana (Old Tehri).

River System	Source Glacier / Origin	Length in UK	Major Tributaries & Confluences
Alaknanda (अलकनंदा)	Satopanth & Bhagirath Kharak (Chamoli)	**195 km**	Saraswati (सरस्वती) (Mana (माना) / Keshavprayag), Western Dhauli (धौली) (Vishnuprayag), Nandakini, Pindar, Mandakini.
Western Ramganga (रामगंगा)	Dudhatoli range (Pauri (पौड़ी)/Chamoli/Almora (अल्मोड़ा))	**155 km**	Gagas, Binau, Deogadh. Traverses Corbett National Park into UP.
Kosi (कोसी) (Kaushiki)	Dharpani Dhar (Kausani, Almora)	**168 km**	Drains Almora, Someshwar, Ramnagar; joins Ramganga in UP.
Yamuna (यमुना)	Yamunotri (यमुनोत्री) Glacier (Bandarpunch (बंदरपूंछ), Uttarkashi)	**136 km**	Rishiganga, Hanuman Ganga, Kamal, Tons (टौंस) (Dakpathar), Giri, Asan (Rampur Mandi (मंडी)).
Tons (Supin)	Bandarpunch / Har-Ki-Doon (Uttarkashi)	**148 km**	Formed by confluence of Rupin and Supin at Netwar. Carries **twice the water volume** of the Yamuna!
Saryu	Sarmool (Bageshwar)	**146 km**	Gomti (Bageshwar), Panar (Rameshwar), Eastern Ramganga (Muawani). Meets Kali at Pancheshwar.
Mandakini (मंदाकिनी)	Chaurabari Glacier (Kedarnath, Rudraprayag)	**72 km**	Vasukiganga (Sonprayag), Madhu Ganga. Meets Alaknanda at Rudraprayag.
Bhilangana	Khatling Glacier (Tehri Garhwal)	—	Balganga, Dharmaganga. Submerged in Tehri Dam (टिहरी बांध) reservoir.

Master Table of High-Altitude Lakes & Tals

Lake / Tal	District & Altitude	High-Yield Prelims Characteristics
Roopkund	**Chamoli** (चमोली) (5,029 m)	**"Mystery Lake" / "Skeleton Lake":** Contains hundreds of 9th-century human skeletal remains (linked to King Jasdhaval of Kannauj (कन्नौज) and Queen Vallabha (वल्लभ)); situated on the Nanda Devi Raj Jat (नंदा देवी राज जात) route.
Hemkund (हेमकुंड) (Lokpal (लोकपाल))	**Chamoli** (4,329 m)	Sacred lake surrounded by **7 snow-clad peaks**; holy Gurudwara of Guru (गुरु) Gobind Singh (गुरु गोबिंद सिंह) and ancient Lakshman Temple; source of Lakshmanganga.
Nainital Lake	**Nainital** (नैनीताल) (1,938 m)	Pear-shaped lake discovered by **P. Barron** in 1841; divided into *Mallital* and *Tallital*; surrounded by 7 hills (China Peak, Naina Peak, Ayarpatta, etc.).
Naukuchiatal	**Nainital**	**Deepest lake** in Kumaon (~40 m); possesses **9 corners**; haven for migratory birds.
Bhimtal	**Nainital**	**Largest lake** in Kumaon region; triangular shaped; features a picturesque island restaurant.
Dodital	**Uttarkashi** (उत्तरकाशी) (3,024 m)	**6-cornered lake**; mythologically associated with the birth of Lord Ganesha; origin of the Asiganga river; famous for Himalayan Golden Trout.
Sahasratal	**Tehri Garhwal**	Group of hundreds of glacial tarns situated on the northern ridge of Khatling glacier; largest tal in Garhwal.
Deoria Tal	**Rudraprayag** (रुद्रप्रयाग) (2,438 m)	Famous for crystal-clear reflection of the **Chaukhamba (चौखंबा) peaks**; mythologically identified with the Yaksha-Yudhisthira dialogue.

Uttarakhand — Topic 3 — Vegetation & Wildlife

Current Affairs (this topic)

- New sanctuary notifications or UNESCO / BR news can refresh fauna (प्राणीजात) lists; keep district–sanctuary pairs stable.
- NP foundation years (Corbett 1936, Rajaji (राजाजी) 1983, Gangotri (गंगोत्री) 1989, Govind 1990) are high-recycle traps.

Quick Revision — Spine Only

Head	Correct
Forest richness	Large forest share; belts from tropical foothill to alpine — **relief + climate** explain it
Altitude stack	Foothill sal → chir → oak–rhododendron → deodar / fir → alpine birch–juniper → **bugyal**
Deodar	Abundant western Himalayan temperate belt **1500–2500 m**; highest among deodar / pine / sal / teak
Birch	Himalayan alpine / moist alpine scrub; writing-bark tradition
Lisa	Resin product from **pine**; turpentine after distillation; rural UK livelihood
Khair	**Sub-tropical** forests (Shiwalik (शिवालिक) / western Himalaya (हिमालय) scrub); also yields **katha**
Nanda Devi BR	UNESCO WHS (with Valley of Flowers (फूलों की घाटी)); fauna includes **snow leopard, musk deer, blue sheep**
Askot sanctuary	**Pithoragarh** (पिथौरागढ़)
Binsar sanctuary	**Almora** (अल्मोड़ा)
Govind Pashu Vihar / NP	**Uttarkashi** (उत्तरकाशी)
Nandhaur sanctuary	**Nainital** (नैनीताल)
Corbett NP	**1936** (India's first NP tradition)
Rajaji NP	**1983**
Gangotri NP	**1989**
Govind NP year	**1990** (not 1989)

Confused Pairs

Pair	Correct
Askot vs Binsar districts	Pithoragarh vs Almora — classic swap
Govind 1989 vs 1990	**1990** is correct; 1989 pairs with Gangotri
Nanda Devi fauna	Snow leopard + musk deer + blue sheep (bharal) all in
A true but R not explain	UKPCS 2025 treats **R as explaining A** for forest richness
Lisa vs Burans	Lisa = **pine** resin; not Burans / deodar / sheesham
Khair forest type	**Sub-tropical** (not sub-alpine / cool temperate)
Deodar vs chir height	Deodar temperate **1500–2500 m**; chir sub-tropical pine lower (**~900–2000 m**)
Birch location	**Himalayas** — not Nilgiris / Aravallis / Vindhyas

3.4 Sanctuary–district map

Sanctuary	District
Askot	Pithoragarh
Binsar	Almora
Govind	Uttarkashi
Nandhaur	Nainital

Common Traps

1. **Govind NP = 1989 → 1990**.
2. **Askot = Almora** → Pithoragarh.
3. **R does not explain forest richness** → UKPCS keys that it does.
4. **Lisa from Burans / Deodar** → from **pine**.
5. **Khair = sub-alpine / temperate → sub-tropical**.
6. **Birch = Nilgiris → Himalayas**.
7. **Highest tree among options = pine / sal → deodar**.
8. **Vembanad = Karnataka → Kerala**.

Uttarakhand — Topic 4 — Minerals, Power & Industry

Current Affairs (this topic)

- Watch new SIIDCUL allotments, hydropower clearances, and any revised mineral-district keys in state releases.
- Tehri (टिहरी) and Alaknanda-belt project news often doubles as power CA.

Quick Revision — Spine Only

Head	Correct
Dolomite	**Dehradun (देहरादून), Pithoragarh (पिथौरागढ़)**
Soapstone (soap stone)	**Chamoli (चमोली), Bageshwar (बागेश्वर)**
Graphite	**Almora (अल्मोड़ा), Nainital (नैनीताल)** (correct pair in UKPCS 2025)
Silica sand	**Dehradun, Nainital, Tehri Garhwal (गढ़वाल)** — **not Pauri (पौड़ी) Garhwal**
Power	Hydropower on Bhagirathi (भागीरथी)–Alaknanda (अलकनंदा)–Yamuna (यमुना) systems; **Tehri** (टिहरी) as flagship
Industry sketch	Forest-based, tourism services, pharma/IT pockets in plains & Dehradun belt

Confused Pairs

Pair	Correct
Silica sand–Pauri	**Wrong** in the 2025 question — use Dehradun / Nainital / Tehri
Soapstone–Pithoragarh	Soapstone card is Chamoli–Bageshwar
Dolomite–Almora	Dolomite card is Dehradun–Pithoragarh
Power vs mineral	Do not answer hydropower questions with mineral districts

Common Traps

1. **Silica sand = Pauri** → wrong.
2. **Soapstone = Dehradun–Pithoragarh** → that is dolomite's pair.
3. **All Himalayan districts have the same minerals** → learn the named pairs.
4. **Graphite = Chamoli–Bageshwar** → that is soapstone; graphite is Almora–Nainital.

Uttarakhand — Topic 5 — Agri, AH & Irrigation

Current Affairs (this topic)

- Fresh land-use releases can shuffle Haridwar (हरिद्वार) vs Udham Singh Nagar (ऊधम सिंह नगर) absolute (निरपेक्ष) ranks — keep both names and prefer the Series B letter when that paper is cited.
- Horticulture and dairy scheme news can appear as CA without rewriting the hill–plain crop map.

Quick Revision — Spine Only

Head	Correct
Hill farming	Terraced cereals, millets, pulses; orchards in mid-hills
Granary belt	Tarai / plains districts lead in cropped area
Highest agri land (2025)	**Haridwar** (हरिद्वार) (Series B provisional key **A**)
Cultivated-area leader	Official Series B (2025) answer is **Haridwar**; many notes also name **Udham Singh Nagar** (ऊधम सिंह नगर) — learn both names
Animal husbandry	**Supplementary** to agriculture (milk, wool, manure)
Sheep / pastures	Environmental-degradation narrative appears in many notes — **2025 Series B accepted Only statement 1**
Irrigation	Canals and tubewells in Tarai; springs / guls / limited canals in hills

Confused Pairs

Pair	Correct
Haridwar vs US Nagar on agri area	**Tarai/plains** lead cropped area vs hills. Exact **rank** varies by source. For **2025 Series B**, mark **Haridwar**; many books still name **Udham Singh Nagar** as the Tarai rival
AH as main vs supplementary	Correct reading is **supplementary** occupation
Q102 both vs only 1	Series B **Only 1** — do not mark both unless an amended key says so
Hill vs plain irrigation	Tubewell–canal plains ≠ spring-fed hill systems

Common Traps

1. **Blindly marking Both on Q102** → Series B is Only 1.
2. **Always US Nagar for max agri land** → 2025 key is Haridwar.
3. **AH replaces agriculture in hills** → it supplements.
4. **Same irrigation system statewide** → plains ≠ hills.

Uttarakhand — Topic 6 — Population, SC/ST & Settlements

Quick Revision — Spine Only

Indicator	Uttarakhand State Value	Highest District	Lowest District
Total Population	**1,00,86,292** (1.01 Crore)	**Haridwar** (हरिद्वार) (18.90 Lakh)	**Rudraprayag** (रुद्रप्रयाग) (2.42 Lakh)
Decadal Growth (2001–11)	**18.81%** (National: 17.70%)	**Udham Singh Nagar** (ऊधम सिंह नगर) (33.45%)	**Pauri (पौड़ी) Garhwal (गढ़वाल)** (**-1.41%** negative!)
Negative Growth Districts	**Pauri (-1.41%)** & **Almora (अल्मोड़ा) (-1.28%)**	Severe out-migration / *Palayan*	Creates "Ghost Villages" (*Bhootiya Gaon*)
Population Density	**189 persons / sq km**	**Haridwar** (801)	**Uttarkashi** (उत्तरकाशी) (41)
Sex Ratio	**963** (National: 943)	**Almora** (अल्मोड़ा) (**1142**)	**Haridwar** (**880**)
Child Sex Ratio (0–6)	**890** (National: 919)	**Almora** (922)	**Pithoragarh** (पिथौरागढ़) (816)
Overall Literacy Rate	**78.82%** (Rank: 17th in India)	**Dehradun** (देहरादून) (84.25%)	**Udham Singh Nagar** (73.10%)
Male Literacy	**87.40%**	**Rudraprayag** (93.90%)	**Haridwar** (81.04%)
Female Literacy	**70.04%**	**Dehradun** (78.54%)	**Uttarkashi** (59.74%)
Scheduled Caste (अनुसूचित जाति) (SC) %	**18.76%** of total pop	**Bageshwar** (बागेश्वर) (27.73%)	**Dehradun** (14.5%)
Scheduled Tribe (अनुसूचित जनजाति) (ST) %	**2.89%** of total pop	**Udham Singh Nagar** (7.46%)	**Rudraprayag** (0.15%)
Urban (नगरीय) Population %	**30.23%** (Rural: 69.77%)	**Dehradun** (55.52%)	**Bageshwar** (3.49%)

Confused Pairs & Common Traps

Confused Pair	Critical Distinction
Almora vs Haridwar (Sex Ratio)	Almora = **highest** sex ratio (1142); Haridwar = **lowest** sex ratio (880).
Haridwar vs Uttarkashi (Density)	Haridwar = **highest** density (801); Uttarkashi = **lowest** density (41).
Pauri vs Udham Singh Nagar (Growth)	US Nagar = **highest decadal growth** (+33.45%); Pauri Garhwal = **lowest/negative** (-1.41%).
Highest SC % vs Highest ST %	Highest SC % = **Bageshwar (27.73%)**; Highest ST % = **Udham Singh Nagar (7.46%)**.
Highest Overall Literacy vs Male Literacy	Highest overall = **Dehradun (84.25%)**; Highest male = **Rudraprayag (93.90%)**.

Census 2011 Master District Rankings (All 13 Districts)

Rank	Population	Density (per \text{km}^2)	Sex Ratio (females/1000)	Literacy Rate (%)
1	**Haridwar** (18.90 L)	**Haridwar** (801)	**Almora** (1142)	**Dehradun** (84.25%)
2	**Dehradun** (16.96 L)	**Dehradun** (549)	**Rudraprayag** (1114)	**Nainital** (नैनीताल) (83.88%)
3	**US Nagar** (16.48 L)	**US Nagar** (425)	**Pauri Garhwal** (1103)	**Chamoli** (चमोली) (82.65%)
4	**Nainital** (9.55 L)	**Nainital** (225)	**Bageshwar** (1090)	**Pithoragarh** (82.25%)
5	**Pauri Garhwal** (6.87 L)	**Tehri Garhwal** (170)	**Tehri Garhwal** (1077)	**Pauri Garhwal** (82.02%)

Rank	Population	Density (per \text{km}^2)	Sex Ratio (females/1000)	Literacy Rate (%)
6	**Tehri Garhwal** (6.18 L)	**Pauri Garhwal** (129)	**Pithoragarh** (1020)	**Rudraprayag** (81.30%)
7	**Almora** (6.22 L)	**Bageshwar** (116)	**Chamoli** (1015)	**Almora** (80.47%)
8	**Pithoragarh** (4.83 L)	**Almora** (198)	**Uttarkashi** (958)	**Champawat** (चंपावत) (79.83%)
9	**Chamoli** (3.91 L)	**Champawat** (147)	**Champawat** (980)	**Bageshwar** (80.01%)
10	**Uttarkashi** (3.30 L)	**Rudraprayag** (122)	**Nainital** (934)	**Tehri Garhwal** (76.36%)
11	**Bageshwar** (2.59 L)	**Pithoragarh** (68)	**US Nagar** (920)	**Uttarkashi** (75.81%)
12	**Champawat** (2.59 L)	**Chamoli** (49)	**Dehradun** (889)	**Haridwar** (73.43%)
13	**Rudraprayag** (2.42 L)	**Uttarkashi** (41)	**Haridwar** (880)	**US Nagar** (73.10%)

Uttarakhand — Topic 7 — Transport, Tourism & Hazards

Quick Revision — Spine Only

Head	Key Fact
Silkyara–Barkot Tunnel	**4.531 km long**; cuts Gangotri (गंगोत्री)–Yamunotri (यमुनोत्री) highway distance by >20 km (UKPCS 2025 Q51)
Airport Mismatch (2025)	**Gauchar** is in **Chamoli** (चमोली) (NOT Rudraprayag (रुद्रप्रयाग)!) (UKPCS 2025 Q103)
Airports in UK	Jolly Grant (Dehradun (देहरादून)), Pantnagar (US Nagar), Naini Saini (Pithoragarh (पिथौरागढ़)), Chinyalisaur (Uttarkashi (उत्तरकाशी)), Gauchar (Chamoli)
Pass: Darma & Byans	**Sinla Pass** (Pithoragarh) — connects Darma valley to Byans valley
Pass: Pithoragarh & Tibet	**Lipulekh (लिपुलेख) Pass** (5,334 m) — traditional Kailash-Mansarovar Yatra route
Pass: Pithoragarh & Bageshwar (बागेश्वर)	**Traill's Pass** (5,312 m) — crossed in 1830 by Commissioner G. W. Traill
Pass: Uttarkashi & HP	**Shringkanth Pass** (connects Uttarkashi to Kinnaur/HP)
Largest Bugyal in UK	**Bedni Bugyal** (Chamoli) — velvet alpine meadow on the Nanda Devi (नंदा देवी) Raj Jat (नंदा देवी राज जात) route
Butter Festival Bugyal	**Dayara Bugyal** (Uttarkashi) — celebrated for **Anduri Utsav / Butter Festival**
State Climate Council (SCCC)	Established in **2011** (Forest & Environment (पर्यावरण) Dept drafts UAPCC, NOT Revenue Dept) (UKPCS 2025 Q106)
Malpa Landslide Tragedy	**17–18 August 1998** (Pithoragarh); killed 200+ pilgrims including dancer Protima Bedi
Kedarnath (केदारनाथ) Flash Deluge	**16–17 June 2013** — Chaurabari glacial lake breach; Mandakini (मंदाकिनी) river deluge
Tapovan/Rishiganga Disaster	**7 February 2021** (Chamoli) — rock-ice avalanche wiped out Rishiganga & Tapovan-Vishnugad hydel plants

Confused Pairs & Common Traps

Confused Pair	Critical Distinction
Gauchar vs Rudraprayag	Gauchar airstrip is in **Chamoli** district (trap distractor: Rudraprayag).
Silkyara Length	Exact length is **4.531 km** (trap distractors: 3.531 km, 6.531 km).
Sinla Pass vs Lipulekh Pass	Sinla Pass = **internal pass** connecting Darma and Byans valleys; Lipulekh = **international pass** to Tibet.
Dayara Bugyal vs Bedni Bugyal	Dayara = **Butter Festival / Anduri Utsav** (Uttarkashi); Bedni = **largest bugyal / Brahma Kamal** (Chamoli).
Malpa (1998) vs Kedarnath (2013)	Malpa = **Landslide / rockfall** on Kali river (Protima Bedi); Kedarnath = **GLOF / cloudburst** on Mandakini.
SCCC Year & Department	SCCC was created in **2011**; UAPCC is managed by the **Forest & Environment Department** (NOT Revenue Dept).

Mountain Passes of Uttarakhand: Master Reference Table

Pass Name	Elevation	Connects / Geographic Span	Key Prelims Features
Lipulekh Pass	5,334 m	**Pithoragarh \longleftrightarrow Tibet (China)**	Landmark pass on the ancient border trade (पण्याध्यक्ष) and annual **Kailash-Mansarovar Yatra**; tri-junction with Nepal.

Pass Name	Elevation	Connects / Geographic Span	Key Prelims Features
Mana (माना) Pass (Dungri-La)	5,608 m	**Chamoli \longleftrightarrow Tibet**	One of the highest motorable roads in the world (NH-58 terminal); ancient Indo-Tibetan trade link.
Niti (नीति) Pass	5,068 m	**Chamoli \longleftrightarrow Tibet**	Last pass in the Dhauliganga (धौलीगंगा) valley; closed following the 1962 war.
Sinla Pass	5,495 m	**Darma Valley \longleftrightarrow Byans Valley**	Inter-valley high pass within Pithoragarh district.
Traill's Pass	5,312 m	**Bageshwar \longleftrightarrow Pithoragarh**	Connects Pindari Glacier to Milam valley; discovered in 1830 by Commissioner G. W. Traill.
Kalindi Pass	5,950 m	**Uttarkashi \longleftrightarrow Chamoli**	Severe glaciated mountaineering pass linking Gangotri/Gaumukh (गौमुख) directly to Badrinath.
Shringkanth Pass	1,500 m	**Uttarkashi \longleftrightarrow Himachal (हिमाचल) Pradesh**	Ancient transit corridor connecting Rawain to Kinnaur.
Thang-La Pass	5,360 m	**Uttarkashi \longleftrightarrow Tibet**	Second highest pass in the Garhwal (गढ़वाल) Himalaya (हिमालय); accessed via Jadhganga valley.
Chorhoti Pass	5,364 m	**Chamoli \longleftrightarrow Pithoragarh**	Connects Niti valley to the Kumaon borderland.
Sunderdhunga Pass	5,520 m	**Bageshwar \longleftrightarrow Chamoli**	Traverses high glacial ridgelines between Maiktoli and Panwali peaks.

Alpine Meadows (Bugyals) of Uttarakhand

District	Prominent Bugyals	High-Yield Prelims Characteristics
Chamoli	**Bedni Bugyal**, **Auli** (औली), **Gorson**, Ali, Panar, Bagji, Chitrakantha, Vedini	**Bedni Bugyal:** Largest bugyal in Uttarakhand; famous for Bedni Kund and rare *Brahma Kamal* (ब्रह्मकमल) flowers. **Auli Bugyal:** Premier skiing destination with 4 km ropeway (*Cable Car*) from Joshimath.
Uttarkashi (उत्तरकाशी)	**Dayara**, **Kush Kalyan**, Panwali Kantha, Manji (मंजी), Devdamini, Har-ki-Doon	**Dayara Bugyal:** Vast 28 sq km meadow; venue of the famous **Anduri Utsav (Butter & Milk Festival)** celebrated on Bhado Sankranti. **Panwali Kantha:** Home to blooming red, pink, and white rhododendrons; long trekking trail.
Rudraprayag (रुद्रप्रयाग)	**Chopta**, Tungnath, Madhyamaheshwar, Kasni Khark	**Chopta:** Hailed as the **"Mini Switzerland of Uttarakhand"**; base camp for Tungnath and Chandrashila treks.
Pithoragarh (पिथौरागढ़)	**Khalia Top**, Chhipla Kot, Byans Bugyal, Ralam, Betulidhar	**Khalia Top:** Near Munsyari; panoramic views of Panchachuli (पंचाचूली) peaks. **Chhipla Kot:** Sacred alpine meadow housing the Chhipla Kedar shrine.
Tehri (टिहरी)	**Kotli Bugyal**, Masartal Bugyal, Sahasratal pastures	Alpine grazing grounds used by Gujjars and local sheep herders.

Natural Hazards & Historical Disaster Timeline

Date / Year	Calamity & Location	Geological Nature & Human/Infrastructure Toll
20 October 1991	**Uttarkashi Earthquake** (6.6 M)	Severe tectonic tremor along the MCT; over **768 fatalities**; flattened thousands of traditional stone houses across Bhatwari and Dunda.

Date / Year	Calamity & Location	Geological Nature & Human/Infrastructure Toll
17–18 August 1998	**Malpa Landslide (Pithoragarh)**	Catastrophic rockfall and debris avalanche in the Kali (काली) valley wiped out the village of Malpa overnight. **221 dead**, including a batch of 60 Kailash-Mansarovar pilgrims and renowned classical dancer **Protima Bedi**.
29 March 1999	**Chamoli Earthquake** (6.8 M)	Epicentre near Chamoli/Gopeshwar; over **100 fatalities**; extensive damage to Gopeshwar, Chamoli, and Rudraprayag.
16–17 June 2013	**Kedarnath Cloudburst & GLOF (Mandakini Deluge)**	Massive multi-day torrential rains coupled with the sudden breach of **Chaurabari Lake (Gandhi (गांधी) Sarovar)** sent a gigantic wall of mud and boulders down the Mandakini river. Devastated Kedarnath temple township, Rambara, Gaurikund, and Sonprayag. Over **5,700 fatalities**; largest Himalayan disaster of modern times. Led to establishment of State Disaster Response Force (SDRF).
7 February 2021	**Chamoli / Tapovan Flash Flood (GLOF & Rockslide)**	A massive hanging glacier and rock mass collapsed from the **Rongti / Nanda Ghunti peak**, creating a colossal flash flood in the Rishi Ganga and Dhauli (धौली) Ganga rivers. Completely destroyed the **Rishi Ganga Hydel Project (13.2 MW)** and severely damaged NTPC's **Tapovan-Vishnugad Project (520 MW)**. Over **204 dead/missing** workers trapped inside tunnels.
January 2023	**Joshimath Land Subsidence**	Severe land sinking, foundational shifting, and deep wall fissures appeared across 9 municipal wards of Joshimath, built upon ancient landslide moraine debris. Hundreds of families evacuated.